A trace costs one quantifier: the exact complexity of perfect play in nonlocal games

Seth Douglas Nidhal Mghirbi

Version 1.2.0, November 2026

Abstract

Given a finite nonlocal game, can the players win with certainty? We prove that the existence of a perfect normal spatial strategy is Σ_2^0-complete, even when restricted to games of value one, so it cannot be decided even with an oracle for the halting problem. This is strictly harder than perfect play in the finite-dimensional and commuting-operator models, which are Σ_1^0-complete and Π_1^0-complete. Within the commuting model, the existence of a perfect normal strategy with a finite or semifinite von Neumann host is Σ_2^0-complete, whereas for every fixed type III_λ subtype it has the same Π_1^0-complete language as unrestricted commuting play. A trace thus costs one existential number quantifier: the hardness constructions force every perfect strategy to choose a finite clock period or a finite tree depth. Spatial Σ_2^0-completeness also holds when finite entanglement entropy is required, and both forms are formally verified in Lean. For every computable $0 < \alpha < 1$, perfect spatial play with $\operatorname{Tr} h^\alpha < \infty$ for the reduced density h is Σ_2^0-complete as well; these conditions form a strict ladder between finite rank and finite entropy. We also prove Σ_2^0-completeness for one-shot zero-error communication with normal entanglement, and construct a finite channel with zero-error codes of a given size each of which needs infinite entanglement entropy.

Opus 5.5, GPT-6.1 Sol and GPT-6 assisted with this work; the authors take responsibility for every claim (see "How this work was done"). Programs and finite checks: https://github.com/Apsiape/perfect-play-arithmetical-hierarchy.

1 Introduction

In a nonlocal game a referee sends each of two separated players a question from a finite list. Each player answers from a finite list, and a fixed rule decides whether they win. Shared entanglement can raise the winning probability. When can the players win with certainty?

For a fixed game this is a yes-or-no question. It has a precise answer only once one says what a quantum strategy is, and only once one separates approaching certainty from attaining it: the supremum of the winning probability can be one while no strategy reaches it. The standard models are finite-dimensional tensor products, normal states on infinite-dimensional tensor products (the spatial model), and commuting operator algebras on one Hilbert space. Inside the commuting model one can further ask which von Neumann algebra hosts the players. This paper finds, for each of these models, the exact level of the arithmetical hierarchy at which perfect play is decided.

The answers are not monotone in the model. As sets of correlations the spatial model sits between the finite-dimensional and the commuting one, yet perfect spatial play is strictly harder to decide than both: it is Σ_2^0-complete, while the other two are Σ_1^0- and Π_1^0-complete. A Σ_2^0 question asks whether some integer passes every one of infinitely many finite tests; no algorithm decides such questions in general, even with an oracle for the halting problem.

Completeness means that perfect spatial play has this form and that every question of this form reduces to it. Our constructions make the integer concrete: a perfect strategy must select a finite clock period or a finite tree depth, and then pass every test attached to that choice. Spatial and semifinite perfection combine an unbounded finite choice with an infinite verification.

1.1 Why a trace costs a quantifier

Commuting perfection is Π_1^0 because the commuting state space is compact. Perfection then equals commuting value one, which a convergent hierarchy of relaxations refutes in finitely many steps whenever it fails. Spatial play has no such compactness: Schmidt mass can escape to infinitely many small coefficients. Our upper bound keeps one Schmidt coefficient from disappearing. Once that coefficient is bounded below and scaled to one, the amplitude operator lies in a norm-compact family, and the losing probabilities become homogeneous zero equations that survive limits. So perfect spatial play has the form

$$\exists \kappa \; \forall d \quad F(G, \kappa, d),$$

with F decidable: for each integer κ, the existence of a perfect strategy whose largest Schmidt probability is at least $1/\kappa$ is a Π_1^0 property (Theorem 3.1).

The same integer appears for every semifinite host. A trace gives a finite corner whose overlap with the state can be bounded below; in type I a rank-one corner recovers the largest Schmidt amplitude (Section 3.5). A type-III factor has no finite corner, and its perfect-play language collapses to the commuting one. In this sense a trace costs exactly one existential quantifier. Finite dimension is the degenerate case: there the whole space is a finite corner, and once the dimension is fixed, perfection is a question of real algebra that Tarski's method decides [2]. Only the search over the dimension remains, which is why that row is Σ_1^0. An operational split between semifinite and type III algebras was known for embezzlement of entanglement [59]; here the split appears as a quantifier.

The hardness proofs show that the quantifier is needed. In each case the existential witness is a finite object that every perfect strategy must choose. In the spatial model it is the order of a clock phase. The Weyl unitaries of the clock commute with the trace-class density, so they act on its eigenspaces, which are finite-dimensional, and there a determinant forces the commutator phase to be a root of unity. In a diffuse semifinite algebra no such determinant exists, but a trace forces a critical binary tree to end at a finite depth. A finite game then checks an infinite sequence of decidable conditions about the chosen object. No computable function of the game bounds a sufficient clock order (Section 7) or a sufficient Schmidt stratum (Theorem 3.1).

1.2 Results

Table 1 collects the answers, and four theorems fill it.

The first theorem completes the standard row of models.

Theorem A (spatial perfection). *Deciding whether a finite rational* $0/1$ *game has a perfect normal spatial strategy is* Σ_2^0*-complete. The same holds when a perfect strategy with finite marginal entropy is required. Hardness holds under the promise* $\omega_q(G) = 1$*, for a family in which no instance has a perfect finite-dimensional strategy.*

The second sorts the commuting model by the type of the players' algebra. A host is an ambient von Neumann algebra containing Alice's measurements, with Bob's measurements in its commutant. Both the host and its normal state may be chosen; the state need not be tracial,

Decision problem	Complete class	Source
Perfect finite-dimensional strategy	Σ_1^0	[34]
Perfect commuting strategy	Π_1^0	[57, 44]
Finite-dimensional value equals one	Π_2^0	[43, 44]
Perfect spatial strategy; also with finite entropy	Σ_2^0	Theorem A
Perfect strategy with finite α-moment, fixed computable $0 < \alpha < 1$	Σ_2^0	Theorem D
Perfect play in type II_1, II_∞ or any semifinite algebra	Σ_2^0	Theorem B
Perfect play in a type III_λ factor, any fixed $\lambda \in [0,1]$	Π_1^0	Theorem B
Commuting perfect play but no semifinite perfect play	Π_2^0	Theorem 14.4
Semifinite perfect play without spatial (or finite-entropy) perfection	$D_2(\Sigma_2^0)$	Theorem 14.4
One-shot zero-error code, normal entanglement; also finite entropy	Σ_2^0	Theorem C
One-shot zero-error code, commuting operators	Π_1^0	Theorem 17.2

Table 1: Finite rational Boolean games and channels; computable many-one reductions. A value is a supremum. A perfect strategy attains one. The first three rows are known; the others are proved here.

and the measurements need not generate the host. Write $\mathrm{PERF}(X)$ for the set of games with a perfect strategy whose host is of class X; finite-host perfection uses a σ-finite finite von Neumann algebra, equivalently one admitting a faithful normal tracial state (Section 2).

Theorem B (perfect play by von Neumann type). *Perfect normal play with a finite von Neumann host, a type* II_1 *host, or a semifinite host is* Σ_2^0*-complete, and so are the type* II_∞ *and type* II *existence languages. For every fixed* $\lambda \in [0,1]$*, the type* III_λ *language equals commuting perfection and is* Π_1^0*-complete.*

Theorem 14.4 completes the type classification: commuting perfection without semifinite perfection is Π_2^0-complete, the semifinite-but-not-spatial differences are $D_2(\Sigma_2^0)$-complete, and every nonempty difference among six models is complete for its class.

The third shows that the same level governs a communication task. A sender uses a finite classical channel once and must convey one of m messages with zero error, helped by shared entanglement. Write $\mathrm{ZE}_t(N, m)$ for the existence of such a code for the channel N in model t (Section 15).

Theorem C (zero-error communication). *For finite rational classical channels,* $\mathrm{ZE}_{\mathrm{qs}}$ *and* $\mathrm{ZE}_{\mathrm{finS}}$ *are* Σ_2^0*-complete. Moreover, there are a finite rational classical channel* N *and an integer* m *for which* $\mathrm{ZE}_{\mathrm{qs}}(N, m)$ *holds, and every normal spatial* m*-message zero-error resource has infinite marginal entropy on both sides.*

The fourth measures how fast the entanglement spectrum of a perfect strategy must decay. For a perfect spatial strategy with reduced density h, finite rank, finite α-moments $\operatorname{Tr} h^\alpha < \infty$ and finite entropy are successively weaker resource conditions.

Theorem D (the entanglement spectrum). *For every fixed computable* $0 < \alpha < 1$*, perfect play with* $\operatorname{Tr} h^\alpha < \infty$*, denoted* $\mathcal{P}_\alpha$*, is* Σ_2^0*-complete. With* $\mathcal{P}_0$ *denoting finite rank,* $\mathcal{P}_1$ *finite entropy and* $\mathcal{P}_{\mathrm{normal}}$ *normal perfection, they form the strict ladder* $\mathcal{P}_0 \subsetneq \mathcal{P}_a \subsetneq \mathcal{P}_b \subsetneq \mathcal{P}_1 \subsetneq \mathcal{P}_{\mathrm{normal}}$ *for* $0 < a < b < 1$*, and all moment orders greater than one collapse to normal perfection.*

Theorem 9.1 adds a single hardness family for all rungs, the completeness of the differences between rungs, and an exact description of the possible critical indices.

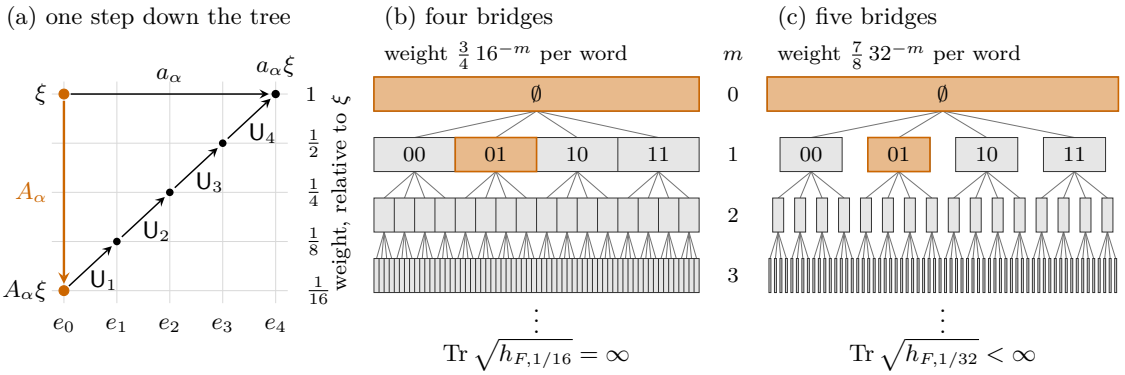


Figure 1: The running example: the tree of four-letter words, with density weight $\frac{3}{4}\,16^{-m}$ on each word of length m (Section 5). (a) Each bridge U_g carries the corner e_{g-1} onto e_g and doubles density weights (Lemma 4.1); the child $A_\alpha\xi = v^* a_\alpha \xi$ of a parent ξ is the vector of e_0 that the four bridges carry onto $a_\alpha\xi$; since a_α keeps weights, the child has one sixteenth of the parent's weight (Section 4.5). (b) Bar widths are proportional to the square roots of the weights, and the shaded bars are a parent and child as in (a); with four bridges the four children exactly tile their parent, so every row has the same width and the sum of all square roots diverges, although the weights sum to one. (c) A fifth bridge gives the ratio $1/32$, each row covers $1/\sqrt{2}$ of the row above, and the sum is finite: this is the zero-error threshold of Section 16.

A running example. Consider the rooted tree of finite words w over four letters; $|w|$ is the word length, and appending a letter selects one of four branches. Give the basis vector $|w\rangle$ density weight $(3/4)16^{-|w|}$, so that each child has one sixteenth of its parent's weight. The weights sum to one and define a normal state of finite entropy. Its half-moment, the sum of the square roots of all weights, diverges: the four child square roots sum to the parent's square root at every level. A *bridge* is a finite game gadget that doubles a density weight between two selected subspaces; four bridges give the factor 16. A fifth changes the ratio to $1/32$ and makes the half-moment converge, because $4/\sqrt{32} = 1/\sqrt{2}$. Figure 1 shows the tree and its bridges. This tree is the root of the spatial hardness game (Section 5), it separates the moment rungs (Section 9), and it sits exactly at the zero-error threshold (Section 16).

Theorem C comes from a correspondence between games and channels due to Mančinska [39], which we extend from finite dimension to trace-class operators. A perfect strategy gives a code exactly when its Schmidt amplitudes are summable. The compiled games of Theorem A sit exactly at this threshold: with four bridges, the amplitude sum diverges for every perfect strategy, and five bridges restore summability. A Catalan version of the construction gives a channel whose game has finite-entropy perfect strategies, while every code of the required size has infinite entropy.

Reading guide. For the trace mechanism, read Sections 2 and 3 (in particular Section 3.5), then Sections 11, 12 and 14, treating the compiler as a black box with the input and output contract stated at the start of Section 4. For the spatial classification, read Section 3, that contract, and Sections 5–7; the implementation in the rest of Section 4 can be deferred, and Section 7 shows how the clock period becomes the existential witness. For communication, read Section 2 and Sections 15–18.

1.3 Further results

The body also proves refinements of Theorems A–D. The largest-Schmidt strata are Π^0_1, and no total $0'$-computable function bounds a sufficient stratum (Theorem 3.1 and Section 3.4). Under a supplied entropy budget, full correlation tables are compact; the entropy busy beaver has Turing degree $0''$ and eventually dominates every total $0'$-computable function (Section 3.4). Exact attainment of a rational Bell bound, of an unspecified supremum, or of a specified value is Σ^0_2-complete (Theorem 8.1). The I_{3322} inequality is an explicit instance. Pál and Vértesi computed its supremum numerically, with finite-dimensional strategies converging to it, and conjectured that no finite dimension attains it [49], and Krohn-Grimberghe has claimed a computer-assisted exact identification of the value [36]; the supremum is not attained in finite dimension [51, 10, 18] but is attained by an infinite-dimensional spatial strategy [10, 18]. The type-I and type-II_1 languages are incomparable, and both are proper subsets of the type-II_∞ language (Section 14). Theorem 14.1 classifies every difference "perfect in X but not in Y" among six models. Finally, one-shot zero-error feasibility with finite-dimensional entanglement is Σ^0_1-complete, a corollary of [39] and [34] (Section 16.7).

1.4 Prior work and attribution

The finite-dimensional row of Table 1 uses perfect completeness in Theorem 12.6 of Ji, Natarajan, Vidick, Wright and Yuen [34], not just a gap statement about a supremum; upper membership follows by a dimension search and real quantifier elimination. Slofstra's embedding theorem [57] and the account of Mousavi, Nezhadi and Yuen [44, Theorem 6.4] give the commuting row. The value-one row was first proved by Mousavi, Nezhadi and Yuen [43]; see also [44, Theorem 6.9]. The commuting upper bound uses the convergent moment hierarchy of Navascués, Pironio and Acín [45]. The operator-solution framework for binary constraint system games originates with Cleve and Mittal [9]; Cleve, Liu and Slofstra gave the commuting-operator extension [8]. Slofstra also established nonclosure of finite-dimensional quantum correlations [56]. Lin's $\mathsf{MIP}^{co} = \mathsf{coRE}$ theorem concerns gap approximation in the commuting model [37]; it does not classify spatial attainment.

Infinite-dimensional spatial correlations were constructed by Coladangelo and Stark [11]. Kim, Paulsen and Schafhauser prove that perfect spatial play collapses to perfect finite-dimensional play for synchronous games [35, Corollary 3.11], and Lupini and coauthors prove the corresponding equivalence for imitation games [38, Corollary 5.4]. Our games keep nonsynchronous bridge questions, so no YES instance of the spatial reduction is synchronous or an imitation game.

Mančinska's flagged magic-square construction supplies the bridge geometry [39]; the magic square itself is due to Mermin and Peres [42, 53]. The normal bridge, shared-label refinement, graph products and Boolean rational-row tests were developed in our entropy-game paper [24, Lemmas 4.1 and 4.2, Theorem B, Eq. (32)]; its Catalan game [24, Theorem C] is used here as a separator, and a four-bridge version of its Catalan assembly gives the infinite-entropy channel. Section 4 restates and proves the versions needed here and extends them to n bridges, arbitrary finite lists of endpoint labels and normal commuting hosts. The standard-form reduction, the commuting bridge and expected-inclusion transfer come from our earlier paper [20, Lemmas 1.6, 2.2 and 3.1]; the type rows of Table 1 answer the question left open in [20, Theorem 7.6]. The separation $\mathrm{PERF}(\mathrm{II}_1) \not\subseteq \mathrm{PERF}(\mathrm{I})$ uses the tracial mechanism of [20, Proposition 7.4] and Slofstra's nonclosure construction; the other direction uses a new one-bridge separator. Our earlier preprints [23] and [19] gave single games with, respectively, finite-energy spatial perfection that no finite-dimensional strategy attains, and a perfect type $\mathrm{III}_{1/2}$ strategy with no injective host.

Group embeddings originate with Higman [32]; the effective form follows Chiodo [7,

Theorem 2.2], with Craig's conversion [15] and the HNN normal form [33]. Subtype transfer uses Connes–Takesaki flow-of-weights theory [14], Takesaki's expectation theorem [58], Connes's invariants [12], and the argument inside Haagerup–Musat [29, Lemma 3.8]. The nearest operational work on host type is the type classification of embezzling states by van Luijk, Stottmeister, Werner and Wilming [59], the commuting self-test for exact embezzlement of Paddock and Schmidt [48], and Harris's Cuntz-algebra self-testing and factor classification [31]. These concern embezzlement rather than the arithmetical classification of host-existence languages.

For communication, Cubitt, Leung, Matthews and Winter established the operator characterization of entanglement-assisted zero-error codes and the pseudo-telepathy construction [16]. Mančinska proved finite-dimensional clique saturation in Lemma 5, the converse and square-root state map in Theorem 6, and the entanglement-assisted independence-number characterization in Corollary 7 of [39]; her zero-error results assume a finite-dimensional shared state. Her Corollary 8 gives the fifty-input flagged-magic-square channel with ten entangled messages and at most nine messages from finite-dimensional maximally entangled resources. Mančinska, Roberson and Varvitsiotis studied perfect strategies and independent-set game reductions [41]; Mančinska and Roberson developed quantum graph homomorphisms [40]; Harris proved universality results for synchronous graph homomorphism games [30]. Bhattacharyya, Mehta and Zhao proved undecidability for a restricted entanglement-assisted one-shot parameter of classical-to-quantum channels [5]; their parameter uses maximal entanglement and projective measurements [5, Definition 2.3]. The finite-dimensional row here removes these restrictions for classical channels and a one-shot threshold; it does not resolve the asymptotic capacity question. For fixed-scenario correlation membership, studied by Fu, Miller and Slofstra [27], see Section 19.

This paper consolidates and supersedes our preprints [25, 22, 21]. Theorem D and the completion of the semifinite-but-not-spatial difference classification are new here. The commuting Π_1^0 classification is known [57, 44]; Theorem B identifies its language with that of every fixed type-III subtype.

1.5 Organization

Section 2 fixes the models, languages and notation. Section 3 proves the Σ_2^0 upper bounds for spatial, finite-entropy, finite and semifinite perfection; Section 9.1 treats moment rungs and Section 16.4 reduces codes to games. Section 4 builds the finite compiler shared by all hardness proofs. Sections 5–7 prove the spatial hardness of Theorem A, and Section 8 the Bell variants. Sections 9–10 prove Theorem D. Sections 11–12, together with Section 13, prove Theorem B, and Section 14 compares the type languages. Sections 15–18 prove Theorem C and the other communication rows. Section 19 discusses open problems.

Contents

1 Introduction **1**

2 Preliminaries **8**

3 Upper bounds: one existential quantifier **12**

4 A finite compiler for normal strategies **20**

5 The rooted tree and computation squares **42**

6 A finite-order clock and its period reader **44**

7 Spatial hardness: the reduction and value one **46**

8 Exact Bell attainment 49

9 The entanglement spectrum and its decision problems 51

10 A uniform family and programmable critical indices 53

11 Semifinite hosts: a critical tree selects a finite depth 58

12 Each type-III subtype has the commuting language 64

13 One selector for spatial and semifinite perfection 65

14 Comparing the type languages 68

15 Zero-error codes and perfect play 74

16 Five bridges and the normal row 77

17 The commuting row 81

18 A channel whose codes require infinite entropy 83

19 Discussion and open problems 86

How this work was done 87

A The nested assembly 88

B The bridge-count and branching variants 94

C Software checks 95

D An exact rotation margin 96

E Results used from other papers 97

F Lean verification of Theorem A 98

2 Preliminaries

2.1 Games, strategies and models

A finite rational Boolean game G is a finite list of questions and answers, a rational distribution $\pi(x, y)$, and a predicate $V(a, b \mid x, y) \in \{0, 1\}$. Perfection means zero probability for every losing event of positive question weight. A spatial strategy consists of positive operator-valued measurements (POVMs) $E_{a|x}$ and $F_{b|y}$ and a normal state, represented by a positive trace-class density ρ of trace one, on $\mathcal{H}_A \otimes \mathcal{H}_B$. Its winning probability is

$$\sum_{xyab} \pi(x, y)V(a, b \mid x, y) \operatorname{Tr} \rho(E_{a|x} \otimes F_{b|y}).$$

We write C_q, C_{qs}, C_{qc} for finite-dimensional, normal spatial, and commuting correlations. For a strategy $\mathcal{S}$, write $\omega(G, \mathcal{S})$ for the displayed winning probability and $\mathrm{loss}_G(\mathcal{S}) = 1 - \omega(G, \mathcal{S})$. The value ω_q is the supremum over C_q. Finite Schmidt truncations show that C_q and C_{qs} have the same closure and the same supremum for every Bell functional.

The set PERF_q consists of games with a perfect finite-dimensional strategy. The set $\mathrm{PERF}_{\mathrm{qs}}$ consists of games with a perfect normal spatial strategy. Its stratum $\mathrm{PERF}_{\mathrm{qs}}(\kappa)$ requires a perfect pure strategy whose largest Schmidt probability is at least $1/\kappa$. The set $\mathrm{PERF}_{\mathrm{finS}}$ requires a perfect strategy with finite marginal entropy. The one-marginal existence convention gives the same game language as requiring both reduced entropies finite, by purification onto the other party and equality of the pure state's marginal spectra; the conventions can differ on a fixed mixed strategy. Commuting perfection, $\mathrm{PERF}_{\mathrm{co}}$, uses two commuting families of POVMs on one Hilbert space and a state.

Hosts. For a class X of von Neumann algebras, $\mathrm{PERF}(X)$ denotes existence of a perfect strategy normal on an ambient von Neumann algebra of class X; Bob acts in its commutant. The ambient algebra need not be generated by the measurements. The state may be chosen along with the host. Finite-host perfection permits non-tracial normal states; this is consistent with $\omega_{\mathrm{tr}}(G_{\mathrm{FMS}}) < 1$ in [39, Theorem 3]. These are existence languages: they do not test a prescribed algebra, state or full correlation table. Write $\mathrm{PERF}_{\mathrm{fin}}$ for hosting in a σ-finite finite von Neumann algebra, equivalently one admitting a faithful normal tracial state, and $\mathrm{PERF}_{\mathrm{sf}}$ for hosting in a semifinite von Neumann algebra. Factor hosting gives the same existence languages as hosting in algebras purely of the indicated type.

We use separable hosts when decomposing. For commuting input this follows from the joint GNS representation of the finitely generated measurement C^*-algebra. For finite or semifinite input the finite-corner overlap construction of Section 3.5 gives a separable host in the same class: generate the two unital C^*-algebras by the finitely many effects and the finite band projection, and take the closure of their orbit of the trace vector. It is separable; Lemma 3.8 preserves a nonzero perfect component and proves its represented host finite or semifinite. One may then standardize Bob and decompose over the center. Every losing probability is an integral of nonnegative probabilities, so the finitely many zero integrals vanish on a common conull set of factor fibres. A finite or semifinite fibre remains in that class; the amplifications in (10) give the required pure type-II factor. For type I use its direct-integral decomposition into spatial factor strategies; in particular the type-I language PERF(I) is the spatial language $\mathrm{PERF}_{\mathrm{qs}}$. The type-III constructions of Section 12 start with an unrestricted commuting factor fibre.

When several languages are compared at once we abbreviate

$$\mathbf{q} = \mathrm{PERF}_q, \quad \mathbf{E} = \mathrm{PERF}_{\mathrm{finS}}, \quad \mathbf{I} = \mathrm{PERF}(\mathrm{I}),$$

$$\mathbf{J} = \mathrm{PERF}(\mathrm{II}_1), \quad \mathbf{U} = \mathrm{PERF}_{\mathrm{sf}}, \quad \mathbf{C} = \mathrm{PERF}_{\mathrm{co}}.$$

Here **q** requires attainment in finite dimension; it is not the value-one language. The inclusions are $\mathbf{q} \subseteq \mathbf{E} \subseteq \mathbf{I} \subseteq \mathbf{U} \subseteq \mathbf{C}$ and $\mathbf{q} \subseteq \mathbf{J} \subseteq \mathbf{U}$; every displayed inclusion is strict (Section 14). We keep PERF strictly for attainment. Nezhadi's 2025 thesis [46] uses "perfectly winnable" for the value-one Π^0_2 problem of [44], so its terminology has a different scope.

2.2 Arithmetical classes

The set Σ^0_2 consists of predicates $\exists j\, \forall t\, \mathcal{R}_{\mathrm{dec}}(e, j, t)$ with decidable $\mathcal{R}_{\mathrm{dec}}$. Completeness means that a problem is itself of this form and every set of this form reduces computably to it. The classes Σ^0_1 and Π^0_1 use a single existential or universal number quantifier, respectively, and Π^0_2 consists of complements of Σ^0_2 sets. We also use the class of differences

$$D_2(\Sigma^0_2) = \{A \setminus B : A, B \in \Sigma^0_2\}.$$

It is a class of differences and should not be confused with the intersection of the pointclasses Σ^0_2 and Π^0_2. The canonical complete language is $\{(e, f) : e \in \mathrm{FIN},\ f \notin \mathrm{FIN}\}$, where FIN is the Σ^0_2-complete set of indices of finite c.e. sets. Indeed reduce the two Σ^0_2 predicates separately to FIN and pair the resulting indices.

All encodings and reductions in this paper are effective finite encodings and computable many-one reductions. Completeness is over all finite scenarios; the numbers of questions and answers are not fixed in advance.

2.3 Conventions

Entropy uses logarithms to base two: $S(h) = -\operatorname{Tr}(h \log_2 h)$ and $H(p) = -\sum_i p_i \log_2 p_i$, with $0 \log_2 0 = 0$. A projection-valued measurement (PVM) is a POVM consisting of orthogonal projections. Write $|\Phi_d\rangle = d^{-1/2} \sum_{j=0}^{d-1} |j, j\rangle$ for a finite maximally entangled vector. Vectorization of an operator T in fixed orthonormal bases is

$$|T\rangle\!\rangle = \sum_{i,j} T_{ij} |i\rangle \otimes |j\rangle, \qquad \| |T\rangle\!\rangle \|^2 = \operatorname{Tr} T^* T,$$

when this sum is square summable. A purification is a unit vector with the given state as its partial trace over an added environment. Normal means represented by a positive trace-class density of trace one; on a spatial tensor product this includes pure vectors and countable mixtures. The centralizer of a density consists of the bounded operators commuting with it; for a faithful normal state φ on a von Neumann algebra M it is $M_\varphi = \{x \in M : \varphi(xy) = \varphi(yx)\ \forall y \in M\}$. For a game written in a code or compiler, YES and NO instances are those with and without the specified perfect strategy or code. Mat_r denotes the scalar r-by-r matrix algebra.

2.4 Supports and synchronization

The compiler of Section 4 uses the following elementary facts, in a spatial and in a standard-form version.

Pure components. If a mixed normal strategy is perfect, write $\rho = \sum_j \lambda_j |\psi_j\rangle\langle\psi_j|$. Each losing event is positive. Its expectation vanishes in every component with $\lambda_j > 0$, so every such component is perfect with the same measurements. Trace-class marginals have separable support. Thus all arguments may be made on separable state supports.

Matching and support. For effects on opposite sides, perfect binary matching gives, on a purification Ω,

$$0 = \langle E + F - 2EF \rangle = \langle (E - F)^2 + (E - E^2) + (F - F^2) \rangle.$$

Each positive term annihilates Ω. If T is the coefficient operator of the pure vector on the chosen Schmidt cut, matching gives $ET = TF^{\mathsf{T}}$; dense coefficient range first makes the local state support reducing. On that support the effects are projections and commute with the marginal density. A positive operator B with $\mathrm{Tr}(hB) = 0$ for faithful h is zero, since $B^{1/2}h^{1/2} = 0$ and $h^{1/2}$ has dense range.

Sharp compressions and joint measurements. If PEP is a projection for an effect E and support projection P, positivity shows that E acts as identity on its range and zero on its kernel inside $P\mathcal{H}$. Hence E reduces the support. A POVM with sharp coarse marginals is their joint spectral measurement: each fine effect is supported in the corresponding marginal ranges, and summation gives the commuting products. These facts apply after compression before projectivity is used.

Synchronization and forbidden events. We call a projection synchronized when its binary question is perfectly matched to the corresponding opposite question. If synchronized projections satisfy $P_A^{\mathrm{sync}}\Omega = P_B^{\mathrm{sync}}\Omega$ and a forbidden event gives $Q_A P_B^{\mathrm{sync}}\Omega = 0$, dense Schmidt range gives the local identity $Q_A P_A^{\mathrm{sync}} = 0$. The arguments work on either party and on purifications of mixed states.

Standard form. A standard-form strategy for a faithful normal state φ on M uses its natural-cone vector ξ_φ in $L^2(M)$, Alice effects in M, and Bob effects in $M' = \mathcal{J}M\mathcal{J}$, where $\mathcal{J}$ is the modular conjugation and σ_t^φ the modular group. Nonfaithful states are interpreted on their support corners. A general normal-marginal commuting strategy has this form without changing probabilities: after joint GNS, compress Bob to $\overline{M\psi}$ using the isometry $x\xi_\varphi \mapsto x\psi$. The compression lies in the standard commutant and is still a POVM. Semifiniteness passes to corners. This is [20, Lemma 1.6]; the argument includes mixed states.

Perfectly matched binary effects are sharp centralizer projections with mirror $\mathcal{J}E\mathcal{J}$. Indeed different matched classes satisfy $E_c F_d \xi_\varphi = 0$. Summing gives $E_c\xi_\varphi = F_c\xi_\varphi$ and $E_c E_d \xi_\varphi = 0$ for $c \neq d$. Separation gives orthogonality, and completeness gives idempotence. The self-adjoint mirror gives $\varphi(E_c x) = \varphi(xE_c)$. A fine positive effect below sharp marginals is their joint projection. These facts replace any finite-dimensional rigidity assumption. On faithful support a synchronized projection is therefore a centralizer projection with its standard commutant mirror.

2.5 Notation

Table 2 collects the notation used across sections. Distinct fonts separate question names (sans serif), languages (bold), automaton symbols (typewriter) and operators. Integer indices such as $i, j, k, \ell, m, n, r, t$ are local to the statement or proof in which they appear, except that n counts bridges in the compiler and in Sections 16–18. Each root presentation (the Fock clock of Sections 5–7, the selector of Section 11, the shift separator of Section 14 and the Catalan presentation of Section 18) has its own labels, introduced where it is defined.

Table 2: Notation.

Symbol	Meaning
Games, models and languages	
$G;\ \pi, V$	finite rational Boolean game; question distribution and predicate
$\mathcal{S}, \omega(G,\mathcal{S}), \omega_q$	strategy, its winning probability, finite-dimensional value
$\mathrm{PERF}_q, \mathrm{PERF}_{\mathrm{qs}};$ $\mathrm{PERF}_{\mathrm{finS}}, \mathrm{PERF}_{\mathrm{co}}$	finite-dimensional, spatial; finite-entropy, commuting perfection
$\mathrm{PERF}(X), \mathrm{PERF}_{\mathrm{fin}}, \mathrm{PERF}_{\mathrm{sf}}$	hosting in class X, in a finite, in a semifinite von Neumann algebra
$\mathbf{q}, \mathbf{E}, \mathbf{I}, \mathbf{J}, \mathbf{U}, \mathbf{C}$	abbreviations for six existence languages
$\kappa;\ \mathrm{PERF}_{\mathrm{qs}}(\kappa), \mathrm{PERF}_{\mathrm{finS}}(m)$	Schmidt stratum integer; stratum and integer entropy-budget stratum
$F_{\mathrm{qs}}, F_{\mathrm{sf}}, F_{\mathrm{Bell}}$	decidable stage predicates of the upper bounds
$\mathcal{R}_{\mathrm{dec}}, \mathcal{A}_{\mathrm{dec}}$	decidable predicate and its Σ_2^0 set
$G_e^{\mathrm{clk}}, G_e^{(n)}, G_e^{\mathrm{sel}}$	Weyl-clock reduction (four bridges), its n-bridge version, selector reduction
$G_{\mathrm{sh}}, G_{\mathrm{lin}}, G_{\mathrm{Cat}}^{(2)}, G_{\mathrm{III}}$	fixed separating games of Section 14
States and hosts	
D_{Sch}, D	largest-amplitude-normalized and $\mathrm{Tr}\, D^2 = 1$ Schmidt operators
$M, \varphi, \sigma_t^{\varphi};\ \mathcal{J}, \xi_{\varphi}$	host, faithful state, modular group; standard conjugation and state vector
τ, h	trace and density
$R_{\mathrm{hyp}}, R_\infty, R_\lambda$	injective II_1, III_1, III_λ factors
The compiler	
$\mathsf{R}_i, \mathsf{C}_j, \mathsf{T}_x, \mathsf{B}_y;\ f, \perp$	flagged magic-square questions; flag and inactive answers
$c_x, r_y;\ P_{\mathrm{mag}}, m_{ab}$	selected column and row triples; magic support and its column-zero projections
$\mathfrak{x}, \mathfrak{z};\ \mathsf{U}$	Pauli axis reflections; calibrated flag-to-magic transport
P^{lab}	synchronized label projection in a refinement
$F_i, F_{ij}, G_{ij}, P_\triangle$	qutrit frame, pair support, graph and triangle projections
$P^{\mathrm{par}}, J_{ab}, s_{\mathrm{par}}$	bound parameter, its joint partition and reflection label
$\Delta, L, V_j;\ k_{\mathrm{row}}$	rational row, denominator, signed reflections; scalar qubits
$\Pi_j^{\pm}, \Pi_t^{\mathrm{row}}, \Pi_d^{\mathrm{ref}}, \mathcal{I}_t$	scalar parity events, row partition, reference projection, column isometries
$d_{\max};\ \varphi_{\mathrm{old}}, h_{\mathrm{old}}$	row-ancilla dimension; state and marginal before the row ancilla
$A_\alpha, P_\circ;\ \nu$	creation contractions and root projection; modular ratio 2^{-n}
$e_0, e_g, m_{g,ab}, m_g$	source and selected corners, magic corners and supports of bridge g
E_g^{act}	activity $e_{g-1} + m_g$ of bridge g
$\mathsf{U}_g, v;\ \mathfrak{c}_n, \mu_n$	bridges and their chain; scaffold cover constant and weights
$\Lambda, u_{r,\ell}, V_{g,\ell}, L_r$	transported encodings, corner copies, block unitaries, decoded copies
$a_\alpha;\ L_{\mathrm{end}}, R_{\mathrm{end}}$	lifted centralizer contractions; endpoint coefficients

Table 2: Notation (continued).

Symbol	Meaning
The Fock clock	
$\mathcal{W}, A_w, \mathcal{K}, h_{F,\nu}, h_F, \rho_{\mathrm{spec}}$	four-letter words, word isometries, root multiplicity, tree densities, spectator
$U_x, U_y; P_x, P_y$	cyclic coordinate shifts; zero-column and zero-row projections
$\Gamma_{\mathrm{CA}}, P_c;\ \mathtt{s}, \mathtt{b}, \mathtt{f}, \mathtt{z}, \mathtt{H}$	automaton alphabet and its projections; start, blank, front, mark, halt
$J_{\mathrm{W}}, K_{\mathrm{W}}, Z_{\mathrm{clk}}, \zeta, d_{\mathrm{clk}}$	Weyl pair, central clock phase, its value and order
$W, C_{\mathrm{ctr}}, P_{\mathrm{ret}}, T_{\mathrm{inv}}$	coordinate clock, counter, return pulse and bounded reader inverse
$h^{[m]}, P_m, t_m$	truncated density, top-depth projection and its mass
The quarter-plane register and comb (Section 10)	
$S, T;\ \Pi_x, \Pi_t;\ C_x, C_t$	shifts, boundary projections and dyadic counters
$Z, J, K, W;\ E_{\mathrm{ret}}, R_{\mathrm{ret}}$	clock labels, return projection and bounded reader
$d_{\mathrm{clk}};\ \rho_{\mathrm{spec}}, \rho_{\mathrm{reg}}$	clock order; spectator and geometric register densities
$V_{\mathrm{sp}}, B_0, B_1;\ Q, P, E_{\mathrm{guard}}$	spine and branch maps; spine, root and guard projections
$C_{\mathrm{sp}}, L;\ E_{C_{\mathrm{sp}}}, E_L, F_{C_{\mathrm{sp}}}, F_L$	comb counters, equality projections and their readers
$k_m, \rho_{\mathrm{comb}}, Z_{\mathrm{comb}}$	bush depths, comb density and its normalizer
$\mathcal{P}_\beta, \alpha_c(h), c(G);\ d_{\mathrm{cut}}$	moment language, witness and game critical indices; cutoff dimension
The nested assembly (Appendix A)	
$X_i;\ a_i, a_i'$	root modular contractions; inner and transported lifts
$f, e, E_{\mathrm{in}}^{\mathrm{act}}, E;\ f', e'$	inner corners and activity, outer corner; transported corners
$u, \mathsf{U};\ \mathsf{U}_0, \mathsf{U}_1;\ t, \ell$	inner/outer transports, outer restrictions, equal-density map and quadratic coefficient
$U_k, V_k, V_k^{\mathrm{blk}}$	inner encodings, outer copies and whole-activity block unitaries
$R_{\mathrm{tar}};\ K_L, K_M, K_R$	target centralizer expression; arbitrary polynomial coefficients
$\epsilon_{xy}, \varsigma;\ J_{\mathrm{sgn}}, Z_{ij}^{\mathrm{gr}}$	FMS signs, scalar axes; sign grading and graph reflection
Communication	
$N, m;\ \mathrm{ZE}_t(N, m)$	finite rational channel and message count; one-shot feasibility
$\Gamma(G);\ \mathcal{Q}_A, \mathcal{Q}_B;\ k$	tagged confusability graph; question sets; number of tagged questions
$R_{i,x}, \rho, \rho_{\mathrm{code}};\ \mathcal{C}_c$	assemblage and code marginal; a clique
$\hat{\mu}_n;\ c_n^{\mathrm{amp}}$	normalized square roots of μ_n; amplitude constant
$h_{\mathrm{Cat}}, k_{\mathrm{Cat}}, C_{\mathrm{Cat}}$	critical Catalan density, its normalized square, Catalan height counter

3 Upper bounds: one existential quantifier

Each tracial language in this paper is a union over one integer of Π_1^0 sets. For spatial play the integer bounds the inverse of the largest Schmidt probability; for semifinite play it bounds the inverse overlap of the state with a finite tracial corner. We first treat the spatial case and prove the following theorem.

Theorem 3.1 (Schmidt strata)**.** *Uniformly for integers $\kappa \geq 1$, the existence of a perfect pure spatial strategy with largest Schmidt probability at least $1/\kappa$ is a Π_1^0 property. Their union is $\mathrm{PERF}_{\mathrm{qs}} \in \Sigma_2^0$. No total computable function $\kappa(G)$ bounds a sufficient stratum for every spatially perfect game. If $\omega_q(G) = 1$ and no perfect spatial strategy exists, the largest Schmidt probability tends to zero along every sequence of pure spatial strategies whose winning probabilities tend to one.*

3.1 Normalize one Schmidt coefficient

For a pure strategy write $\psi = \sum_i s_i |i,i\rangle$ with $s_1 \geq s_2 \geq \cdots$ and $\sum_i s_i^2 = 1$. Set $d_i = s_i/s_1$ and $D_{\mathrm{Sch}} = \mathrm{diag}(d_i)$. For some integer κ,

$$d_1 = 1, \qquad d_i \downarrow 0, \qquad \operatorname{Tr} D_{\mathrm{Sch}}^2 \leq \kappa. \tag{1}$$

For positive effects E, F, vectorization gives

$$p(E,F) = \frac{\|\sqrt{E}\, D_{\mathrm{Sch}} \sqrt{F^{\mathsf{T}}}\|_{\mathrm{HS}}^2}{\operatorname{Tr} D_{\mathrm{Sch}}^2}, \qquad p(E,F) = 0 \iff E D_{\mathrm{Sch}} F^{\mathsf{T}} = 0.$$

To prove the reverse implication, $E D_{\mathrm{Sch}} F^{\mathsf{T}} = 0$ sends $D_{\mathrm{Sch}} \operatorname{ran} F^{\mathsf{T}}$ into $\ker E$. The closures of the ranges of F^{T} and $\sqrt{F^{\mathsf{T}}}$ agree, and $\ker E = \ker \sqrt{E}$. Continuity gives the square-root sandwich identity. The forward implication follows by multiplying the zero square-root sandwich by the square roots.

Let $P_{d_{\mathrm{cut}}}$ be the first d_{cut} coordinate projection. Monotonicity in (1) gives

$$\|D_{\mathrm{Sch}} - P_{d_{\mathrm{cut}}} D_{\mathrm{Sch}} P_{d_{\mathrm{cut}}}\| = d_{d_{\mathrm{cut}}+1} \leq \sqrt{\kappa/(d_{\mathrm{cut}}+1)}. \tag{2}$$

Consequently the family in (1) is compact in operator norm: take coordinatewise limits and use this common tail estimate; Fatou preserves the squared-sum bound. Finite families of POVMs on ℓ^2 are compact in the weak operator topology. Positivity and the finite sum-to-identity relations survive these limits.

Lemma 3.2 (Compact sandwiches)**.** *If $D_{\mathrm{Sch},j} \to D_{\mathrm{Sch}}$ in norm with compact D_{Sch}, and uniformly bounded E_j, F_j converge weakly to E, F, then $E_j D_{\mathrm{Sch},j} F_j$ converges weakly to $E D_{\mathrm{Sch}} F$.*

Proof. For fixed v, compactness makes $D_{\mathrm{Sch}} F_j v \to D_{\mathrm{Sch}} F v$ in norm. Pair this with the weakly convergent vectors $E_j^* u$. The replacement of D_{Sch} by $D_{\mathrm{Sch},j}$ has norm error tending to zero. □

3.2 An effective finite predicate

Let $\mathcal{L}_G$ be the losing tuples with $\pi(x,y) > 0$. Define $F_{\mathrm{qs}}(G, \kappa, d_{\mathrm{cut}})$ by the existence of d_{cut}-dimensional POVMs and real numbers $1 = d_1 \geq \cdots \geq d_{d_{\mathrm{cut}}} \geq 0$, $\sum_i d_i^2 \leq \kappa$, such that

$$|(E_{a|x} D_{\mathrm{Sch},d_{\mathrm{cut}}} F_{b|y}^{\mathsf{T}})_{ij}|^2 \leq \kappa/(d_{\mathrm{cut}}+1) \quad ((x,y,a,b) \in \mathcal{L}_G;\ i,j \leq d_{\mathrm{cut}}). \tag{3}$$

This is semialgebraic over the rationals. Splitting complex entries into real and imaginary parts and using principal minors for positivity makes real quantifier elimination a decision procedure [2]. Decidability is more than the upper bound needs. All variables lie in $[-1,1]$, where each constraint is Lipschitz in the maximum norm, so F_{qs} fails exactly when, at some grid resolution, every grid point violates some constraint by more than its Lipschitz constant times the spacing. Thus F_{qs} is Π_1^0 by compactness alone; Appendix F checks this route.

A perfect strategy with $s_1^2 \geq 1/\kappa$ supplies all these witnesses. Compress its POVMs. Since $P_{d_{\mathrm{cut}}}$ commutes with D_{Sch} and $E D_{\mathrm{Sch}} F^{\mathsf{T}} = 0$, the finite sandwich is $-P_{d_{\mathrm{cut}}} E (D_{\mathrm{Sch}} - D_{\mathrm{Sch},d_{\mathrm{cut}}}) F^{\mathsf{T}} P_{d_{\mathrm{cut}}}$, with norm bounded by (2).

Conversely choose a witness for every d_{cut}. Extend $D_{\mathrm{Sch},d_{\mathrm{cut}}}$ by zero and each POVM by a fixed deterministic POVM on the complement. Pass to a subsequence with $D_{\mathrm{Sch},d_{\mathrm{cut}}} \to D_{\mathrm{Sch}}$ in norm and all effects weakly convergent. For every fixed matrix entry, (3) and Lemma 3.2 give $E D_{\mathrm{Sch}} F^{\mathsf{T}} = 0$. We retain

$$1 \leq \operatorname{Tr} D_{\mathrm{Sch}}^2 \leq \kappa, \qquad d_1 = 1.$$

The normalized vectorization of D_{Sch} is a perfect normal strategy with largest Schmidt probability $1/\operatorname{Tr} D_{\mathrm{Sch}}^2 \geq 1/\kappa$. Therefore

$$G \in \mathrm{PERF}_{\mathrm{qs}}(\kappa) \iff \forall d_{\mathrm{cut}}\ F_{\mathrm{qs}}(G, \kappa, d_{\mathrm{cut}}), \qquad \mathrm{PERF}_{\mathrm{qs}} = \bigcup_{\kappa \geq 1} \mathrm{PERF}_{\mathrm{qs}}(\kappa). \tag{4}$$

This proves the upper membership in Theorems A and 3.1.

3.3 No effective choice of stratum; vanishing largest weight

If a total computable $\kappa(G)$ were sufficient for every YES game, (4) would put spatial perfection in Π_1^0. Perfect completeness of [34, Theorem 12.6] already makes it Σ_1^0-hard. NO instances of that map have spatial value at most $1/2$, because the finite-dimensional and spatial suprema agree. Thus the same map reduces the halting set to spatial perfection. The halting set cannot be Π_1^0, so such a function does not exist.

Now suppose pure spatial strategies approach perfect play while $p_{\max} \geq 1/\kappa$ on a subsequence. Normalize their largest Schmidt amplitudes. For a losing event,

$$\|E D_{\mathrm{Sch}} F^{\mathsf{T}}\|_{\mathrm{HS}} \leq \|\sqrt{E} D_{\mathrm{Sch}} \sqrt{F^{\mathsf{T}}}\|_{\mathrm{HS}} = \sqrt{\operatorname{Tr} D_{\mathrm{Sch}}^2\, p(E, F)}.$$

Positive question weights force every such probability to zero. The same compact limit gives a perfect spatial strategy. Contraposition proves the final assertion of Theorem 3.1, including sequences of finite-dimensional pure strategies. Equivalently their Schmidt min-entropy tends to infinity.

3.4 Entropy budgets

This subsection treats entropy budgets and their busy-beaver profiles; readers heading for the trace mechanism may skip to Section 3.5.

Fix a finite Bell scenario and let K_H be its correlations realized by normal spatial strategies with $S(\rho_A) \leq H$, for $0 \leq H < \infty$. Purify onto Bob, the unconstrained party: Alice's marginal and every correlation are preserved. Thus pure strategies suffice, with the same budget, for entire correlation tables as well as winning probabilities.

Proposition 3.3 (Entropy-budget compactness)**.** *The set K_H is compact. It is an effectively closed (Π_1^0) subset of $[0,1]^t$ uniformly in the scenario and H: for rational budgets this is an ordinary effective presentation; for real budgets it is uniform relative to a name of H.*

Proof. For sorted Schmidt probabilities p_i, $p_i \leq 1/i$, so

$$\delta_n = \sum_{i>n} p_i \leq H/\log_2(n+1). \tag{5}$$

This gives tightness in ℓ^1, and finite partial entropy sums give lower semicontinuity. Hence the sorted distributions of entropy at most H are compact in ℓ^1. Their Schmidt vectors are compact in ℓ^2, since $\|\sqrt{p} - \sqrt{p'}\|_2^2 \leq \|p - p'\|_1$. Compress the POVMs to the Schmidt supports and absorb the basis changes into them, obtaining common copies of ℓ^2 (if the Schmidt rank is finite, extend each POVM by a fixed deterministic answer on the complement in ℓ^2). Finite POVM tuples are weak-operator sequentially compact there. Tensor products of convergent effects converge weakly, first on simple tensors and then by boundedness on all vectors. The norm-convergent Schmidt vectors therefore give convergence of every correlation entry. This is compactness of *correlations*; the original states on fixed local spaces need not form a compact set. This argument is the bounded-entropy compactness of [24, Section 9.1, "Divergence at bounded entropy"].

Here are explicit outer approximations. Fix an integer $\bar{H} \geq 1$ and work with $0 \leq H \leq \bar{H}$. Choose positive rational $\eta_{\bar{H},n} > \sqrt{\bar{H}/\log_2(n+1)}$ tending to zero. Compute rational polynomials h_n with

$$\sup_{p \in \mathsf{S}_n} |h_n(p) - S(p)| \leq 1/n, \qquad \mathsf{S}_n = \{p_i \geq 0 : \textstyle\sum_i p_i = 1\}.$$

For example, Bernstein approximation to $-t \log_2 t$ on $[0,1]$, with rationally rounded coefficients and error at most n^{-2} per summand, gives these polynomials. Let $A^{\bar{H}}_{H,n}$ consist of tables $c \in [0,1]^t$ (t is the number of entries) for which there are sorted $p \in \mathsf{S}_n$, nonnegative s_i with $s_i^2 = p_i$, and n-dimensional POVMs (extended by a fixed deterministic answer on the complement in ℓ^2) satisfying

$$h_n(p) \leq H + 1/n, \qquad |c_{ab|xy} - \langle \psi_s, E_{a|x} \otimes F_{b|y} \psi_s \rangle| \leq \eta_{\bar{H},n}, \quad \psi_s = \sum_i s_i |i,i\rangle. \tag{6}$$

Split complex entries into real and imaginary parts, impose positivity by principal minors, and impose the POVM sums exactly. These are compact semialgebraic outer sets, uniformly decidable by real quantifier elimination [2, Chapter 14] for rational input data. The joint sets in (H, c) are semialgebraic over the rationals as well.

The normalized first-n truncation majorizes p, so its entropy does not increase. Its pure-state trace distance from the original state is $\sqrt{\delta_n}$; compressing the POVMs thus changes each entry by at most this amount. Consequently $K_H \subseteq A^{\bar{H}}_{H,n}$ for every n. Conversely witnesses in (6) have entropy at most $H + 2/n$. Along any sequence $n \to \infty$, the preceding tail and POVM argument gives a subsequential limit with entropy at most H, and the entrywise slack vanishes. Hence

$$K_H = \bigcap_{n \geq 1} A^{\bar{H}}_{H,n}.$$

Compact semialgebraic sets with rational data are effectively closed; the displayed intersection proves the uniform Π^0_1 presentation. It also gives a joint effectively closed set on $[0, \bar{H}] \times [0,1]^t$, which explains the assertion for real H. □

Corollary 3.4 (Exact tables and the vanishing-loss entropy limit)**.** *Membership of a prescribed rational correlation table in the class realizable with some finite marginal entropy is in Σ^0_2, uniformly over finite scenarios. The same holds for real-algebraic tables. For a finite rational game G, set*

$$E_G(\varepsilon) = \inf\{S(\rho_A) : \omega(G, \mathcal{S}) \geq 1 - \varepsilon\}, \qquad H_{\mathrm{ex}}(G) = \inf\{S(\rho_A) : \omega(G, \mathcal{S}) = 1\},$$

where the infima range over normal spatial strategies and $\inf \varnothing = \infty$*. In particular $E_G(\varepsilon) = \infty$ when $\varepsilon < 1 - \omega_q(G)$. Then*

$$\lim_{\varepsilon \downarrow 0} E_G(\varepsilon) = H_{\mathrm{ex}}(G), \qquad G \in \mathrm{PERF}_{\mathrm{finS}} \iff \sup_{\varepsilon > 0} E_G(\varepsilon) < \infty.$$

If $H_{\mathrm{ex}}(G)$ is finite, a perfect strategy attains it.

Proof. Exact membership in K_m is uniformly Π^0_1 for integer $m \geq 0$; take the union over m. The perfect-value constraint is closed too, so the games $\mathrm{PERF}_{\mathrm{finS}}(m)$ with perfect play of entropy at most m form a uniform Π^0_1 stratum, and $\mathrm{PERF}_{\mathrm{finS}} \in \Sigma^0_2$. Purification preserves the supplied marginal entropy, and a pure state's two marginal entropies agree.

The increasing limit $L = \lim_{\varepsilon \downarrow 0} E_G(\varepsilon)$ is at most $H_{\mathrm{ex}}(G)$. If $L < \infty$, choose strategies of loss at most $1/j$ and entropy at most $L + 1/j$. Proposition 3.3, with its lower-semicontinuity argument, gives a perfect limit of entropy at most L. Thus $H_{\mathrm{ex}}(G) \leq L$, and equality follows, also in the extended reals. A sequence of perfect strategies whose entropies decrease to a finite $H_{\mathrm{ex}}(G)$ gives attainment by the same argument. □

Lemma 3.5 (Entropy of clock witnesses)**.** *Suppose $e \in \mathcal{A}_{\mathrm{dec}}$, and let j_* be the least witness to $\forall t\, \mathcal{R}_{\mathrm{dec}}(e, j, t)$; if $e \notin \mathcal{A}_{\mathrm{dec}}$, Theorem 7.1 gives no perfect strategy. Every perfect normal spatial strategy for G_e^{clk} has full Alice-marginal entropy at least $\log_2 p_{j_*+1}$.*

Proof. This lemma and the next corollary use the hardness construction of Sections 4–7, which does not use them. Purify onto Bob as in (N1) of Section 4; this preserves Alice's full marginal h and all correlations. Work on its support. Lemma 4.5 extracts a faithful normalized source density $h_0 = 121e_0he_0$. By (32), $h_0 = h_F \otimes \rho_{\mathrm{spec}}$. Every positive eigenspace of ρ_{spec} splits into Z_{clk} phase sectors. The soundness proof of Theorem 7.1 applies to each nonzero sector: its word descendants persist at every depth, so an invalid order or a false check would contradict $P_{\mathtt{H}} = 0$. Thus each phase order is p_{j+1} for a valid witness $j \geq j_*$. Equation (41) makes the dimension of that sector divisible by its order. Consequently every positive eigenvalue of ρ_{spec} has multiplicity at least p_{j_*+1}, and hence is at most $1/p_{j_*+1}$. Therefore $S(\rho_{\mathrm{spec}}) \geq \log_2 p_{j_*+1}$.

To pass to the full marginal, use the entire synchronized scaffold cover, rather than an entropy inequality for a compression. Set $v_0 = e_0$ and $v_g = \mathsf{U}_g \cdots \mathsf{U}_1$ for $1 \leq g \leq 4$. Let $t_{g,ab} = \mathfrak{x}_g^a \mathfrak{z}_g^b e_g$ be the synchronized magic matrix units, and put $T_{g,ab} = t_{g,ab} v_g$. These partial isometries all have initial projection e_0 and final projections $m_{g,ab}$, respectively. Together with e_0, their final projections are orthogonal and sum to the identity on the full support. Moreover $ht_{g,ab} = t_{g,ab}h$ and $hv_g = 2^g v_g h$, so

$$T_{g,ab}^* h T_{g,ab} = 2^g e_0 h e_0.$$

Write $T_0 = e_0$. Since each corner commutes with h and the corners are orthogonal, $T_k^* h T_l = 0$ for distinct members $k \neq l$ of the seventeen-element list $(T_0; T_{g,ab})$. Hence $U(\delta_k \otimes \xi) = T_k \xi$ is a unitary from $\mathbb{C}^{17} \otimes e_0 P_A \mathcal{H}_A$ onto $P_A \mathcal{H}_A$, and by (23)

$$h \simeq \mathrm{diag}(\mu_4) \otimes h_0, \qquad \mu_4 = 121^{-1}(1; 2^{\oplus 4}; 4^{\oplus 4}; 8^{\oplus 4}; 16^{\oplus 4}).$$

Here h_0 is the density of the extracted realization on the whole source corner $e_0 P_A \mathcal{H}_A$; by (31) any further tensor factor present there lies in $\mathcal{K}$ and is counted in ρ_{spec}. Nonnegative spectral sums now give, also when an entropy is infinite,

$$S(h) = S(\mathrm{diag}(\mu_4)) + S(h_F) + S(\rho_{\mathrm{spec}}) \geq \log_2 p_{j_*+1}.$$

□

Corollary 3.6 (The entropy busy beaver)**.** *Fix a binary encoding of finite rational games, with decidable validity, and write $|G|$ for its bit length. Put*

$$m_*(G) = \min\{m \in \mathbb{N} : G \in \mathrm{PERF}_{\mathrm{finS}}(m)\},$$
$$B(n) = \max(\{m_*(G) : |G| \leq n,\ m_*(G) < \infty\} \cup \{0\}),$$

where $\mathbb{N}$ includes zero and an empty minimum is infinity. For finite $m_*(G)$, attainment gives $m_*(G) = \lceil H_{\mathrm{ex}}(G) \rceil$. There is no total $0'$-computable upper bound on B, and

$$B \equiv_T 0'', \qquad B \oplus 0' \equiv_T 0''.$$

Moreover B eventually strictly dominates every total $0'$-computable function, and hence exceeds each such function infinitely often. This is the profile of Radó's busy beaver function [54] relativized to $0'$, the "super Busy Beaver" BB_1 of [1, Section 3], with Beeping Busy Beavers giving an earlier natural example of this growth [1, Section 5.10]; the content of the corollary is that a resource of perfect play realizes it.

Proof. The fixed-budget test is uniformly Π^0_1, hence decidable in $0'$; at budget zero it reduces to the decidable product-state test. If a total $0'$-computable $b(n)$ bounded $B(n)$, testing the single budget $b(|G|)$ would decide $\mathrm{PERF}_{\mathrm{finS}}$ in $0'$, contradicting its Σ^0_2-completeness in Theorem A. Using $0''$, decide which of the finitely many games of size at most n lie in $\mathrm{PERF}_{\mathrm{finS}}$, then search their budgets using $0'$ to compute $B(n)$. Conversely $B \oplus 0'$ decides $\mathrm{PERF}_{\mathrm{finS}}$ by testing budget $B(|G|)$, so computes $0''$. Finally, an eventual upper bound could be patched at finitely many inputs to a total $0'$-computable upper bound, which has just been excluded.

For the reverse reduction $0' \leq_T B$, use the clock games for the decidable predicate "the eth machine on input e halts within j steps." If it halts at time T_e, the valid witnesses are exactly $j \geq T_e$, and Lemma 3.5 gives

$$T_e < p_{T_e+1} \leq 2^{m_*(G_e^{\mathrm{clk}})} \leq 2^{B(|G_e^{\mathrm{clk}}|)}.$$

Thus simulating $2^{B(|G_e^{\mathrm{clk}}|)}$ steps decides the halting set from B: a computation that has not halted by then never halts. Combining this with $B \oplus 0' \equiv_T 0''$ proves $B \equiv_T 0''$.

For eventual domination, fix a computable array $g(a, n, s)$ obtained by finite-stage simulations of oracle machines with the standard finite approximations to $0'$. Every total $0'$-computable function occurs as $n \mapsto \lim_s g(a, n, s)$ for some a. Compile the decidable predicates

$$\mathcal{R}_{\mathrm{dec}}(a, n, \langle s_0, v\rangle, t) \iff \forall s \in [s_0, t]\ g(a, n, s) = v$$

into clock games $G_{a,n}$, with hardwired input $e = \langle a, n\rangle$, using a computable pairing with $\langle s_0, v\rangle \geq v$. Pad by harmless always-accepted questions so that $|G_{a,n}| \geq n + 1$. A perfect strategy for a padded game restricts to one for the unpadded game with the same state, and conversely extends by arbitrary answers; padding therefore changes neither h nor the bound of Lemma 3.5. Define the computable, nondecreasing, unbounded size envelope

$$\ell(n) = \max_{a,k \leq n} |G_{a,k}|.$$

For a given total $0'$-computable $F : \mathbb{N} \to \mathbb{N}$, the limit lemma [55] supplies an index a_0 with

$$\lim_s g(a_0, n, s) = f(n), \qquad f(n) = 2^{1+\max_{m \leq \ell(n+1)} F(m)}.$$

The envelope was defined using all indices before selecting a_0, so there is no circular size definition. Each $G_{a_0,n}$ is a YES instance, and every witness index is at least $f(n)$. For $n \geq a_0$ and $\ell(n) \leq m < \ell(n + 1)$, Lemma 3.5 therefore gives

$$B(m) \geq m_*(G_{a_0,n}) \geq \log_2 p_{f(n)+1} > \log_2 f(n) > F(m).$$

The intervals cover every sufficiently large integer since ℓ is nondecreasing and unbounded. This proves eventual strict domination. □

Remark (Finite strata and jump bounds). Let $n \geq 2$ and suppose a Σ^0_n-complete attainment language has $e \in L \iff \exists m\, R(e, m)$ with R uniformly Π^0_{n-1}. No total $0^{(n-2)}$-computable function bounds a sufficient stratum index on all YES inputs; in fact no total $0^{(n-1)}$-computable function does. Indeed the latter oracle decides each stratum test, so a bounded finite search would put L in Δ^0_n, contradicting completeness. The entropy busy beaver is the case $n = 2$. The Schmidt resource has the full profile as well.

Corollary 3.7 (The Schmidt busy beaver)**.** *For the same encoding, put*

$$\kappa_{\min}(G) = \min\{\kappa \geq 1 : G \in \mathrm{PERF}_{\mathrm{qs}}(\kappa)\},$$
$$K(n) = \max\big(\{\kappa_{\min}(G) : |G| \leq n,\ G \in \mathrm{PERF}_{\mathrm{qs}}\} \cup \{0\}\big).$$

An empty minimum is infinity. For each YES clock game,

$$\kappa_{\min}(G_e^{\mathrm{clk}}) \geq \frac{121}{12} p_{j_*+1}.$$

Moreover $K \equiv_T 0''$, and K eventually strictly dominates every total $0'$-computable function.

Proof. For every perfect pure strategy, the full-marginal equivalence in Lemma 3.5 and its spectral-multiplicity argument give

$$\|h\| \leq \|\operatorname{diag}(\mu_4)\| \, \|h_F\| \, \|\rho_{\mathrm{spec}}\| \leq \frac{16}{121} \frac{3}{4} \frac{1}{p_{j_*+1}} = \frac{12}{121 p_{j_*+1}}.$$

The largest Schmidt probability is $\|h\|$. A strategy in stratum κ therefore requires $\kappa \geq (121/12)p_{j_*+1}$, proving the bound. Every spatially perfect game lies in some finite stratum by Theorem 3.1. Using $0''$, decide which of the finitely many valid games of length at most n are spatially perfect, then use $0'$ to test their uniform Π_1^0 strata and search for each least one. This computes K. Conversely $K \oplus 0'$ decides spatial perfection by testing the single stratum $K(|G|)$ (or stratum one if that value is zero), and hence computes $0''$ by Theorem A.

To remove the extra $0'$ oracle, use the halting clock games from the proof of Corollary 3.6. If machine e halts at time T_e, its least valid witness is T_e, so

$$T_e < p_{T_e+1} \leq \frac{12}{121} \kappa_{\min}(G_e^{\mathrm{clk}}) \leq \frac{12}{121} K(|G_e^{\mathrm{clk}}|) \leq K(|G_e^{\mathrm{clk}}|).$$

Simulating $K(|G_e^{\mathrm{clk}}|)$ steps therefore decides whether it halts: if it has not halted by then, it never halts. Thus $0' \leq_T K$, and the preceding reductions give $K \equiv_T 0''$.

Finally use the same computable array, padded games and nondecreasing unbounded envelope ℓ as in Corollary 3.6; padding preserves the state and hence the Schmidt probabilities. Given total $0'$-computable F, the limit lemma supplies a_0 for $f(n) = 2^{1+\max_{r \leq \ell(n+1)} F(r)}$. Every witness for $G_{a_0,n}$ is at least $f(n)$. For $n \geq a_0$ and $\ell(n) \leq m < \ell(n+1)$,

$$K(m) \geq \kappa_{\min}(G_{a_0,n}) \geq \frac{121}{12} p_{f(n)+1} > f(n) > F(m).$$

The intervals cover all sufficiently large integers, proving eventual strict domination. □

Remark (Least entropy and the least witness). For a YES clock game, let $d_{\max}(e)$ be the computable maximum row ancilla dimension in its compiler. Then

$$H_{\mathrm{ex}}(G_e^{\mathrm{clk}}) = \log_2 p_{j_*+1} + H(\mu_4) + S(h_F) + \delta_e, \qquad 0 \leq \delta_e \leq \log_2(3 d_{\max}(e)).$$

Indeed the full-marginal identity and multiplicity bound in Lemma 3.5 give the lower bound $S(h) \geq H(\mu_4) + S(h_F) + \log_2 p_{j_*+1}$ for every perfect normal strategy, including mixed strategies by purification onto Bob. For the upper bound choose the least witness in the explicit pure strategy of Theorem 7.1: its marginal is the tensor product of $\operatorname{diag}(\mu_4)$, h_F, $I_{p_{j_*+1}}/p_{j_*+1}$, $I_3/3$ and $I_{d_{\max}(e)}/d_{\max}(e)$. Entropy additivity gives exactly the same lower-bound expression plus $\log_2(3d_{\max}(e))$. Taking the infimum proves the display. In particular the least entropy determines $\log_2 p_{j_*+1}$ up to a computable additive error.

3.5 A finite tracial corner

The spatial upper bound fixes a nonzero Schmidt coefficient. For a general semifinite host, the corresponding parameter is a nonzero overlap with a finite tracial corner. The existential integer bounds this overlap, and the quantifier pattern is the same.

Lemma 3.8 (Finite-corner overlap)**.** *Let $\mathcal{A}, \mathcal{B}$ be commuting unital C^*-algebras containing a game's effects. Suppose unit vectors ψ, η and $e = e^2 = e^* \in \mathcal{A}$ satisfy: ψ is perfect, $e\eta = \eta$, $\langle \eta, \psi \rangle \neq 0$, and the vector state of η is tracial on $e\mathcal{A}e$. Then the game has a perfect normal semifinite strategy. If $e = 1$, it has a perfect normal strategy in a finite von Neumann algebra.*

Proof. Let $\mathcal{H}_\eta = \overline{\operatorname{span} \mathcal{A}\mathcal{B}\eta}$. It reduces both algebras because they are self-adjoint and commute. Put $M = (\mathcal{A}|_{\mathcal{H}_\eta})''$ and $p = e|_{\mathcal{H}_\eta}$. The vector functional of η is normal on M and tracial on pMp: Kaplansky density approximates each corner element by a bounded net from $p\mathcal{A}p$, and separate ultraweak continuity extends $\tau_\eta(xy) = \tau_\eta(yx)$ one variable at a time. Moreover

$$p\mathcal{H}_\eta = \overline{\operatorname{span} \mathcal{B}(pMp)\eta}. \tag{7}$$

Indeed $pab\eta = bpap\eta$, and bounded strong approximation supplies the reverse closure. If $x \geq 0$ in pMp has $\tau_\eta(x) = 0$, then, for $a \in pMp$,

$$\|x^{1/2}a\eta\|^2 = \tau_\eta(a^*xa) = \tau_\eta(x^{1/2}aa^*x^{1/2}) \leq \|a\|^2\tau_\eta(x) = 0.$$

Since x commutes with $\mathcal{B}$, (7) gives $x = 0$. Thus pMp has a faithful normal finite trace. If a central projection z of M annihilates p, it annihilates η, and hence $\mathcal{A}\mathcal{B}\eta$ and $\mathcal{H}_\eta$; so $z = 0$. Thus p has full central support. A type-III summand has no nonzero finite projection, so its intersection with p is zero and fullness excludes that summand. Consequently M is semifinite.

The orthogonal projection onto $\mathcal{H}_\eta$ commutes with both parties. $\zeta = P_{\mathcal{H}_\eta}\psi$ is nonzero by its overlap with η. Each losing positive product annihilates ψ and hence ζ. Normalize ζ. Its state is normal on M and Bob's compressed effects commute with M. If $e = 1$, the faithful finite trace is on M. □

Conversely standardize a normal semifinite strategy and write $\psi = h^{1/2} \in L^2(M, \tau)$, with $\tau(h) = 1$. Some band $e = 1_{[a,b]}(h)$, $0 < a < b < \infty$, is nonzero. It has $0 < \tau(e) \leq 1/a$. For $\eta = e/\sqrt{\tau(e)}$,

$$\langle \eta, \psi \rangle = \frac{\tau(eh^{1/2}e)}{\sqrt{\tau(e)}} \geq \sqrt{a}\sqrt{\tau(e)} > 0. \tag{8}$$

Its corner state is tracial. In a finite von Neumann algebra take a faithful normalized trace and $e = 1$; then $\tau(h^{1/2}) > 0$. Adjoin e to Alice's algebra if necessary.

Theorem 3.9. $\mathrm{PERF}_{\mathrm{fin}}$ *and* $\mathrm{PERF}_{\mathrm{sf}}$ *belong to* Σ_2^0.

Proof. For $k \geq 1$ seek moments $m_{ij}(w) = \langle \xi_i, w\xi_j \rangle$, where $\xi_0 = \psi, \xi_1 = \eta$. Letters are the self-adjoint effects and Alice's extra projection e. Impose positivity for all polynomial vector lists, localizing positivity for each effect and its complement, POVM normalization, Alice–Bob commutation *including* e, $e^2 = e$, $e\eta = \eta$, unit norms, zero losing expectations on ψ, and

$$\mathrm{Re}\langle \eta, \psi \rangle \geq 1/k, \qquad \langle \eta, eue\, eve\, \eta \rangle = \langle \eta, eve\, eue\, \eta \rangle$$

for all Alice words u, v, including e. A vector equality means vanishing squared norm. These form a recursive list of closed rational polynomial constraints. Impose $|m_{ij}(w)| \leq 1$ directly. All genuine contraction moments obey this bound, and the ambient product of closed disks is compact. Localizing positivity gives bounded GNS effects $0 \leq E \leq 1$. One can also use the certificate $E - E^2 = (I - E)E(I - E) + E(I - E)E$.

At stage m take the first m constraints and their coordinates. Feasibility $F_{\mathrm{sf}}(G, k, m)$ is a non-strict rational semialgebraic question, decidable by real quantifier elimination [2]. Its solution set is a closed cylinder in the compact product of all moment disks. The cylinders are nested because stages retain previous constraints. If every stage is feasible, their intersection is nonempty: otherwise their open complements have a finite subcover, already making one

stage empty. A point of the intersection gives a positive matrix-valued functional. Quotient polynomial pairs by its null space and complete. Left multiplication yields the bounded GNS effects and the two vectors. Relations hold on the polynomial dense space and extend by boundedness. Corner traciality holds on the linear span of eue and then in norm on $e\mathcal{A}e$. Apply Lemma 3.8. Conversely (8) supplies some k for every semifinite strategy. Thus

$$G \in \mathrm{PERF}_{\mathrm{sf}} \iff \exists k\, \forall m\, F_{\mathrm{sf}}(G, k, m). \tag{9}$$

Set $e = 1$ for finite von Neumann algebras. □

Tensoring a finite factor host with R_{hyp} gives a II_1 host. Tensoring a semifinite factor host with $R_{\mathrm{hyp}}\bar{\otimes}B(\ell^2)$ gives a II_∞ host. Product-state expectations, or standard-form tensoring, preserve the probabilities. Factor decomposition as in Section 2 therefore gives

$$\mathrm{PERF}_{\mathrm{fin}} = \mathrm{PERF}(\mathrm{II}_1), \qquad \mathrm{PERF}_{\mathrm{sf}} = \mathrm{PERF}(\mathrm{II}) = \mathrm{PERF}(\mathrm{II}_\infty). \tag{10}$$

These are equalities of existence languages.

Remark (One parameter for both corners). In a type I factor $B(\mathcal{H})$ with its trace, the state vector is $\psi = h^{1/2}$. If e is a rank-one subprojection of the spectral projection of h for its largest eigenvalue, then $\langle e, \psi\rangle = \mathrm{Tr}(eh^{1/2}) = s_1$ is the largest Schmidt amplitude. This rank-one choice recovers the largest Schmidt amplitude from the corner overlap. General finite-corner overlaps need not equal it, and the parameters κ and k need not agree. Both upper bounds put one existential quantifier on that amount and one universal quantifier on finite stages. Without a trace there is no such corner. Commuting perfection is in Π_1^0 by the convergent hierarchy of Navascués, Pironio and Acín [45]: a NO instance eventually has an infeasible perfect-value relaxation, and feasibility at every order gives a commuting strategy by compactness. Section 12 shows that every type-III subtype has exactly this commuting language.

4 A finite compiler for normal strategies

From a presentation to a game

The spatial, semifinite and communication hardness proofs compile operator relations into one finite game. This section builds that compiler once, for normal spatial and for normal commuting strategies; the finite predicates are the same in both models. The proofs follow in Sections 4.1–4.7.

Input. An admissible presentation (Section 4.4) is a finite rational list: four contraction symbols A_α, a root projection $P_\circ$, finitely many typed centralizer labels and finitely many rational $*$-polynomial equations. Its intended solutions are normal realizations: bounded operators on a Hilbert space $\mathcal{H}_0$ with a faithful trace-class density h_0, such that the labels commute with h_0, the equations hold, and $h_0 A_\alpha = A_\alpha h_0/16$.

Output. A finite game with a rational question distribution and a 0/1 predicate. Its perfect normal spatial strategies and the normal realizations of the presentation correspond in both directions (Lemma 4.5).

Construction.

1. *Labels become unitaries.* A projection P is encoded by $2P - I$, a positive contraction T_+ by a unitary u_+ with $T_+ = (2I + u_+ + u_+^*)/4$, and a general contraction by an average of two unitaries.
2. *A scaffold carries the factor* 16. A synchronized event commutes with the marginal density, so it cannot change a density eigenvalue. Conditional flagged magic-square bridges, whose sign questions are not synchronized, supply that change: seventeen

synchronized corners and four such bridges certify partial isometries U_g with $h\mathsf{U}_g = 2\mathsf{U}_g h$ (Lemma 4.1). Their product v maps the source corner e_0 onto e_4 with $hv = 16vh$, and the cover fixes the source mass $1/121$.

3. *Labels travel through the bridges.* Corner copies, block unitaries and shared-question refinements (Lemma 4.2) force $L_4 v = vL_0$ for every transported label L.
4. *The modular operators disappear.* Each creation symbol is written $A_\alpha = v^* a_\alpha$ with a centralizer contraction a_α from e_0 to e_4. Every application equation becomes an equation among centralizer labels in which v does not occur, so every word passed to the graph compiler lies in the density centralizer.
5. *Words become projections.* A qutrit frame, graph projections and triangle projections certify actual products of unitary labels. A rational equation among words becomes a rational linear equation among synchronized projections (Section 4.2).
6. *Linear equations become Boolean tests.* Each such equation is enforced by finitely many orthogonality tests against a certified scalar matrix algebra (Lemma 4.3).
7. *Assembly.* All questions are kept, the question sets are padded, and question pairs are sampled uniformly; unspecified pairs are accepted.

Soundness. In a perfect normal strategy every losing event of positive weight has probability zero. This makes each synchronized event a projection on the state support that commutes with the marginal density, and it turns each imposed equation into an operator identity on that support. The bridges then give the realization on the source corner, with density $h_0 = 121 e_0 h e_0$.

Completeness. A realization is copied into the scaffold density (27). Contractions are written as averages of unitaries inside the finite-dimensional positive eigenspaces of that density. A qutrit frame and a finite maximally entangled row ancilla are appended, and Bob uses transposes in an eigenbasis of the density. Every test then holds exactly. Only finite-dimensional factors are added, so finite entropy is preserved.

Size and approximation. Section 4.7 bounds the loss when the application equations hold only approximately. Section 7 bounds the size of the reduction game and counts the questions of one full instance. Appendix C describes a small compiled instance with four bridges, one modular isometry and no transported labels.

Normal states, supports and exact cancellation

We first restate the bridge, label and rational-row tools from [24, Lemmas 4.1 and 4.2, Theorem B, Eq. (32)], with their proofs, and extend them from spatial strategies to normal commuting strategies, where separation in standard form replaces Schmidt-range cancellation. These arguments concern arbitrary normal supports and finite word lists. We then give the n-bridge assembly.

The proofs use only the following facts, which hold on arbitrary, possibly nonseparable, Hilbert spaces. When a marginal density has infinite rank, its inverse on its range is unbounded. Cancellations therefore go through the dense range of the marginal: by (N3), an operator that kills the state vector vanishes on the whole support.

(N1) Normal states and purifications. A normal state on $\mathcal{H}_A \otimes \mathcal{H}_B$ is a positive trace-class operator ρ of trace one. By the spectral theorem for compact operators, $\rho = \sum_j \lambda_j |\psi_j\rangle\langle\psi_j|$ with countably many $\lambda_j > 0$. Let J be the countable index set, put $\mathcal{H}_{\mathrm{env}} = \ell^2(J)$, and put $\Omega = \sum_j \lambda_j^{1/2} \psi_j \otimes |j\rangle$. Then

$$\operatorname{Tr} \rho(E \otimes F) = \langle \Omega, (E \otimes F \otimes I_{\mathrm{env}})\Omega \rangle$$

for all bounded E, F, and no measurement acts on $\mathcal{H}_{\mathrm{env}}$. The marginals $h_A = \operatorname{Tr}_B \rho$ and $h_B = \operatorname{Tr}_A \rho$ are positive trace-class operators of trace one. Write P_A, P_B for their support

projections, the projections onto the closures of their ranges; these closures are separable. The restriction of h_A to $P_A\mathcal{H}_A$ is injective with dense range. Its positive eigenspaces are finite-dimensional and span a dense subspace of $P_A\mathcal{H}_A$. The same holds for h_B.

(N2) Coefficient operators. Put $\mathcal{H}_{B,\mathrm{env}} = \mathcal{H}_B \otimes \mathcal{H}_{\mathrm{env}}$. Fix orthonormal bases (b_j) of $\mathcal{H}_B$ and (e_ℓ) of $\mathcal{H}_{\mathrm{env}}$, and use their tensor-product basis $(f_{j\ell}) = (b_j \otimes e_\ell)$ of $\mathcal{H}_{B,\mathrm{env}}$, and let $Y^{\mathsf{T}} = CY^*C$ be the transpose of $Y \in B(\mathcal{H}_{B,\mathrm{env}})$, where C is the complex conjugation fixing every $f_{j\ell}$. Bob's transpose F^{T} is defined in the basis (b_j). Transposition preserves norms, adjoints, positivity and projections, reverses products, and satisfies $(F \otimes I_{\mathrm{env}})^{\mathsf{T}} = F^{\mathsf{T}} \otimes I_{\mathrm{env}}$. The map $T \mapsto \sum_{j,\ell} Tf_{j\ell} \otimes f_{j\ell}$ is a unitary from the Hilbert–Schmidt operators $\mathcal{H}_{B,\mathrm{env}} \to \mathcal{H}_A$ onto $\mathcal{H}_A \otimes \mathcal{H}_{B,\mathrm{env}}$. If it sends T_Ω to Ω, it sends $XT_\Omega Y^{\mathsf{T}}$ to $(X \otimes Y)\Omega$, and $T_\Omega T_\Omega^* = h_A$. We call T_Ω the coefficient operator. Identities such as $ET = TF^{\mathsf{T}}$ below are coefficient forms of vector identities, and we state both forms where it helps. For a mixed state this is the cut $A \mid B\,\mathrm{env}$, and every Bob operator enters as $F \otimes I_{\mathrm{env}}$.

(N3) Exact cancellation. For $X \in B(\mathcal{H}_A)$,

$$\|(X \otimes I)\Omega\|^2 = \mathrm{Tr}(h_A X^* X) = \|Xh_A^{1/2}\|_{\mathrm{HS}}^2.$$

Hence $(X \otimes I)\Omega = 0$, equivalently $XT_\Omega = 0$, holds exactly when $XP_A = 0$. Indeed the range of $h_A^{1/2}$ is dense in $P_A\mathcal{H}_A$, and a bounded operator that vanishes on a dense subspace vanishes on its closure. Likewise $(I \otimes F \otimes I_{\mathrm{env}})\Omega = 0$, equivalently $T_\Omega(F \otimes I_{\mathrm{env}})^{\mathsf{T}} = 0$, holds exactly when $FP_B = 0$, because its squared norm is $\mathrm{Tr}(h_B F^* F)$. These are the dense-range cancellations used throughout. The same statements hold for any vector in place of Ω, with the supports of its own marginals.

(N4) Mirrors. Call $X \in B(\mathcal{H}_A)$ and $Y \in B(\mathcal{H}_B)$ *mirrors* if $(X \otimes I)\Omega = (I \otimes Y \otimes I_{\mathrm{env}})\Omega$, that is, $XT_\Omega = T_\Omega(Y \otimes I_{\mathrm{env}})^{\mathsf{T}}$.

(a) If X_k and Y_k are mirrors for $k = 1, \ldots, n$, then $X_1 \cdots X_n$ and $Y_n \cdots Y_1$ are mirrors. Operators on different tensor factors commute, so $(X_1X_2 \otimes I)\Omega = (X_1 \otimes Y_2 \otimes I)\Omega = (I \otimes Y_2Y_1 \otimes I)\Omega$; induct. Linear combinations of mirror pairs are mirror pairs.

(b) If X and X' are both mirrors of Y, then $(X - X')P_A = 0$ by (N3). If Y and Y' are both mirrors of X, then $(Y - Y')P_B = 0$.

(c) Suppose X, Y are mirrors and X^*, Y^* are mirrors. This holds for self-adjoint mirrors, and for unitary mirrors because applying $X^* \otimes Y^*$ to their mirror identity gives $(I \otimes Y^* \otimes I)\Omega = (X^* \otimes I)\Omega$. Then $Xh_A = h_A X$ and $Yh_B = h_B Y$. For $Z \in B(\mathcal{H}_A)$,

$$\mathrm{Tr}(h_A ZX) = \langle (I \otimes Y^* \otimes I)\Omega, (Z \otimes I)\Omega \rangle = \langle (X^* \otimes I)\Omega, (Z \otimes I)\Omega \rangle = \mathrm{Tr}(h_A XZ).$$

Cyclicity of the trace, valid because h_A is trace class and X, Z are bounded, gives $\mathrm{Tr}((Xh_A - h_A X)Z) = 0$. Taking Z of rank one shows that the trace-class operator $Xh_A - h_A X$ is zero. Bob's statement is symmetric.

(d) An operator commuting with h_A commutes with P_A and with every spectral projection of h_A. In particular it maps each finite-dimensional positive eigenspace of h_A into itself.

(N5) Perfect matching. Let E be an Alice effect and F a Bob effect whose two disagreement events $E \otimes (I - F)$ and $(I - E) \otimes F$ have probability zero. By the matching identity, the sum of these two probabilities is the expectation of

$$(E \otimes I - I \otimes F)^2 + (E - E^2) \otimes I + I \otimes (F - F^2),$$

a sum of three positive operators. Each expectation vanishes, so

$$(E \otimes I)\Omega = (I \otimes F \otimes I)\Omega, \qquad ((E - E^2)^{1/2} \otimes I)\Omega = 0, \qquad (I \otimes (F - F^2)^{1/2} \otimes I)\Omega = 0.$$

Thus E and F are mirrors. Next, E reduces P_A. Since $((I - P_A) \otimes I)\Omega = 0$,

$$((I - P_A)E \otimes I)\Omega = (I \otimes F \otimes I)((I - P_A) \otimes I)\Omega = 0,$$

so $(I - P_A)EP_A = 0$ by (N3), and self-adjointness gives $EP_A = P_AEP_A = P_AE$. Then (N3) gives $(E - E^2)P_A = 0$, so EP_A is a projection, and E commutes with h_A by (N4c). The same holds for F on Bob's side. We call EP_A a *synchronized projection* and mean this support-restricted operator in all local identities. Reduction of the support is proved before projectivity is used.

(N6) Compressions and joint measurements.

(a) If $0 \le Y \le Z$ with Z a projection, then $Y = ZYZ$. Indeed $0 \le (I - Z)Y(I - Z) \le (I - Z)Z(I - Z) = 0$, so $Y^{1/2}(I - Z) = 0$.

(b) *Projective compression.* If P is a projection, E an effect and PEP a projection, then E reduces $P\mathcal{H}$. For $u \in \mathrm{ran}(PEP)$, $\langle u, (I - E)u\rangle = 0$ and positivity give $Eu = u$. For $u \in P\mathcal{H}$ with $PEPu = 0$, $\langle u, Eu\rangle = 0$ gives $Eu = 0$. Thus E maps $P\mathcal{H}$ into itself, and self-adjointness gives reduction.

(c) *Only compressions matter.* Since $(P_A \otimes I)\Omega = \Omega = (I \otimes P_B \otimes I)\Omega$, every probability equals $\langle \Omega, (P_AEP_A \otimes P_BFP_B \otimes I)\Omega\rangle$. Identities may therefore be proved first for compressed effects. Changing effects outside the supports, while keeping them POVMs, changes no probability.

(d) *Sharp marginals.* Let (E_c) be a finite POVM indexed by tuples $c = (c_1, \dots, c_m)$ whose compressions to a reducing subspace $P\mathcal{H}$ have projections as coarse marginals $Q^k_s = \sum_{c:\, c_k = s} E_c$. By (a), each compressed E_c satisfies $E_c = Q^k_{c_k} E_c Q^k_{c_k}$, hence $E_c Q^k_s = \delta_{s,c_k} E_c$. Summing, $Q^1_{s_1} Q^2_{s_2} = \sum_{c_1 = s_1,\, c_2 = s_2} E_c$, which is self-adjoint, so the marginals commute. Iterating, $Q^1_{s_1} \cdots Q^m_{s_m}$ is the compressed effect of the outcome $(s_1, \dots, s_m)$, or zero if there is no such outcome. Each compressed fine effect is thus a product of commuting projections, hence a projection, and (b) makes the original fine effect reduce $P\mathcal{H}$.

(N7) Forbidden events. Let Q be an Alice effect and R a Bob effect whose compression $R_B = P_BRP_B$ is a synchronized projection with Alice mirror R_A. If the event $Q \otimes R$ has probability zero, then $QR_AP_A = 0$. If Q also reduces P_A, then $R_AQP_A = 0$ as well. Indeed $0 = \langle \Omega, (Q \otimes R_B \otimes I)\Omega\rangle = \|(Q^{1/2} \otimes R_B \otimes I)\Omega\|^2$, and $(Q^{1/2} \otimes R_B \otimes I)\Omega = (Q^{1/2}R_A \otimes I)\Omega$, so (N3) gives $Q^{1/2}R_AP_A = 0$. The second claim is the adjoint of the first, because P_A commutes with Q and R_A.

(N8) Canonical purifications for completeness. Let h be a faithful positive trace-class operator of trace one on $\mathcal{H}$. By (N1), $\mathcal{H}$ is separable and has an orthonormal basis (φ_i) with $h\varphi_i = \lambda_i\varphi_i$ and $\lambda_i > 0$. Put $\Omega_h = \sum_i \lambda_i^{1/2} \varphi_i \otimes \varphi_i$, and let X^{T} denote the transpose in (φ_i). Both marginals of Ω_h equal h. If X commutes with h, it commutes with $h^{1/2}$, and the rule of (N2) gives

$$(I \otimes X^{\mathsf{T}})\Omega_h = (X \otimes I)\Omega_h,$$

since both vectors correspond to $h^{1/2}X = Xh^{1/2}$. The map $X \mapsto X^{\mathsf{T}}$ is linear, reverses products, and preserves adjoints, positivity and projections; and X^{T} commutes with $h^{\mathsf{T}} = h$. Hence a centralizer projection X for Alice and X^{T} for Bob are perfectly matched: their disagreement probability is $\langle \Omega_h, ((X(I - X) + (I - X)X) \otimes I)\Omega_h\rangle = 0$. For centralizer effects Q of Alice and R^{T} of Bob, the joint probability is $\mathrm{Tr}(hQR)$, so an event with $QR = 0$ has probability zero.

4.1 The normal flagged magic-square bridge

The magic-square relations originate with Mermin and Peres [42, 53]. We use Mančinska's flagged magic-square (FMS) geometry [39]. We spell out the predicate and prove its normal extension because the published finite-dimensional statement does not suffice here. All FMS question and answer indices start at zero; frame indices below start at one.

4.1.1 Questions, calibration and exact bindings

Alice has $\mathsf{R}_0, \mathsf{R}_1, \mathsf{R}_2, \mathsf{T}_0, \mathsf{T}_1$ and Bob has $\mathsf{C}_0, \mathsf{C}_1, \mathsf{C}_2, \mathsf{B}_0, \mathsf{B}_1$. A row question has a flag answer f or a triple of product $+1$. A column question has f or a triple of product $+1, +1, -1$ according to its column. The magic square is

$$\begin{pmatrix} XI & IX & XX \\ IZ & ZI & ZZ \\ XZ & ZX & YY \end{pmatrix}. \tag{11}$$

Here X, Y, Z denote the Pauli matrices, I the 2×2 identity, and adjacent letters their tensor product. The displayed matrices describe a canonical realization; the predicate only checks the corresponding signs. Select column-zero outcomes $c_0 = (+++)$, $c_1 = (+--)$ and row-zero outcomes $r_0 = (+++)$, $r_1 = (+--)$. Question T_x has answers $+, -$ and the three column-zero triples other than c_x; B_y analogously retains all row-zero triples other than r_y.

Table 3 and Figure 2 summarize the rules. On $\mathsf{R}_i, \mathsf{C}_j$, reject flag/nonflag mismatch and otherwise check the intersecting cell; two flags are accepted. On $\mathsf{T}_x, \mathsf{C}_0$, a sign is allowed only against f or c_x; a retained triple must equal the opposite triple. On $\mathsf{R}_0, \mathsf{B}_y$ use the analogous rule with r_y. On $\mathsf{T}_x, \mathsf{B}_y$, reject two signs $a, b \in \{+1, -1\}$ unless $ab = (-1)^{xy}$; all pairs with a nonsign answer are accepted. Other question pairs are accepted. For a conditional instance add an inactive answer $\perp$, accept two inactive answers, and reject exactly one inactive answer. Bind each noninactive event to its synchronized active projection and all row/column flags to the synchronized flag.

Let P_{mag} be the nonflag magic support. Its four column-zero projections $m_{00}, m_{01}, m_{10}, m_{11}$ correspond respectively to $(+++), (+--), (-+-), (--+)$. On Alice, m_{00} is read from the retained $(+++)$ answer of T_1 and the other three from T_0. On Bob all four come from C_0. Each of these chosen occurrences is matched to its opposite synchronized binary question. The positive nonflag ZI event is read from Alice's R_1 component 1 and Bob's C_1 component 1. The positive IX event comes from Alice's R_0 component 1 and Bob's C_1 component 0. Bind both positive nonflag events to synchronized binary questions, denoted $E_{ZI,+}$ and $E_{IX,+}$. Put

$$\mathfrak{x} = I - 2P_{\mathrm{mag}} + 2E_{ZI,+}, \quad \mathfrak{z} = I - 2P_{\mathrm{mag}} + 2E_{IX,+}, \quad t_{ab} = \mathfrak{x}^a \mathfrak{z}^b m_{00}.$$

These are reflections extended by identity outside P_{mag}, and $t_{ab}^* t_{ab} = m_{00}$, $t_{ab} t_{ab}^* = m_{ab}$.

Questions	Answers	Tested rule
$\mathsf{R}_i, \mathsf{C}_j$	flag or parity triple	flags match; nonflags agree at (i, j)
$\mathsf{T}_x, \mathsf{C}_0$	sign or retained triple	signs pair only with f, c_x; retained triples match
$\mathsf{R}_0, \mathsf{B}_y$	triple or sign	symmetric calibration at r_y
$\mathsf{T}_x, \mathsf{B}_y$	signs or triples	two signs satisfy $ab = (-1)^{xy}$
Conditional copies	also inactive $\perp$	activity matches; flags and events bind to named projections

Table 3: Flagged magic-square questions, answers and rules. Unspecified pairs are accepted.

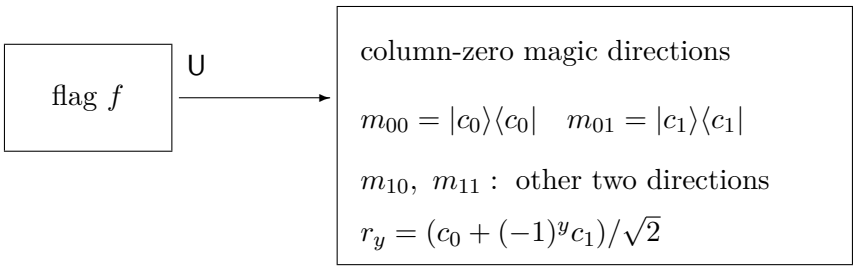


Figure 2: Selected column and row directions. The flag transports to c_0; m_{ab} denote projections.

4.1.2 Normal bridge theorem

Lemma 4.1. *A perfect normal conditional FMS strategy has on its active supports the decomposition*

$$\mathcal{H}_{\text{active}} = \mathcal{H}_f \oplus (\mathbb{C}^4 \otimes \mathcal{H}_{\text{mag}}), \qquad h = h_f \oplus (I_4 \otimes h_{\text{mag}}).$$

The calibrated transports $\mathsf{U}_A, \mathsf{U}_B$ from the flag to the selected c_0 corner satisfy $\mathsf{U}^*\mathsf{U} = f$, $\mathsf{U}\mathsf{U}^* = m_{00}$ and $h\mathsf{U} = 2\mathsf{U}h$. On a purification their matched components satisfy

$$\Omega_{c_0 c_0} = \sqrt{2}(\mathsf{U}_A \otimes \mathsf{U}_B \otimes I_{\text{env}})\Omega_{ff}. \tag{12}$$

The flag answer and its projection are both denoted f. The other three magic directions obey the transported versions of this equation.

The proof first extracts the magic matrix algebra from the synchronized cells. Calibration then identifies the two binary sign supports. The four sign equations force an off-diagonal unitary between the flag and a magic corner, and commutation with the calibrated marginal gives the density ratio two.

Proof. Fix a purification Ω as in (N1). The predicate makes the following pairs of events perfectly matched in the sense of (N5): an inactive answer on one side and the inactive answer on the other; each noninactive event and the opposite synchronized activity projection; each row or column flag and the opposite synchronized flag; and, for every cell (i, j) and sign s, Alice's event "R_i is not a flag and its entry j equals s" and Bob's event "C_j is not a flag and its entry i equals s". The last pair is matched because the rule on $\mathsf{R}_i, \mathsf{C}_j$ rejects a flag against a nonflag and checks the intersecting cell otherwise. By (N5), each coarse effect in this list reduces its party's support, its compression is a projection commuting with that party's marginal density, and matched compressions are mirrors. Let a_A and f_A be Alice's activity and flag projections, put $N_A = a_A - f_A$, and let Q^s_{ij} be the compression of the cell event above, so that $Q^+_{ij} + Q^-_{ij} = N_A$. The cell observables $\mathsf{M}^A_{ij} = Q^+_{ij} - Q^-_{ij}$ are self-adjoint unitaries on $N_A P_A \mathcal{H}_A$. Each is a mirror of Bob's operator M^B_{ij}, built in the same way from C_j.

For a nonflag answer $t = (t_0, t_1, t_2)$ of R_i, the compressed effect lies below $Q^{t_j}_{ij}$ for each j, since the other nonflag effects of R_i with entry j equal to t_j are positive. By (N6d) it equals $Q^{t_0}_{i0} Q^{t_1}_{i1} Q^{t_2}_{i2}$, the three cell observables of row i commute, and the original fine effect reduces P_A. Summing over the triples of product $+1$ gives $\mathsf{M}^A_{i0}\mathsf{M}^A_{i1}\mathsf{M}^A_{i2} = N_A$ on $P_A\mathcal{H}_A$. The same argument for Bob's columns gives commuting $\mathsf{M}^B_{0j}, \mathsf{M}^B_{1j}, \mathsf{M}^B_{2j}$ with product $\epsilon_j N_B$, where $\epsilon_0 = \epsilon_1 = 1$ and $\epsilon_2 = -1$. By (N4a), $\mathsf{M}^A_{0j}\mathsf{M}^A_{1j}\mathsf{M}^A_{2j}$ is a mirror of $\mathsf{M}^B_{2j}\mathsf{M}^B_{1j}\mathsf{M}^B_{0j} = \epsilon_j N_B$, which is a mirror of $\epsilon_j N_A$. Then (N4b) gives $\mathsf{M}^A_{0j}\mathsf{M}^A_{1j}\mathsf{M}^A_{2j} = \epsilon_j N_A$ on $P_A\mathcal{H}_A$. In the same way $\mathsf{M}^A_{ij}\mathsf{M}^A_{i'j}$ and $\mathsf{M}^A_{i'j}\mathsf{M}^A_{ij}$ are mirrors of the equal operators $\mathsf{M}^B_{i'j}\mathsf{M}^B_{ij}$ and $\mathsf{M}^B_{ij}\mathsf{M}^B_{i'j}$, so they agree on $P_A\mathcal{H}_A$. The nine cell observables therefore satisfy every row and column relation of (11) as operator identities on $N_A P_A \mathcal{H}_A$. The symmetric argument gives the same for Bob on $N_B P_B \mathcal{H}_B$.

Next comes the Pauli-pair calculation. Write $a = \mathsf{M}^A_{00}, b = \mathsf{M}^A_{01}, c = \mathsf{M}^A_{10}, d = \mathsf{M}^A_{11}$ for the four top-left cell observables of the square. Context relations give

$$[a,b] = [c,d] = [a,c] = [b,d] = 0, \qquad (ac)(bd) = -(ab)(cd).$$

Cancel the outside a, d to obtain $cb = -bc$. Commutation of ac and bd then gives $da = -ad$. Thus (a,d) and (b,c) are anticommuting pairs, and each commutes with the other pair. Put $\mathcal{N}_A = N_A P_A \mathcal{H}_A$, and let $\mathcal{N}_\pm$ be the ± 1 eigenspaces of a, so that $\mathcal{N}_A = \mathcal{N}_+ \oplus \mathcal{N}_-$. Since $da = -ad$, d maps $\mathcal{N}_\pm$ into $\mathcal{N}_\mp$, and $d^2 = I$ makes $d|_{\mathcal{N}_+}$ unitary onto $\mathcal{N}_-$. This identifies $\mathcal{N}_A$ with $\mathbb{C}^2 \otimes \mathcal{N}_+$, with $a = Z \otimes I$ and $d = X \otimes I$. The operators b, c commute with a and d, hence with the algebra $\mathrm{Mat}_2 \otimes I$ they generate, so $b = I \otimes b'$ and $c = I \otimes c'$ for anticommuting self-adjoint unitaries b', c'. Repeating the argument for c', b' gives $\mathcal{N}_+ = \mathbb{C}^2 \otimes \mathcal{H}_{\mathrm{mag}}$. The space $\mathcal{H}_{\mathrm{mag}}$ may be infinite-dimensional; it is separable because it lies in $P_A \mathcal{H}_A$.

For later use we record the identification through the units t_{ab}. The projection m_{00} is the joint $(+1,+1)$ eigenprojection of the column-zero cells $a = \mathsf{M}_{00}$ and $c = \mathsf{M}_{10}$, and m_{ab} is their joint $((-1)^a, (-1)^b)$ eigenprojection; this is the stated correspondence with $(++), (+-), (-+), (--)$. On $\mathcal{N}_A$ the reflection $\mathfrak{x}$ equals $d = \mathsf{M}_{11}$, which anticommutes with a and commutes with c, and $\mathfrak{z}$ equals $b = \mathsf{M}_{01}$, which commutes with a and anticommutes with c. Hence $t_{ab} = \mathfrak{x}^a \mathfrak{z}^b m_{00}$ maps $m_{00}\mathcal{N}_A$ isometrically onto $m_{ab}\mathcal{N}_A$, and

$$U_A : \mathbb{C}^4 \otimes \mathcal{H}^A_{\mathrm{mag}} \to \mathcal{N}_A, \qquad U_A(u_{ab} \otimes \xi) = t_{ab}\xi, \qquad \mathcal{H}^A_{\mathrm{mag}} = m_{00}\mathcal{N}_A,$$

is unitary. With $u_{ab} = |a\rangle \otimes |b\rangle$, these coordinates give $m_{ab} = |u_{ab}\rangle\langle u_{ab}| \otimes I$, $a = Z \otimes I$, $c = I \otimes Z$, $d = X \otimes I$ and $b = I \otimes X$. The operators $\mathfrak{x}, \mathfrak{z}, m_{00}$ are built from synchronized projections, so they commute with h_A by (N5), and so does each t_{ab}. Therefore $U_A^* h_A U_A = I_4 \otimes h^A_{\mathrm{mag}}$, where h^A_{mag} is the restriction of h_A to $\mathcal{H}^A_{\mathrm{mag}}$. Together with $[f_A, h_A] = 0$ this is the decomposition in the lemma, with $\mathcal{H}_f = f_A P_A \mathcal{H}_A$. Both blocks are faithful, because h_A is faithful on $P_A\mathcal{H}_A$ and they are its restrictions to reducing subspaces. The same construction on Bob's side gives U_B and h^B_{mag}.

Now the coefficient blocks. Flags and activity are matched, so $(f_A \otimes I)\Omega = (I \otimes f_B \otimes I)\Omega$ and $(N_A \otimes I)\Omega = (I \otimes N_B \otimes I)\Omega$. Put $\Omega_{ff} = (f_A \otimes f_B \otimes I)\Omega$ and $\Omega_{NN} = (N_A \otimes N_B \otimes I)\Omega$. The mixed components vanish; for example $(f_A \otimes N_B \otimes I)\Omega = (f_A N_A \otimes I)\Omega = 0$. Each of $\mathfrak{x}, \mathfrak{z}, m_{00}$ is a linear combination of matched events, hence a mirror of its Bob counterpart by (N4a); for $\mathfrak{x}$ the events are the activity, the flag and the positive ZI event, which the rule on $\mathsf{R}_1, \mathsf{C}_1$ matches. The reflections are self-adjoint, so (N4a) makes t^A_{ab} a mirror of $(t^B_{ab})^*$, and the matrix unit $t^A_{ab}(t^A_{a'b'})^*$ a mirror of $t^B_{a'b'}(t^B_{ab})^*$. In the coordinates $U_A \otimes U_B$ write $\Omega_{NN} = \sum_{k,l} u_k \otimes u_l \otimes \omega_{kl}$, where k, l range over the four labels ab and $\omega_{kl} \in \mathcal{H}^A_{\mathrm{mag}} \otimes \mathcal{H}^B_{\mathrm{mag}} \otimes \mathcal{H}_{\mathrm{env}}$. The mirror identity for the units $|u_k\rangle\langle u_l|$ and $|u_l\rangle\langle u_k|$ reads

$$\sum_{l'} u_k \otimes u_{l'} \otimes \omega_{ll'} = \sum_{k'} u_{k'} \otimes u_l \otimes \omega_{k'k},$$

so $\omega_{ll'} = 0$ for $l' \neq l$ and $\omega_{ll} = \omega_{kk}$. Hence

$$\Omega_{NN} = \sum_k u_k \otimes u_k \otimes \omega_{\mathrm{mag}}$$

for a single vector ω_{mag}, whose two marginals are h^A_{mag} and h^B_{mag}. In coefficient form, the coefficient operator of Ω_{NN} is $I_4 \otimes T_{\mathrm{mag}}$, where T_{mag} is that of ω_{mag}, and T_f is that of Ω_{ff}. Here $T_{\mathrm{mag}} : \mathcal{H}^B_{\mathrm{mag}} \otimes \mathcal{H}_{\mathrm{env}} \to \mathcal{H}^A_{\mathrm{mag}}$ and $T_f : \mathcal{H}^B_f \otimes \mathcal{H}_{\mathrm{env}} \to \mathcal{H}^A_f$ are the coefficient maps in the fixed product coordinates. They satisfy $T_{\mathrm{mag}} T^*_{\mathrm{mag}} = h^A_{\mathrm{mag}}$ and $T_f T_f^* = h^A_f$, and have dense range. All cancellations below use (N3) on the corresponding original local support.

Now the sign questions. For a column-zero triple c, let Π_c be the joint spectral projection of Alice's three cell reflections in column zero, and Π_r the joint projection of Bob's row-zero cells; put $\epsilon_{xy} = (-1)^{xy}$. The rule on $\mathsf{T}_x, \mathsf{C}_0$ matches Alice's coarse sign event of T_x with Bob's event $\{f, c_x\}$ of C_0, and each retained triple of T_x with the same triple of C_0. Bob's C_0 projection for a triple is the joint projection of his column-zero cells, and by (N4a) its mirror is Alice's joint projection Π_c of her cells $\mathsf{M}^A_{00}, \mathsf{M}^A_{10}, \mathsf{M}^A_{20}$. Thus (N4b) gives $S^A_x = f_A + \Pi_{c_x}$ on $P_A\mathcal{H}_A$, and the retained effects of T_x are the other column-zero projections. Symmetrically, the rule on $\mathsf{R}_0, \mathsf{B}_y$ gives $S^B_y = f_B + \Pi_{r_y}$, where Π_{r_y} is Bob's joint projection of his row-zero cells. In the coordinates above, $c_0 = u_{00}$ and $c_1 = u_{01}$, while $r_y = (u_{00} + (-1)^y u_{01})/\sqrt{2}$, because row zero consists of $a = Z \otimes I$, $b = I \otimes X$ and ab. Hence $\langle c_x, r_y\rangle = \epsilon_{xy}/\sqrt{2}$, and

$$\Omega_{xy} := (S^A_x \otimes S^B_y \otimes I)\Omega = \Omega_{ff} + \frac{\epsilon_{xy}}{\sqrt{2}}\, c_x \otimes r_y \otimes \omega_{\mathrm{mag}}.$$

The maps $V^A_x(\xi \oplus \eta) = \xi + c_x \otimes \eta$ and $V^B_y(\xi' \oplus \eta') = \xi' + r_y \otimes \eta'$ identify $\mathcal{H}_f \oplus \mathcal{H}^A_{\mathrm{mag}}$ and $\mathcal{H}^B_f \oplus \mathcal{H}^B_{\mathrm{mag}}$ with the two sign supports. Under them Ω_{xy} becomes $\Omega_{ff} + \epsilon_{xy}\omega_{\mathrm{mag}}/\sqrt{2}$, with the two terms in the flag-flag and the magic-magic blocks; its coefficient operator is

$$W_{xy} = \mathrm{diag}(T_f, \epsilon_{xy} T_{\mathrm{mag}}/\sqrt{2}).$$

Its Alice marginal is $h_f \oplus h^A_{\mathrm{mag}}/2$ and its Bob marginal is $h^B_f \oplus h^B_{\mathrm{mag}}/2$; both are faithful.

Let $E'_a = P_A E_a P_A$ and $F'_b = P_B F_b P_B$ be the compressed sign effects of T_x and B_y. Then $0 \le E'_a \le S^A_x$ and $0 \le F'_b \le S^B_y$, so $E'_a = E'_a S^A_x$ and $F'_b = F'_b S^B_y$ by (N6a). If $ab \neq \epsilon_{xy}$, the pair is rejected, so

$$0 = \langle \Omega, (E'_a \otimes F'_b \otimes I)\Omega\rangle = \|(E'^{1/2}_a \otimes F'^{1/2}_b \otimes I)\Omega\|^2,$$

whence $(E'_a \otimes F'_b \otimes I)\Omega = 0$; in coefficient form, $E^{1/2}_a W_{xy} (F^T_b)^{1/2} = 0$ and $E_a W_{xy} F^T_b = 0$. Writing $\epsilon = \epsilon_{xy}$ and summing the complete binary effects,

$$(E'_a \otimes I)\Omega_{xy} = (E'_a \otimes S^B_y \otimes I)\Omega = (E'_a \otimes F'_{\epsilon a} \otimes I)\Omega = (S^A_x \otimes F'_{\epsilon a} \otimes I)\Omega = (I \otimes F'_{\epsilon a} \otimes I)\Omega_{xy}.$$

Next, $(E'_a E'_{-a} \otimes I)\Omega_{xy} = (E'_a \otimes F'_{-\epsilon a} \otimes I)\Omega_{xy} = (E'_a \otimes F'_{-\epsilon a} \otimes I)\Omega = 0$. Since $E'_{-a} = S^A_x - E'_a$, this says $((E'_a - E'^2_a) \otimes I)\Omega_{xy} = 0$. The Alice marginal of Ω_{xy} is faithful on $S^A_x P_A \mathcal{H}_A$, so (N3), applied to Ω_{xy}, gives $E'^2_a = E'_a$. This is where dense range enters. Bob's case is symmetric. By (N6b) the original sign effects reduce the supports. The differences $\mathcal{X}_x = E'_+ - E'_-$ and $\mathcal{Y}_y = F'_+ - F'_-$ are self-adjoint unitaries on $S^A_x P_A \mathcal{H}_A$ and $S^B_y P_B \mathcal{H}_B$, and

$$(\mathcal{X}_x \otimes I)\Omega_{xy} = \epsilon_{xy}(I \otimes \mathcal{Y}_y \otimes I)\Omega_{xy}.$$

In coefficient form this reads

$$\mathcal{X}_x W_{xy} = \epsilon_{xy} W_{xy} \mathcal{Y}^T_y. \tag{13}$$

Set $T_{\mathrm{cal}} = \mathrm{diag}(T_f, T_{\mathrm{mag}}/\sqrt{2})$ and $J_{\mathrm{sgn}} = \mathrm{diag}(I, -I)$, so that $W_{xy} = T_{\mathrm{cal}}$ for $(x, y) \neq (1, 1)$ and $W_{11} = J_{\mathrm{sgn}} T_{\mathrm{cal}}$. Put $\widehat{\mathcal{X}}_x = V^{A*}_x \mathcal{X}_x V^A_x$ and $\widehat{\mathcal{Y}}_y = V^{B*}_y \mathcal{Y}_y V^B_y$, and let $\Omega_{\mathrm{cal}} = \Omega_{ff} + \omega_{\mathrm{mag}}/\sqrt{2}$, whose coefficient operator is T_{cal}. For $(x, y) \neq (1, 1)$ the vector Ω_{xy} becomes Ω_{cal}, and for $(1, 1)$ it becomes $(J_{\mathrm{sgn}} \otimes I)\Omega_{\mathrm{cal}} = (I \otimes J_{\mathrm{sgn}} \otimes I)\Omega_{\mathrm{cal}}$. The cases $(0, 0)$ and $(1, 0)$ give $((\widehat{\mathcal{X}}_0 - \widehat{\mathcal{X}}_1) \otimes I)\Omega_{\mathrm{cal}} = 0$, and the cases $(0, 0)$ and $(0, 1)$ give $(I \otimes (\widehat{\mathcal{Y}}_0 - \widehat{\mathcal{Y}}_1) \otimes I)\Omega_{\mathrm{cal}} = 0$. Both marginals of Ω_{cal}, namely $h_f \oplus h^A_{\mathrm{mag}}/2$ and $h^B_f \oplus h^B_{\mathrm{mag}}/2$, are faithful, so (N3) gives $\widehat{\mathcal{X}}_0 = \widehat{\mathcal{X}}_1 = \mathcal{X}$ and $\widehat{\mathcal{Y}}_0 = \widehat{\mathcal{Y}}_1 = \mathcal{Y}$. These are applications of (N3) to the two original local marginals. The case $(1, 1)$ then gives

$$(\mathcal{X} J_{\mathrm{sgn}} \otimes I)\Omega_{\mathrm{cal}} = -(I \otimes \mathcal{Y} \otimes I)(J_{\mathrm{sgn}} \otimes I)\Omega_{\mathrm{cal}} = -(J_{\mathrm{sgn}} \mathcal{X} \otimes I)\Omega_{\mathrm{cal}},$$

so $\mathcal{X}J_{\rm sgn} + J_{\rm sgn}\mathcal{X} = 0$ by (N3). A self-adjoint unitary anticommuting with $J_{\rm sgn}$ has zero diagonal blocks, so

$$\mathcal{X} = \begin{pmatrix} 0 & \beta^* \\ \beta & 0 \end{pmatrix},$$

and $\mathcal{X}^2 = I$ gives $\beta^*\beta = I_{\mathcal{H}_f}$ and $\beta\beta^* = I_{\mathcal{H}_{\rm mag}}$: the block $\beta : \mathcal{H}_f \to \mathcal{H}_{\rm mag}$ is unitary. The operators $\mathcal{X}, \mathcal{Y}$ are self-adjoint mirrors for $\Omega_{\rm cal}$, so (N4c) gives $[\mathcal{X}, h_f \oplus h^A_{\rm mag}/2] = 0$. In coefficient form: from $\mathcal{X}T_{\rm cal} = T_{\rm cal}\mathcal{Y}^T$, the operator $\mathcal{X}T_{\rm cal}T^*_{\rm cal} = T_{\rm cal}\mathcal{Y}^T T^*_{\rm cal}$ is self-adjoint, and $T_{\rm cal}T^*_{\rm cal} = h_f \oplus h^A_{\rm mag}/2$ with $h_f = T_f T_f^*$ and $h^A_{\rm mag} = T_{\rm mag}T^*_{\rm mag}$. Its off-diagonal block is $\beta h_f = \frac{1}{2}h^A_{\rm mag}\beta$, so $h^A_{\rm mag}\beta = 2\beta h_f$: the corner transport has the required ratio. On Bob's side the same argument gives a unitary β_B with $h^B_{\rm mag}\beta_B = 2\beta_B h^B_f$.

The actual Alice extraction is $\mathsf{U}_A = \Pi_{c_0}\mathcal{X}_0 f$. Bob's r_0 projection is the retained $(+++)$ answer of B_1, equivalently the sign support of B_0 minus its flag. His extraction to the same c_0 direction is

$$\mathsf{U}_B = \sqrt{2}\Pi_{c_0}\Pi_{r_0}\mathcal{Y}_0 f.$$

The coefficient $\sqrt{2}$ enters only the extraction; the predicate stays rational.

In the coordinates V_0^A, U_A is the map $\xi \mapsto c_0 \otimes \beta\xi$ from $\mathcal{H}_f$ into $m_{00}P_A\mathcal{H}_A$. Thus $\mathsf{U}_A^*\mathsf{U}_A = f_A$ and $\mathsf{U}_A\mathsf{U}_A^* = m_{00}$ on the support, and $h_A\mathsf{U}_A = 2\mathsf{U}_A h_A$, because h_A acts as h_f on the flag and as $h^A_{\rm mag}$ on $c_0 \otimes \mathcal{H}^A_{\rm mag}$. On Bob's side $\mathcal{Y}_0 f_B$ is $\xi' \mapsto r_0 \otimes \beta_B\xi'$, and $\sqrt{2}\Pi_{c_0}\Pi_{r_0}$ restricts to the isometry $r_0 \otimes \eta \mapsto c_0 \otimes \eta$ from $\Pi_{r_0}P_B\mathcal{H}_B$ onto $m_{00}P_B\mathcal{H}_B$, since $\langle c_0, r_0\rangle = 2^{-1/2}$. So $\mathsf{U}_B = c_0 \otimes \beta_B$ has the same three properties. The components in $\mathcal{H}^A_{\rm mag} \otimes \mathcal{H}^B_f \otimes \mathcal{H}_{\rm env}$ of the two sides of $(\mathcal{X} \otimes I)\Omega_{\rm cal} = (I \otimes \mathcal{Y} \otimes I)\Omega_{\rm cal}$ give $(\beta \otimes I)\Omega_{ff} = (I \otimes \beta_B^* \otimes I)\omega_{\rm mag}/\sqrt{2}$, that is, $\omega_{\rm mag} = \sqrt{2}(\beta \otimes \beta_B \otimes I)\Omega_{ff}$. Since $\Omega_{c_0c_0} = c_0 \otimes c_0 \otimes \omega_{\rm mag}$, this is (12). The formula for Ω_{NN} also gives $\Omega_{m_{ab}m_{ab}} = (t^A_{ab} \otimes t^B_{ab} \otimes I)\Omega_{c_0c_0}$, and the components for two different magic directions vanish. These are the transported versions for the other three directions.

Mixed states are covered by the same calculation. It used the purification Ω on the cut $A \mid B\,\mathrm{env}$, with every Bob operator of the form $F \otimes I_{\rm env}$. The synchronized opposite effects reduce the Schmidt support, and the sign effects do so once their compressed sharpness is proved, so every compressed product above is the restriction of an original Bob-only product tensored with $I_{\rm env}$. Bob's identities come from the cut $B \mid A\,\mathrm{env}$, the Bob form of (N3). The supports P_A, P_B are those of $\mathrm{Tr}_B\,\rho$ and $\mathrm{Tr}_A\,\rho$, so every identity holds on the original marginal supports, and (12) holds on the original purification with $\mathsf{U}_A \otimes \mathsf{U}_B \otimes I_{\rm env}$. The partial trace over $\mathcal{H}_{\rm env}$ of the rank-one operators of its two sides gives $(m_{00} \otimes m_{00})\rho(m_{00} \otimes m_{00}) = 2(\mathsf{U}_A \otimes \mathsf{U}_B)(f \otimes f)\rho(f \otimes f)(\mathsf{U}_A \otimes \mathsf{U}_B)^*$. □

4.1.3 The commuting form

On a normal commuting host the same predicate calibrates a modular eigenoperator. We recall the arbitrary-normal bridge argument of [20, Lemma 2.2]. Matching makes the cells sharp centralizer effects. Write the upper-left four cell reflections as a, b, c, d. The row/column relations give $[a, b] = [a, c] = [b, d] = [c, d] = 0$, $(ac)(bd) = -(ab)(cd)$, and $[ac, bd] = 0$. Canceling the outside a, d gives $cb = -bc$; substitution into the last commutator gives $ad = -da$. The two commuting Pauli pairs therefore generate centralizer Mat_4. In $\mathbb{C}^2 \otimes \mathbb{C}^2$ the calibrated unit vectors are $\chi_x = |+\rangle \otimes |x\rangle$ and $\rho^{\rm dir}_y = |+\rangle \otimes |(-1)^y\rangle$, where $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and $|(-1)^y\rangle = (|0\rangle + (-1)^y|1\rangle)/\sqrt{2}$. Choose a minimal projection $m_{\rm min}$ of the magic matrix algebra Mat_4 and centralizer partial isometries u^A_x, u^B_y with initial projection $m_{\rm min}$ and ranges the projections onto $\chi_x, \rho^{\rm dir}_y$ (tensored with the multiplicity identity). Choose their phases so that their overlap is $(u^A_x)^* u^B_y = (-1)^{xy} m_{\rm min}/\sqrt{2}$. If Σ^A_x, Σ^B_y are the original sign differences, their pull-backs on the standard corner $(f + m_{\rm min})\mathcal{J}(f + m_{\rm min})\mathcal{J}L^2(M)$ are

$$\mathcal{X}_x = (f + (u^A_x)^*)\Sigma^A_x(f + u^A_x), \qquad \mathcal{Y}_y = \mathcal{J}(f + (u^B_y)^*)\mathcal{J}\Sigma^B_y\mathcal{J}(f + u^B_y)\mathcal{J}.$$

They are commuting self-adjoint contractions. Their joint-support vector is

$$\theta_{xy} = (f + (-1)^{xy} m_{\min}/\sqrt{2})\xi_\varphi.$$

Forbidden sign events give $\mathcal{X}_x\theta_{xy} = (-1)^{xy}\mathcal{Y}_y\theta_{xy}$ and $\mathcal{X}_x\mathcal{Y}_y\theta_{xy} = (-1)^{xy}\theta_{xy}$. The multiplier $f \pm m_{\min}/\sqrt{2}$ is bounded invertible in the corner centralizer, so θ_{xy} separates both corner algebras. Commuting the opposite contraction through the first identity gives $\mathcal{X}_x^2\theta_{xy} = \theta_{xy}$ and $\mathcal{Y}_y^2\theta_{xy} = \theta_{xy}$; separation makes them reflections. Put $c_{\rm cal} = f + m_{\min}/\sqrt{2}$. The three positive cells give $\mathcal{X}c_{\rm cal}\xi_\varphi = \mathcal{Y}c_{\rm cal}\xi_\varphi$ and identify $\mathcal{X}_0 = \mathcal{X}_1 = \mathcal{X}$ and $\mathcal{Y}_0 = \mathcal{Y}_1 = \mathcal{Y}$ by separation on $c_{\rm cal}\xi_\varphi$. The fourth vector is $J_{\rm sgn}c_{\rm cal}\xi_\varphi$, $J_{\rm sgn} = f - m_{\min}$. Its equation is $\mathcal{X}J_{\rm sgn}c_{\rm cal}\xi_\varphi = -\mathcal{Y}J_{\rm sgn}c_{\rm cal}\xi_\varphi = -J_{\rm sgn}\mathcal{Y}c_{\rm cal}\xi_\varphi = -J_{\rm sgn}\mathcal{X}c_{\rm cal}\xi_\varphi$, since opposite algebras commute. Separation gives $\mathcal{X}J_{\rm sgn} = -J_{\rm sgn}\mathcal{X}$. Thus $\mathcal{X} = \mathsf{U} + \mathsf{U}^*$, $\mathsf{U} = m_{\min}\mathcal{X}f$, $\mathsf{U}^*\mathsf{U} = f$, $\mathsf{U}\mathsf{U}^* = m_{\min}$. The mirror equation means that $\mathcal{X}$ centralizes $\omega(\cdot) = \varphi(d_{\rm cal}\,\cdot)$, $d_{\rm cal} = f + m_{\min}/2$. The bounded centralizer perturbation formula of Pedersen–Takesaki [52, Lemma 4.5] $\sigma_t^\omega(x) = d_{\rm cal}^{it}\sigma_t^\varphi(x)d_{\rm cal}^{-it}$ gives on the $m_{\min}$–f block

$$\sigma_t^\varphi(\mathsf{U}) = 2^{it}\mathsf{U}, \qquad \varphi(m_{\min}) = 2\varphi(f). \tag{14}$$

The last equality is KMS. Multiplication by the magic matrix units gives all four transports. If one sector vanished, the off-diagonal reflection would kill the other; hence on a nonzero active corner both sectors are nonzero.

4.1.4 A shared question fixes transported labels

Lemma 4.2 (Label refinement)**.** *Retain the original FMS questions. For a synchronized projection $P^{\rm lab}$, add Alice's $\mathsf{T}_0[P^{\rm lab}]$ with answers (a, z). Ignore z against the unchanged $\mathsf{B}_0, \mathsf{B}_1, \mathsf{C}_0$ and use the original predicates; match z to Bob's binary $P^{\rm lab}$. Add the symmetric $\mathsf{B}_0[P^{\rm lab}]$ against unchanged $\mathsf{T}_0, \mathsf{T}_1, \mathsf{R}_0$. Also retain activity matching. In every perfect normal strategy the coarse signs equal the original signs and commute with $P^{\rm lab}$ on their state supports. Conversely commuting labels give these refinements. Distinct labels require no mutual commutation.*

Proof. Let $q_a = \sum_z q_{a,z}$ be the compressed coarse effects of $\mathsf{T}_0[P^{\rm lab}]$. Against C_0 the original rule is applied to a, so the matching argument in the proof of Lemma 4.1 gives, by (N5) and (N4b), $q_t = \Pi_t$ for each retained triple t and $q_+ + q_- = R_A = f_A + \Pi_{c_0}$. Against $\mathsf{B}_0, \mathsf{B}_1$ the forbidden sign pairs are the original ones. The calculation in that proof which gave $(E'_a \otimes I)\Omega_{0y} = (I \otimes F'_a \otimes I)\Omega_{0y}$ and $E'^2_a = E'_a$ used only these pairs and the projection R_A, so it applies verbatim to $q_\pm$. With $\mathcal{X}' = q_+ - q_-$ it gives

$$(\mathcal{X}' \otimes I)\Omega_{00} = (I \otimes \mathcal{Y}_0 \otimes I)\Omega_{00} = (\mathcal{X}_0 \otimes I)\Omega_{00},$$

the vector form of $\mathcal{X}'W_{00} = W_{00}\mathcal{Y}_0^T$. The Alice marginal of Ω_{00} is faithful on $R_AP_A\mathcal{H}_A$, so (N3) gives $\mathcal{X}' = \mathcal{X}_0$ and $q_\pm = (R_A \pm \mathcal{X}_0)/2$.

The z-marginal $\sum_a Q_{a,1}$ is perfectly matched with Bob's binary question for $P^{\rm lab}$. By (N5) its compression is a projection and a mirror of Bob's synchronized projection, whose Alice mirror is $P_A^{\rm lab}$; (N4b) gives equality. For every a, positivity gives $0 \le q_{a,1} \le P_A^{\rm lab}$ and $0 \le q_{a,0} \le P_A - P_A^{\rm lab}$. By (N6a), $q_{a,1} = P_A^{\rm lab}q_{a,1}P_A^{\rm lab}$ and $q_{a,0} = (P_A - P_A^{\rm lab})q_{a,0}(P_A - P_A^{\rm lab})$, so $P_A^{\rm lab}q_a = q_{a,1} = q_aP_A^{\rm lab}$. All of these are identities on $P_A\mathcal{H}_A$. For the converse, if the original effects E_a commute with $P^{\rm lab}$, then $E_aP^{\rm lab}$ and $E_a(I - P^{\rm lab})$ are positive. Summing over z gives E_a, and summing the $z = 1$ effects over a gives $P^{\rm lab}$. Every probability involving only a is an original probability, and the z-test is the original perfect matching of $P^{\rm lab}$. By (N6c) one may first replace $P^{\rm lab}$ by its support-restricted projection. No two different labels are measured jointly, so no commutation between them enters either direction. □

If a label is block diagonal with flag block P_f^{lab} and identical magic blocks P_m^{lab}, the commutator says $\beta P_f^{\mathrm{lab}} = P_m^{\mathrm{lab}}\beta$. In particular, for a graph projection of a block unitary it certifies transport of that same unitary by the calibrated U. It does not supply a fresh bridge with an independent gauge. Indeed, with respect to $R_A = f \oplus \Pi_{c_0}$ the commutator $[\mathcal{X}_0, P^{\mathrm{lab}}]$ has the (c_0, f) block $\beta P_f^{\mathrm{lab}} - P_m^{\mathrm{lab}}\beta$.

Proof in the commuting model. Calibration fixes the retained effects and the sign support. Pull the new coarse sign difference $\mathcal{X}'$ to the same corner. Against the unchanged opposite sign it satisfies $\mathcal{X}'\theta_{0y} = \mathcal{Y}_y\theta_{0y} = \mathcal{X}_0\theta_{0y}$. The separating vector of Section 4.1.3 gives $\mathcal{X}' = \mathcal{X}_0$. Thus its coarse effects E_a are the old projections. For the fine effects $Q_{a,z}$, label matching gives $\sum_a Q_{a,1} = P^{\mathrm{lab}}$ and $\sum_a Q_{a,0} = I - P^{\mathrm{lab}}$ on faithful support. Positivity gives $0 \le Q_{a,1} \le E_a, P^{\mathrm{lab}}$ and $0 \le Q_{a,0} \le E_a, I - P^{\mathrm{lab}}$. Hence $E_a P^{\mathrm{lab}} = Q_{a,1} = P^{\mathrm{lab}} E_a$, using completeness of each marginal. This proves commutation, and the symmetric argument handles Bob. Conversely use the joint products $E_a P^{\mathrm{lab}}, E_a(I - P^{\mathrm{lab}})$. Each label has its own refined question with the same unchanged opposite signs, so there is no cross-label commutation constraint. □

4.2 Rational projection equations that certify products

All projections in this section are synchronized. We give the finite rules used in the compiler. The projection-partition viewpoint is standard [26]; the formulas here provide the needed actual multiplication identities. See also the graph-projection encoding in [6, Section 4].

4.2.1 Graphs and triangles

Let F_i, F_j be orthogonal, put $F_{ij} = F_i + F_j$, and introduce projections $G, H, G^\perp, H^\perp$ with

$$G + G^\perp = F_{ij}, \quad H + H^\perp = F_{ij}, \quad 5H + F_{ij} = 3F_i + 4G. \tag{15}$$

On F_{ij}, put $Z_{ij}^{\mathrm{gr}} = F_i - F_j$, $\Xi_{ij} = 2G - F_{ij}$. These are reflections, and $2H - F_{ij} = (3Z_{ij}^{\mathrm{gr}} + 4\Xi_{ij})/5$. Squaring forces $Z_{ij}^{\mathrm{gr}}\Xi_{ij} + \Xi_{ij}Z_{ij}^{\mathrm{gr}} = 0$. Thus

$$\Xi_{ij} = \begin{pmatrix} 0 & u_{ij} \\ u_{ij}^* & 0 \end{pmatrix}, \qquad u_{ij}^* u_{ij} = F_j, \quad u_{ij} u_{ij}^* = F_i.$$

The equations thus certify the graph blocks. Conversely any such unitary corner map gives the four projections in (15).

For a partition $F_1 + F_2 + F_3 = I$ impose the graph tests on all three pairs and add a projection $P_\triangle$ with

$$3P_\triangle = 2(G_{12} + G_{23} + G_{13}) - I. \tag{16}$$

Its block matrix has diagonal $I/3$ and off-diagonal $u_{ij}/3$. The $(1,3)$ block of $P_\triangle^2 = P_\triangle$ gives $2u_{13} + u_{12}u_{23} = 3u_{13}$, hence $u_{13} = u_{12}u_{23}$. Conversely this product identity makes the entire matrix a projection.

Start with a reference triangle for identity. Its scalar matrix algebra is called the *frame*. It gives scalar matrix units and $\mathcal{H} = \mathbb{C}^3 \otimes \mathcal{H}_{\mathrm{base}}$. For every unitary label Θ, use graphs in all three positions. Triangles $(\Theta, 1, \Theta)$ and $(1, \Theta, \Theta)$ identify the placements. A triangle $(\Theta, \Theta_2, \Theta_3)$ then certifies $\Theta_3 = \Theta\Theta_2$; a triangle with product identity certifies inverse labels. Build each word from its proper prefix and final token. Finally a real rational equation $\sum_w q_w \Theta_w = 0$ is precisely

$$2\sum_w q_w G_{12}(\Theta_w) - \Big(\sum_w q_w\Big)(F_1 + F_2) = 0. \tag{17}$$

Its two off-diagonal blocks are the equation and its adjoint. All products therefore refer to actual common operators. Synchronization makes the extracted unitaries commute with the base marginal density.

4.2.2 Binding prescribed parameters

A prescribed synchronized projection P^{par} is linked to the frame by projections $P^{\mathrm{par}}F_i, NF_i$ with

$$P^{\mathrm{par}}F_i + NF_i = F_i, \quad \sum_i P^{\mathrm{par}}F_i = P^{\mathrm{par}}, \quad \sum_i NF_i = I - P^{\mathrm{par}}.$$

These symbols are names for projections before the equations identify their products; the names NF_i denote complements of $P^{\mathrm{par}}F_i$ inside F_i. For every pair ij, introduce $J_{11}, J_{10}, J_{01}, J_{00}$ and impose

$$\begin{aligned} J_{11} + J_{10} &= P^{\mathrm{par}}F_i + P^{\mathrm{par}}F_j, & J_{00} + J_{01} &= NF_i + NF_j, \\ J_{11} + J_{01} &= G_{ij}(1), & G_{ij}(s_{\mathrm{par}}) &= J_{11} + J_{00}. \end{aligned}$$

Projection sums give the indicated joint partitions. The symbol s_{par} is a unitary parameter label. These equations force P^{par} to commute with the frame and identify the extracted s_{par} with $2e - I$, where $P^{\mathrm{par}} = I_3 \otimes e$. In particular each parameter is the specified FMS event itself. The four-bridge assembly uses twenty-five such parameters.

These rules are a deterministic finite compiler for the finite word list. Cancellation of adjacent inverse tokens is performed before taking proper prefixes. All coefficients stay rational. Conversely a centralizer unitary assignment with the prescribed parameter projections gives every graph, complement, H, triangle and joint projection simultaneously by these formulas.

Scope in infinite dimensions. All projections in this section are support-restricted synchronized projections on Alice's support $P_A\mathcal{H}_A$, as in (N5), and every displayed equation holds there as an operator identity by Lemma 4.3. The derivations are block-matrix calculations for orthogonal decompositions of $P_A\mathcal{H}_A$. Projections whose sum is a projection are mutually orthogonal, and a self-adjoint unitary anticommuting with $\mathrm{diag}(I, -I)$ has the off-diagonal form used for Ξ_{ij}. The frame identification is the unitary

$$\mathbb{C}^3 \otimes \mathcal{H}_{\mathrm{base}} \to P_A\mathcal{H}_A, \qquad e_i \otimes \xi \mapsto u_{i1}\xi, \qquad \mathcal{H}_{\mathrm{base}} = F_1 P_A \mathcal{H}_A,$$

where $u_{11} = F_1$, $u_{21} = u_{12}^*$, $u_{31} = u_{13}^*$, and $u_{ij} = F_i(2G_{ij}(1) - F_i - F_j)F_j$. The frame projections and the maps u_{ij} commute with h_A by (N5), so $h_A = I_3 \otimes h_{\mathrm{base}}$ in these coordinates. An extracted label Θ is defined by $F_1(2G_{12}(\Theta) - F_1 - F_2)F_2 = e_{12} \otimes \Theta$, and it commutes with h_{base}.

The binding equations are solved in the same way. Write $\pi_i^{\mathrm{par}} = P^{\mathrm{par}}F_i$ and $\eta_i^{\mathrm{par}} = NF_i$. These are projections below F_i with $\pi_i^{\mathrm{par}} + \eta_i^{\mathrm{par}} = F_i$, and $\sum_i \pi_i^{\mathrm{par}} = P^{\mathrm{par}}$, so $P^{\mathrm{par}}F_i = \pi_i^{\mathrm{par}} = F_i P^{\mathrm{par}}$ and P^{par} commutes with every F_i. For a pair ij the four projections J_{ab} sum to F_{ij} and are therefore mutually orthogonal. Both $\pi^{\mathrm{par}} = \pi_i^{\mathrm{par}} + \pi_j^{\mathrm{par}}$ and $G_{ij}(1)$ are sums of them, so they commute and $\pi^{\mathrm{par}}G_{ij}(1) = J_{11}$. In frame coordinates write $\pi_i^{\mathrm{par}} = e_{ii} \otimes p_i$. Commutation of π^{par} with $\Xi_{ij}(1) = (e_{ij} + e_{ji}) \otimes I$ gives $p_i = p_j$; over all pairs, $P^{\mathrm{par}} = I_3 \otimes e$. Finally $J_{00} = (F_{ij} - \pi^{\mathrm{par}})(F_{ij} - G_{ij}(1))$, so

$$G_{ij}(s_{\mathrm{par}}) = \pi^{\mathrm{par}}G_{ij}(1) + (F_{ij} - \pi^{\mathrm{par}})(F_{ij} - G_{ij}(1)) = \tfrac{1}{2}\big(F_{ij} + (e_{ij} + e_{ji}) \otimes (2e - I)\big),$$

which identifies s_{par} with $2e - I$.

4.3 Rational rows as Boolean winning conditions

We implement each rational row by a subspace orthogonality test. This is the general rational-row theorem of [24, Theorem B], restated here to specify the extension and its quantitative loss.

Lemma 4.3 (Boolean rational rows)**.** *Let a finite strategy specification name finitely many synchronized binary projections and finitely many rational linear identities among them. There is a finite extension by Boolean tests with the following properties.*

Soundness: *in every perfect normal spatial strategy for the extended tests, the identities hold as operator identities for the named support-restricted projections on each local state support.*

Completeness: *any normal spatial strategy with these synchronized projections, satisfying the identities on its supports, extends to satisfy all the added tests perfectly after adjoining a finite maximally entangled state. The old measurements act trivially on the new factor. The rest of the old strategy, including nonsynchronous questions, is retained; if its old tests were perfect, the whole extension is perfect.*

Remark. The same proof, paragraph "The commuting form of rational rows", gives the commuting form used in Section 4.6.

4.3.1 A certified scalar matrix algebra

Choose $d_{\max} = 2^{2r}$ large enough for every row construction below. Use r synchronized magic squares, binding each context's cell marginals to synchronized binary cells. Add four-answer joint questions for all pairs of axes from distinct squares, joint questions between every named old projection and every scalar axis, and global Z and global X measurements with all bit outcomes, binding each bit marginal to its axis. These steps concern only synchronized projections, never an FMS sign.

The Pauli-pair calculation in the proof of Lemma 4.1, without flags, gives two anticommuting pairs of axes per square. A four-answer joint question for axes ς, ς' of distinct squares has its two binary marginals bound to synchronized cells, so the axes commute on $P_A\mathcal{H}_A$ by (N6d). Iterating the eigenspace argument of that proof over the $2r$ mutually commuting anticommuting pairs identifies $P_A\mathcal{H}_A$ with $\mathbb{C}^{d_{\max}} \otimes \mathcal{H}'$, the axes generating $\mathrm{Mat}_{d_{\max}} \otimes I$; the space $\mathcal{H}'$ may be infinite-dimensional. The axes commute with h_A by (N5), so h_A lies in $(\mathrm{Mat}_{d_{\max}} \otimes I)' = I \otimes B(\mathcal{H}')$: the marginal is $I_{d_{\max}}/d_{\max} \otimes h$ for a density h on $\mathcal{H}'$. A joint question between a named projection E_i and an axis makes them commute by (N6d). The axes generate $\mathrm{Mat}_{d_{\max}} \otimes I$ as an algebra, so E_i lies in its commutant, and $E_i = I_{d_{\max}} \otimes e_i$ for a projection e_i on $\mathcal{H}'$. The global Z and X measurements have sharp bit marginals, so by (N6d) they are the joint spectral measures of the commuting Z axes and of the commuting X axes.

For a row dimension $d = 2^{k_{\mathrm{row}}} \le d_{\max}$ use the first k_{row} qubits; the rest are spectators. For each nonzero $j \in \mathbb{F}_2^{k_{\mathrm{row}}}$, choose a basis $\ell_1, \dots, \ell_{k_{\mathrm{row}}-1}$ of $j^\perp$. Add a joint context for $X^j, Z^{\ell_1}, \dots, Z^{\ell_{k_{\mathrm{row}}-1}}$, binding its marginals to the corresponding global parity events. These commute because $\ell \cdot j = 0$. The zero-Z-syndrome outcomes with X^j sign $\pm$ are

$$\Pi_j^{\pm} = \tfrac{1}{2}(|0\rangle \pm |j\rangle)(\langle 0| \pm \langle j|) \otimes I_{\mathrm{spectator}}.$$

Write $|+_d\rangle = d^{-1/2} \sum_{j=0}^{d-1} |j\rangle$. The reference $\Pi_d^{\mathrm{ref}} = |+_d\rangle\langle +_d| \otimes I_{\mathrm{spectator}}$ is the event that the first k_{row} global X bits are positive. It includes all spectator outcomes.

4.3.2 Signed reflections and column graphs

For one row $\Delta = \alpha_0 I + \sum_i \alpha_i e_i$, let L clear all coefficient denominators and set

$$s_i = 2e_i - I, \quad c_i = L\alpha_i, \quad c_0 = 2L\alpha_0 + \sum_i L\alpha_i.$$

Then $2L\Delta = c_0 I + \sum_i c_i s_i$. Expand this into signed reflections with integer repetitions, add a canceling $I, -I$ pair with $V_0 = I$, and pad by further such pairs to a power of four d. The list

has even length because its coefficient sum is even. Thus

$$\sum_{j=0}^{d-1} V_j = 2L\Delta, \qquad V_0 = I, \qquad V_j \in \{\pm I,\ \pm(2e_i - I)\}. \tag{18}$$

For each $j > 0$ use a six-answer joint question of e_i and the three-way scalar measurement $\Pi_j^+, \Pi_j^-, I - \Pi_j^+ - \Pi_j^-$. For $V_j = \pm I$, bind its binary bit to a synchronized deterministic anchor: its 1-projection is I and its 0-projection is zero. Matching both marginals identifies all joint effects. Appropriate coarse events are the two-coordinate graph projections

$$G_j^{\pm} = \frac{1}{2}\begin{pmatrix} I & \pm V_j \\ \pm V_j & I \end{pmatrix}$$

on coordinates $0, j$, zero elsewhere. A negative reflection sign exchanges the two coarse events.

Add a synchronized d-answer question with projections Π_t^{row}, indexed by $t \in \mathbb{F}_2^{k_{\mathrm{row}}}$. Put $\chi_t(j) = (-1)^{t\cdot j}$. Forbid Π_t^{row} together with G_j^- when $\chi_t(j) = 1$, or G_j^+ when $\chi_t(j) = -1$. Define

$$\mathcal{I}_t\xi = d^{-1/2}(\chi_t(j)V_j\xi)_{j=0}^{d-1}, \qquad \mathcal{P}_t = \mathcal{I}_t\mathcal{I}_t^*.$$

Unitarity and character orthogonality give $\mathcal{I}_t^*\mathcal{I}_u = \delta_{tu}I$ and $\sum_t \mathcal{P}_t = I$. Neither equation requires different V_j to commute. Finally forbid Π_0^{row} jointly with the opposite Π_d^{ref}.

For soundness, the forbidden pairs are handled by (N7). Take Q to be the coarse event $G_j^{\mp}$ of the party that asks it and R the opposite synchronized Π_t^{row}. Both reduce the support, so $G_j^{\mp}\Pi_t^{\mathrm{row}} = \Pi_t^{\mathrm{row}}G_j^{\mp} = 0$ on that party's support. The events $G_j^{\pm}$ and Π_t^{row} are products of mirrored projections, so (N4a) and (N3) transfer this identity to the other party. In particular it holds on $P_A\mathcal{H}_A$. Write a vector of $P_A\mathcal{H}_A = \mathbb{C}^d \otimes \mathbb{C}^{d_{\max}/d} \otimes \mathcal{H}'$ as $(v_j)_{j\in\mathbb{F}_2^{k_{\mathrm{row}}}}$, and regard each V_j as a support-restricted reflection, a self-adjoint unitary on $\mathbb{C}^{d_{\max}/d} \otimes \mathcal{H}'$. If $\chi_t(j) = 1$, the equation $G_j^- v = 0$ says $v_j = V_j v_0$. If $\chi_t(j) = -1$, the equation $G_j^+ v = 0$ says $v_j = -V_j v_0$. Thus $\operatorname{ran}\Pi_t^{\mathrm{row}} \subseteq \operatorname{ran}\mathcal{I}_t$, that is, $\Pi_t^{\mathrm{row}} \le \mathcal{P}_t$. The two families are complete, so $\sum_t(\mathcal{P}_t - \Pi_t^{\mathrm{row}}) = 0$ with positive terms, and $\Pi_t^{\mathrm{row}} = \mathcal{P}_t$. The forbidden pair $(\Pi_0^{\mathrm{row}}, \Pi_d^{\mathrm{ref}})$ gives $\Pi_d^{\mathrm{ref}}\mathcal{P}_0 = 0$ by (N7), and $\Pi_d^{\mathrm{ref}}\mathcal{I}_0 = \Pi_d^{\mathrm{ref}}\mathcal{P}_0\mathcal{I}_0 = 0$. Finally $(\langle +_d| \otimes I)\mathcal{I}_0 = d^{-1}\sum_j V_j = (2L/d)\Delta$, so

$$\|\Pi_d^{\mathrm{ref}}\mathcal{I}_0\xi\|^2 = \frac{4L^2}{d^2}\|\Delta\xi\|^2$$

for every ξ in the support. Hence $\Pi_d^{\mathrm{ref}}\mathcal{I}_0 = 0$ holds exactly when Δ vanishes on the support. This proves soundness on arbitrary normal supports.

For completeness, let the old strategy have purification Ω and supports P_A, P_B. By (N6c) we may replace every named synchronized effect by its support-restricted projection, completed by a fixed outcome off the support; no probability changes. Adjoin $\Phi_{d_{\max}}$, so that the new purification is $\Omega' = \Omega \otimes \Phi_{d_{\max}}$ and the new state is again normal. Alice's new effects are the displayed operators: scalar Paulis, joint products $E_i\varsigma$ and $E_i\Pi_j^{\pm}$ of commuting projections, and the column projections $\mathcal{P}_t = \mathcal{I}_t\mathcal{I}_t^*$, all defined on the support and completed by a fixed outcome off it. For an Alice operator $\sum_k z_k \otimes p_k(e)$, with scalar factors z_k and products $p_k(e)$ of named projections, Bob uses $\sum_k z_k^{\mathsf{T}} \otimes p_k^{\mathrm{rev}}(e^B)$, where $p_k^{\mathrm{rev}}(e^B)$ is the reversed product of his matched projections; equivalently, Bob uses transposes in canonical Schmidt coordinates. The order matters when the e_i do not commute. By (N4a) and $(z \otimes I)\Phi_{d_{\max}} = (I \otimes z^{\mathsf{T}})\Phi_{d_{\max}}$, it is a mirror of the Alice operator. This assignment commutes with adjoints, so the counterpart Y of an Alice projection P is self-adjoint. Moreover $(I \otimes (Y^2 - Y) \otimes I)\Omega' = ((P^2 - P) \otimes I)\Omega' = 0$, so Y is a projection on Bob's support by (N3), and the counterparts of the column projections sum to the identity. Every matching test therefore has disagreement probability zero. Every forbidden pair has probability zero because the corresponding Alice product vanishes, and the

row-zero pair does so because $\Delta = 0$ on the support. The old measurements act as $E \otimes I_{d_{\max}}$, so all old tests, including nonsynchronous ones, keep their probabilities. Since the argument runs on a purification, it covers mixed original states.

Write h_{old} for the trace-one marginal before the row ancilla is appended (including the qutrit frame when present). For the honest column construction, without assuming $\Delta = 0$, the sole row-zero event has probability

$$p(\Pi_0^{\mathrm{row}}, \Pi_d^{\mathrm{ref}}) = \frac{4L^2}{d^3} \operatorname{Tr}(h_{\mathrm{old}} \Delta^2). \tag{19}$$

All column-graph and matching tests remain exact even when the row is nonzero. This formula is [24, Eq. (32)] in the general row setting. To prove it, note that Bob's reference effect is the transpose of Π_d^{ref} on the scalar factor and is a mirror of Π_d^{ref}. The probability is therefore $\|(\mathcal{P}_0 \otimes (\Pi_d^{\mathrm{ref}})^{\mathsf{T}})\Omega'\|^2 = \operatorname{Tr}(h' \Pi_d^{\mathrm{ref}} \mathcal{P}_0 \Pi_d^{\mathrm{ref}})$, where $h' = (I_d/d) \otimes (d/d_{\max}) I_{\mathrm{spectator}} \otimes h_{\mathrm{old}}$. Since $(\langle +_d| \otimes I)\mathcal{I}_0 = (2L/d)\Delta$, we have $\Pi_d^{\mathrm{ref}} \mathcal{P}_0 \Pi_d^{\mathrm{ref}} = |+_d\rangle\langle +_d| \otimes I_{\mathrm{spectator}} \otimes (4L^2/d^2)\Delta^2$. The trace against h' is (19): the factor $1/d$ is $\langle +_d | I_d/d | +_d \rangle$, and the spectator trace is one. The calculation uses only that the V_j are self-adjoint unitaries, so it holds for every row, zero or not.

The commuting form of rational rows. On a normal commuting host the same finite extension is used. The Pauli calculation of Section 4.1.3 identifies the tensor matrix algebra, and old projections are $I \otimes e_i$ with centralizer scalar matrix units. A forbidden pair with mirror $\mathcal{J}G\mathcal{J}$ gives $\Pi_t^{\mathrm{row}} G \xi_\varphi = 0$, since $\mathcal{J}G\mathcal{J}\xi_\varphi = G\xi_\varphi$; separation gives $\Pi_t^{\mathrm{row}} G = 0$. The remaining soundness argument is unchanged. Conversely amplify by $\mathrm{Mat}_{d_{\max}}$ with normalized trace. Use the actual scalar Paulis, joint products and projections $\mathcal{P}_t$. All are centralizer projections. Bob uses $\mathcal{J}_{\mathrm{new}} X \mathcal{J}_{\mathrm{new}}$ for each projection X. For a centralizer word xy, $\mathcal{J}(xy)\mathcal{J}\xi_\varphi = (xy)^* \xi_\varphi$; adjunction supplies the reversed order. Thus Bob's products are the actual transferred products, even when the old projections do not commute. Every structural forbidden event is zero, and the final event is zero precisely for a zero row. Old measurements act trivially on the new matrix factor, preserving all old probabilities. Finite matrix amplification preserves all indicated factor types. This proves Lemma 4.3.

4.4 Admissible presentations

Definition 4.4 (Admissible presentation; normal realization)**.** Fix an integer $n \geq 3$ and put $\nu = 2^{-n}$. The four-isometry normal model requires $4\nu < 1$; the generic chained-bridge statement of Lemma 4.6 separately permits $n \geq 1$. An admissible presentation is a finite rational list with four contraction symbols A_α ($\alpha \in \{0,1\}^2$), a centralizer projection label $P_\circ$ for the root, and finitely many centralizer labels, each typed as a unitary, projection, positive contraction ($0 \leq T \leq I$), or general contraction ($\|T\| \leq 1$). Its application equations (the identities enforced by the final row tests) are rational $*$-polynomials in the centralizer labels, the creation and range equations

$$A_\alpha^* A_\beta = \delta_{\alpha\beta} I, \qquad \sum_\alpha A_\alpha A_\alpha^* = I - P_\circ,$$

and endpoint equations $L_{\mathrm{end}} A_\alpha = A_\beta R_{\mathrm{end}}$ or $L_{\mathrm{end}} A_\alpha = 0$. Finite sums $\sum_\iota L_\iota A_{\alpha_\iota} L'_\iota = 0$ are allowed. Here $L_{\mathrm{end}}, R_{\mathrm{end}}, L_\iota, L'_\iota$ are centralizer polynomial expressions. Apart from the displayed creation and range equations, no monomial contains more than one creation symbol. Every label used on a left endpoint is included in the transported unitary list Λ defined in the proof of Lemma 4.5, and the root projection is transported even when it occurs in no endpoint equation.

A normal realization consists of a nonzero Hilbert space $\mathcal{H}_0$, a faithful positive trace-class operator h_0 with $\operatorname{Tr} h_0 = 1$, and bounded operators of the declared types on $\mathcal{H}_0$. Every centralizer label, including $P_\circ$, commutes with h_0. All application equations hold as operator identities on $\mathcal{H}_0$, and

$$h_0 A_\alpha = \nu A_\alpha h_0. \tag{20}$$

Thus all identities are imposed on the state support, where the density is faithful. A positive contraction need not be bounded below. Finite entropy of a realization means $S(h_0) < \infty$. The ratio is imposed by the assembly, rather than by a real coefficient in the Boolean game. The spatial hardness proof of Sections 5–8 uses $n = 4$, so $\nu = 1/16$; Section 16 uses $n = 5$.

4.5 The n-bridge compiler

Each FMS bridge forces a density ratio of two between consecutive selected corners (Figure 3). A chain of n bridges therefore gives the ratio 2^n from source to far corner, and transporting a source-to-far creation operator back to the source gives $\nu = 2^{-n}$. Set

$$\mathfrak{c}_n = 1 + 4\sum_{g=1}^{n} 2^g = 2^{n+3} - 7, \qquad \mu_n = \mathfrak{c}_n^{-1}(1; 2^{\oplus 4}; 4^{\oplus 4}; \ldots; (2^n)^{\oplus 4}). \tag{21}$$

The notation $t^{\oplus 4}$ denotes four entries equal to t.

Lemma 4.5 (n-bridge compiler)**.** *Every admissible presentation effectively yields a finite rational Boolean game $G^{(n)}$. Every perfect normal spatial strategy yields a normal realization in the sense of Definition 4.4, on the nonzero source corner with faithful trace-one density $h_0 = \mathfrak{c}_n e_0 h e_0$, where h is the marginal density. Every such normal realization extends to a perfect normal spatial strategy after adjoining the scaffold and finite ancillary factors (scaffold, qutrit and row factors). Finite marginal entropy is preserved by the completeness construction.*

The assembly parameters are

quantity	value
orthogonal scaffold corners	$4n + 1$
frame-bound parameters	$6n + 1$
cover constant	$\mathfrak{c}_n = 2^{n+3} - 7$
chain covariance	$hv = 2^n vh$
normalized source density	$\mathfrak{c}_n e_0 h e_0$
far-corner weight relative to source	2^n

For $n = 4$ there are seventeen corners and twenty-five parameters, $\mathfrak{c}_4 = 121$, $hv = 16vh$ and $h_0 = 121 e_0 h e_0$.

Proof. We give the assembly and both directions.

Corners and parameters. Use $4n + 1$ orthogonal synchronized corners

$$e_0; \quad m_{g,00}, m_{g,01}, m_{g,10}, m_{g,11} \quad (1 \le g \le n).$$

Put $e_g = m_{g,00}$ and $m_g = \sum_{ab} m_{g,ab}$, and impose the cover $e_0 + \sum_g m_g = I$. Conditional bridge g has flag e_{g-1}, magic support m_g and activity $E_g^{\mathrm{act}} = e_{g-1} + m_g$. Use all event bindings described above in both orientations, including the flag/four-magic/inactive partition. Bind

to the qutrit frame the $4n+1$ corners and the $2n$ positive ZI, IX events. These are the $6n+1$ frame-bound parameters. Introduce $2n$ axis labels $\mathfrak{x}_g, \mathfrak{z}_g$ with the explicit equations

$$\mathfrak{x}_g = I - 2m_g + 2E_{g,ZI,+}, \qquad \mathfrak{z}_g = I - 2m_g + 2E_{g,IX,+},$$

where $E_{g,ZI,+}, E_{g,IX,+}$ are the positive nonflag events of bridge g, bound to synchronized binary questions as above.

Let a perfect normal spatial strategy be given, with purification and supports as in (N1), and write $h = h_A$. By (N5) every corner, frame and graph question yields a synchronized projection on $P_A\mathcal{H}_A$ commuting with h, and by Section 4.2 and Lemma 4.3 every imposed equation, including the cover, holds on $P_A\mathcal{H}_A$. Projections with sum I are mutually orthogonal, so the $4n+1$ corners partition $P_A\mathcal{H}_A$. Bridge g is a perfect conditional FMS strategy with activity $e_{g-1} + m_g$, flag e_{g-1} and magic support m_g, whose magic directions are the bound corners $m_{g,ab}$. Lemma 4.1 gives calibrated partial isometries U_g with $\mathsf{U}_g^*\mathsf{U}_g = e_{g-1}$, $\mathsf{U}_g\mathsf{U}_g^* = e_g$ and $h\mathsf{U}_g = 2\mathsf{U}_g h$ on the support; the other three magic corners are identified with e_g by synchronized Pauli matrix units and have the same density block. Moving h through the n factors gives

$$v = \mathsf{U}_n \cdots \mathsf{U}_1, \quad v^*v = e_0, \quad vv^* = e_n, \quad hv = 2^n vh. \tag{22}$$

For the masses, h is trace class and U_g is bounded, so cyclicity gives

$$\operatorname{Tr}(he_g) = \operatorname{Tr}(h\mathsf{U}_g\mathsf{U}_g^*) = \operatorname{Tr}(\mathsf{U}_g^* h\mathsf{U}_g) = 2\operatorname{Tr}(\mathsf{U}_g^*\mathsf{U}_g h) = 2\operatorname{Tr}(he_{g-1}).$$

Each t_{ab} of bridge g commutes with h, so $\operatorname{Tr}(hm_{g,ab}) = \operatorname{Tr}(he_g)$: the four magic masses are twice the preceding flag mass. Summing over the cover, with $\operatorname{Tr} h = 1$, gives

$$\operatorname{Tr}(he_0) = 1/\mathfrak{c}_n; \qquad 121 = \mathfrak{c}_4 = 1 + 4(2 + 4 + 8 + 16). \tag{23}$$

In particular the source corner is nonzero.

$$e_0 \xrightarrow{\mathsf{U}_1} \boxed{m_1 \supset e_1} \xrightarrow{\mathsf{U}_2} \boxed{m_2 \supset e_2} \xrightarrow{\mathsf{U}_3} \boxed{m_3 \supset e_3} \xrightarrow{\mathsf{U}_4} \boxed{m_4 \supset e_4}$$

corner	e_0	e_1	e_2	e_3	e_4
$121\operatorname{Tr}(he_g)$	1	2	4	8	16
$121\operatorname{Tr}(hm_g)$	–	8	16	32	64

Figure 3: Four calibrated bridges ($n = 4$). Each m_g has four equal-mass magic corners; its selected e_g is the next flag. For n bridges the selected masses are $2^g/\mathfrak{c}_n$, and returning along the chain gives the creation ratio $\nu = 2^{-n}$.

Every transported label. Encode a unitary directly, a projection P by $2P - I$, a positive contraction T_+ by a unitary u_+ with $T_+ = (2I + u_+ + u_+^*)/4$, and a general contraction by an average of two unitaries. Let Λ be the resulting finite unitary label list that must be transported; it contains the encoding of $P_\circ$ and all unitary encodings required at left endpoints. For every $\ell \in \Lambda$, $r = 0, \ldots, n$, introduce a unitary $u_{r,\ell}$ and impose

$$(u_{r,\ell} - I)(I - e_r) = 0. \tag{24}$$

The unitary reduces e_r and is identity on its complement. For a typed label L, its corner copy L_r is explicitly decoded as

type of L	L_r
unitary	$e_r u_{r,L}$
projection	$e_r(I + u_{r,L})/2, \quad u_{r,L}^2 = I$
positive contraction	$e_r(2I + u_{r,L} + u_{r,L}^*)/4$
general contraction	$e_r(u_{r,L,+} + u_{r,L,-})/2$

Thus $(P_\circ)_n$ denotes the decoded root projection at e_n. The two encodings of a general contraction are transported separately when that label occurs at a left endpoint.

For each $g = 1, \dots, n$ and every $\ell \in \Lambda$ introduce a unitary $V_{g,\ell}$ with

$$V_{g,\ell} = I - E_g^{\mathrm{act}} + e_{g-1}u_{g-1,\ell} + \sum_{a,b} \mathfrak{x}_g^a \mathfrak{z}_g^b e_g u_{g,\ell} \mathfrak{z}_g^{-b} \mathfrak{x}_g^{-a}. \tag{25}$$

For this bridge, apply Lemma 4.2 to the six projections

$$F_1, F_2, F_3, G_{12}(1), G_{13}(1), G_{23}(1)$$

and the $|\Lambda|$ graph projections $G_{12}(V_{g,\ell})$. Thus there are $6 + |\Lambda|$ refinements of each retained sign question per bridge, on each party. Each label has its own duplicated question; the unchanged opposite questions are always retained. There are $(n+1)|\Lambda|$ support equations and $n|\Lambda|$ block equations.

We show that the commutator of the original sign with $G_{12}(V_{g,\ell})$, in its (c_0, f) block, is precisely

$$\mathsf{U}_g u_{g-1,\ell} = u_{g,\ell} \mathsf{U}_g, \tag{26}$$

with the same actual U_g for every label; distinct transported labels need not commute with each other. Equation (24) says $u_{r,\ell}(I - e_r) = I - e_r$. Multiplying by $u_{r,\ell}^*$ gives $u_{r,\ell}^*(I - e_r) = I - e_r$, and the adjoint of this is $(I - e_r)u_{r,\ell} = I - e_r$. Hence $u_{r,\ell}$ commutes with e_r and is the identity on $(I - e_r)P_A\mathcal{H}_A$. On the active space of bridge g, the (a, b) term of (25) is $t_{ab}(e_g u_{g,\ell} e_g)t_{ab}^*$, because $\mathfrak{x}_g, \mathfrak{z}_g$ are reflections and $\mathfrak{x}_g^a \mathfrak{z}_g^b e_g = t_{ab}$. So $V_{g,\ell}$ is block diagonal for $e_{g-1} \oplus \bigoplus_{ab} m_{g,ab}$, with the block $u_{g-1,\ell}$ on the flag and transported copies of $u_{g,\ell}$ on the magic corners; in particular it reduces e_{g-1} and e_g. The six frame refinements make the sign $\mathcal{X}_0$ of bridge g commute with F_1, F_2, F_3 and $G_{ij}(1)$. These generate $\mathrm{Mat}_3 \otimes I$ in the frame coordinates of Section 4.2, whose commutant is $I_3 \otimes B(\mathcal{H}_{\mathrm{base}})$, and the corners are frame-bound; hence $\mathsf{U}_g = I_3 \otimes \widetilde{\mathsf{U}}_g$. The refinement by $G_{12}(V_{g,\ell}) = \frac{1}{2}(F_1 + F_2 + e_{12} \otimes V_{g,\ell} + e_{21} \otimes V_{g,\ell}^*)$ makes $\mathcal{X}_0$ commute with it, and the $(1, 2)$ frame block gives $[\widetilde{\mathcal{X}}_0, V_{g,\ell}] = 0$ on the base. With respect to $e_{g-1} \oplus e_g$,

$$\widetilde{\mathcal{X}}_0 = \begin{pmatrix} 0 & \beta^* \\ \beta & 0 \end{pmatrix}, \qquad V_{g,\ell} = \begin{pmatrix} u_{g-1,\ell} & 0 \\ 0 & u_{g,\ell} \end{pmatrix},$$

and the lower-left block of the commutator is $\beta u_{g-1,\ell} - u_{g,\ell}\beta$. This is (26). Induction over g gives $u_{n,\ell}v = vu_{0,\ell}$. Multiplying on the left by $u_{n,\ell}^*$ and on the right by $u_{0,\ell}^*$ gives $u_{n,\ell}^* v = v u_{0,\ell}^*$. The decoded copies in the table are polynomials in $e_r, u_{r,L}, u_{r,L}^*$ with the same coefficients for every r, and $e_n v = v = v e_0$. Hence $L_n v = v L_0$ and $L_n^* v = v L_0^*$ for every decoded transported label, where L_r is the decoded copy at e_r with corner identity e_r, and the same holds for sums and products of decoded labels.

Extraction and exact translation. Introduce four centralizer contractions $a_\alpha = (a_{\alpha,+} + a_{\alpha,-})/2$ and impose $a_\alpha = e_n a_\alpha e_0$. On the root corner extract $A_\alpha = v^* a_\alpha$. Supply the following polynomials to the graph compiler:

Root identity	Centralizer identity
$A_\alpha^* A_\beta = \delta_{\alpha\beta} I$	$a_\alpha^* a_\beta = \delta_{\alpha\beta} e_0$
$\sum_\alpha A_\alpha A_\alpha^* = I - P_\circ$	$\sum_\alpha a_\alpha a_\alpha^* = e_n - (P_\circ)_n$
$L_{\mathrm{end}} A_\alpha = A_\beta R_{\mathrm{end}}$	$(L_{\mathrm{end}})_n a_\alpha = a_\beta (R_{\mathrm{end}})_0$
$L_{\mathrm{end}} A_\alpha = 0$	$(L_{\mathrm{end}})_n a_\alpha = 0$
$\sum_\iota L_\iota A_{\alpha_\iota} L'_\iota = 0$	$\sum_\iota (L_\iota)_n a_{\alpha_\iota} (L'_\iota)_0 = 0$
Root-only polynomial	Same polynomial in source copies

Expand the finite words, bind their parameters, apply graphs and triangles, and then Lemma 4.3. Retain all FMS and refinement questions. Pad the two question sets if necessary and sample uniformly from all pairs, accepting every unspecified pair. The result has rational question distribution and a Boolean predicate. The support arguments prove soundness on a nonzero corner, using Alice alone; mirror equations are unnecessary.

Put $\mathcal{H}_0 = e_0 P_A \mathcal{H}_A$ and let h_0 be the restriction of $\mathfrak{c}_n e_0 h_A e_0$ to $\mathcal{H}_0$. It is positive and trace class, it is faithful because h_A is faithful on $P_A\mathcal{H}_A$ and commutes with e_0, and it has trace one by (23); in particular $\mathcal{H}_0 \neq 0$. Every source copy L_0 maps $\mathcal{H}_0$ into itself and commutes with h_0, because $u_{0,L}$ commutes with e_0 and with h_A. The declared types hold: $e_0 u_{0,L}$ is unitary on $\mathcal{H}_0$; $e_0(I + u_{0,L})/2$ is a projection when $u_{0,L}^2 = I$; $e_0(2I + u + u^*)/4$ is a positive contraction; and an average of two unitaries is a contraction. The contraction $a_\alpha = e_n a_\alpha e_0$ commutes with h_A, so $A_\alpha = v^* a_\alpha$ maps $\mathcal{H}_0$ into itself, and

$$h_0 A_\alpha = \mathfrak{c}_n\, e_0 h_A v^* a_\alpha = 2^{-n} \mathfrak{c}_n\, v^* h_A a_\alpha = 2^{-n} \mathfrak{c}_n\, v^* a_\alpha e_0 h_A e_0 = 2^{-n} A_\alpha h_0,$$

using $e_0 v^* = v^*$, the adjoint $h_A v^* = 2^{-n} v^* h_A$ of (22), $a_\alpha = a_\alpha e_0$ and $[e_0, h_A] = 0$; this is $h_0 A_\alpha = \nu A_\alpha h_0$. The translations are exact. For the creation equations, $A_\alpha^* A_\beta = a_\alpha^* v v^* a_\beta = a_\alpha^* e_n a_\beta = a_\alpha^* a_\beta$. For the range sum, $\sum_\alpha A_\alpha A_\alpha^* = v^*(\sum_\alpha a_\alpha a_\alpha^*)v$. If $\sum_\alpha a_\alpha a_\alpha^* = e_n - (P_\circ)_n$, this equals $e_0 - v^*(P_\circ)_n v = e_0 - (P_\circ)_0$; conversely, conjugation by v recovers the centralizer form, because $vv^* = e_n$ and $a_\alpha = e_n a_\alpha$. For an endpoint equation, $v^* L_n = L_0 v^*$ gives $L_0 A_\alpha - A_\beta R_0 = v^*(L_n a_\alpha - a_\beta R_0)$. The bracket has range in $e_n P_A \mathcal{H}_A$, where v^* is isometric, so either side vanishes exactly when the other does. Sums $\sum_\iota L_\iota A_{\alpha_\iota} L'_\iota$ translate in the same way, with $(L_\iota)_n$ on the left. Root-only polynomials involve only source copies. Every application equation therefore holds as an operator identity on $\mathcal{H}_0$.

One simultaneous realization. Let $(\mathcal{H}_0, h_0)$ be a normal realization, and put $\mathcal{H}_{\mathrm{sc}} = \mathbb{C}^{4n+1} \otimes \mathcal{H}_0$ with the density

$$h_{\mathrm{sc}} = \mathrm{diag}(\mu_n) \otimes h_0 = \frac{1}{\mathfrak{c}_n} \mathrm{diag}(h_0;\, 2h_0^{\oplus 4};\, 4h_0^{\oplus 4};\, \ldots;\, 2^n h_0^{\oplus 4}). \tag{27}$$

It is faithful and trace class, of trace one by the definition of $\mathfrak{c}_n$. Use identity corner identifications in the bridges and $a_\alpha = |e_n\rangle\langle e_0| \otimes A_\alpha$; its commutation with h_{sc} is exactly the modular relation. The positive eigenspaces of h_{sc} are spanned by corner vectors tensored with positive eigenvectors of h_0, so they are finite-dimensional, and a bounded operator commuting with h_{sc} maps each of them into itself by (N4d). On such a block $\mathcal{E}$ the contraction $T = a_\alpha|_{\mathcal{E}}$ has a polar decomposition $T = V|T|$. Because $\dim\ker T = \dim\ker T^*$ in finite dimension, V extends to a unitary V_{pol} with $T = V_{\mathrm{pol}}|T|$, and $|T| \pm i(I - |T|^2)^{1/2}$ are commuting unitaries, so

$$T = (U_+ + U_-)/2, \qquad U_\pm = V_{\mathrm{pol}}(|T| \pm i\sqrt{I - |T|^2}).$$

The direct sums of the resulting unitaries over all blocks are unitaries commuting with h_{sc}; they give centralizer encodings for all four a_α at once. Finite-dimensional blocks are needed here: a nonunitary isometry, such as a root isometry A_α, is an extreme point of the unit ball and is therefore not an average of two unitaries. That is why A_α is never encoded directly and the contraction completion takes place on the scaffold. Encode each source contraction inside the source density blocks, choose $2P - I$ for projections, and use

$$u_+ = 2T_+ - I + 2i\sqrt{T_+(I - T_+)}$$

for a positive contraction T_+. These encodings are functions of operators commuting with h_0, so they commute with h_0. Copy these same choices through the corner identifications and extend by identity outside the corners: the copies $u_{r,\ell} = |e_r\rangle\langle e_r| \otimes U_\ell + (I - |e_r\rangle\langle e_r|) \otimes I$ satisfy

(24), the block unitaries (25) act as U_ℓ on every corner of bridge g and as I elsewhere, and $\mathsf{U}_g = |e_g\rangle\langle e_{g-1}| \otimes I$ gives (26). Thus the block equations hold simultaneously. The translated equations hold because, for example, $L_n a_\alpha - a_\beta R_0 = |e_n\rangle\langle e_0| \otimes (LA_\alpha - A_\beta R)$. Append the scalar qutrit frame, use the graph formulas, and take actual joint products for all refinements; their block unitaries commute with the coarse FMS effects, and separate labels need not commute. Finally append the finite maximally entangled row factor of dimension $d_{\max}$. The scaffold and ancillary dimensions are finite, so they add only finite entropy.

All tests hold with one state and one assignment; the strategy is explicit. Take $\mathcal{H}_A = \mathcal{H}_B = \mathbb{C}^3 \otimes \mathcal{H}_{\mathrm{sc}} \otimes \mathbb{C}^{d_{\max}}$, the density $h = (I_3/3) \otimes h_{\mathrm{sc}} \otimes (I_{d_{\max}}/d_{\max})$ and the pure state Ω_h of (N8), formed in a product eigenbasis. Every synchronized question of Alice measures a projection commuting with h: corner, frame, graph, triangle and binding projections built from the centralizer unitaries by the formulas of Section 4.2; the FMS cell, activity and corner projections; and the row projections of Lemma 4.3. Bob measures their transposes. By (N8) every matching test passes. Every forbidden pair between centralizer events has probability $\mathrm{Tr}(hQR) = 0$, because the defining identities give $QR = 0$. The FMS sign questions are not in the centralizer. For bridge g, Alice's sign observable for T_x is the reflection $|f\rangle\langle c_x| + |c_x\rangle\langle f|$ on the span of the flag corner e_{g-1} and the corner c_x, tensored with the identity of $\mathcal{H}_0$. Bob's sign observable for B_y is $|f\rangle\langle r_y| + |r_y\rangle\langle f|$, with $r_y = (c_0 + (-1)^y c_1)/\sqrt{2}$. Their retained answers are the remaining column-zero and row-zero projections. If μ is the mass of e_{g-1}, each magic corner has mass 2μ, and $\langle c_x, r_y\rangle = \epsilon_{xy}/\sqrt{2}$. So, up to the frame and row factors, the compressed state is $\Omega_{xy} = \mu^{1/2}(|f, f\rangle + \epsilon_{xy}|c_x, r_y\rangle) \otimes \Omega_{h_0}$. Substitution gives $(\mathcal{X}_x \otimes I)\Omega_{xy} = \epsilon_{xy}(I \otimes \mathcal{Y}_y)\Omega_{xy}$, so every forbidden sign pair has probability zero; the calibrated rules against C_0 and R_0 hold by construction. The FMS effects are corner operators tensored with the identity of $\mathcal{H}_0$. They commute with every block unitary $V_{g,\ell}$ and with the frame, on Alice's side and, after transposition in the product eigenbasis, on Bob's. The refinements $E_a P^{\mathrm{lab}}$ of Lemma 4.2 are therefore POVMs. The marginal entropy is $\log_2 3 + S(h_{\mathrm{sc}}) + \log_2 d_{\max}$, where $S(h_{\mathrm{sc}}) = H(\mu_n) + S(h_0)$. □

4.6 Chained bridges on normal commuting hosts

The same assembly works on arbitrary normal commuting supports. Here the edges may be any finite family and the range equation may carry a transported defect.

Lemma 4.6 (Chained bridges)**.** *For any fixed $n \geq 1$, the conditional FMS predicates, individual label refinements and rational rows above compile the following finite root specifications on arbitrary normal commuting supports. The centralizer labels have prescribed types (projection, positive contraction, contraction or unitary). Each edge A_α has modular ratio 2^{-n}, with isometry and range equations and endpoint relations $L_{\mathrm{end}} A_\alpha = A_\beta R_{\mathrm{end}}$ or $L_{\mathrm{end}} A_\alpha = 0$; $L_{\mathrm{end}}, R_{\mathrm{end}}$ are real rational polynomials in the typed centralizer labels (for example $R_{\mathrm{end}} = C^2/2$). Other relations are root centralizer polynomials with real rational coefficients. Every perfect strategy extracts that root specification on a nonzero corner. Every faithful normal root realization extends simultaneously to a perfect strategy after finite amplification, preserving its factor type and, for a spatial finite-entropy model, finiteness of entropy. Only labels occurring at a left endpoint (including a range-equation label) need transport; transporting further labels is permitted and changes neither direction.*

Proof. Use the synchronized partition

$$e_0 + \sum_{g=1}^{n} m_g = I, \qquad m_g = \sum_{\epsilon,\delta \in \{0,1\}} m_{g,\epsilon\delta}, \qquad e_g = m_{g,00}.$$

For bridge g use a conditional FMS copy with flag e_{g-1}, magic support m_g, and activity $E_g^{\mathrm{act}} = e_{g-1} + m_g$. The inactive effect is $I - E_g^{\mathrm{act}}$. Match activity on every conditional question

and bind the flag and four chosen magic events to their named binary projections, in both party orientations. The flag/four-magic/inactive column partition then has precisely these six marginals. Different copies have disjoint question names; no agreement between their nonsynchronous signs is imposed.

The commuting bridge of Section 4.1.3 applies in each $E_g^{\mathrm{act}} M E_g^{\mathrm{act}}$. Indeed E_g^{act} is centralizer, so its normalized restriction has the restricted modular group. It gives $\mathsf{U}_g^* \mathsf{U}_g = e_{g-1}$, $\mathsf{U}_g \mathsf{U}_g^* = e_g$ and $\sigma_t^{\varphi}(\mathsf{U}_g) = 2^{it}\mathsf{U}_g$. These equations extend by zero off E_g^{act} and imply

$$v = \mathsf{U}_n \cdots \mathsf{U}_1, \quad v^* v = e_0, \quad v v^* = e_n, \quad \sigma_t^{\varphi}(v) = 2^{nit} v.$$

This is an extracted operator, never a centralizer word passed to the graph compiler. KMS and the magic matrix units give

$$\varphi(m_{g,\epsilon\delta}) = 2^g \varphi(e_0), \qquad \varphi(e_0) = \mathfrak{c}_n^{-1}.$$

If an active corner vanishes its flag and all four magic corners vanish together; otherwise the local bridge makes all five nonzero. The recurrence therefore holds also before nonvanishing is known, and the cover proves that $e_0 \neq 0$.

Bind every corner and every positive Pauli parameter to the same qutrit frame. Let Λ be the unitary encodings of the labels to be transported, including every label used on a left endpoint: use a unitary directly, a reflection for a projection, the displayed functional calculus for a positive contraction, and two unitaries for a general contraction. Introduce the axis labels $\mathfrak{x}_g, \mathfrak{z}_g$ of each bridge as in Lemma 4.5. For each $\ell \in \Lambda$ and $r = 0, \ldots, n$, impose $(u_{r,\ell} - I)(I - e_r) = 0$, and introduce the unitary label $V_{g,\ell}$ of (25). The orthogonal block supports make this a unitary. At bridge g apply Lemma 4.2 separately to the six frame projections and to each $G_{12}(V_{g,\ell})$. There is one refinement of each retained sign question per label per bridge, on each party, always against the unchanged opposite questions. There is no joint refinement of distinct labels. The frame refinements give $\mathsf{U}_g = I_3 \otimes \widetilde{\mathsf{U}}_g$. In the selected magic–flag block the graph commutator is exactly

$$\mathsf{U}_g u_{g-1,\ell} = u_{g,\ell} \mathsf{U}_g.$$

Induction gives $(L_{\mathrm{end}})_n v = v (L_{\mathrm{end}})_0$ for every decoded transported label and its adjoints and words. Labels used only at the source need one source encoding and no refinement.

For each edge introduce a centralizer contraction $a_\alpha = e_n a_\alpha e_0$ and extract $A_\alpha = v^* a_\alpha$. Compile the centralizer equations

$$a_\alpha^* a_\beta = \delta_{\alpha\beta} e_0, \qquad \sum_\alpha a_\alpha a_\alpha^* = e_n - (P_\circ)_n, \qquad (L_{\mathrm{end}})_n a_\alpha = a_\beta (R_{\mathrm{end}})_0$$

for the corresponding root equations; the zero endpoint uses zero on the right. The equations with a specified initial defect use that defect in the first right side. More generally a range equation $\sum_\alpha A_\alpha A_\alpha^* = R_{\mathrm{ran}}$, with R_{ran} a transported centralizer polynomial, is compiled as $\sum_\alpha a_\alpha a_\alpha^* = (R_{\mathrm{ran}})_n$. For the Catalan input of [24, Section 6] this is $R_{\mathrm{ran}} = I - BB^* - P$; its centralizer contraction B and projection P are already in the transported list. Multiplying by v or its adjoint proves both directions, since its initial and final projections are e_0, e_n. Every root-only polynomial stays at e_0. In particular the edge has modular ratio 2^{-n} and all labels refer to the same extracted root operators. Graphs, triangles and Lemma 4.3 enforce these finitely many centralizer polynomials without changing the conditional tests. This proves commuting soundness.

For completeness take a faithful normal root (M_0, φ_0) and on $\mathrm{Mat}_{1+4n} \bar{\otimes} M_0$ put

$$\varphi_{\mathrm{sc}} = \mathrm{diag}(\mu_n) \otimes \varphi_0 = \mathfrak{c}_n^{-1} \,\mathrm{diag}(1;\, 2^{\oplus 4};\, 4^{\oplus 4};\, \ldots;\, (2^n)^{\oplus 4}) \otimes \varphi_0.$$

Choose the actual matrix corner maps for U_g, the ideal real FMS effects on E_g^{act}, and $I - E_g^{\mathrm{act}}$ for every inactive answer. Take $a_\alpha = |e_n\rangle\langle e_0| \otimes A_\alpha$. Its modular ratio is $2^n 2^{-n} = 1$, so it lies in the finite centralizer. Its polar partial isometry therefore has a unitary completion, giving its two-unitary encoding. Choose the root centralizer encodings once and copy them identically to the required corners. All $V_{g,\ell}$ then commute with the FMS effects at their own bridge, even though adjacent activities overlap at e_g. Use their actual joint products for the separate refinements. Append the common scalar frame and row factors and use standard commutant mirrors as in Lemma 4.3. All old conditional probabilities and every added row are simultaneously correct. Only finite matrix factors have been appended. Their state entropy is finite; in the spatial case it adds to the root entropy. This proves completeness and the stated preservation properties. □

4.7 A dimension-independent completeness estimate

The exact construction gives a useful quantitative statement for the four-bridge compiler ($n = 4$). Keep the label types, centralizer conditions, contractivity and modular covariance exact, but allow residuals r_j in the application equations of Definition 4.4. Write $\|T\|_{2,h_0}^2 = \mathrm{Tr}(h_0 T^* T)$. The constructed strategy satisfies

$$1 - \omega(G, \mathcal{S}) \le C_{\mathrm{comp}} \sum_j \|r_j\|_{2,h_0}^2, \tag{28}$$

where C_{comp} depends only on the finite presentation and the compiler.

Here is the proof, including the density factors. The same scaffold, exact unitary encodings and actual products make all structural tests perfect. Only the final application row-zero tests may lose. Let p_j be the lifted polynomial residual. For a source or root-polynomial row it is the source copy of r_j; for an endpoint row it is vr_j; for a range row it is $vr_j v^*$. Consequently

$$\|p_j\|_{2,h_{\mathrm{sc}}}^2 = \frac{\beta_j}{121}\|r_j\|_{2,h_0}^2, \qquad \beta_j = \begin{cases} 16, & \text{range row at } e_4, \\ 1, & \text{otherwise.} \end{cases}$$

For the endpoint case use $v^* v = e_0$. For the range case use $v^* h_{\mathrm{sc}} v = 16 h_0/121$.

The qutrit graph row Δ_j has off-diagonal blocks p_j, p_j^*. Since p_j commutes with h_{sc},

$$\mathrm{Tr}(h_{\mathrm{old}}\Delta_j^2) = \frac{1}{3}\,\mathrm{Tr}[h_{\mathrm{sc}}(p_j^* p_j + p_j p_j^*)] = \frac{2\beta_j}{363}\|r_j\|_{2,h_0}^2.$$

Apply (19), with L_j and d_j the denominator and slot dimension of row j. An orientation of the row-zero test is a choice of which party receives the row question, and each orientation has probability (19). If the forbidden pair is imposed in both ordered question pairs, as in the formal compiler of Appendix F, both contributions count. For uniform Q-by-Q sampling one may therefore take

$$C_{\mathrm{comp}} = \max_j \frac{16 L_j^2 \beta_j}{363 d_j^3 Q^2}. \tag{29}$$

With one oriented row-zero test per row, the convention of (54), the coefficient 16 may be replaced by 8. Summing event probabilities gives an upper bound even if rejecting rules overlap. The dimension, clock order, density spectrum and spectator dimensions do not enter this constant. The estimate applies to the constructed strategies, whose structural tests are exact.

5 The rooted tree and computation squares

Sections 5–7 prove the hardness half of Theorem A. The game is the four-bridge compilation of one admissible presentation, the Fock clock. Its root carries a four-ary tree, each depth of the tree is a finite square, and a deterministic computation runs on the squares. A Weyl clock supplies the existential witness in Section 6.

5.1 Normality removes every rootless sector

Require four isometries and a root projection with

$$A_{ab}^{*}A_{cd}=\delta_{ac}\delta_{bd}I,\qquad \sum_{ab}A_{ab}A_{ab}^{*}=I-P_{\circ},\qquad h_0A_{ab}=A_{ab}h_0/16. \tag{30}$$

Here h_0 is a faithful density on the extracted root-corner Hilbert space. For $w=w_1\cdots w_m$ set $A_w=A_{w_1}\cdots A_{w_m}$ (the empty product is I) and put $Q_m=\sum_{|w|=m}A_wA_w^{*}$. Orthogonality and trace cyclicity give $\operatorname{Tr}(h_0Q_m)=4^{-m}$. The projections Q_m decrease. Their strong limit has zero h_0-weight, hence is zero by faithfulness. Also $Q_m-Q_{m+1}=\sum_{|w|=m}A_wP_{\circ}A_w^{*}$. Therefore

$$\mathcal{H}=\bigoplus_{w}A_w\mathcal{K}\simeq\ell^2(\mathcal{W})\otimes\mathcal{K},\qquad \mathcal{K}=P_{\circ}\mathcal{H}\neq 0. \tag{31}$$

Here $\mathcal{W}$ denotes finite words in the four-letter alphabet $\{0,1\}^2$. The modular equation makes h_0 commute with every $A_wP_{\circ}A_w^{*}$. Since $\operatorname{Tr}(h_0P_{\circ})=3/4$, transporting its root block gives

$$h_0=h_F\otimes\rho_{\mathrm{spec}},\qquad h_F=\frac{3}{4}\sum_{w}16^{-|w|}|w\rangle\langle w|,\qquad \operatorname{Tr}\rho_{\mathrm{spec}}=1. \tag{32}$$

In particular

$$S(h_F)=\log_2(4/3)+4/3. \tag{33}$$

Indeed the depth-m mass is $(3/4)4^{-m}$ and its mean depth is $1/3$; the log probability at each word is $\log_2(4/3)+4m$. This proves the formula and excludes a rootless normal summand. Section 16.2 repeats this calculation at a general ratio; there $h_F=h_{F,1/16}$.

5.2 Each depth is a finite square

For $w=(a_1,b_1)\cdots(a_m,b_m)$ define

$$x(w)=\sum_{j=1}^{m}a_j2^{j-1},\qquad y(w)=\sum_{j=1}^{m}b_j2^{j-1}.$$

The first letter is the least significant bit. Prefixing A_{ab} maps (x,y) to $(2x+a,2y+b)$. Depth m is a $2^m\times 2^m$ square. Require centralizer unitaries U_x,U_y and projections P_x,P_y satisfying, for $a,b\in\{0,1\}$,

$$U_xA_{0b}=A_{1b},\qquad U_xA_{1b}=A_{0b}U_x,\qquad U_xP_{\circ}=P_{\circ}, \tag{34}$$
$$U_yA_{a0}=A_{a1},\qquad U_yA_{a1}=A_{a0}U_y,\qquad U_yP_{\circ}=P_{\circ}, \tag{35}$$
$$P_xA_{0b}=A_{0b}P_x,\qquad P_xA_{1b}=0,\qquad P_xP_{\circ}=P_{\circ}, \tag{36}$$
$$P_yA_{a0}=A_{a0}P_y,\qquad P_yA_{a1}=0,\qquad P_yP_{\circ}=P_{\circ}. \tag{37}$$

Induction on word length identifies U_x and U_y as cyclic increments of x and y respectively, acting identically on $\mathcal{K}$. The same induction makes P_x,P_y the indicators of $x=0,y=0$. In particular U_x,U_y commute, and

$$P_{x,\mathrm{last}}=U_x^{*}P_xU_x,\qquad P_{y,\mathrm{last}}=U_y^{*}P_yU_y$$

mark the final column and row. These identities hold on the full root multiplicity space.

5.3 A finite deterministic tableau

Choose a deterministic radius-one cellular automaton on the half-line. Its finite alphabet Γ_{CA} includes start `s`, blank `b` and halt `H`, and its update is $\Phi(\ell, c, r, \varepsilon)$ with left boundary symbol # and external pulse $\varepsilon \in \{0, 1\}$. The initial row is $\mathtt{s}, \mathtt{b}, \mathtt{b}, \ldots$. We use these conventions:

- $\Phi(\mathtt{b}, \mathtt{b}, \mathtt{b}, \varepsilon) = \mathtt{b}$ for both pulse values.
- The initial start state ignores the pulse at time zero. Later waiting-front states respond to positive-time pulses.
- Cells passed by the waiting front keep their unary marks. Frozen cells and all states of the later input-checking computation ignore subsequent pulses.
- A transition using the pulse at time y produces row $y + 1$. Thus the pulse at time d_{clk} records the front at coordinate d_{clk} in that next row.

Here is an explicit front layer, with alphabet $\{\mathtt{s}, \mathtt{b}, \mathtt{f}, \mathtt{z}, \mathtt{H}\}$; `f` is the front state. The right neighbour is unused in these six rules:

condition	next centre
$c = \mathtt{s}$	`b`
$c = \mathtt{f}$	`z` if $\varepsilon = 1$, `b` if $\varepsilon = 0$
$c = \mathtt{b},\ \ell = \mathtt{s}$	`f`
$c = \mathtt{b},\ \ell = \mathtt{f}$	`f` if $\varepsilon = 0$, `b` if $\varepsilon = 1$
$c = \mathtt{z}$	`z`
$c = \mathtt{H}$	`H`

All remaining blank centres stay blank. At time $y \geq 1$, until a return, the unique front is at $x = y$. The first positive pulse at d_{clk} writes `z` at $(d_{\mathrm{clk}}, d_{\mathrm{clk}} + 1)$ and suppresses the front's successor. The frozen mark's distance from the left boundary is the unary input d_{clk}; blank cells between the boundary and the mark represent its unary positions. Pulse-dependence is confined to neighbourhoods containing the front. In particular a blank right of `f` may read the pulse.

Add a persistent unary bit $u \in \{0, 1\}$, initially zero, with $u' = 1$ when the old front centre is `s` or `f`, and $u' = u$ otherwise. Thus passed cells retain their unary marks on a separate track, even though their front-layer component is blank. Take the product alphabet $\{\mathtt{s}, \mathtt{b}, \mathtt{f}, \mathtt{z}, \mathtt{H}\} \times \{0, 1\} \times \mathcal{T}_{\mathrm{TM}}$, with a finite TM alphabet $\mathcal{T}_{\mathrm{TM}} = \mathcal{A}_{\mathrm{tape}} \times (Q_{\mathrm{TM}} \cup \{\text{no head}\})$ for the fixed one-tape half-line program in Section 7. All TM cells start blank without a head. The transition $\mathtt{f} \to \mathtt{z}$ initializes its head in the start state at that cell. Thereafter a centre carrying (a, p) writes the symbol prescribed by $\delta(p, a, u)$ and retains a head only for a stationary move. A head moving right (left) is received by the right (left) neighbour, whose old tape symbol is retained. A centre receiving no head retains its symbol without a head. Invalid multiple incoming heads may be sent to a fixed error state; they never occur from this initialization. The left boundary # is read as a tape end marker and no transition moves left of it. The frozen front track is read-only input for this TM. A halting head writes a designated halt symbol `H` in the combined alphabet and keeps it. These prescriptions give a finite radius-one lookup table effectively from the finite TM transition table; all subsequent TM transitions ignore the pulse. Denote this finite product alphabet, with the halt convention, by Γ_{CA}. In the combined alphabet, `s` and `b` abbreviate the start and fully blank product states. Collapse any product state with a halting head to the single absorbing symbol `H`.

Blank quiescence holds before a front or head arrives. The `s` rule ignores the time-zero pulse and seeds `f` at time one via its right neighbour. The mark is permanent and later pulses cannot restart a head. Finally, all disturbances start at the origin, or at the marked front within its radius-one cone, so no nonblank cell can occur at $x > y$. These checks give all four conventions and the blank right ghost used below.

Let $(P_c)_{c\in\Gamma_{\mathrm{CA}}}$ be a centralizer PVM and P_{ret} a centralizer projection, with $P_{\mathrm{ret},0} = I - P_{\mathrm{ret}}$, $P_{\mathrm{ret},1} = P_{\mathrm{ret}}$. Require

$$P_{\mathtt{s}}P_xP_y = P_xP_y, \qquad P_{\mathtt{b}}(I-P_x)P_y = (I-P_x)P_y, \qquad P_{\mathtt{H}} = 0. \tag{38}$$

Define centralizer polynomials

$$\begin{aligned} L_{\#} &= P_x, & L_c &= U_xP_cU_x^*(I-P_x),\\ N_r &= U_x^*P_rU_x(I-P_{x,\mathrm{last}}) + \mathbf{1}_{\{r=\mathtt{b}\}}P_{x,\mathrm{last}}. \end{aligned}$$

For every $a \in \Gamma_{\mathrm{CA}}$ impose

$$\left[U_y^*P_aU_y - \sum_{\Phi(\ell,c,r,\varepsilon)=a} L_\ell P_c N_r P_{\mathrm{ret},\varepsilon}\right](I - P_{y,\mathrm{last}}) = 0. \tag{39}$$

The factors have the displayed order. They need not commute globally. The factor $I - P_{y,\mathrm{last}}$ removes a spurious transition from the final row to row zero. The ghost to the right of the last column is blank.

Lemma 5.1 (Definite computation)**.** *Suppose a root vector ξ belongs to a sector where the clock pulse on every coordinate has a specified scalar value. Then every P_c acts on $A_w\xi$ as the indicator of the genuine classical computation at $(x(w), y(w))$.*

Proof. Work at a fixed depth m and induct on time, for every spatial coordinate at once. Equation (38) gives row zero. On an induction vector every factor $L_\ell, P_c, N_r, P_{\mathrm{ret},\varepsilon}$ acts by its classical zero or one. Their ordered product therefore selects exactly one update. Equation (39) gives $U_y^*P_aU_y$ with the correct scalar on that vector. Multiplying by U_y gives the next-row assertion.

The half-line and the square agree throughout this induction. At time y no disturbance has travelled beyond coordinate y. In particular the ghost cell 2^m is blank for all times $y \le 2^m - 1$, by radius one and quiescence under either pulse. The left ghost is always #. This proves the genuine-prefix assertion. A halting event at finite coordinates and time occurs in every sufficiently large square and contradicts $P_{\mathtt{H}} = 0$. □

The tableaux at distinct depths need not intertwine with the creation operators. Determinism forces each of them to the same finite prefix; soundness needs only one sufficiently large square. Every depth has positive normal weight by (32).

5.4 A worked front and mask example

Table 4 shows the depth-3 square for a clock of order three; the verification software regenerates it (Appendix C). Its `d` denotes a passed cell and `m` a frozen mark. These persistent annotations differ from the minimal front layer above, whose passed cells are blank and whose mark is `z`; both put the mark at $(3, 4)$. The start state `s` ignores the pulse at time zero. The pulse at $y = 3$ produces the mark in row 4; the later pulse at $y = 6$ has no effect.

6 A finite-order clock and its period reader

6.1 A determinant supplies a finite witness

Add centralizer unitaries $J_{\mathrm{W}}, K_{\mathrm{W}}, Z_{\mathrm{clk}}$ with

$$\begin{gathered} J_{\mathrm{W}}K_{\mathrm{W}} = Z_{\mathrm{clk}}K_{\mathrm{W}}J_{\mathrm{W}}, \qquad [Z_{\mathrm{clk}}, J_{\mathrm{W}}] = [Z_{\mathrm{clk}}, K_{\mathrm{W}}] = 0,\\ [J_{\mathrm{W}}, A_{ab}] = [K_{\mathrm{W}}, A_{ab}] = [Z_{\mathrm{clk}}, A_{ab}] = 0. \end{gathered} \tag{40}$$

y	pulse	0	1	2	3	4	5	6	7	update mask
0	1	s	b	b	b	b	b	b	b	1
1	0	d	f	b	b	b	b	b	b	1
2	0	d	d	f	b	b	b	b	b	1
3	1	d	d	d	f	b	b	b	b	1
4	0	d	d	d	m	b	b	b	b	1
5	0	d	d	d	m	b	b	b	b	1
6	1	d	d	d	m	b	b	b	b	1
7	0	d	d	d	m	b	b	b	b	0

Table 4: Depth $m = 3$, period $q = 3$: the decoder's 8×8 toy tableau. P_x marks column 0, $P_{x,\mathrm{last}}$ column 7, P_y row 0 and $P_{y,\mathrm{last}}$ row 7. The last-column neighbour is replaced by blank; the last-row mask suppresses the toroidal $7 \to 0$ update.

A unitary commuting with an isometry also commutes with its adjoint: its inverse commutes with the isometry, and taking adjoints gives the assertion. Thus these labels reduce $\mathcal{K} = P_\circ \mathcal{H}$ and commute with ρ_{spec}. Their action on every word sector is their root action.

Every positive eigenspace of ρ_{spec} is finite-dimensional. It reduces $J_{\mathrm{W}}, K_{\mathrm{W}}, Z_{\mathrm{clk}}$ and decomposes into Z_{clk} eigenspaces. On a nonzero such eigenspace of dimension d, with phase ζ, the Weyl relation gives

$$\det(J_{\mathrm{W}} K_{\mathrm{W}}) = \zeta^d \det(K_{\mathrm{W}} J_{\mathrm{W}}), \qquad \zeta^d = 1. \tag{41}$$

Since the positive eigenspaces exhaust $\mathcal{K}$, Z_{clk} has pure point spectrum there and every phase is a root of unity. At least one sector is nonzero. Conversely a d_{clk}-dimensional Weyl pair realizes every primitive d_{clk}th root, with density $I_{d_{\mathrm{clk}}}/d_{\mathrm{clk}}$.

6.2 The clock and a counter without a positive lower bound

Take a centralizer unitary W and a positive centralizer contraction C_{ctr} with

$$W P_\circ = P_\circ, \qquad W A_{ab} = A_{ab} Z_{\mathrm{clk}}^b W^2, \tag{42}$$

$$C_{\mathrm{ctr}} P_\circ = P_\circ, \qquad C_{\mathrm{ctr}} A_{ab} = 2^{-b} A_{ab} C_{\mathrm{ctr}}^2. \tag{43}$$

Also require Z_{clk} to commute with $U_x, U_y, P_x, P_y, W, C_{\mathrm{ctr}}, P_{\mathrm{ret}}, T_{\mathrm{inv}}$ and all P_c. Induction on word length, using the coordinate recurrence, gives

$$W|_{(x,y)} = Z_{\mathrm{clk}}^y, \qquad C_{\mathrm{ctr}}|_{(x,y)} = 2^{-y} I. \tag{44}$$

In particular C_{ctr} is injective even though its spectrum accumulates at zero.

For a centralizer projection P_{ret} and contraction T_{inv}, impose

$$(W - I) P_{\mathrm{ret}} = 0, \qquad T_{\mathrm{inv}}(W - I) = \tfrac{1}{2} C_{\mathrm{ctr}}(I - P_{\mathrm{ret}}). \tag{45}$$

The first equation puts $\operatorname{ran} P_{\mathrm{ret}}$ in $\ker(W - I)$. For ξ in that kernel the second gives $C_{\mathrm{ctr}}(I - P_{\mathrm{ret}})\xi = 0$. Injectivity of C_{ctr} gives $P_{\mathrm{ret}}\xi = \xi$. Thus

$$P_{\mathrm{ret}} = \operatorname{proj} \ker(W - I). \tag{46}$$

On a root phase sector of order d_{clk}, the pulse is exactly $\mathbf{1}_{\{d_{\mathrm{clk}} | y\}}$ at every spatial coordinate. It cannot vanish spuriously or postpone the first return.

6.3 A bounded inverse on every prime clock

Lemma 6.1 (A good primitive phase)**.** *For every prime d_{clk} there is a primitive d_{clk}th root ζ such that*

$$|\zeta^t - 1| \geq 2^{-t-1} \quad (t \geq 1,\ d_{\mathrm{clk}} \nmid t).$$

Proof. For $1 \leq r < d_{\mathrm{clk}}$,

$$|e^{2\pi i r/d_{\mathrm{clk}}} - 1| \geq 4 \min(r, d_{\mathrm{clk}} - r)/d_{\mathrm{clk}}.$$

Fix $1 \leq t < d_{\mathrm{clk}}$. Multiplication by t permutes nonzero residues. A phase $e^{2\pi i k/d_{\mathrm{clk}}}$ violating the required inequality must satisfy $\min(tk \bmod d_{\mathrm{clk}}, d_{\mathrm{clk}} - (tk \bmod d_{\mathrm{clk}})) < d_{\mathrm{clk}} 2^{-t-3}$. There are at most $d_{\mathrm{clk}} 2^{-t-2}$ such residues. Summing over t excludes fewer than $d_{\mathrm{clk}}/4$ choices, while $d_{\mathrm{clk}}/4 < d_{\mathrm{clk}} - 1$. One of the $d_{\mathrm{clk}} - 1$ nontrivial phases works. For $t \geq d_{\mathrm{clk}}$ reduce to a nonzero residue $r < d_{\mathrm{clk}}$; its lower bound 2^{-r-1} is at least 2^{-t-1}. □

For this phase set

$$T_{\mathrm{inv}}|_{(x,y)} = \begin{cases} 2^{-y-1}(\zeta^y - 1)^{-1} I, & d_{\mathrm{clk}} \nmid y, \\ 0, & d_{\mathrm{clk}} \mid y. \end{cases} \tag{47}$$

Lemma 6.1 gives $\|T_{\mathrm{inv}}\| \leq 1$. Equations (44) and (46) then verify (45). The phase belongs to the strategy, so the game coefficients remain rational.

7 Spatial hardness: the reduction and value one

Fix any decidable predicate $\mathcal{R}_{\mathrm{dec}}(e, j, t)$ and the set $\mathcal{A}_{\mathrm{dec}} = \{e : \exists j \, \forall t \ \mathcal{R}_{\mathrm{dec}}(e, j, t)\}$. We effectively choose the cellular automaton in Section 5 to run the following program, with e hardwired.

Read a positive-time return. At time y a waiting front is at position y. Ignore the time-zero pulse. At the first positive-time pulse, freeze the front and retain its coordinate as unary input d_{clk}. With the transition convention above the mark is written into row $d_{\mathrm{clk}} + 1$ at position d_{clk}. Test d_{clk}. If $d_{\mathrm{clk}} = 1$ or d_{clk} is composite, halt. Otherwise compute the unique $j \geq 0$ with $d_{\mathrm{clk}} = p_{j+1}$, the $(j+1)$st prime. Then evaluate $\mathcal{R}_{\mathrm{dec}}(e, j, 0), \mathcal{R}_{\mathrm{dec}}(e, j, 1), \ldots$ and halt at the first false answer. All later computation states ignore the external pulse.

For a Σ_2^0-complete choice of $\mathcal{A}_{\mathrm{dec}}$, there is no computable bound on a sufficient clock order as a function of the input game: such a bound would leave only finitely many possible j in this reduction, making $\exists j \, \forall t \ \mathcal{R}_{\mathrm{dec}}(e, j, t)$ a Π_1^0 predicate. Indeed one tests, for every finite time cutoff, whether some one of those finitely many candidates survives all checks so far; if none survives forever, all have failed by a common finite cutoff. This contradicts Σ_2^0-hardness.

This finite program has an effective finite radius-one simulation. The initial-front rule, frozen unary track and ordinary tape/head tracks give the stated boundary conventions. We do not require a bound on its running time. Each completed false check causes the halt symbol after finitely many updates.

The clock presentation and its labels. Include the Fock, coordinate, clock, reader and tableau equations above, all typing relations, and the stated Z_{clk} commutators. They form an admissible finite presentation with $n = 4$. In the four-bridge compiler of Lemma 4.5 the transported typed list is exactly

$$P_\circ,\ U_x,\ U_y,\ P_x,\ P_y,\ W,\ C_{\mathrm{ctr}},\ Z_{\mathrm{clk}},\ J_{\mathrm{W}},\ K_{\mathrm{W}}. \tag{48}$$

Table 5 records the label types. Here $P_\circ, P_x, P_y$ are projections, C_{ctr} is positive, and the other six are unitary. Thus one unitary encoding per typed label suffices, so $|\Lambda| = 10$. There are fifty support equations, forty block equations and sixteen refinements per bridge per party. Source-only labels $P_{\mathrm{ret}}, T_{\mathrm{inv}}, P_c$ do not occur at left endpoints of an A_α relation and need no transport. For example the transported polynomials include

$$W_4 a_{ab} = a_{ab} Z_{\mathrm{clk},0}^b W_0^2, \qquad C_{\mathrm{ctr},4} a_{ab} = 2^{-b} a_{ab} C_{\mathrm{ctr},0}^2, \qquad J_{\mathrm{W},4} a_{ab} = a_{ab} J_{\mathrm{W},0}.$$

The source-only return equations and tableau equations are centralizer polynomials. Lemma 4.5 therefore computes a finite rational Boolean game G_e^{clk}.

Label	Type	Transported?	Use
$P_\circ$	projection	yes	Fock root
U_x, U_y	unitary	yes	coordinate increments
P_x, P_y	projection	yes	boundaries
W	unitary	yes	clock
C_{ctr}	positive contraction	yes	counter
$Z_{\mathrm{clk}}, J_{\mathrm{W}}, K_{\mathrm{W}}$	unitary	yes	Weyl clock
P_{ret}	projection	no	pulse
T_{inv}	general contraction	no	reader inverse
P_c	projection	no	tableau
a_α	general contraction	scaffold only	lifted creation

Table 5: Typed application labels of the clock presentation. Transported labels are copied through all bridges for use at a left endpoint.

$$\begin{gathered}
\text{Presentation} \xrightarrow{\text{Lemma 4.5}} \text{finite game } G_e^{\mathrm{clk}} \\
\text{Perfect normal strategy} \longrightarrow \text{normal realization} \\
\longrightarrow \text{Fock tree} \longrightarrow \text{finite clock sector} \\
\longrightarrow \text{deterministic tableau} \longrightarrow \forall t\ \mathcal{R}_{\mathrm{dec}}(e,j,t) \\
\text{Witness } j \longrightarrow \text{Fock/Weyl model} \\
\longrightarrow \text{scaffold, frame, row ancilla} \longrightarrow \text{perfect strategy}
\end{gathered}$$

Figure 4: Dependencies of the spatial hardness proof. Compactness and Bell attainment have separate upper-bound arguments.

Theorem 7.1. *The map $e \mapsto G_e^{\mathrm{clk}}$ satisfies*

$$G_e^{\mathrm{clk}} \in \mathrm{PERF}_{\mathrm{qs}} \iff G_e^{\mathrm{clk}} \in \mathrm{PERF}_{\mathrm{finS}} \iff \exists j\, \forall t\ \mathcal{R}_{\mathrm{dec}}(e,j,t).$$

No G_e^{clk} has a perfect finite-dimensional strategy.

Proof. A perfect normal strategy extracts a nonzero faithful root realization. The determinant argument supplies a nonzero root phase sector of finite order d_{clk}. By (40) its word descendants persist at all depths. Equations (42)–(46) force the exact return sequence, with its first positive return at d_{clk}. Lemma 5.1 needs only that the initial root vector lie in this Z_{clk}-eigenspace: the clock pulse is then scalar on all its word descendants. The imposed $[Z_{\mathrm{clk}}, P_c] = 0$ relations are redundant for this induction. An invalid d_{clk} halts. For valid $d_{\mathrm{clk}} = p_{j+1}$ any false $\mathcal{R}_{\mathrm{dec}}(e,j,t)$ also halts. A sufficiently large square would contain that halt, contrary to $P_{\mathtt{H}} = 0$. Hence the extracted j witnesses membership in $\mathcal{A}_{\mathrm{dec}}$.

Conversely take a witness j and set $d_{\mathrm{clk}} = p_{j+1}$. Use (32) with $\rho_{\mathrm{spec}} = I_{d_{\mathrm{clk}}}/d_{\mathrm{clk}}$, a good primitive phase and its Weyl pair, the canonical coordinate permutations, (44), (46), (47),

and the classical nonhalting tableau. Every required operator is bounded and commutes with the density when so specified. This is a simultaneous realization. On $\mathbb{C}^{d_{\mathrm{clk}}}$ one may take $J_{\mathrm{W}}|i\rangle = \zeta^i|i\rangle$, $K_{\mathrm{W}}|i\rangle = |i+1 \bmod d_{\mathrm{clk}}\rangle$, and $Z_{\mathrm{clk}} = \zeta I$. Its root entropy and largest Schmidt probability are

$$S(h) = \log_2 d_{\mathrm{clk}} + \log_2(4/3) + 4/3, \qquad p_{\max} = 3/(4d_{\mathrm{clk}}). \tag{49}$$

The compiler adds finite factors, and the resulting perfect strategy can be written explicitly. Let $\mathcal{H}_F = \ell^2(\mathcal{W})$ and let $\mathcal{H}_{\mathrm{sc}} = \mathbb{C}^{17}$ index the scaffold corners. Take

$$\mathcal{H}_A = \mathcal{H}_B = \mathcal{H}_{\mathrm{sc}} \otimes \mathcal{H}_F \otimes \mathbb{C}^{d_{\mathrm{clk}}} \otimes \mathbb{C}^3 \otimes \mathbb{C}^{d_{\max}}.$$

With $\mu_4 = (1,2,2,2,2,4,4,4,4,8,8,8,8,16,16,16,16)/121$, use the pure vector

$$|\psi\rangle = |\sqrt{\mu_4}\rangle\!\rangle \otimes |\sqrt{h_F}\rangle\!\rangle \otimes |\Phi_{d_{\mathrm{clk}}}\rangle \otimes |\Phi_3\rangle \otimes |\Phi_{d_{\max}}\rangle, \quad |\sqrt{\mu_4}\rangle\!\rangle = \sum_i \sqrt{\mu_{4,i}}|i,i\rangle.$$

The Fock factor is $\sum_w \sqrt{(3/4)16^{-|w|}}|w,w\rangle$. Alice uses the scaffold FMS effects, actual graph projections, refinements and column partitions of Lemma 4.5; Bob uses the prescribed opposite FMS effects and the transposed transferred centralizer products. The row factor is the finite maximally entangled ancilla of Lemma 4.3. These simultaneous effects were checked in the compiler proof. The marginal entropy is $H(\mu_4) + S(h_F) + \log_2 d_{\mathrm{clk}} + \log_2 3 + \log_2 d_{\max} < \infty$.

Finally finite-dimensional perfect play would extract four isometries with mutually orthogonal ranges on one nonzero finite-dimensional root space. Their range dimensions would sum to four times the dimension of that space, which is impossible. The argument applies to every input, whether or not it has a witness. □

Together with Section 3, this proves both completeness statements in Theorem A. The value-one promise is established next. For the promise version, the reduction maps every input into that promise; the unrestricted Σ_2^0 membership predicate restricts to it, giving promise-problem completeness. The construction can be made syntactically nonsynchronous and non-imitation on every input by adding a fresh binary question to each side, accepting every pair involving it. Either answer then has the same nonempty allowed-answer support against every opposite question, which violates the imitation separation condition. This addition does not change perfection or the value-one property.

Size of the finite game. Here is a coarse order-of-magnitude bound, independent of the witness d_{clk}. Let $|\Gamma_{\mathrm{CA}}|$ be the alphabet size, let N_e^{proj} count the named graph projections, rows and word prefixes after endpoint translation, and let b_e bound the bit length of their rational coefficients. The lookup table has at most $2(|\Gamma_{\mathrm{CA}}|+1)|\Gamma_{\mathrm{CA}}|^2$ entries, so unshared explicit polynomials give $N_e^{\mathrm{proj}} = \mathrm{poly}(|\Gamma_{\mathrm{CA}}|)$ for this fixed list of operator relations. Denominator clearing and signed repetitions give row dimension $d_{\max} \le 2^{O(N_e^{\mathrm{proj}}(b_e + \log(N_e^{\mathrm{proj}}+1)))}$. The scalar contexts, column questions and marginal bindings are polynomial in $N_e^{\mathrm{proj}}, d_{\max}$, so the padded question count Q_e and maximum answer count obey the same coarse exponential bound. An explicit predicate table is polynomial in those two counts and has the same exponential order. Thus the effective compilation depends on the hardwired program, never on its chosen prime or running time. For scale, the small compiled example of Appendix C, with one modular isometry and no transported labels, already has 748 rows and 27 991 questions per party. For the full ten-label instance with $e = 0$ we counted the questions without writing out the game. Expanding the tableau polynomials literally gives 6 191 638 087 questions per party. Sharing bounded-circuit gates among the eight update rows gives 8 516 580; this encoding adds source-only gates and keeps the soundness and completeness translation. Both counts use an explicit eight-symbol scan automaton for $e = 0$, with alphabet `s`, `b`, `f`, `d`, `z0`, `z`, `h`, `H`, which halts unless the clock order is two; it replaces the generic product alphabet of Section 5.3.

7.1 Every constructed game has value one

The finite-dimensional approximations do not need a witness for e. Truncate the Fock tree to depths $0, \dots, m$ and choose a prime $d_{\mathrm{clk}} > 2^m$ with a good phase (Lemma 6.1). Use the normalized density

$$h^{[m]} = \frac{\sum_{|w|\le m} 16^{-|w|} |w\rangle\langle w|}{\sum_{i=0}^{m} 4^{-i}} \otimes \frac{I_{d_{\mathrm{clk}}}}{d_{\mathrm{clk}}}.$$

Every represented time is less than d_{clk}. The decoder therefore remains in the waiting stage in every square and never halts. Let P_m denote the top-depth projection. Its mass is

$$t_m = \mathrm{Tr}(h^{[m]} P_m) = \frac{3}{4^{m+1} - 1}. \tag{50}$$

Every endpoint identity $L_{\mathrm{end}} A_\alpha = A_\beta R_{\mathrm{end}}$ remains exact. Below top depth it is the canonical identity; on top depth both sides vanish because the endpoint expression R_{end} preserves depth. The modular equations, the range-sum identity and all off-diagonal $A_\alpha^* A_\beta = 0$ also remain exact. The only failed original relations are the four diagonal isometry equations:

$$A_\alpha^* A_\alpha - I = -P_m \quad (\alpha = 00, 01, 10, 11). \tag{51}$$

All graph and triangle projections are defined from the actual products. Every structural graph, synchrony, binding, FMS, refinement and column test stays exact.

Here is the full row accounting. For each α let ℓ_α be the denominator-clearing integer and d_α the column dimension of the row obtained by translating $a_\alpha^* a_\alpha - e_0 = 0$ into graph projections. They depend only on the finite presentation, so they are independent of m and d_{clk}. In the scalar qutrit frame its defect Δ_α has off-diagonal blocks $-(P_m)_0$ and adjoint and zero diagonal blocks. The frame density is $I_3/3$ and the source scaffold mass is $1/121$, so

$$\mathrm{Tr}(h_{\mathrm{old}} \Delta_\alpha^2) = \frac{2}{3} \frac{1}{121} t_m. \tag{52}$$

The same calculation applies separately to each of the four rows. Their row-zero events have distinct column questions and occur in no other failed rule. If the game has Q_e equally padded questions per party, (19) gives

$$1 - \omega(G_e^{\mathrm{clk}}, \mathcal{S}_m) \le C_e \frac{3}{4^{m+1} - 1}, \tag{53}$$

$$C_e = \frac{1}{Q_e^2} \frac{2}{363} \sum_{\alpha \in \{00,01,10,11\}} \frac{4\ell_\alpha^2}{d_\alpha^3}. \tag{54}$$

If the four row constructions have common ℓ, d, this is $C_e = 32\ell^2/(363 Q_e^2 d^3)$. The constant counts one oriented row-zero test per row; testing both orientations would double it.

Since the density factor $I_{d_{\mathrm{clk}}}/d_{\mathrm{clk}}$ has trace one, no factor of d_{clk} enters the loss. Thus $\omega_{\mathrm{q}}(G_e^{\mathrm{clk}}) = 1$ for every e, proving the promise in Theorem A.

8 Exact Bell attainment

The same compactness decides when a rational Bell functional attains a value exactly. A Bell functional generalizes a game: its coefficients may be any rationals, and attainment of its supremum replaces perfection.

Theorem 8.1 (Bell attainment)**.** *For rational Bell functionals, each of the following is Σ_2^0-complete in the normal spatial model: attainment of a supplied rational global upper bound; attainment of the supremum, with no bound supplied; and attainment of a supplied rational scalar value. In the first problem, validity of the supplied bound can either be promised or required as part of the decision problem.*

The Bell functional I_{3322} is an explicit instance of the second problem in which finite dimension does not suffice. Pál and Vértesi computed its supremum numerically, with finite-dimensional strategies converging to it, and conjectured that no finite dimension attains it [49], and Krohn-Grimberghe has claimed a computer-assisted exact identification of the value [36]. The supremum is attained by no finite-dimensional strategy [51, 10, 18], and it is attained by an infinite-dimensional spatial strategy [10, 18].

Let $L(p) = \sum_{xyab} c_{xyab} p(a, b \mid x, y)$ have rational coefficients and let $C_{\mathrm{abs}} = \sum |c_{xyab}|$. Its value on any strategy lies in $[-C_{\mathrm{abs}}, C_{\mathrm{abs}}]$. Finite Schmidt truncation shows that its supremum over spatial strategies equals its supremum over finite-dimensional strategies.

8.1 A specified global bound

Assume first that $\beta \in \mathbb{Q}$ is a global upper bound. For fixed measurements the Bell operator

$$B(E, F) = \sum c_{xyab} E_{a|x} \otimes F_{b|y}$$

satisfies $B \leq \beta I$. A normal state attains β only if every positive-weight pure component does so. For a pure vector this is equivalent to $B\psi = \beta\psi$, because $\beta I - B$ is positive. Compress to its Schmidt supports and normalize the largest amplitude as before. Vectorization yields the homogeneous equation

$$\sum c_{xyab} E_{a|x} D_{\mathrm{Sch}} F_{b|y}^{\mathsf{T}} = \beta D_{\mathrm{Sch}}. \tag{55}$$

At stage d_{cut} use the variables of (3) and require every squared entry of the residual in (55) to be at most $C_{\mathrm{abs}}^2 \kappa/(d_{\mathrm{cut}} + 1)$. Compression of a true witness satisfies this bound: only the compact tails of the sandwiched terms contribute, with coefficient sum C_{abs}. Conversely the compact-sandwich lemma gives an exact nonzero limit solution. Its vectorization is an attaining strategy. Thus attainment is $\exists\kappa\, \forall d_{\mathrm{cut}}$ of a decidable predicate.

If validity is not promised, impose also $\omega_q(L) \leq \beta$. This is Π_1^0: for each finite dimension, the universal assertion that all states and POVMs give value at most β is decidable by real quantifier elimination. Finite-dimensional density states suffice for this validity test. The conjunction remains Σ_2^0.

8.2 No bound supplied

The quantifier pattern is explicitly $\exists\kappa \geq 1\, \forall d_{\mathrm{cut}} \geq 1\ F_{\mathrm{Bell}}(L, \kappa, d_{\mathrm{cut}})$, with F_{Bell} decidable as follows. At stage d_{cut} introduce also a real variable $\beta \in [-C_{\mathrm{abs}}, C_{\mathrm{abs}}]$, use the same residual bound, and require $\beta \geq \omega_{d_{\mathrm{cut}}}(L)$, where $\omega_{d_{\mathrm{cut}}}$ is the maximum in local dimension d_{cut}. This requirement is a universal semialgebraic formula in β, so no numerical value of $\omega_{d_{\mathrm{cut}}}(L)$ is needed. Full real quantifier elimination decides the stage formula with its existential and universal real variables.

An attaining strategy supplies all stages. Conversely pass to a subsequence with $\beta_{d_{\mathrm{cut}}} \to \beta_*$, $D_{\mathrm{Sch},d_{\mathrm{cut}}} \to D_{\mathrm{Sch}}$ in norm and effects weakly convergent. For every fixed dimension d, all sufficiently late stages have $\beta_{d_{\mathrm{cut}}} \geq \omega_d(L)$. Hence $\beta_* \geq \sup_d \omega_d(L)$. The limit equation (55) gives a unit vector attaining β_*. Its expectation cannot exceed that supremum, so equality holds. This proves Σ_2^0 membership for attainment of the unspecified supremum.

8.3 A specified scalar value

Let $t \in \mathbb{Q}$ be supplied without a bound promise. A value strictly between the infimum and supremum has finite-dimensional values on either side, by equality of the finite and spatial extrema. A finite direct-sum mixture then attains t. Existence of a finite-dimensional strategy with value exactly t is Σ_1^0, by dimension search and quantifier elimination. Every remaining attainable value is an endpoint. Upper endpoints are handled above; lower endpoints are upper endpoints for $-L$. This finite union is Σ_2^0.

For hardness in all three cases use the winning-probability functional of G_e^{clk}. Its coefficients are rational, its global bound and finite-dimensional supremum are both one, and attaining the specified value one is precisely perfect spatial play. Theorem 7.1 and Section 7.1 therefore prove Theorem 8.1. In its promised-bound version every reduction output has valid bound 1, and the upper predicate restricts to the valid-bound domain, so the same map establishes promise-problem completeness.

9 The entanglement spectrum and its decision problems

A finite entropy bound controls the tail of the Schmidt distribution. A positive moment gives stronger control, and varying the moment order reveals a strict ladder of existence languages. We first prove compactness and the upper predicates, then give the spectral separators. The common hardness family and the realization of all possible critical indices follow in Section 10, using the nested compiler proved in Appendix A.

For $0 < \alpha < 1$ let $\mathcal{P}_\alpha$ require a perfect pure spatial strategy with $\operatorname{Tr} h^\alpha < \infty$. Put $\mathcal{P}_0 = \mathbf{q}$, $\mathcal{P}_1 = \mathbf{E}$, and $\mathcal{P}_{\mathrm{normal}} = \mathbf{I}$. For moments of order $\alpha > 1$ the finiteness condition imposes no restriction: $0 < \operatorname{Tr} h^\alpha \leq 1$ for every density, so the Rényi entropy of order α is finite, and the largest eigenvalue is positive, so the order-infinity entropy is finite too. Thus all these languages equal $\mathcal{P}_{\mathrm{normal}}$. Here $\mathcal{P}_{\mathrm{normal}}$ is a language label; it does not denote a moment of infinite order. Mixed witnesses can be purified on Bob's side preserving Alice's marginal. Pure witnesses therefore suffice for these existence languages.

9.1 Moment budgets are compact and effective

For sorted probabilities with $\sum_i p_i^\alpha \leq K$, one has

$$t_D := \sum_{i>D} p_i \leq K^{1/\alpha}(D+1)^{-(1-\alpha)/\alpha}. \tag{56}$$

Indeed $p_{D+1}^\alpha \leq K/(D+1)$ and $t_D \leq p_{D+1}^{1-\alpha} \sum_{i>D} p_i^\alpha$. This gives uniform tails. Coordinate limits preserve total mass and converge in ℓ^1, while Fatou preserves the moment bound. The Schmidt vectors converge in norm, since $\|\sqrt{p} - \sqrt{p'}\|_2^2 \leq \|p - p'\|_1$. Bounded POVMs have a common weakly convergent subsequence on fixed copies of ℓ^2. Their tensor products converge weakly on finite simple tensors and then on all vectors by uniform boundedness. Thus all game probabilities converge. The maximum winning probability $v_G^\alpha(K)$ under the supplied budget is attained. The same proof uses (5) at $\alpha = 1$. Trace one alone supplies no uniform tail, as I_N/N demonstrates.

Write $v_{G,D}^\alpha(K)$ for the maximum winning probability with local dimension at most D and moment budget K. Normalize the first D probabilities by their retained mass. This truncation majorizes the full probability vector, so does not increase the concave moment sum or entropy. Compression of the POVMs therefore gives

$$0 \leq v_G^\alpha(K) - v_{G,D}^\alpha(K) \leq \sqrt{t_D}.$$

For rational α, the finite-dimensional budget constraints are semialgebraic using nonnegative root variables. For a fixed computable irrational order, approximate x^α uniformly on $[0,1]$ by rational polynomials. Compact outer approximations give upper semicomputability. For lower approximation, move a nonpure boundary spectrum slightly towards $(1,0,\dots,0)$: the resource strictly decreases and probabilities change continuously. Rational spectra and feasible POVMs with strict slack are dense. At moment budget $K=1$, only a pure marginal is possible and the value is the finite classical value. The analogous entropy exception is budget zero. These arguments give computable budget values for computable orders, and in particular the upper semicomputability needed here. Equivalently, compact rational outer feasibility tests, with vanishing rejection slack and uniform constraint error, give

$$G \in \mathcal{P}_\alpha \iff \exists K \geq 1\, \forall D \geq 1\ R_\alpha(G,K,D)$$

with R_α decidable. Explicitly, compute rational polynomials f_D with $\sup_{x\in[0,1]} |f_D(x) - x^\alpha| \leq D^{-2}$. Choose rational bounds $u_D \geq K^{1/\alpha}(D+1)^{-(1-\alpha)/\alpha}$ tending to zero and rational $\epsilon_D \geq 4\sqrt{u_D}$ tending to zero. If $u_D \geq 1$, let the stage be true. Otherwise require ordered $p_i = s_i^2$, $s_i \geq 0$, two D-dimensional POVM assignments, and

$$1 - u_D \leq \sum_{i=1}^{D} p_i \leq 1, \qquad \sum_{i=1}^{D} f_D(p_i) \leq K + D^{-1}, \qquad \mathrm{loss}_G(s,E,F) \leq \epsilon_D.$$

These are rational semialgebraic constraints. A perfect budget-K witness supplies every stage by compression. Conversely, the true moment of each stage is at most $K + 2/D$; the same uniform-tail compactness argument and Fatou give a normalized perfect limit with moment at most K. The rational-order version is written explicitly in Section 10.

9.2 Tree spectra separate the rungs

For b orthogonal isometries of common ratio ν and faithful normal density h, the word-sector argument of Section 5 and its b-ary version in Appendix B gives

$$h = \tau_{b,\nu} \otimes \rho_{\mathrm{spec}}, \qquad \tau_{b,\nu} = (1-b\nu)\sum_w \nu^{|w|} |w\rangle\langle w|, \quad b\nu < 1.$$

One can also see exhaustion by pulling a positive density eigenvector back through adjoint edges: every pullback multiplies its eigenvalue by ν^{-1}, so it must terminate before exceeding $\|h\|$. Consequently

$$\operatorname{Tr} h^\alpha = \frac{(1-b\nu)^\alpha}{1-b\nu^\alpha} \operatorname{Tr} \rho_{\mathrm{spec}}^\alpha \tag{57}$$

when $b\nu^\alpha < 1$, and the moment diverges otherwise. The bare-tree critical index is $\log_2 b/n$ at $\nu = 2^{-n}$. Its finite compiler is the n-bridge construction with a finite list of b edge labels, as specified in Appendix B. For four branches the normality requirement is $n > 2$; for $0 < \alpha \leq 1$, $n = \lfloor 2/\alpha \rfloor + 1$ ensures both normality and moment summability. For larger orders the requirement $n > 2$ remains.

For any $0 < a < b < 1$, choose integers n, j with $a < \log_2 j/n < b$. Such choices exist by taking n large and an integer between the corresponding exponential bounds. The bare j-ary tree has finite b-moment, infinite a-moment in every realization, and finite entropy in its canonical realization. A comb with critical index one and finite entropy, constructed next, separates every order below one from entropy. The Catalan game of Section 14 separates entropy from arbitrary normal play. The uniform family of Section 10 separates finite rank from every positive rung. Thus

$$\mathcal{P}_0 \subsetneq \mathcal{P}_a \subsetneq \mathcal{P}_b \subsetneq \mathcal{P}_1 \subsetneq \mathcal{P}_{\mathrm{normal}} \quad (0 < a < b < 1). \tag{58}$$

Compression to the positive source corner preserves finite moments; finite compiler factors preserve them in completeness. These separators therefore apply to every perfect strategy.

9.3 Complexity, differences, and critical indices

The next theorem collects the refinements of Theorem D. The rungs are the languages $\mathcal{P}_0, \mathcal{P}_\alpha, \mathcal{P}_1, \mathcal{P}_{\text{normal}}$ defined at the start of this section.

Theorem 9.1 (Refinements of Theorem D)**.**

(a) One computable hardness family has a single perfect witness with all positive moments finite on every YES input, value one on every input, and no finite-dimensional perfect instance.

(b) Every nonempty difference between distinct positive computable rungs is $D_2(\Sigma_2^0)$-complete; a positive rung minus finite rank is Σ_2^0-complete.

(c) For a perfect witness with marginal h, its critical index is $\alpha_c(h) = \inf\{\alpha > 0 : \operatorname{Tr} h^\alpha < \infty\}$. For a spatially perfect game G, put

$$c(G) = \inf\{\alpha_c(h) : h \text{ is the marginal of a perfect spatial witness for } G\}.$$

The possible values of $c(G)$ (an infimum, attained by the realizing games) are exactly the right-c.e.-in-$0'$ reals in $[0, 1]$, already among games with finite-entropy perfect strategies.

Proof. The construction in Section 10 gives one computable family H_e such that

$$e \in \mathcal{A}_{\text{dec}} \iff H_e \in \mathbf{I} \iff H_e \text{ has one perfect state with every positive moment finite.}$$

Every instance has $\omega_q = 1$ and no finite-dimensional perfection. This proves (a). Together with the upper predicates it proves Σ_2^0-completeness for every fixed positive computable rung, including entropy and normal play, as stated in Theorem D. The finite-rank rung is Σ_1^0-complete.

Exact conjunction and disjunction preserve these resources. For conjunction tensor the witnesses; moment sums multiply. For disjunction, the globally matched branch projections reduce the marginal and give $h = \bigoplus_j p_j h_j$. Hence $\operatorname{Tr} h^\alpha = \sum_j p_j^\alpha \operatorname{Tr} h_j^\alpha$ and $S(h) = H(p) + \sum_j p_j S(h_j)$; each positive branch inherits the resource. Use the full-support branch agreement proved in Section 14. If $\mathcal{P} \subsetneq \mathcal{Q}$ are distinct positive rungs, fix $G_{\text{sep}} \in \mathcal{Q} \setminus \mathcal{P}$. Then

$$H_e \wedge (G_{\text{sep}} \vee H_f) \in \mathcal{Q} \setminus \mathcal{P} \iff e \in \mathcal{A}_{\text{dec}} \text{ and } f \notin \mathcal{A}_{\text{dec}}.$$

All nonempty such differences are therefore $D_2(\Sigma_2^0)$-complete, and a positive rung minus finite rank is Σ_2^0-complete, since no H_e has finite-dimensional perfection. This proves (b). Reverse differences, and differences between the languages of moment order greater than one, are empty. The property that NO instances have no normal perfect strategy holds for H_e; it need not hold for every NO instance of the difference family. Both hardness promises persist under these operations.

For (c), unused entanglement can increase an individual strategy's index; the invariant is the infimum $c(G)$. Section 10 proves that its possible values are exactly the right-c.e.-in-$0'$ reals in $[0, 1]$, and that every such value has a realizing game with a finite-entropy witness attaining the minimum. The computability characterization uses *limsup* of computable rational sequences, following Becher, Bugeaud and Slaman [3, Lemma 4]. □

10 A uniform family and programmable critical indices

The construction separates the spectrum from the computation that verifies it. Two commuting one-sided shifts first force quarter-plane coordinates and a geometric density. A finite clock and a deterministic tableau then read an integer witness. For programmable critical indices, a comb supplies the desired spectral threshold while an independent quarter-plane register verifies its branching rule. The nested compiler translates the quadratic shift relations into finite game tests.

10.1 The quarter-plane register and its finite presentation

Use two modular contractions S, T and impose

$$S^*S = T^*T = I, \quad ST = TS, \quad S^*T = TS^*. \tag{59}$$

Name projections Π_x, Π_t with $SS^* = I - \Pi_x$, $TT^* = I - \Pi_t$. Name positive contraction counters C_x, C_t and impose their commutation, $C_x\Pi_x = \Pi_x$, $C_t\Pi_t = \Pi_t$, and

$$C_xS = \tfrac{1}{2}SC_x,\ C_xT = TC_x, \quad C_tS = SC_t,\ C_tT = \tfrac{1}{2}TC_t. \tag{60}$$

All clock and computation labels commute with both counters. For the clock use centralizer unitaries Z, J, K, W with

$$JK = ZKJ,\ [Z, J] = [Z, K] = 0,$$

Z, J, K constant along both shifts, and

$$WS = SZW,\ WT = TW,\ W\Pi_x = \Pi_x.$$

For a projection E_{ret} and a contraction R_{ret} use

$$(W - I)E_{\mathrm{ret}} = 0, \qquad R_{\mathrm{ret}}(W - I) = \tfrac{1}{2}C_x(I - E_{\mathrm{ret}}). \tag{61}$$

Every row is either centralizer polynomial, degree-one endpoint, or one of the quadratic rows of the translation table in Appendix A. In particular the mixed row $S^*T = TS^*$ of (59) is compiled as $a_S'^* a_T' = t a_T a_S^* t^*$, in the notation of that appendix; no modular transport is synchronized.

To specify a verifier, supply a finite radius-one deterministic one-sided cellular automaton, with alphabet $\mathcal{A}$, interior rule $F : \mathcal{A}^3 \to \mathcal{A}$, and left-boundary rule $G : \mathcal{A}^2 \to \mathcal{A}$. For every d name four projection families P_d, L_d, R_d, N_d, each summing to identity and commuting with C_x, C_t. Their shift equations are

$$L_dS = SP_d, \qquad P_dS = SR_d, \qquad P_dT = TN_d. \tag{62}$$

Thus $L_d(x, t) = P_d(x - 1, t)$ for $x > 0$, $R_d(x, t) = P_d(x + 1, t)$, and $N_d(x, t) = P_d(x, t + 1)$. The unused $L_d(0, t)$ is arbitrary. Submit the exact finite update polynomials

$$N_d = (I - \Pi_x) \sum_{F(a,b,c)=d} L_aP_bR_c + \Pi_x \sum_{G(b,c)=d} P_bR_c. \tag{63}$$

Specify every initial track on Π_t: the return marker from E_{ret}, the head from Π_x, and the blank work tracks. If ι maps these finitely many input bits to an alphabet symbol, the initial row for d is $\Pi_tP_d = \Pi_t \sum_{\iota(b)=d} \prod_i Q_i^{b_i}$, where $Q_i^1 = Q_i$, $Q_i^0 = I - Q_i$. The input projections commute: they are scalar coordinate markers in the normal representation proved next. Equivalently one may add their finite commutator rows. For each rejecting symbol impose $P_d = 0$. Equations (62) have modular degree one; (63) has modular degree zero, regardless of its centralizer polynomial degree. This is the precise admissibility distinction needed for the tableau.

Use the Fibonacci convention $F_1 = F_2 = 1$, $F_{k+2} = F_{k+1} + F_k$. For a total computable predicate $R(e, j, s)$, a finite machine first scans for the first positive return marker, verifies that its period is F_{j+3}, decodes j, and searches for s with $\neg R(e, j, s)$. Its ordinary finite transition table gives such a cellular automaton: a symbol records a tape letter and optionally a head with its control state. A head writes its new letter, and its indicated neighbor receives the new head at the next step; a boundary rule handles the missing left neighbor. Invalid head collisions can be assigned a rejecting symbol. Multiple tapes can be interleaved with finitely many tracks and finite control, with extra time steps as needed. Only the unique initialized one-head computation is used by the soundness induction. This construction is effective in a supplied program for R and in e. Concrete word and row counts depend on that table; the reduction family is parametric. Appendix C describes the compiled validation instances.

10.2 Normal classification, the clock, and all positive moments

Cyclicity gives $\mathrm{Tr}(h_0 S^k S^{*k}) = 2^{-k}$ and the same for T. Hence normality removes both unitary Wold parts. Double commutation makes the two wandering decompositions compatible and gives

$$\mathcal{H}_0 = \ell^2(\mathbb{N}^2) \otimes \mathcal{K}, \quad h_0 = \rho_{\mathrm{reg}} \otimes \rho_{\mathrm{reg}} \otimes \rho_{\mathrm{spec}}, \quad \rho_{\mathrm{reg}} = \sum_{x \geq 0} 2^{-x-1} |x\rangle\langle x|. \tag{64}$$

Indeed the mutually orthogonal spaces $S^x T^t \Pi_x \Pi_t \mathcal{H}_0$ exhaust the support; their density blocks are 2^{-x-t} times the joint root block. Normalization makes that block $\rho_{\mathrm{spec}}/4$. Equation (60) then forces $C_x = 2^{-x}$ and $C_t = 2^{-t}$, by induction from each boundary. Commutation with these counters prevents labels from mixing coordinates of equal total density weight.

The constant Z, J, K act on $\mathcal{K}$ and commute with the compact ρ_{spec}. Pick a positive finite-dimensional ρ_{spec} eigenspace and then a nonzero $Z = zI$ subspace. It is invariant under J, K, and determinants in $JK = zKJ$ give $z^d = 1$. Let its finite order be d_{clk}. The equations for W force $W(x,t) = Z^x$. Equation (61) forces $E_{\mathrm{ret}}(x,t) = I$ exactly when d_{clk} divides x on this sector: its range lies in $\ker(W - I)$; on that kernel the second equation and $C_x > 0$ force it to be identity. On each chosen clock sector the initialized tableau labels are scalars. Equations (62)–(63) force each next row to be the unique classical successor; the induction also proves invariance of the sector for those projections. Thus rejecting symbols cannot be absent unless $\forall s\, R(e,j,s)$ for the decoded integer.

Conversely a true witness uses $d_{\mathrm{clk}} = F_{j+3}$, $a = F_{j+2}$, $z = \exp(2\pi i a/d_{\mathrm{clk}})$, $Z = zI_{d_{\mathrm{clk}}}$, and Weyl matrices on $\mathbb{C}^{d_{\mathrm{clk}}}$. With $\rho_{\mathrm{spec}} = I_{d_{\mathrm{clk}}}/d_{\mathrm{clk}}$ the honest initialized tableau is diagonal and all the relations hold. Before the finite compiler factors,

$$\mathrm{Tr}(h_0^\alpha) = 4^{-\alpha} {d_{\mathrm{clk}}}^{1-\alpha} (1 - 2^{-\alpha})^{-2} \quad (\alpha > 0), \qquad S(h_0) = 4 + \log_2 d_{\mathrm{clk}}. \tag{65}$$

This is one common witness for all positive moments.

A bounded reader at every nonreturn. Put $\theta = (\sqrt{5} - 1)/2$. If a is a nearest integer to $x\theta$, then $\delta = x\theta - a$ has nonzero integer norm $\delta(x\theta' - a)$, where $\theta' = -(1 + \sqrt{5})/2$. The second factor has magnitude at most $\sqrt{5}x + 1/2 < 3x$, giving $\|x\theta\| > 1/(3x)$. Also $F_k\theta - F_{k-1} = (-1)^{k-1}\theta^k$ and $F_k\theta^k < 1$, so the Fibonacci approximation error is less than F_k^{-2}. These elementary bounds give, when $x \leq d_{\mathrm{clk}}/3$,

$$\|xa/d_{\mathrm{clk}}\| > 1/(3x) - x/{d_{\mathrm{clk}}}^2 \geq 2/(9x).$$

For every nonreturn $x \geq 2d_{\mathrm{clk}}/9$, the coprime residue bound gives $\|xa/d_{\mathrm{clk}}\| \geq 1/d_{\mathrm{clk}} \geq 2/(9x)$. These ranges overlap and cover every positive nonreturn, including $x > d_{\mathrm{clk}}$. Concavity of sine on $[0, \pi/2]$ gives $|z^x - 1| \geq 4\|xa/d_{\mathrm{clk}}\| \geq 8/(9x)$. Consequently the canonical reader

$$R_{\mathrm{ret}}(x) = \begin{cases} 2^{-x-1}/(z^x - 1), & d_{\mathrm{clk}} \nmid x, \\ 0, & d_{\mathrm{clk}} \mid x \end{cases}$$

has norm at most $\sup_{x \geq 1} 9x/2^{x+4} = 9/32$. Returns, including $x = 0$, are checked separately and give zero on both sides of (61).

Value one. For any e, take arbitrarily large Fibonacci periods and the genuine computation even if it eventually rejects. All equations except the forbidden-reject row remain exact. The first scan prevents rejection before time d_{clk}. Its root probability is at most $\sum_{t \geq d_{\mathrm{clk}}} 2^{-t-1} = 2^{-d_{\mathrm{clk}}}$, so (130) gives *loss at most* $C_{\mathrm{nest}} 2^{-d_{\mathrm{clk}}}$; the constant contains the compilation factors of the row tests. Since C_{q} and C_{qs} have the same closure (Section 2), the finite-dimensional value is one. Finite-dimensional perfection is impossible: $S^*S = I$ would make S unitary, whereas $h_0 S = S h_0/2$ and trace cyclicity would give $1 = 1/2$.

10.3 The comb and the independent checker

The spectral threshold will come from the number of branches attached at each spine position. A separate geometric register verifies the branching guard. Keeping its computation time in that separate factor allows arbitrarily slow total programs without changing the programmed critical index. First normality establishes coordinates; only then does the deterministic verifier force the guard on each coordinate fibre.

Let Q, P, E_{guard} be centralizer projections, $P \le Q$, and let C_{sp}, L be commuting positive contractions, commuting with Q, P. Use modular contractions $V_{\mathrm{sp}}, B_0, B_1$ with

$$\begin{aligned} {V_{\mathrm{sp}}}^* V_{\mathrm{sp}} &= Q, & V_{\mathrm{sp}} {V_{\mathrm{sp}}}^* &= Q - P, \\ B_i^* B_j &= \delta_{ij} E_{\mathrm{guard}}, & B_0 B_0^* + B_1 B_1^* &= I - Q, \\ C_{\mathrm{sp}} P &= P/2, & C_{\mathrm{sp}} V_{\mathrm{sp}} &= V_{\mathrm{sp}} C_{\mathrm{sp}}/2, \\ C_{\mathrm{sp}} B_i &= B_i C_{\mathrm{sp}}, & LQ &= Q, \qquad LB_i = B_i L/2. \end{aligned}$$

All modular symbols have ratio $1/2$. These are inner-product, range, and endpoint rows of the translation table. In particular a range sum is compiled as a sum $t a_i a_i^* t^*$, together with the transported range projection. Every range product XX^* used as a root centralizer label comes with its defining translated row.

Add an independent quarter-plane pair S, T satisfying (59)–(60). For every comb modular symbol $X \in \{V_{\mathrm{sp}}, B_0, B_1\}$ impose both $XS = SX$, $X^*S = SX^*$ and the corresponding equations with T. For Q, P, C_{sp}, L impose shift commutation by endpoint rows. These relations give the tensor decomposition (64) with the entire comb algebra acting on its multiplicity space. The checker has its own C_x, C_t, Π_x, Π_t and a separate computation time.

The guard preserves the coordinate fibres. Since $E_{\mathrm{guard}} = B_0^* B_0$, the counter relations and their adjoints give

$$E_{\mathrm{guard}} C_{\mathrm{sp}} = B_0^* C_{\mathrm{sp}} B_0 = C_{\mathrm{sp}} E_{\mathrm{guard}}, \qquad E_{\mathrm{guard}} L = 2 B_0^* L B_0 = L E_{\mathrm{guard}}. \tag{66}$$

Ordinary and double commutation with checker shifts similarly give commutation of E_{guard} with those shifts and adjoints, so adding these commutator rows to the presentation is harmless.

The four orthogonal projections $P, V_{\mathrm{sp}} {V_{\mathrm{sp}}}^*, B_0 B_0^*, B_1 B_1^*$ sum to identity. Starting from an h eigenvector of eigenvalue $p > 0$, each nonzero pullback through ${V_{\mathrm{sp}}}^*, B_i^*$ doubles its eigenvalue. After finitely many steps it would exceed $\|h\|$ and must vanish. Thus descendants of root vectors span the support. Since $V_{\mathrm{sp}} B_i = 0$, surviving words have form $B_w {V_{\mathrm{sp}}}^{m-1} P$, with $m \ge 1$. The counter relations force $C_{\mathrm{sp}} = 2^{-m}$ and $L = 2^{-|w|}$ on their ranges. Their span is dense, so the joint dyadic eigenspaces $\mathcal{H}_{m,\ell}$ exhaust the support. Equation (66) shows E_{guard} preserves each such eigenspace.

Name equality projections $E_{C_{\mathrm{sp}}}, E_L$ and contraction readers $F_{C_{\mathrm{sp}}}, F_L$. Impose

$$(C_{\mathrm{sp}} - C_x) E_{C_{\mathrm{sp}}} = 0, \quad F_{C_{\mathrm{sp}}} (C_{\mathrm{sp}} - C_x) = \tfrac{1}{2} C_{\mathrm{sp}} C_x (I - E_{C_{\mathrm{sp}}}), \tag{67}$$

and the analogous rows for L. For two unequal dyadic values $u, v \in (0, 1]$, $|uv/[2(u - v)]| \le 1$: the larger is at least twice the smaller. On equal fibers the second equation forces the equality projection to be identity; on unequal fibers the first forces it to be zero. The canonical reader is $uv/[2(u - v)]$, with value zero at equality. Thus the initial checker tracks mark exactly $x = m$ and $x = \ell$, and receive the bit E_{guard}. Require every checker label to commute with $C_{\mathrm{sp}}, L, E_{\mathrm{guard}}, C_x, C_t$, and initialize all tracks and the head. Its total program computes k_m and rejects exactly when the bit is different from $\mathbf{1}_{\{\ell < k_m\}}$. Apply the same finite (62)–(63) presentation and forbid rejection.

On a joint $(m, \ell, E_{\mathrm{guard}})$ fiber the initial symbols are scalars. The deterministic tableau induction therefore forces the actual finite computation of this total program. A wrong guard fiber would produce a nonzero rejecting projection, contradicting faithfulness and the forbidden row. Hence

$$E_{\mathrm{guard}}|_{\mathcal{H}_{m,\ell}} = \mathbf{1}_{\{\ell<k_m\}} I. \tag{68}$$

Now, and only now, legal word maps are isometries on the root with mutually orthogonal ranges. They exhaust the support by the previous pullback argument, and yield

$$\begin{aligned} h_{\mathrm{comb}} &\cong \rho_{\mathrm{comb}} \otimes \rho_{\mathrm{spec}}, \\ \rho_{\mathrm{comb}} &= Z_{\mathrm{comb}}^{-1} \sum_{m\geq 1,\ |w|\leq k_m} 2^{-(m+|w|)} |m, w\rangle\langle m, w|, \\ Z_{\mathrm{comb}} &= \sum_{m\geq 1} (k_m + 1) 2^{-m}. \end{aligned} \tag{69}$$

The complete checker adds $\rho_{\mathrm{reg}} \otimes \rho_{\mathrm{reg}}$, independently of how long the program takes to verify a particular fiber.

10.4 Programming and classifying the minimum index

For a computable rational sequence a_m with $\limsup a_m = r \in [0, 1]$, put

$$u_m = \min\left(\max(a_m, 0), \frac{m}{m+1}\right), \qquad k_m = \left\lfloor \frac{m u_m}{1 - u_m} \right\rfloor.$$

Then $0 \leq k_m \leq m^2$, and rounding changes $k_m/(m + k_m)$ from u_m by less than $1/m$. Thus its limsup is r, and $1 \leq Z_{\mathrm{comb}} \leq 7$. For $0 < \alpha < 1$ the moment of (69) converges precisely when

$$\sum_m 2^{(1-\alpha)k_m - \alpha m} < \infty.$$

The exponent equals $(k_m/(m + k_m) - \alpha)(m + k_m)$. If $\alpha > r$ the tail decays geometrically; if $\alpha < r$ a subsequence of terms grows. Therefore $\alpha_c(\rho_{\mathrm{comb}}) = r$, without a claim about convergence at the endpoint $\alpha = r$. The entropy is bounded by

$$\log_2 7 + \sum_{m\geq 1} 2^{-m} \left[(m^2 + 1)m + \frac{m^2(m^2 + 1)}{2} \right] = 106 + \log_2 7.$$

Every realization contains the comb factor, so extra multiplicity cannot lower its critical index. Root multiplicity one gives the canonical witness attaining minimum index r, with finite entropy.

For necessity, restrict the moment order to rational $0 < \beta < 1$. For sorted Schmidt probabilities with $\sum_i p_i^\beta \leq K$, use the tail bound (56) with $\alpha = \beta$ and $D = d_{\mathrm{cut}}$. Choose computable rational upper bounds $t_{d_{\mathrm{cut}}}^{\mathrm{bd}}$ for this tail, tending to zero, and rational $\epsilon_{d_{\mathrm{cut}}} \geq 4\sqrt{t_{d_{\mathrm{cut}}}^{\mathrm{bd}}}$, also tending to zero. For fixed $G, \beta, K, d_{\mathrm{cut}}$ let $\mathcal{R}(G, \beta, K, d_{\mathrm{cut}})$ say there exist two local d_{cut}-dimensional POVM assignments and a subnormalized Schmidt vector with ordered probabilities $p_1, \ldots, p_{d_{\mathrm{cut}}}$ such that

$$1 - t_{d_{\mathrm{cut}}}^{\mathrm{bd}} \leq \sum_{i\leq d_{\mathrm{cut}}} p_i \leq 1, \qquad \sum_{i\leq d_{\mathrm{cut}}} p_i^\beta \leq K, \qquad \Pr(\mathrm{reject}) \leq \epsilon_{d_{\mathrm{cut}}}.$$

When $t_{d_{\mathrm{cut}}}^{\mathrm{bd}} \geq 1$ take this stage to be true. At the other stages this is a decidable semialgebraic feasibility predicate: rational powers are encoded by nonnegative root variables, and POVM positivity by principal minors. A perfect witness compresses to these assignments; its truncated

vector differs in norm by at most $\sqrt{t^{\mathrm{bd}}_{d_{\mathrm{cut}}}}$, giving the displayed slack. Conversely the uniform tail bound makes these ordered vectors compact in ℓ^2. A diagonal subsequence of the bounded POVM matrix entries gives positive limiting POVMs with the same finite outcome sums. Finite coordinate tests followed by the uniform vector tail pass their tensor probabilities to the limit. The limiting vector has norm one, zero rejection and β-moment at most K by Fatou's lemma. Consequently the quantifiers are explicitly

$$G \in \mathcal{P}_\beta \quad \Longleftrightarrow \quad \exists K \in \mathbb{N}_{\geq 1}\ \forall d_{\mathrm{cut}} \in \mathbb{N}_{\geq 1}\ \mathcal{R}(G, \beta, K, d_{\mathrm{cut}}).$$

For rational $0 < a \leq 1$, $c(G) < a$ is equivalent to this condition for some rational $0 < \beta < a$, when $c(G)$ denotes the infimum of critical indices of perfect witnesses and is finite in $[0, 1]$. Cuts above 1 are automatic on this domain. The rational order and integer budget can be paired in one existential integer, leaving the same $\exists K\, \forall d_{\mathrm{cut}}$ form. Consequently the strict upper cut is c.e. in $0'$. Together with the characterization of Becher, Bugeaud and Slaman [3, Lemma 4] by limsups of computable rationals, this gives exactly the right-c.e.-in-$0'$ indices in $[0, 1]$. The input program computes k_m, and contains no oracle.

11 Semifinite hosts: a critical tree selects a finite depth

In a diffuse semifinite algebra the Weyl determinant of Section 6 is unavailable, so the spatial witness cannot be used. A trace still forces a finite object. A critical binary tree with modular ratio $1/2$ must have a nonzero leaf sector at some finite depth, and a group word chosen by that depth carries the Σ^0_2 witness. This section builds the resulting reduction and proves the semifinite half of Theorem B.

11.1 The critical-tree selector

Work on a faithful normal support. Fix integers $d \geq 2$ and a real $0 < \gamma < 1$. Let P, Q be centralizer projections, $0 \leq C \leq I$ a centralizer contraction, and let $B_1, \ldots, B_d$ satisfy

$$B_i^* B_j = \delta_{ij}(I - P), \qquad \sum_i B_i B_i^* = I - Q, \tag{70}$$

$$\sigma^\varphi_t(B_i) = d^{-it} B_i, \tag{71}$$

$$CQ = Q, \qquad CB_i = \gamma B_i C, \qquad [C, P] = 0. \tag{72}$$

These labels $P, Q, C, B_1, \ldots, B_d$ belong to the selector presentation only.

Lemma 11.1 (Critical-tree selector)**.** *For branching d, modular ratio $1/d$ and depth ratio γ, in a normal semifinite realization, $P \neq 0$, $\ker C = 0$, and $E_n = 1_{\{\gamma^n\}}(C)$ satisfy $\sum_{n \geq 0} E_n = I$ strongly. Some finite n has $PE_n \neq 0$.*

Proof. The modular eigenoperators are entire analytic. KMS gives $\varphi(B_i B_i^*) = \varphi(B_i^* B_i)/d$. Summing gives

$$\varphi(P) = \varphi(Q). \tag{73}$$

If $P = 0$, faithfulness makes $Q = 0$, giving a full Cuntz family. Write $\varphi(x) = \tau(hx)$, h injective, $\tau(h) = 1$. The covariance is $h^{it} B_i = B_i (h/d)^{it}$, hence $f(h) B_i = B_i f(h/d)$ for bounded Borel f. For bounded $g \geq 0$ with compact support in $(0, \infty)$, $hg(h)$ is positive integrable, and

$$\varphi(g(h)) = \sum_i \tau(B_i^* h g(h) B_i) = \tfrac{1}{d} \sum_i \tau(h g(h/d)) = \varphi(g(h/d)).$$

The trace exchange is $\tau(T^{1/2} v v^* T^{1/2}) = \tau(v^* T v)$ for positive integrable T. Use indicators of $(d^k, d^{k+1}]$, or monotone approximations to them. All these windows have equal φ-mass and partition $(0, \infty)$, whose mass is one. Countable additivity is contradicted. Thus $P \neq 0$.

Let $z = 1_{\{0\}}(C)$. Covariance and its adjoint give $zB_i = B_i z$ by functional calculus at zero. Also $Qz = 0$ and $[P, z] = 0$. Since z is a centralizer projection, zMz is modularly invariant. The KMS calculation there gives $\varphi(Pz) = 0$, hence $Pz = 0$. If $z \neq 0$, the normalized restriction of φ is faithful normal on a semifinite corner with a full critical Cuntz family. The same window argument excludes this. Hence $z = 0$.

Put $\mathcal{T}(X) = \sum_i B_i X B_i^*$. Telescoping gives

$$I = \sum_{j=0}^{m-1} \mathcal{T}^j(Q) + \mathcal{T}^m(I). \tag{74}$$

For a length-m word v, $CB_v = \gamma^m B_v C$. Bounded spectral calculus puts its range in $1_{[0,\gamma^m]}(C)\mathcal{H}$. Thus

$$0 \leq \mathcal{T}^m(I) \leq 1_{[0,\gamma^m]}(C) \longrightarrow 0$$

strongly by injectivity of C. Each $\mathcal{T}^j(Q)$ is supported in the γ^j eigenspace of C. These positive terms sum to I and have mutually orthogonal spectral supports. Compressing the sum to each eigenspace gives

$$E_j = \mathcal{T}^j(Q), \qquad \sum_{j \geq 0} E_j = I. \tag{75}$$

There is no residual spectral subspace since the sum is I. As $[P, C] = 0$ and $P \neq 0$, some PE_n is nonzero. In particular $Q = E_0$. □

The selector of this section uses $d = 2$ and $\gamma = 1/2$, with edges reindexed B_0, B_1; the coupled construction uses $d = 16$ and the same depth ratio.

11.2 Finite models and a small example

For words w of length at most n, let B_i append i unless w is a leaf, when it vanishes. Let Q select the empty word and P the depth-n words. Put

$$C|w\rangle = 2^{-|w|}|w\rangle, \qquad h_n|w\rangle = \frac{2^{-|w|}}{n+1}|w\rangle. \tag{76}$$

Each nonleaf has two distinct children and each nonroot a unique parent. This proves (70), including the cross term. The child-to-parent density ratio is $1/2$, proving (71); the same ratio for C proves (72). Each depth has state mass $1/(n+1)$, whence

$$\varphi_n(P) = \varphi_n(Q) = \frac{1}{n+1}, \qquad S(h_n) = \log_2(n+1) + \frac{n}{2}. \tag{77}$$

For $n = 2$, the seven word densities are $1/3$, two copies of $1/6$, and four copies of $1/12$. Their C labels are $1, 1/2, 1/4$. The leaf rank is four but its state mass is $1/3$. A finite host's normalized trace gives $\tau(P) \geq 1/2$ from (70); this does not bound $\varphi(P)$ away from zero uniformly over states.

11.3 A finite presentation verifies the depth

Fix a total recursive predicate $\mathcal{R}_{\mathrm{dec}}(e, n, t)$ and $\mathcal{A}_{\mathrm{dec}} = \{e : \exists n \, \forall t \, \mathcal{R}_{\mathrm{dec}}(e, n, t)\}$. With $[x, y] = xyx^{-1}y^{-1}$ set

$$\begin{aligned} \Gamma_e = \langle a, b, c \mid {r^{\mathrm{grp}}}_n^2 = 1 \ (n \geq 0), \\ {r^{\mathrm{grp}}}_n = 1 \text{ if } \exists t \, \neg\mathcal{R}_{\mathrm{dec}}(e, n, t)\rangle, \end{aligned} \qquad {r^{\mathrm{grp}}}_n = [b, a^n c a^{-n}]. \tag{78}$$

Lemma 11.2. *${r^{\mathrm{grp}}}_n$ is nontrivial in Γ_e exactly when $\forall t\, \mathcal{R}_{\mathrm{dec}}(e,n,t)$; a nontrivial ${r^{\mathrm{grp}}}_n$ has order two. Uniformly there is a finite presentation $\mathcal{G}_e$ and explicit words for embedded a,b,c preserving this property.*

Proof. Let $\mathcal{D}_{\mathrm{good}} \subseteq \mathbb{N}$ be the good integers. Take two copies of $\mathbb{V} = \bigoplus_{m\in\mathbb{Z}} \mathbb{F}_2\varepsilon_m$ and $\mathbb{W} = \bigoplus_{k\in\mathcal{D}_{\mathrm{good}}} \mathbb{F}_2 z_k$. Define $\mathfrak{b}(\varepsilon_m,\varepsilon_\ell) = z_{\ell-m}$ if $\ell - m \in \mathcal{D}_{\mathrm{good}}$, zero otherwise. On $\mathbb{V}\oplus\mathbb{V}\oplus\mathbb{W}$ put

$$(x,y,z)(x',y',z') = (x+x', y+y', z+z'+\mathfrak{b}(x,y')). \tag{79}$$

Bilinearity makes the third coordinate in either association $z+z'+z''+\mathfrak{b}(x,y')+\mathfrak{b}(x,y'')+\mathfrak{b}(x',y'')$. The identity is $(0,0,0)$; the inverse of (x,y,z) is $(x,y,z+\mathfrak{b}(x,y))$. Simultaneously shifting both $\mathbb{V}$ indices preserves $\mathfrak{b}$; adjoin this automorphism as a. Put $b = (\varepsilon_0,0,0)$ and $c = (0,\varepsilon_0,0)$. Multiplication gives ${r^{\mathrm{grp}}}_n = (0,0,z_n)$ for $n \in \mathcal{D}_{\mathrm{good}}$, identity otherwise. All relators hold, and every good word has a nontrivial order-two image. Bad words are killed by listed relators, proving the assertion.

For precision we give a uniform Craig conversion [15] of the c.e. relator list to a decidable literal-word list. Chiodo's convention already permits recursive enumeration, so this extra normalization is optional for the cited embedding theorem. Here is a Craig conversion that also preserves the old generator words. Adjoin d with relator $d = 1$. For an enumerated relator r at time t, replace that occurrence by the literal word rd^k, where distinct increasing codes k are chosen effectively with $k \geq t$. Given a word of length ℓ, membership in the resulting literal relator set is decidable by simulating the first ℓ enumeration steps, constructing those coded words of length at most ℓ, and comparing literally. Use freely reduced old words r, which contain no d; the final d^k therefore survives free reduction. Include d itself as a relator. Modulo $d = 1$, the coded relations are exactly the original r. The group is therefore unchanged and the old generator images remain explicit. Higman embedding [32], in the effective uniform formulation of [7, Theorem 2.2], gives a finite presentation with an explicit injective embedding. Injectivity preserves the nontrivial ${r^{\mathrm{grp}}}_n$. The model (79) need not itself be computable. □

Represent the generators of $\mathcal{G}_e$ by centralizer unitaries satisfying its finite relators. Add centralizer unitaries T, L. Require every group generator and T to commute with B_0, B_1, C. Impose

$$LQ = cQ, \quad LB_i = B_i aLa^*, \quad [L,C] = [L,P] = 0, \tag{80}$$

and the centralizer polynomial

$$P\big(T[b,L]T^* + [b,L]\big) = 0. \tag{81}$$

Lemma 11.3 (Level reader)**.** *In every semifinite realization $LE_n = a^n c a^{-n} E_n$. The terminal equation then implies that some n is good.*

Proof. Group generators commute with C, hence with $Q = E_0$. In particular $a^*Q = Qa^*$; this is needed in the induction. For a word v of length n, the endpoint equation and commutation of a with the B_i give $LB_vQ = B_v a^n L a^{-n} Q = B_v a^n c a^{-n} Q$. By (75) these ranges span $E_n\mathcal{H}$, proving the level identity. A unitary commuting with B_i also commutes with B_i^*: commute its inverse, then take adjoints. It therefore commutes with P. Thus PE_n reduces the group generators, T and L. Choose a nonzero such sector by Lemma 11.1. On it (81) becomes $T{r^{\mathrm{grp}}}_n T^* = -{r^{\mathrm{grp}}}_n$. A bad n has relator ${r^{\mathrm{grp}}}_n = I$, giving $I = -I$ on a nonzero space. This contradiction proves goodness. □

Lemma 11.4 (Finite realization)**.** *A good n gives a realization of all relations in a finite von Neumann algebra.*

Proof. In $\mathcal{G}_e \times \langle j \mid j^2 = 1\rangle$, the subgroup $\langle {r^{\mathrm{grp}}}_n, j\rangle$ is C_2^2. The map ${r^{\mathrm{grp}}}_n \mapsto j{r^{\mathrm{grp}}}_n$, $j \mapsto j$ is its automorphism. Form the HNN extension Γ_n^{HNN} with stable letter T:

$$T{r^{\mathrm{grp}}}_n T^{-1} = j{r^{\mathrm{grp}}}_n, \qquad TjT^{-1} = j. \tag{82}$$

The HNN normal-form theorem [33] embeds the base group. Consequently $j \neq 1$, and j commutes with the base and with T, so is central in Γ_n^{HNN}. This is the group-extension mechanism used in the nonlocal-game setting of [57]. In $L(\Gamma_n^{\mathrm{HNN}})$, $p_- = (1-\lambda(j))/2$ is central and has trace $1/2$. The corner $M_{\mathrm{HNN}} = p_- L(\Gamma_n^{\mathrm{HNN}})$ with trace 2τ is finite. Compressed group unitaries are unitaries of M_{HNN}, and (82) becomes $T{r^{\mathrm{grp}}}_n T^* = -{r^{\mathrm{grp}}}_n$.

Tensor M_{HNN} with the depth-n tree, with density $h_n \otimes 1$ relative to matrix trace times the normalized trace of M_{HNN}. Group generators and T are constant across vertices. Set $L = a^\ell c a^{-\ell}$ on depth ℓ. It is a unitary centralizer label commuting with P, C. On a nonleaf, $LB_i = B_i a L a^*$ because depth increases by one; both sides vanish on a leaf. At the root $LQ = cQ$. The terminal equation is the sign-flip identity on the leaves and is vacuous elsewhere. The tree relations hold by (76). Tensor with R_{hyp} and select a factor fibre to obtain a II_1 host. The presentation itself never needs to find the good n. □

For a small example, at good depth $n = 1$ the tree is $\{\emptyset, 0, 1\}$ and the blocks of L are c, aca^{-1}, aca^{-1}. The terminal word is ${r^{\mathrm{grp}}}_1 = [b, aca^{-1}]$. The model (79) with good set $\{1\}$ sends it to $z_1 \neq 1$, while ${r^{\mathrm{grp}}}_0, {r^{\mathrm{grp}}}_2$ vanish. The HNN step flips the chosen involution after compression to $j = -1$. One T is not required to flip all good words.

11.4 One bridge compiles the selector presentation

Use five orthogonal synchronized corners $e_0, m_{00}, m_{01}, m_{10}, m_{11}$ covering I, with one FMS copy whose flag is e_0 and magic support is their other four corners. Put $e_1 = m_{00}$. The actual bridge $v = \mathsf{U}_1 : e_0 \to e_1$ obeys (14). KMS and the cover give $\varphi(e_0) = 1/9$ and $\varphi(m_{ab}) = 2/9$; here $9 = \mathfrak{c}_1$. These masses are independent of the root model.

$$\underbrace{e_0}_{1/9} \xrightarrow[\sigma_t^{\varphi}(v) = 2^{it} v]{v} \boxed{\underbrace{m_{00} = e_1}_{2/9} \quad m_{01} \quad m_{10} \quad m_{11}} \qquad \text{each magic corner has mass } 2/9.$$

Figure 5: One calibrated bridge. All copies of transported labels use this same v; the box is the four-dimensional magic support.

Bind the five corners and the two positive ZI, IX parameters to the qutrit frame. Introduce the two axis unitaries $\mathfrak{x}, \mathfrak{z}$ with the explicit formulas of Section 4.1. A *typed label* is an operator name together with its required class (projection, positive contraction or unitary). The transported typed label list is exactly

$$Q,\ C,\ T,\ L,\ \{U_g : g \text{ is a generator of } \mathcal{G}_e\}. \tag{83}$$

Encode a projection by a reflection, a positive contraction by $u_C = 2C - I + 2i\sqrt{C(I-C)}$, and a unitary directly. A *unitary name* is a formal generator for the chosen unitary encoding, including separate names for separate summands when needed. For each resulting unitary name ℓ introduce copies $u_{r,\ell}$, $r = 0, 1$, with $(u_{r,\ell} - I)(I - e_r) = 0$. The *decoded copies* are the actual corner operators recovered from these encodings by the following formulas, respectively: $e_r(I + u_{r,Q})/2$, $e_r(2I + u_{r,C} + u_{r,C}^*)/4$, and $e_r u_{r,\ell}$. Require $u_{r,Q}^2 = I$. Introduce the block unitary $V_{1,\ell}$ of (25); since the cover is the whole space it reads

$$V_{1,\ell} = e_0 u_{0,\ell} + \sum_{a,b} \mathfrak{x}^a \mathfrak{z}^b e_1 u_{1,\ell} \mathfrak{z}^{-b} \mathfrak{x}^{-a}.$$

Refine the retained FMS signs separately for the six frame projections

$$F_1, \quad F_2, \quad F_3, \quad G_{12}(1), \quad G_{13}(1), \quad G_{23}(1),$$

and for every $G_{12}(V_{1,\ell})$. Lemma 4.2 first makes v preserve the frame, so $v = I_3 \otimes \widetilde{v}$. The graph-label commutators in the target–source block give

$$v u_{0,\ell} = u_{1,\ell} v. \tag{84}$$

Thus the same actual calibrated transport works for every label. Transport of decoded adjoints and polynomials follows too. P is root-only, encoded by $P_0 = e_0(I + u_P)/2$ with $u_P^2 = I$ and $(u_P - I)(I - e_0) = 0$.

Introduce two centralizer contractions $b_i = e_1 b_i e_0$. For compilation encode each as an average of two global unitaries. Extract $B_i = v^* b_i$ on $e_0 M e_0$. Their modular eigenvalue is $1/2$ because that of b_i is one. The complete endpoint translation is

Root relation	Centralizer polynomial
$B_i^* B_j = \delta_{ij}(I - P)$	$b_i^* b_j = \delta_{ij}(e_0 - P_0)$
$\sum_i B_i B_i^* = I - Q$	$\sum_i b_i b_i^* = e_1 - Q_1$
$C B_i = \frac{1}{2} B_i C$	$C_1 b_i = \frac{1}{2} b_i C_0$
$U_g B_i = B_i U_g$	$(U_g)_1 b_i = b_i (U_g)_0$
$T B_i = B_i T$	$T_1 b_i = b_i T_0$
$L B_i = B_i a L a^*$	$L_1 b_i = b_i a^{[0]} L_0 (a^{[0]})^*$

Every remaining relation is a root centralizer polynomial: the finite relators of $\mathcal{G}_e$, $C_0 Q_0 = Q_0$, $[C_0, P_0] = 0$, $[U_g, C_0] = [T_0, C_0] = 0$, $L_0 Q_0 = c^{[0]} Q_0$, $[L_0, C_0] = [L_0, P_0] = 0$, and the terminal equation. Expand its root projection as $P = (I + s_P)/2$. Here $a^{[0]}, c^{[0]}$ mean the source words for the embedded a, c. If $W_{\mathrm{ter}} = [b, L]$, its non-self-adjoint polynomial is explicitly

$$p = \tfrac{1}{2}\big(T W_{\mathrm{ter}} T^* + W_{\mathrm{ter}} + s_P T W_{\mathrm{ter}} T^* + s_P W_{\mathrm{ter}}\big).$$

Use (17) for these words and their prefixes. In global scaffold coordinates multiply by e_0 and expand $e_0 = (I + s_{e_0})/2$ as well. The two off-diagonal blocks impose $p = 0$ and $p^* = 0$. Expand all other typed polynomials in the same manner, apply graphs and triangles, then Lemma 4.3. Retain all FMS and refinement questions. Sample all padded question pairs uniformly, reject the listed events, and accept unspecified pairs. This is an effective finite rational Boolean game, denoted G_e^{sel}.

Soundness and simultaneous completeness. Soundness uses the nonzero root corner with normalized state $9\varphi|_{e_0 M e_0}$. Equations (84) and the table give all root relations as bounded operator identities. For example $C_0 v^* = v^* C_1$ gives $C_0 B_i = v^* C_1 b_i = B_i C_0/2$. Conjugating the range equation by v^* uses $v^* Q_1 v = Q_0$. The corner of a semifinite host is semifinite, so Lemmas 11.1 and 11.3 prove soundness.

Conversely take any faithful normal root realization (M, φ). On $\mathrm{Mat}_5 \bar{\otimes} M$ use the state

$$\varphi_{\mathrm{sc}} = \mathrm{diag}(1, 2, 2, 2, 2)/9 \ \otimes \varphi. \tag{85}$$

Use the identity corner maps $v = |e_1\rangle\langle e_0| \otimes I$, the ideal real FMS effects, and $b_i = |e_1\rangle\langle e_0| \otimes B_i$. The product modular group makes each b_i centralizer because the two modular eigenvalues 2 and $1/2$ cancel. The whole centralizer is finite, with its faithful trace φ_{sc}. For any contraction x in a finite von Neumann algebra, the polar partial isometry extends to a unitary u: equivalent initial and final projections have equivalent complements in a finite von Neumann algebra. Then

$$u_\pm = u(|x| \pm i\sqrt{I - |x|^2}), \qquad x = (u_+ + u_-)/2.$$

This supplies the exact encodings of both b_i at once. Use $2Q - I, 2P - I$ and the displayed functional-calculus encoding of C, copying the same choices to each required corner. Amplify by the scalar qutrit frame. All block labels commute with the ideal FMS signs, so use their actual joint products for refinement. Separate labels use separate questions. Finally apply the standard-form construction in Lemma 4.3, with Bob's effects defined by its conjugation $\mathcal{J}_{\text{new}}$. Formula (85) and the ideal real FMS effects give exactly the five-dimensional FMS probabilities, including the nonsynchronous signs. All graph and endpoint identities hold simultaneously. A root in a finite von Neumann algebra stays finite; after tensoring with R_{hyp} a II_1 host exists. A $\text{III}_{1/2}$ root stays $\text{III}_{1/2}$ under the finite amplifications.

11.5 Both uniform promises and the hardness theorem

Theorem 11.5 (Uniform semifinite reduction)**.** *Uniformly from a total recursive $\mathcal{R}_{\text{dec}}$ and an input e, the compiler gives a finite rational Boolean game G_e^{sel} such that*

$$G_e^{\text{sel}} \in \text{PERF}(\text{II}_1) \iff G_e^{\text{sel}} \in \text{PERF}_{\text{sf}} \iff \exists n\, \forall t\, \mathcal{R}_{\text{dec}}(e, n, t).$$

For every input e, G_e^{sel} has a perfect $\text{III}_{1/2}$ strategy and $\omega_q(G_e^{\text{sel}}) = 1$.

Proof. For the equivalence, soundness follows from the compiler and the selector and level-reader lemmas. A good n gives a finite model by Lemma 11.4, which the compiler turns into perfect finite-host play and hence a II_1 host.

For the first uniform promise take the uniform KMS representation of $\mathcal{O}_2$ with generators s_i satisfying $s_i^* s_j = \delta_{ij} I$, $\sum_i s_i s_i^* = I$ and $\sigma_t(s_i) = 2^{-it} s_i$. Its weak closure is $R_{1/2}$; see the Cuntz/KMS identification in [47] for the gauge KMS state and [31, Theorem 4.9 and Example 4.13] for this factor identification, also used in [20, Section 3.1]. Set $B_i = s_i$, $P = Q = C = 0$, and take all group generators, T, L equal to I. Every root equation holds: $LQ = cQ$ and the terminal row are vacuous. In the scaffold (85), $b_i = |e_1\rangle\langle e_0| \otimes s_i$ is centralizer. Its exact two-unitary completion exists in the finite centralizer, as proved above. The encoding of $C = 0$ is $u_C = -I$ on each corner; the zero projections use reflection $-I$ there as well. Their transported block labels commute with the ideal signs. Thus every compiler gadget has a simultaneous perfect realization. The host, including all finite ancillary matrices, is $\text{III}_{1/2}$.

For the second promise take the finite-tree strategy $\mathcal{V}_N$ with the depth-N finite tree and trivial group labels. Every root equation is exact except the terminal row. Its residual is

$$R_N^{\text{ter}} = 2P, \qquad \varphi_N(R_N^{\text{ter}*} R_N^{\text{ter}}) = 4\varphi_N(P) = \frac{4}{N+1}. \tag{86}$$

Build all graph, prefix, joint and column measurements by their exact formulas even for this nonzero application row. All their structural tests remain exact. Only its final row-zero event can lose, with probability given by (19). The lifted root residual is $p = e_0 R_N^{\text{ter}}$, and its graph row has off-diagonal blocks p, p^*. The qutrit trace and root mass give

$$\varphi_{\text{old}}(\Delta_{\text{graph}}^2) = \tfrac{1}{3}\varphi_{\text{sc}}(p^*p + pp^*) = \tfrac{2}{27}\varphi_N(R_N^{\text{ter}*} R_N^{\text{ter}}).$$

The last equality uses traciality on the centralizer. The *padded question count* is the common size of the two question sets after adding always-accepted fresh questions. An *orientation* is a choice of which party receives the row question and which receives the scalar reference question. Let $m_e \geq 1$ be the finite number of retained terminal-test copies in the chosen encoding, determined solely by e. If Q_e is the padded question count and ℓ_e, d_e are the row denominator and slot dimension, then, for each retained terminal-test copy, union bounding its orientations gives a loss of at most $64\ell_e^2/(27 d_e^3 Q_e^2 (N+1))$. Thus the total loss satisfies

$$1 - \omega(G_e^{\text{sel}}, \mathcal{V}_N) \leq m_e \frac{64\ell_e^2}{27 d_e^3 Q_e^2} \frac{1}{N+1} = \frac{C_e^{\text{sel}}}{N+1}.$$

Take

$$C_e^{\mathrm{sel}} = m_e \frac{64\ell_e^2}{27 d_e^3 Q_e^2}.$$

Thus C_e^{sel} depends only on the finite compiled specification, denominators, row dimensions and the fixed root mass $1/9$. It is independent of N, the density spectrum and the finite tree dimension. Letting $N \to \infty$ proves $\omega_q(G_e^{\mathrm{sel}}) = 1$. □

Choose $\mathcal{A}_{\mathrm{dec}}$ to be Σ_2^0-complete. Theorem 11.5 and (10), together with Theorem 3.9, prove Σ_2^0-completeness of every claimed semifinite existence language in Theorem B. Furthermore every G_e^{sel} is commuting perfect and has quantum value one, and

$$G_e^{\mathrm{sel}} \in \mathrm{PERF}_{\mathrm{co}} \setminus \mathrm{PERF}_{\mathrm{sf}} \iff e \notin \mathcal{A}_{\mathrm{dec}}.$$

This proves Π_2^0-hardness of $\mathrm{PERF}_{\mathrm{co}} \setminus \mathrm{PERF}_{\mathrm{sf}}$ within that promise, a stronger statement than hardness of intersection with the value-one language. Membership follows from $\mathrm{PERF}_{\mathrm{co}} \in \Pi_1^0$ and $\mathrm{PERF}_{\mathrm{sf}}^c \in \Pi_2^0$: their intersection is Π_2^0.

Our earlier finite game [19] also has a perfect strategy on a type $\mathrm{III}_{1/2}$ factor and quantum value one, and no perfect strategy whose marginal is normal on an injective von Neumann algebra. The exclusions differ: every G_e^{sel} has a perfect strategy on a matrix amplification of the injective factor $R_{1/2}$, and the NO instances exclude every semifinite host instead.

12 Each type-III subtype has the commuting language

A type-III factor has no nonzero finite projection, so the corner of Section 3.5 does not exist. This section shows that nothing replaces it: every fixed subtype hosts exactly the commuting-perfect games. The state and host may both be chosen.

Background lemma (Expected-inclusion transfer; [58, 20])**.** *If $E : N \to M$ is a faithful normal conditional expectation, every faithful standard-form correlation on M is realized on N.*

Proof. For the state $\varphi_N = \varphi_M \circ E$, put $\mathcal{H}_{\mathrm{inc}} = \overline{M\xi_{\varphi_N}} \subset L^2(N)$. The map $x\xi_{\varphi_M} \mapsto x\xi_{\varphi_N}$ is an isometry. Takesaki's expectation theorem [58] makes M invariant under the modular group. The Tomita operators agree on the common core $M\xi$, and their conjugations restrict as $\mathcal{J}_N|_{\mathcal{H}_{\mathrm{inc}}} = \mathcal{J}_M$. Thus for left effects A and right effects $\mathcal{J}_M B \mathcal{J}_M$,

$$\langle \xi_{\varphi_N}, A\mathcal{J}_N B \mathcal{J}_N \xi_{\varphi_N} \rangle = \langle \xi_{\varphi_M}, A\mathcal{J}_M B \mathcal{J}_M \xi_{\varphi_M} \rangle.$$

Positivity, sums and all probabilities are preserved. Support reduction handles nonfaithful states. This is also [20, Lemma 3.1]. □

Lemma 12.1. *For any factor N_{in} with separable predual, $N_{\mathrm{in}} \bar{\otimes} R_\infty$ is III_1, and contains N_{in} with a faithful normal expectation.*

Proof. The expectation is $\mathrm{id} \otimes \rho$ for a faithful normal state ρ on R_∞. We follow the argument of [29, Lemma 3.8]. For each $0 < \lambda < 1$, choose $0 < \mu < 1$ with $\log \lambda / \log \mu$ irrational. The injective factors satisfy $R_\infty \cong R_\lambda \bar{\otimes} R_\mu$ and $R_\infty \bar{\otimes} R_\lambda \cong R_\infty$. Consequently $(N_{\mathrm{in}} \bar{\otimes} R_\infty) \bar{\otimes} R_\lambda \cong N_{\mathrm{in}} \bar{\otimes} R_\infty$. The asymptotic-ratio argument of Connes [12, Section 3.6], used in that proof, gives $\lambda \in r_\infty(N_{\mathrm{in}} \bar{\otimes} R_\infty)$. Connes's implication [12, Theorem 3.6.1] gives $r_\infty \setminus \{0\} \subseteq S$. Every $0 < \lambda < 1$ therefore belongs to the S-invariant, and the factor is III_1. These steps apply to every factor N_{in} with separable predual. Hyperfiniteness enters [29, Lemma 3.8] only through its final identification with the injective III_1 factor, which this lemma does not need. □

Theorem 12.2. *For each fixed* $\lambda \in [0,1]$, $\mathrm{PERF}(\mathrm{III}_\lambda) = \mathrm{PERF}_{\mathrm{co}}$.

Proof. The inclusion into commuting perfection is immediate. For the converse, standardize a commuting-perfect strategy and choose a factor host N_{in} by central decomposition as in Section 2. Put $N_{\mathrm{abs}} = N_{\mathrm{in}} \bar{\otimes} R_\infty$. By expected-inclusion transfer and Lemma 12.1, this III_1 factor hosts the game. Retain it for $\lambda = 1$.

Fix a faithful normal state φ on N_{abs}. For a countable subgroup $\Upsilon \subset \mathbb{R}$, given the discrete topology, form $N_\Upsilon = N_{\mathrm{abs}} \rtimes_{\sigma^\varphi|_\Upsilon} \Upsilon$. The Fourier expectation onto N_{abs} is faithful normal, so transfers the game. It is a factor: $\mathrm{T}(N_{\mathrm{abs}}) = \{0\}$ for a III_1 factor, hence every nonzero modular time is outer. If a central element has Fourier coefficient x_g, commutation with N_{abs} makes x_g an intertwiner between id and σ_g^φ. A nonzero such coefficient for $g \neq 0$ would, by polar decomposition in a factor, implement that automorphism innerly. Thus only $g = 0$ remains, and x_0 is scalar. Faithfulness of Fourier coefficients proves factoriality.

The continuous core is the crossed product by the modular $\mathbb{R}$-action; its dual action restricted to the center is the *flow of weights*. For factors this flow distinguishes the type-III subtypes [14]. Here is the core computation, with the dual flow specified. The modular group of the dual state fixes the canonical unitaries u_g of the discrete crossed product. Writing $\lambda(t)$ for the continuous-core implementing unitaries, it follows that

$$\begin{aligned} c(N_\Upsilon) &\cong c(N_{\mathrm{abs}}) \rtimes_{\mathrm{Ad}\,\lambda(g)} \Upsilon \\ &\cong c(N_{\mathrm{abs}}) \bar{\otimes} L(\Upsilon). \end{aligned} \tag{87}$$

For the second isomorphism the unitaries $y_g = \lambda(g)^* u_g$ commute with $c(N_{\mathrm{abs}})$ and satisfy the group law of Υ; the inverse sends u_g to $\lambda(g) \otimes y_g$. Since N_{abs} is III_1, $c(N_{\mathrm{abs}})$ is a factor, so the center is $L(\Upsilon) = L^\infty(\widehat{\Upsilon})$. The dual action obeys $\theta_s(\lambda(t)) = e^{-ist}\lambda(t)$ and $\theta_s(u_g) = u_g$; therefore $\theta_s(y_g) = e^{isg} y_g$. Its action on the center is translation by the character $\chi_s(g) = e^{isg}$.

For $0 < \lambda < 1$, take $\Upsilon = t_\lambda \mathbb{Z}$, $t_\lambda = 2\pi/(-\log\lambda)$. The resulting flow is rotation on the circle with least period $-\log\lambda$, which is the flow of a III_λ factor [14]. For $\lambda = 0$ take $\Upsilon = \mathbb{Z} + \sqrt{2}\mathbb{Z}$. Its dual is the two-torus, and the flow is $s \mapsto (e^{is}, e^{i\sqrt{2}s})$. Fourier characters show ergodicity: an invariant character indexed by (m,n) requires $m + n\sqrt{2} = 0$, hence $m = n = 0$. Every orbit is a dense one-dimensional subgroup coset of Haar measure zero (its bounded segments have zero area). Thus the flow is not transitive. The Connes–Takesaki flow classification [14] gives subtype III_0. This is the same modular crossed-product mechanism as [29, Example 3.9], without requiring injectivity of N_{abs}.

For clarity non-semifiniteness is a separate conclusion. The core flow of a semifinite factor is translation on $L^\infty(\mathbb{R})$; it is transitive and admits no invariant normal probability state. Each of our circle or torus flows preserves Haar probability, so cannot be that semifinite flow. For N_{abs} itself non-semifiniteness was already proved by its S-invariant. □

Slofstra's commuting classification [57], also proved in [44, Theorem 6.4], now proves Π_1^0-completeness for every fixed subtype, including zero. With Section 11 this completes the proof of Theorem B.

13 One selector for spatial and semifinite perfection

The critical selector can choose the same witness in spatial and semifinite models. The construction uses two edge families on one scaffold: the selector is critical, while the four-letter computation tree is subcritical. The key point is that a nonzero selected terminal sector has positive intersection with the computation root. This supplies a finite witness without using a density eigenvector in a diffuse host.

Lemma 13.1 (Several edge families on a common chain)**.** *Fix $n \geq 1$ and finitely many edge families $X_i^{(r)}$. Each family has its own Gram equations, initial defect and range equation, and its own within-family endpoint equations with typed centralizer coefficients. There are no prescribed cross-family Gram equations. If all edges have modular ratio 2^{-n}, one finite rational Boolean game compiles these systems on a common nonzero source corner. It extracts them from every perfect normal commuting strategy, and every faithful normal realization extends simultaneously, preserving host type under finite amplification and finite entropy for spatial realizations.*

Proof. Use the corners, activities, bound axes, and common chain $v = \mathsf{U}_n \cdots \mathsf{U}_1$ of Lemma 4.6. Its source mass is $\mathfrak{c}_n^{-1}$. For each edge, independently introduce a supported centralizer contraction $a_i^{(r)} = e_n a_i^{(r)} e_0$. A family with initial defect D_r and range R_r is compiled as

$$a_i^{(r)*} a_j^{(r)} = \delta_{ij}(e_0 - (D_r)_0), \qquad \sum_i a_i^{(r)} a_i^{(r)*} = (R_r)_n.$$

Only that family's own equations are imposed. Transport the union of all labels appearing at left endpoints or in range equations, with one separate graph refinement per encoding and bridge. The common original signs force $L_n v = v L_0$ for every label in this union. Decode $X_i^{(r)} = v^* a_i^{(r)}$. Multiplication by v, v^* proves the Gram, range and endpoint translations in both directions; for example $L_n a_i^{(r)} = a_j^{(r)} (R_{\mathrm{end}})_0$ becomes $L_0 X_i^{(r)} = X_j^{(r)} R_{\mathrm{end}}$. The centralizer condition and the modular eigenvalue of v give ratio 2^{-n} for every extracted edge. All equations concern the same faithful corner and state.

Conversely use $\mathrm{diag}(\mu_n) \otimes \varphi_0$ and $a_i^{(r)} = |e_n\rangle\langle e_0| \otimes X_i^{(r)}$ for every family. Every lifted contraction lies in the finite centralizer and has its own two-unitary encoding. Choose the root encodings once and copy them through the identity corner maps. The resulting block labels commute with their bridge's canonical signs. Actual joint products give all separate refinements, followed by the common qutrit and row extensions. There is one state and one assignment for all families. The construction adds only finite factors and imposes no mixed orthogonality. □

13.1 The coupled root presentation

Use four edges A_{ab} satisfying (30) at modular ratio 1/16, with root $P_\circ$, and sixteen different edges B_i with

$$B_i^* B_j = \delta_{ij}(I - P), \quad \sum_{i=1}^{16} B_i B_i^* = I - Q, \quad \sigma_t^{\varphi}(B_i) = 16^{-it} B_i.$$

For a positive centralizer contraction C_{sel} impose

$$C_{\mathrm{sel}} Q = Q, \quad C_{\mathrm{sel}} B_i = \tfrac{1}{2} B_i C_{\mathrm{sel}}, \quad [C_{\mathrm{sel}}, P] = 0, \quad [P, A_{ab}] = [C_{\mathrm{sel}}, A_{ab}] = 0.$$

Lemma 11.1 above uses modular ratio 1/16 and depth ratio 1/2. Retain the coordinate and boundary equations (34)–(37) and the counter equations (43), with $[C_{\mathrm{ctr}}, C_{\mathrm{sel}}] = 0$. Replace the clock by

$$(2C_{\mathrm{ctr}} - C_{\mathrm{sel}}) P_{\mathrm{ret}} = 0, \qquad T_{\mathrm{inv}}(2C_{\mathrm{ctr}} - C_{\mathrm{sel}}) = \tfrac{1}{8} C_{\mathrm{ctr}} C_{\mathrm{sel}} (I - P_{\mathrm{ret}}). \tag{88}$$

Here T_{inv} is a general contraction. The tableau has the same initial and update equations, but its halt prohibition is $P_{\mathtt{H}} P = 0$. Its machine reads the first positive pulse d_{clk}, sets $j = d_{\mathrm{clk}} - 1$, and tests $\mathcal{R}_{\mathrm{dec}}(e, j, 0), \mathcal{R}_{\mathrm{dec}}(e, j, 1), \ldots$, halting at the first false answer.

Lemma 13.1 with $n = 4$ compiles the two separate Gram systems. The transported typed list is exactly

$$P_\circ, U_x, U_y, P_x, P_y, C_{\mathrm{ctr}}, P, C_{\mathrm{sel}}, Q.$$

The return projection, its inverse and the tableau projections are source-only. In particular no relation $A_{ab}^* B_i = 0$ is imposed. Denote this finite game by F_e.

13.2 The terminal sector meets the root

Lemma 13.2. *Let orthogonal isometries $A_1, \dots, A_r$ have modular ratio ν on a faithful normal support. If a centralizer projection p commutes with them and $r\nu < 1$, then*

$$\varphi\left(p\left(I - \sum_i A_i A_i^*\right)\right) = (1 - r\nu)\varphi(p).$$

Proof. Commutation with a self-adjoint projection also gives commutation with the adjoints. KMS gives $\varphi(A_i p A_i^*) = \nu\varphi(p A_i^* A_i) = \nu\varphi(p)$. Subtract the sum of these r terms from $\varphi(p)$. □

In a semifinite realization, Lemma 11.1 with $d = 16$ gives some nonzero $p_j = P 1_{\{2^{-j}\}}(C_{\mathrm{sel}})$. It commutes with all A_{ab} and their adjoints. Hence

$$\varphi(p_j P_\circ) = \tfrac{3}{4}\varphi(p_j) > 0. \tag{89}$$

Choose $0 \neq \xi \in p_j P_\circ \mathcal{H}$. Its word descendants remain in $p_j \mathcal{H}$, are nonzero, and are orthogonal at every fixed word length. The coordinate inductions identify every finite square on these vectors.

For self-adjoint X, Y, with Y injective, the two equations $(2X - Y)R = 0$ and $T(2X - Y) = \kappa XY(I - R)$, $\kappa \neq 0$, force $R = 1_{\{0\}}(2X - Y)$. Indeed $R \leq K = 1_{\{0\}}(2X - Y)$. For $\zeta \in K\mathcal{H}$ put $\eta = (I - R)\zeta \in K\mathcal{H}$. The second equation gives $XY\eta = 0$, and $Y\eta = 2X\eta$ gives $2X^2\eta = 0$. Self-adjointness gives $X\eta = 0$, then $Y\eta = 0$, then $\eta = 0$. Apply this to (88), using injectivity of C_{sel} from the selector lemma. On the chosen descendants, $C_{\mathrm{ctr}} = 2^{-y}$ and $C_{\mathrm{sel}} = 2^{-j}$. Thus the pulse is exactly $y = j + 1$, including no time-zero pulse.

Lemma 13.3 (Definite computation on a terminal sector)**.** *Suppose $0 \neq \xi \in P_\circ \mathcal{H}$, every $A_w \xi$ belongs to $P\mathcal{H}$, and P_{ret} has the specified classical scalar on each $A_w \xi$. With the coordinate, initial and update equations above, every symbol projection acts on these descendants as the genuine classical symbol. If the presentation imposes $P_{\mathtt{H}} P = 0$, that classical computation cannot halt.*

Proof. At each fixed depth, the coordinate recursions give cyclic shifts and boundary masks on the word descendants. Row zero follows from (38) without its global halt equation. Induct on time at all spatial coordinates simultaneously. Each factor in the displayed ordered update product acts as its classical zero or one on the induction vector: left and right factors refer to the preceding row, and the pulse is scalar by hypothesis. Exactly the prescribed successor survives. Multiplication by U_y gives the next-row assertion. The last-row mask removes wraparound; radius one and blank quiescence make the right ghost blank throughout the required prefix. This proves the half-line computation on every sufficiently large square. If a halt occurred, one nonzero descendant $\eta \in P\mathcal{H}$ would satisfy $P_{\mathtt{H}}\eta = \eta$, contradicting $P_{\mathtt{H}}\eta = P_{\mathtt{H}} P \eta = 0$. □

Theorem 13.4 (A shared family)**.** *The family F_e satisfies*

$$F_e \in \mathbf{E} \iff F_e \in \mathbf{I} \iff F_e \in \mathbf{U} \iff \exists j\, \forall t\, \mathcal{R}_{\mathrm{dec}}(e, j, t), \qquad F_e \notin \mathbf{J}.$$

Proof. Semifinite soundness follows from the selected sector, the exact pulse and Lemma 13.3. A false check at that selected j would halt, so j is a witness.

For completeness let j be a witness. On the finite sixteen-letter tree of words v of length at most j, use prefix maps B_i, zero on leaves, with P the leaves, Q the root and

$$h_{\mathrm{sel}}|v\rangle = (j+1)^{-1}16^{-|v|}|v\rangle, \qquad C_{\mathrm{sel}}|v\rangle = 2^{-|v|}|v\rangle.$$

Tensor it with the four-letter Fock density h_F of (32). At selector depth ℓ, put $P_{\mathrm{ret}} = 1_{\{y=\ell+1\}}$ and

$$T_{\mathrm{inv}}(y,\ell) = \begin{cases} [8(2^{\ell+1}-2^{y})]^{-1}, & y \neq \ell+1, \\ 0, & y = \ell+1. \end{cases}$$

The nonzero denominator before the factor eight is an integer. The norm is exactly $1/8$, attained at $\ell = 0, y = 0$. Negative values occur. Use the genuine classical tableau for input ℓ on each such fibre; all symbol projections are its diagonal indicator projections. Coordinate permutations preserve Fock depth, and every other centralizer label is diagonal. Both edge families have ratio $1/16$. Computations at depths $\ell < j$ may halt; the terminal equation excludes only depth j. This gives every label and equation simultaneously, also at $j = 0$, where $B_i = 0$, $P = Q = C_{\mathrm{sel}} = I$.

The selector entropy is $\log_2(j+1) + 2j$, so the root product entropy is

$$\log_2(j+1) + 2j + \log_2(4/3) + 4/3 < \infty.$$

The compiler adds finite entropy. The inclusions $\mathbf{E} \subseteq \mathbf{I} \subseteq \mathbf{U}$ finish the equivalence. In a finite host the four extracted isometries would have orthogonal range traces summing to four under a normalized faithful trace, but their sum is at most the identity. Thus every instance is outside $\mathbf{J}$. □

14 Comparing the type languages

Equal arithmetical complexity does not mean equal languages. This section shows that the type-I and type-II_1 languages are incomparable and classifies the differences "perfect in X but not in Y" among six models.

14.1 A fixed one-bridge separator

Apply the complete one-bridge assembly of Section 11.4 to the fixed root presentation

$$A^*A = I, \qquad AA^* = I - P_\circ, \qquad \sigma_t^{\varphi}(A) = 2^{-it}A, \tag{90}$$

where $P_\circ$ is a centralizer projection. Call the resulting finite rational Boolean game G_{sh}. Here the transported label list contains just $P_\circ$. Encode its copies by reflections, introduce a centralizer contraction $a = e_1 a e_0$, and impose exactly

$$a^*a = e_0, \qquad aa^* = e_1 - (P_\circ)_1.$$

Use the six frame refinements and the graph-label refinement for $P_\circ$ from Section 11.4; apply the same graphs, triangles and rational rows. Extraction is $A = v^*a$, and $P = (P_\circ)_0$. Thus $A^*A = e_0$, $AA^* = e_0 - P$, and the modular ratio is $1/2$. Conversely every root model of (90) extends by that assembly, with $a = |e_1\rangle\langle e_0| \otimes A$.

For its type-I model take $\ell^2(\mathbb{N}_0)$, $A|m\rangle = |m+1\rangle$, $P_\circ = |0\rangle\langle 0|$, and

$$h|m\rangle = 2^{-m-1}|m\rangle, \qquad S(h) = \sum_{m\geq 0}(m+1)2^{-m-1} = 2 \text{ bits}.$$

The relation $hA = Ah/2$ verifies modular covariance. The scaffold and the finite maximally entangled compiler factors add finite entropy, so G_{sh} is perfect spatially with finite entropy.

In any perfect finite-host strategy, extraction takes place in the nonzero finite corner $e_0 M e_0$. Its isometry A must be unitary: a faithful finite trace gives $\tau(e_0 - AA^*) = 0$. Consequently $AA^* = e_0$ forces $P = 0$. On the other hand KMS for the original faithful state gives

$$\varphi(AA^*) = \tfrac{1}{2}\varphi(A^*A) = \tfrac{1}{2}\varphi(e_0), \qquad \varphi(P) = \tfrac{1}{2}\varphi(e_0) > 0,$$

a contradiction. Therefore G_{sh} has no finite host and witnesses $\mathrm{PERF(I)} \not\subseteq \mathrm{PERF(II_1)}$. In particular it has no finite-dimensional perfect strategy. Our earlier binary-scoring game [23] is another finite game that is perfect in infinite dimension with finite entanglement entropy but never perfect in finite dimension.

14.2 What happens to the Weyl-clock family

The spatial reduction of Sections 5–7 uses the four Fock isometries A_{ab} of (30), with h_F of (32), the coordinate unitaries and boundary projections of (34)–(37), the Weyl relation (40), the clock and counter values (44), and the reader equations (45). Spatial soundness uses finite positive density eigenspaces and a determinant to force a finite-order phase. This determinant is unavailable in a diffuse finite factor.

Here is a normal II_∞ model of *every* input of that family. Put $\vartheta = (\sqrt{5}-1)/2$, $\zeta = e^{2\pi i\vartheta}$. In the finite rotation factor $L^\infty(\mathbb{T}) \rtimes_\vartheta \mathbb{Z}$, take unitaries $J_{\mathrm{W}}, K_{\mathrm{W}}$ with commutator ζI and put $Z_{\mathrm{clk}} = \zeta I$. The invariant integral and Fourier coefficient at zero give a faithful finite trace. Irrational rotation is free and ergodic, so a central Fourier coefficient vanishes except at zero and its zero coefficient is constant; the algebra is a factor. It is infinite-dimensional and hence II_1. It is the hyperfinite factor R_{hyp} by amenability of this crossed product and injective-factor classification [13]. The construction itself only needs that it is finite.

Use the tensor density $h_F \otimes 1$ on $B(\ell^2(\mathcal{W}))\bar{\otimes}R_{\mathrm{hyp}}$. Set $W = \zeta^y$, $C_{\mathrm{ctr}} = 2^{-y}$, $P_{\mathrm{ret}} = 1_{\{y=0\}}$, and

$$T_{\mathrm{inv}}|_{(x,y)} = \begin{cases} 0, & y = 0, \\ 2^{-y-1}(\zeta^y - 1)^{-1}, & y > 0. \end{cases}$$

The norm bound follows from the quadratic irrationality estimate. For an integer p nearest to $y\vartheta$, the nonzero integer $p^2 + py - y^2$ factors as $(p - y\vartheta)(p - y\vartheta')$, $\vartheta' = -(1+\sqrt{5})/2$. Since $|p - y\vartheta| \le 1/2$,

$$\|y\vartheta\| \ge \frac{1}{\sqrt{5}\,y + 1/2}, \qquad |\zeta^y - 1| = 2\sin(\pi\|y\vartheta\|) \ge \frac{4}{\sqrt{5}\,y + 1/2} > 2^{-y-1}.$$

The sine inequality uses concavity, $\sin(\pi t) \ge 2t$ for $0 \le t \le 1/2$. In particular it supplies the explicit all-integer bound $|\sin(\pi y\vartheta)| \ge 8/(11y)$, since $\sqrt{5} < 9/4$ and $y \ge 1$. For the last inequality it suffices to replace $\sqrt{5}$ by $9/4$: $2^{y+3} > 9y/4 + 1/2$ for $y = 1$, and doubling proves it inductively. Thus T_{inv} is a contraction and (45) holds. The binary coordinate recurrences $y \mapsto 2y + b$ verify the clock and counter endpoint equations.

In the tableau of Sections 5 and 7 a waiting front advances one cell per time step, at $x = y$ at time y. The first positive-time return pulse P_{ret} freezes it, and the frozen coordinate becomes the unary input of the check of the hardwired predicate; the time-zero pulse is ignored. Here P_{ret} selects $y = 0$ only, so no positive-time pulse occurs. The front never freezes, and the decoder never starts a checking computation, irrespective of the hardwired input. Its finite deterministic tableau relations are satisfied by that classical nonhalting tableau in every finite square. The coordinate, boundary and tableau projections are diagonal on word sectors and commute with $h_F \otimes 1$. All endpoint labels have the required centralizer type. Lemma 4.6 with $n = 4$ applies to exactly the conditional copies and transported

labels of Lemma 4.5 and to the endpoint rows of Sections 5–7. Its left-endpoint labels are $P_\circ, U_x, U_y, P_x, P_y, W, C_{\mathrm{ctr}}, Z_{\mathrm{clk}}, J_{\mathrm{W}}, K_{\mathrm{W}}$; source-only labels are $P_{\mathrm{ret}}, T_{\mathrm{inv}}$ and the tableau projections. The activity corners and separate refinements coincide with that compiler. Its root mass is $1/121$. The resulting ambient host is II_∞ for every input. The supplementary exact margin certificate is in Appendix D.

Lemma 4.6, with $n = 4$, also extracts the four isometries of every Weyl-clock game in any normal commuting host. Two isometries with orthogonal ranges cannot lie in a nonzero finite corner. Hence every Weyl-clock game has no finite host. This additional fact is independent of the separator G_{sh} above.

14.3 A tracial game with no spatial perfection

For the other direction use a linear-system game with a perfect commuting strategy and no perfect finite-dimensional strategy, provided by [56]. The solution-group criterion [8, Theorems 4–5] says its distinguished central involution j is nontrivial. The central corner $(1 - \lambda(j))L(\Gamma_{\mathrm{sol}})/2$ of the solution group Γ_{sol} has normalized trace and a perfect left/right regular strategy: equation observables commute in each constraint, square to I, and their product is the specified sign because $j = -I$ in the corner. Use their joint spectral projections on the left and the commutant mirrors of variable observables on the right. Consistency and each equation are then exact. A factor fibre is a finite factor host; tensoring with R_{hyp} gives II_1. This is the tracial mechanism in [20, Proposition 7.4]. Call this game G_{lin}. Gao, Yu and Zhi give an explicit linear-system game with a perfect commuting strategy and finite-dimensional value strictly below one [28]; the same construction places it in $\mathrm{PERF}(\mathrm{II}_1)$, and its value gap excludes spatial perfection.

Cleve–Mittal [9] and [8, Theorem 5] characterize the finite-dimensional case. Spatial perfect play also implies finite-dimensional perfect play for linear-system games; Slofstra [56] uses this implication in the nonclosure construction. Here is the normal spatial argument. Each variable appearing in a constraint has an Alice involution with a self-adjoint Bob mirror on the state vector, by perfect consistency. Equality of those vectors makes Alice's trace-class marginal commute with every such involution. The within-constraint products and parity equations hold on the marginal support, by the zero-loss equations and support separation. A positive density eigenvalue has a nonzero finite-dimensional eigenspace reducing all variable involutions. Restriction to it preserves the products and all constraint equations. Its maximally entangled vector with transposed Bob variables is a finite-dimensional perfect strategy, a contradiction. For a mixed spatial state first purify; the same Alice eigenprojection argument applies. Type-I hosting decomposes into spatial factor fibres, so it is also excluded. Thus $\mathrm{PERF}(\mathrm{II}_1) \not\subseteq \mathrm{PERF}(\mathrm{I})$.

Every type-I factor host tensored with R_{hyp} has type II_1 in finite dimension and type II_∞ in infinite dimension; a further $B(\ell^2)$ amplification handles the former case. Every II_1 host similarly amplifies to II_∞. The two witnesses just constructed prove

$$\mathrm{PERF}(\mathrm{I}) \text{ and } \mathrm{PERF}(\mathrm{II}_1) \text{ are incomparable}, \qquad \mathrm{PERF}(\mathrm{I}),\ \mathrm{PERF}(\mathrm{II}_1) \subsetneq \mathrm{PERF}(\mathrm{II}_\infty).$$

Equal arithmetical completeness does not imply equal languages.

14.4 Exact conjunction and disjunction

We now compare the six languages $\mathbf{q}, \mathbf{E}, \mathbf{I}, \mathbf{J}, \mathbf{U}, \mathbf{C}$ of Section 2. Equation (10) identifies $\mathbf{U}$ with both $\mathrm{PERF}(\mathrm{II})$ and $\mathrm{PERF}(\mathrm{II}_\infty)$; Theorem 12.2 identifies $\mathbf{C}$ with each fixed type-III language.

Background lemma (Exact game operations)**.** *Uniform effective operations $G \wedge G'$ and $G \vee G'$ satisfy, for every $\mathbf{X} \in \{\mathbf{q}, \mathbf{E}, \mathbf{I}, \mathbf{J}, \mathbf{U}, \mathbf{C}\}$,*

$$G \wedge G' \in \mathbf{X} \iff G, G' \in \mathbf{X}, \qquad G \vee G' \in \mathbf{X} \iff G \in \mathbf{X} \text{ or } G' \in \mathbf{X}.$$

The same statements hold for each of the type languages in this paper.

Proof. First pad question and answer alphabets to common finite alphabets. For an original zero-weight question pair accept everything; preserve its original predicate on positive-weight pairs. Give every padded question pair positive rational weight, accept all pairs involving fresh questions, and reject a padded answer on an original question. This changes no perfect-play language. It permits fixed common question alphabets with full support; no implication below relies on an untested pair.

For conjunction, send both players the same tag and questions for that tagged game, and test its predicate. A perfect strategy restricts to either tag in the same ambient algebra and state. Conversely, tensor two perfect strategies and on a tagged question measure the appropriate factor. Type I is closed under tensor products, as are finite factors, type II_∞ factors, and semifinite algebras. Amplification gives the stated II existence languages when needed. For **E** use pure spatial witnesses: marginal entropies add, so their sum is finite. Finite-dimensional witnesses stay finite. Commuting strategies tensor as well. For III_λ use its language equality with **C** from Theorem 12.2; no closure of that subtype under direct tensoring is asserted.

For disjunction, ask the common questions and require each answer to name a branch $s \in \{G, G'\}$ as well as an answer in that branch. Reject mismatched branches and otherwise test the named predicate. A strategy for either game wins by always naming that branch. For soundness let $A^x_{s,a}, B^y_{s,b}$ be the effects and put $P^x_s = \sum_a A^x_{s,a}$, $Q^y_s = \sum_b B^y_{s,b}$. Work first on the faithful standard-form support. The matching argument of Section 2.4 makes each branch marginal a projection, with $P^x_s \xi = Q^y_s \xi$ for every x, y. Fixing y and using separation makes $P^x_s = P_s$ independent of x; likewise $Q^y_s = Q_s$. These are opposite mirrors, $P_s\xi = Q_s\xi =: \xi_s$. At least one ξ_s is nonzero. Since $0 \le A^x_{s,a} \le P_s$, we have $A^x_{s,a}P_s = P_s A^x_{s,a} = A^x_{s,a}$, and the same holds on Bob's side. Complete each branch POVM to the identity by adding $I - P_s$ to one Alice answer and $I - Q_s$ to one Bob answer. These added effects annihilate ξ_s. A losing branch product annihilates ξ and, by the displayed support identities, also ξ_s. Normalizing ξ_s therefore wins branch s. All effects and the vector remain on the same Hilbert space, in the same ambient M, M'; the selected vector need not be faithful. Thus its host type is preserved. This also handles mixed strategies by purification and standardization.

For finite entropy one can work in the original spatial purification. Orthogonality of Bob's branch projections gives $\rho_A = \sum_s p_s \rho_{A,s}$, $p_s = \|\xi_s\|^2$. Entropy concavity, valid also by finite-rank truncation for infinite entropy, gives $\sum_s p_s S(\rho_{A,s}) \le S(\rho_A) < \infty$. Every positive-weight branch has finite entropy. This proves the last model-specific assertion and completes both directions. □

14.5 Four fixed separators and two uniform families

Choose both reductions G^{clk}_e and G^{sel}_e for the same predicate, $e \in \mathrm{FIN}$, and write $\mathrm{Fin}(e)$ for its truth value. Theorems 7.1 and 11.5, including Lemma 11.4, give

$$G^{\mathrm{clk}}_e \in \mathbf{E} \iff G^{\mathrm{clk}}_e \in \mathbf{I} \iff \mathrm{Fin}(e), \tag{91}$$

$$G^{\mathrm{sel}}_e \in \mathbf{J} \iff G^{\mathrm{sel}}_e \in \mathbf{U} \iff \mathrm{Fin}(e), \qquad G^{\mathrm{sel}}_e \in \mathbf{C}. \tag{92}$$

Every G^{clk}_e has no finite-dimensional perfect strategy; every G^{sel}_e is commuting perfect with $\omega_q = 1$. The chained lemma additionally excludes G^{clk}_e from **J** and places it in **U** (Section 14.2). The reductions below use its displayed spatial coordinates and use G_{sh} for finite-host exclusion.

Take the fixed one-bridge game G_{sh} of Section 14.1. Its shift model has root entropy 2 bits and its finite compiler factors preserve finite entropy. The finite-corner KMS contradiction there proves that it has no finite host. Thus its complete profile uses only the one-bridge assembly.

The tracial solution-group game G_{lin} is in $\mathbf{J} \setminus \mathbf{I}$; spatial perfection would imply finite perfection. Let $G^{(2)}_{\mathrm{Cat}}$ be the compiled Catalan game of [24, Theorem C and Sections 3 and 6]. Its perfect spatial strategies have infinite marginal entropy, so it is in $\mathbf{I} \setminus \mathbf{E}$. Lemma 4.6 with $n = 2$ applies to its nine-corner assembly: activity $e_{g-1} + m_g$, root mass $1/25$, transported encodings b_+, b_-, p, q, c (names of [24, Section 6]), and the source-only polynomials of [24, Section 6]. Thus its extracted isometry A has modular ratio $1/4$ also on a commuting normal host. In a finite root corner A would be unitary, and KMS would give $1 = 1/4$; hence it is outside $\mathbf{J}$ as well. Finally choose a fixed NO instance of G^{sel} and call it G_{III}. Theorem 11.5 gives $G_{\mathrm{III}} \in \mathbf{C} \setminus \mathbf{U}$ and $\omega_q(G_{\mathrm{III}}) = 1$. Their complete profiles are

	q	**E**	**I**	**J**	**U**	**C**
G_{sh}	0	1	1	0	1	1
G_{lin}	0	0	0	1	1	1
$G^{(2)}_{\mathrm{Cat}}$	0	0	1	0	1	1
G_{III}	0	0	0	0	0	1

In particular $G_{\mathrm{sh}} \wedge G_{\mathrm{lin}}$ is in $\mathbf{U}$ but in neither $\mathbf{I}$ nor $\mathbf{J}$, so even $\mathbf{I} \cup \mathbf{J} \subsetneq \mathbf{U}$. The remaining inclusions asserted above are strict: G_{sh} separates $\mathbf{q}$ from $\mathbf{E}$, $G^{(2)}_{\mathrm{Cat}}$ separates $\mathbf{E}$ from $\mathbf{I}$, G_{lin} separates $\mathbf{I}$ from $\mathbf{U}$, and G_{III} separates $\mathbf{U}$ from $\mathbf{C}$. Both $\mathbf{E}$ and $\mathbf{I}$ are incomparable with $\mathbf{J}$.

14.6 The table, with reductions for every nonempty entry

Theorem 14.1 (Directional differences)**.** *The following table gives row minus column. Each suffix c means computable many-one completeness for the displayed pointclass. Here D_2 abbreviates $D_2(\Sigma^0_2)$. Every nonempty cell is complete for its displayed class.*

	q	**E**	**I**	**J**	**U**	**C**
q	$\varnothing$	$\varnothing$	$\varnothing$	$\varnothing$	$\varnothing$	$\varnothing$
E	Σ^0_2-c	$\varnothing$	$\varnothing$	D_2-c	$\varnothing$	$\varnothing$
I	Σ^0_2-c	D_2-c	$\varnothing$	D_2-c	$\varnothing$	$\varnothing$
J	Σ^0_2-c	D_2-c	D_2-c	$\varnothing$	$\varnothing$	$\varnothing$
U	Σ^0_2-c	D_2-c	D_2-c	D_2-c	$\varnothing$	$\varnothing$
C	Π^0_1-c	Π^0_2-c	Π^0_2-c	Π^0_2-c	Π^0_2-c	$\varnothing$

Proof. All empty entries follow from the stated inclusions. For upper bounds, $\mathbf{q} \in \Sigma^0_1$, $\mathbf{E}, \mathbf{I}, \mathbf{J}, \mathbf{U} \in \Sigma^0_2$, and $\mathbf{C} \in \Pi^0_1$. A difference between two middle languages is therefore in D_2. For a middle $\mathbf{X}$, $\mathbf{X} \setminus \mathbf{q} \in \Sigma^0_2$: write its predicate as $\exists k \forall m$ for $\mathbf{X}$ and conjoin the Π^0_1 predicate for nonmembership in $\mathbf{q}$. Also $\mathbf{C} \setminus \mathbf{q} \in \Pi^0_1$ and $\mathbf{C} \setminus \mathbf{X} \in \Pi^0_2$. It remains to prove the matching lower bounds.

For the four incomparable entries use $G^{\mathrm{clk}}_e \wedge G_{\mathrm{sh}}$ as the $\mathbf{E}$- or $\mathbf{I}$-side family, and $G^{\mathrm{sel}}_f \wedge G_{\mathrm{lin}}$ as the $\mathbf{J}$-side family. Their disjunction has $\mathbf{E}$ and $\mathbf{I}$ coordinates exactly $\mathrm{Fin}(e)$ and $\mathbf{J}$ coordinate exactly $\mathrm{Fin}(f)$. Thus its membership in either $\mathbf{E} \setminus \mathbf{J}$ or $\mathbf{I} \setminus \mathbf{J}$ is $\mathrm{Fin}(e) \wedge \neg\mathrm{Fin}(f)$. Swap e, f between the two branches for the reverse differences. The canonical D_2 problem reduces in all four cases; in particular $\mathbf{I} \setminus \mathbf{J}$ is $D_2(\Sigma^0_2)$-complete.

For the $\mathbf{q}$ column, use G^{clk}_e for $\mathbf{E}$ and $\mathbf{I}$; use $G^{\mathrm{sel}}_e \wedge G_{\mathrm{lin}}$ for $\mathbf{J}$ and $\mathbf{U}$. Their larger coordinate is $\mathrm{Fin}(e)$ and their finite-dimensional coordinate is identically zero. Each such entry is therefore Σ^0_2-hard.

For the four rightward entries of the $\mathbf{C}$ row, use $G_e^{\mathrm{clk}} \vee G_{\mathrm{III}}$ when $\mathbf{X} = \mathbf{E}, \mathbf{I}$, and $G_e^{\mathrm{sel}} \vee G_{\mathrm{III}}$ when $\mathbf{X} = \mathbf{J}, \mathbf{U}$. The commuting coordinate is always one and the $\mathbf{X}$ coordinate is exactly $\mathrm{Fin}(e)$, giving Π_2^0-hardness. These instances all have $\omega_q = 1$, since choosing branch G_{III} approximates perfection in finite dimension. For $\mathbf{C} \setminus \mathbf{q}$, take a Σ_1^0-complete finite-perfection reduction G_e^{MIP} with $G_e^{\mathrm{MIP}} \in \mathbf{q} \iff e$ halts, available by perfect completeness in [34, Theorem 12.6] and finite-dimensional feasibility. Then $G_e^{\mathrm{MIP}} \vee G_{\mathrm{Cat}}^{(2)}$ is always commuting perfect, has $\omega_q = 1$ because $G_{\mathrm{Cat}}^{(2)}$ is spatial perfect, and is in $\mathbf{q}$ exactly when e halts. Nonhalting reduces to the difference, proving Π_1^0-hardness even under the promise.

For the two remaining complete nested differences use a shared family. In general suppose $\mathbf{Y} \subset \mathbf{X}$ and a computable family $\mathcal{F}_e$ has both coordinates equal to $\mathrm{Fin}(e)$. Fix $G_{\mathrm{sep}} \in \mathbf{X} \setminus \mathbf{Y}$. For

$$\mathcal{D}(e, f) = \mathcal{F}_e \wedge (G_{\mathrm{sep}} \vee \mathcal{F}_f) \tag{93}$$

the $\mathbf{X}$ coordinate is $\mathrm{Fin}(e)$ and the $\mathbf{Y}$ coordinate is $\mathrm{Fin}(e) \wedge \mathrm{Fin}(f)$. The difference coordinate is therefore $\mathrm{Fin}(e) \wedge \neg\mathrm{Fin}(f)$, proving D_2-hardness. Apply this first with $(\mathbf{X}, \mathbf{Y}) = (\mathbf{I}, \mathbf{E})$, $\mathcal{F}_e = G_e^{\mathrm{clk}}$ and $G_{\mathrm{sep}} = G_{\mathrm{Cat}}^{(2)}$, using (91). Apply it again with $(\mathbf{X}, \mathbf{Y}) = (\mathbf{U}, \mathbf{J})$, $\mathcal{F}_e = G_e^{\mathrm{sel}}$ and $G_{\mathrm{sep}} = G_{\mathrm{sh}}$, using (92). Thus $\mathbf{I} \setminus \mathbf{E}$ and $\mathbf{U} \setminus \mathbf{J}$ are $D_2(\Sigma_2^0)$-complete. The required YES witness has finite entropy by Theorem 7.1.

For $\mathbf{U} \setminus \mathbf{I}$, $\mathbf{U} \setminus \mathbf{E}$, and the stronger language $\mathbf{U} \setminus (\mathbf{I} \cup \mathbf{J})$, take the shared family F_e of Theorem 13.4 for FIN and fix $H = G_{\mathrm{lin}} \in \mathbf{J} \setminus \mathbf{I}$. Set $\Lambda(e, f) = F_e \wedge (H \vee F_f)$. Its coordinates are

$$\mathbf{U}(\Lambda) = \mathrm{Fin}(e), \quad \mathbf{I}(\Lambda) = \mathbf{E}(\Lambda) = \mathrm{Fin}(e) \wedge \mathrm{Fin}(f), \quad \mathbf{J}(\Lambda) = 0.$$

Each difference therefore has predicate $\mathrm{Fin}(e) \wedge \neg\mathrm{Fin}(f)$. All three are in $D_2(\Sigma_2^0)$, completing the table and proving the additional simultaneous exclusion. Replace each F_e by $F_e \vee G_{\mathrm{III}}$ to obtain commuting perfection and $\omega_q = 1$ on every input, with the same four lower coordinates. The two conjuncts then both have value one. □

Corollary 14.2. *The language of games with perfect spatial play but no finite-entropy perfect spatial strategy is $D_2(\Sigma_2^0)$-complete. In particular it belongs to neither Σ_2^0 nor Π_2^0.*

Proof. The identification is $\mathbf{I} \setminus \mathbf{E}$ in the theorem. A D_2-complete language is hard for both Σ_2^0 and Π_2^0 by fixing one of the canonical FIN indices to a YES or NO index. Membership in either pointclass would collapse these two pointclasses, contradicting strictness of the arithmetical hierarchy. □

The corollary concerns exact attainment; it asserts no rate at which entropy must diverge for near-optimal strategies.

For differences between middle models, value-one padding can be made explicit. Replacing G by $G \vee G_{\mathrm{III}}$ preserves all lower coordinates and sets the commuting coordinate to one, while forcing $\omega_q = 1$. To remove finite-dimensional perfect play as well, use $G \wedge (G_{\mathrm{sh}} \vee G_{\mathrm{lin}})$; the second factor has all five infinite-model coordinates equal to one and its $\mathbf{q}$ coordinate zero. In particular (93) for $\mathbf{I} \setminus \mathbf{E}$ already has no finite-dimensional perfect instance; adding G_{III} also makes every instance value one. The $\mathbf{C}$-row hardness families above already have value one, including the separate $\mathbf{C} \setminus \mathbf{q}$ construction.

Corollary 14.3 (A natural uniqueness problem)**.** *Deciding whether a graph on $\mathbb{N}$ with primitive recursive adjacency has exactly one finite connected component is $D_2(\Sigma_2^0)$-complete, and hence computably many-one equivalent to $\mathbf{U} \setminus \mathbf{I}$.*

Proof. For fixed j, at least j finite components is Σ_2^0: existentially specify j disjoint nonempty finite connected sets and universally prohibit an edge leaving any one. Exactly one is at least one minus at least two.

For hardness, from c.e. sets W_a, W_b construct three stars with centres $0, 1, 2$. At stage s add vertices $3s+3, 3s+4, 3s+5$. Attach the first to centre zero if W_a gains a new distinct element, otherwise to two. Attach the second to centre one if either set gains a new element, otherwise to two. Always attach the third to two. Bounded enumeration simulation makes adjacency primitive recursive. The three components are finite precisely under the respective conditions W_a finite, both W_a, W_b finite, and never. Thus exactly one is finite precisely when $a \in \mathrm{FIN}$ and $b \notin \mathrm{FIN}$. The natural statement still has a difference in its logical form. □

14.7 The refinements of Theorem B

The following theorem collects the difference results for the type languages.

Theorem 14.4 (Refinements of Theorem B)**.** *Commuting perfection without semifinite perfection is Π^0_2-complete, with hardness on a family of commuting-perfect games all having $\omega_q = 1$. The two languages* semifinite but not spatial *and* semifinite but not finite-entropy spatial *are $D_2(\Sigma^0_2)$-complete, also with no finite-host perfection. Every nonempty directional difference among the six models in Theorem 14.1 is complete for its displayed class.*

Proof. The first assertion is proved at the end of Section 11.5: every G_e^{sel} is commuting perfect with $\omega_q(G_e^{\mathrm{sel}}) = 1$, and $G_e^{\mathrm{sel}} \in \mathrm{PERF}_{\mathrm{co}} \setminus \mathrm{PERF}_{\mathrm{sf}}$ exactly when $e \notin \mathcal{A}_{\mathrm{dec}}$. The two languages of the second assertion are $\mathbf{U} \setminus \mathbf{I}$ and $\mathbf{U} \setminus \mathbf{E}$. The proof of Theorem 14.1 reduces the canonical $D_2(\Sigma^0_2)$-complete problem to both through the family $\Lambda(e, f)$, whose $\mathbf{J}$ coordinate is identically zero, so no instance has finite-host perfection. The third assertion is Theorem 14.1. □

15 Zero-error codes and perfect play

A sender uses a finite noisy classical channel once. Beforehand, sender and receiver share an entangled state. The sender encodes one of m messages into the channel input, and the receiver must identify that message with certainty. We ask whether such a code exists. The channel and the message count remain finite; only the shared resource is unrestricted. Zero-error communication depends on exact zeros, rather than on how closely a protocol approaches perfect transmission. This makes attainment decisive, and the question becomes a question about perfect play.

For a finite rational stochastic matrix N and an integer $m \geq 1$, write $\mathrm{ZE}_t(N, m)$ for this feasibility problem. The model q uses finite-dimensional tensor products. The model qs uses a normal density operator on a spatial tensor product, with bounded finite-outcome POVMs. The model qc uses commuting local operator algebras and a state. The entropy restriction finS means that both resource marginals have finite von Neumann entropy. We use countably infinite Schmidt decompositions.

15.1 Channels and trace-class assemblages

For channel inputs x, x', write $x \sim x'$ when some output y has $N(y \mid x)N(y \mid x') > 0$. This is a reflexive relation: it includes $x = x'$. Its restriction to distinct inputs is the confusability graph. Every finite graph X has a rational realizing channel N_{X}: a nonisolated vertex outputs a uniformly random incident edge, and an isolated vertex outputs its own private symbol.

We first extend the finite-dimensional assemblage characterization of [16]. All supports below are closed range supports.

Lemma 15.1 (Normal assemblages)**.** *An m-message normal zero-error code exists if and only if there are positive trace-class operators $R_{i,x}$ and a density ρ such that*

$$\sum_x R_{i,x} = \rho \quad (1 \leq i \leq m), \qquad R_{i,x} R_{j,x'} = 0 \quad (i \neq j,\ x \sim x'). \tag{94}$$

The assemblage is realizable with the pure resource $|\sqrt{\rho}\rangle\!\rangle$*. The equivalence includes mixed resources.*

Proof. Given a mixed resource ω and Alice's encoding effect $E_{i,x}$, define

$$R_{i,x} = \mathrm{Tr}_A[(E_{i,x}^{1/2} \otimes I)\omega(E_{i,x}^{1/2} \otimes I)].$$

The sum is Bob's marginal, independently of i. For output y, Bob must distinguish the positive operators $T_i^y = \sum_x N(y \mid x) R_{i,x}$ perfectly. If F_i^y is a decoding POVM, perfect decoding gives $F_i^y = I$ on $\operatorname{supp} T_i^y$ and $F_j^y = 0$ there for $j \neq i$. Thus these supports are orthogonal. Positivity of the sum puts $\operatorname{supp} R_{i,x}$ in $\operatorname{supp} T_i^y$ whenever $N(y \mid x) > 0$. This proves the product condition, including equal inputs.

Conversely restrict to $\operatorname{supp} \rho$, so ρ is faithful. Each $0 \leq R \leq \rho$ has a unique bounded effect K with

$$R = \rho^{1/2} K \rho^{1/2}, \qquad 0 \leq K \leq I. \tag{95}$$

Indeed Douglas' factorization lemma [17] applied to $R \leq \rho$ gives $R^{1/2} = \rho^{1/2} V_0$ for a contraction V_0, and $K = V_0 V_0^*$. Uniqueness follows from the dense range of $\rho^{1/2}$. The sum identities imply $\sum_x K_{i,x} = I$, by the same dense-range argument.

In a spectral basis for ρ, let Alice measure $K_{i,x}^{\mathsf{T}}$ on the purification $|\sqrt{\rho}\rangle\!\rangle$. Bob's conditional operator is $R_{i,x}$. Equation (94) makes the T_i^y have orthogonal supports, so their support projections, with any unused subspace assigned to one message, form a decoding POVM. This constructs a normal pure code.

Alternatively, any normal mixed perfect code decomposes spectrally into pure perfect codes with the same measurements: every forbidden probability is nonnegative, so it vanishes on every positive-weight component. This observation will also be used for entropy. □

15.2 Saturation of a clique partition

Beigi's Lovász-theta bound implies the finite-dimensional clique-cover bound [4]. The following trace-class saturation statement extends [39, Lemma 5].

Lemma 15.2. *If* X *is partitioned into* k *cliques* $\mathcal{C}_c$*, every normal code has at most* k *messages. If it has exactly* k*, put*

$$B_{i,c} = \sum_{x \in \mathcal{C}_c} R_{i,x}, \qquad P_{i,c} = \operatorname{supp} B_{i,c}.$$

On $\operatorname{supp} \rho$ *one has*

$$B_{i,c} = \rho P_{i,c} = P_{i,c} \rho, \qquad \sum_i P_{i,c} = I, \qquad \sum_c P_{i,c} = I. \tag{96}$$

Each row and column of $(P_{i,c})$ *is an orthogonal projection partition.*

Proof. For fixed c, distinct i give orthogonal $B_{i,c}$, because every pair of inputs in $\mathcal{C}_c$ is confusable, including identical inputs. Since $B_{i,c} \leq \rho$,

$$\sum_i \mathrm{Tr}\, B_{i,c} \leq \sum_i \mathrm{Tr}(P_{i,c}\rho) \leq 1.$$

Summing over c gives $m \leq k$. If $m = k$, all inequalities are equalities. Faithfulness of ρ gives $\sum_i P_{i,c} = I$. For each i, c, positivity and $\mathrm{Tr}[P_{i,c}(\rho - B_{i,c})] = 0$ imply $(\rho - B_{i,c})^{1/2} P_{i,c} = 0$. Hence $\rho P_{i,c} = B_{i,c} = P_{i,c}\rho$. Summing over c yields $\rho \sum_c P_{i,c} = \rho$; injectivity gives the row sum. A finite family of projections summing to identity is pairwise orthogonal: compress the sum by one member and use positivity of all other compressed members. All traces used here are finite traces of positive trace-class operators. □

A code is *saturating* for this partition if its message count equals the clique bound, $m = k$.

15.3 The normal correspondence

Let G have question sets $\mathcal{Q}_A, \mathcal{Q}_B$, finite nonempty answer sets, and a Boolean losing predicate on the question pairs of positive sampling probability. Its graph $\Gamma(G)$ has vertices (L, u, a) and (R, v, b). Distinct answers to the same tagged question form a clique; cross edges are precisely losing tuples. There are no other edges. Set $k = |\mathcal{Q}_A| + |\mathcal{Q}_B|$, the number of tagged local questions; Mančinska's question count q_{M} is our k.

For a pure strategy, use Schmidt form $|D\rangle\!\rangle$ with $D \geq 0$ and $\operatorname{Tr} D^2 = 1$. For bounded positive effects, zero probability is equivalent to $EDF^{\mathsf{T}} = 0$, as in Section 3: zero probability is the vanishing of the Hilbert–Schmidt norm of $\sqrt{E}D\sqrt{F^{\mathsf{T}}}$; multiplication by the square roots gives one implication, and equality of the closures of the ranges of an effect and its square root gives the other.

Theorem 15.3 (Trace-class correspondence)**.** *For every finite game G,*

$$\mathrm{ZE}_{\mathrm{qs}}(N_{\Gamma(G)}, k) \iff G \text{ has a perfect pure spatial strategy with } \operatorname{Tr} D < \infty.$$

For a specified code marginal ρ, the reverse strategy has $D = \rho/\|\rho\|_2$. For a specified game strategy D with finite trace, the forward pure code has marginal $\rho = D/\operatorname{Tr} D$. The game need not be synchronous.

Proof. In the forward construction, store Bob's effects in transposed form, and denote all local effects by $E_{c,a}$ for tagged questions c. Put $\rho = D/\operatorname{Tr} D$ and

$$R_{c,(c,a)} = \rho^{1/2} E_{c,a} \rho^{1/2},$$

with zero operators off the designated clique. Each message family sums to ρ. For cross losing edges, the perfect-game sandwich $E_{c,a}DE_{c',b} = 0$ gives the required zero product. The remaining products between distinct messages vanish by their clique assignments. Lemma 15.1 realizes the code. The trace condition is exactly what normalizes its marginal.

For the converse, as in [39, Theorem 6], apply Lemma 15.2 to an arbitrary saturating code, whether or not it has the displayed form. Aggregate

$$T_{c,a} = \sum_i R_{i,(c,a)}.$$

Each family sums to ρ. If $(c, a), (c', b)$ is a cross losing edge, every product $R_{i,(c,a)}R_{j,(c',b)}$ is zero. For $i \neq j$ this is a code condition. For $i = j$ its two factors are supported on the orthogonal row projections $P_{i,c}, P_{i,c'}$. Thus $T_{c,a}T_{c',b} = 0$.

Use (95) to write $T_{c,a} = \rho^{1/2} E_{c,a} \rho^{1/2}$ with bounded POVMs. The product equality becomes

$$\rho^{1/2} E_{c,a} \rho E_{c',b} \rho^{1/2} = 0.$$

The injectivity and dense range of $\rho^{1/2}$ cancel its two outside factors, giving $E_{c,a}\rho E_{c',b} = 0$. Use the left-tagged effects on Alice and the transposes of the right-tagged effects on Bob. With $D = \rho/\|\rho\|_2$ these are a perfect spatial strategy. Since $\operatorname{Tr}\rho = 1$, $\operatorname{Tr} D = 1/\|\rho\|_2 < \infty$.

Only the coarse clique projections must commute with ρ. The fine outcomes need not do so; this is why the argument retains nonsynchronous games. Restricting every step to finite dimension recovers [39, Theorem 6]. □

Remark. For the game marginal $h = D^2$, the condition $\operatorname{Tr} D < \infty$ is finite Rényi entropy of order $1/2$, namely $2\log_2 \operatorname{Tr}\sqrt{h}$. It is stronger than finite von Neumann entropy. Replacing the forward marginal by D^2 generally destroys zero-error orthogonality; Appendix C includes an explicit check on the flagged magic square.

15.4 The resource correspondence at every order

Write $h = D^2$ for the game marginal, $Z = \operatorname{Tr}\sqrt{h}$, and $\rho_{\mathrm{code}} = \sqrt{h}/Z$ for the code marginal. A normal code first requires $Z < \infty$. Whenever it exists,

$$\operatorname{Tr}\rho_{\mathrm{code}}^{\beta} = Z^{-\beta}\operatorname{Tr}h^{\beta/2}, \qquad S(\rho_{\mathrm{code}}) = \log_2 Z + (2Z)^{-1}\operatorname{Tr}[\sqrt{h}\log_2(1/h)].$$

The resulting correspondence is

Code resource	Required game spectrum
Finite rank	finite rank
Finite Rényi entropy, $0 < \beta < 1$	$\operatorname{Tr}h^{\beta/2} < \infty$
Finite von Neumann entropy	$\operatorname{Tr}[\sqrt{h}\log_2(1/h)] < \infty$
Finite Rényi entropy, $1 < \beta \le \infty$	$\operatorname{Tr}\sqrt{h} < \infty$
Arbitrary normal resource	$\operatorname{Tr}\sqrt{h} < \infty$

The logarithmic condition implies summability of the square roots: only finitely many eigenvalues exceed any fixed positive threshold, and below such a threshold the logarithm is bounded below. It cannot be replaced by the half-moment alone. The squared-Catalan channel of Section 18 has a summable half-moment and requires infinite code entropy. For $\beta > 1$, the four-bridge tree has finite $\beta/2$-moment but no normal saturating code. At the spectrum level $p_j \asymp j^{-2}(\log j)^{-4}$ has summable square roots and a divergent logarithmic half-moment. These examples explain both endpoints of the table.

16 Five bridges and the normal row

We apply the trace-class correspondence to the compiler. The four-bridge games of Section 7 lie exactly at the summability threshold: none of their channels admits a saturating normal code, even on YES inputs. Adding a fifth bridge changes the modular ratio and restores summability. The same five-bridge YES codes have finite marginal entropy. The threshold is visible in the four-letter tree density $h_{F,\nu} = (1-4\nu)\sum_w \nu^{|w|}|w\rangle\langle w|$ of Section 16.2:

$$\operatorname{Tr}\sqrt{h_{F,\nu}} = \sqrt{1-4\nu}\sum_{\ell=0}^{\infty}(4\sqrt{\nu})^{\ell}, \qquad \nu = 2^{-n}.$$

There are four branches per level, so summability requires $n \ge 5$.

16.1 An exact spectrum identity

The FMS transports need not commute with the row factor $\mathrm{Mat}_{d_{\max}}$, because the sign questions are not synchronized. Removing the row factor in soundness therefore uses the following elementary cancellation instead of a partial trace over an infinite space.

Finite-amplification cancellation. If two finite-dimensional tuples of unitaries become unitarily equivalent after $d_{\max}$ identical copies, the original tuples are unitarily equivalent. Indeed they are finite-dimensional representations of the universal $*$-algebra on the tuple. The image in their direct sum is a finite-dimensional C^*-algebra. Decompose its two representations into irreducibles; amplification multiplies every multiplicity by $d_{\max}$, so equality of amplified multiplicities cancels $d_{\max}$.

Lemma 16.1 (Exact spectrum identity)**.** *For every pure perfect strategy of $G^{(n)}$, with full marginal h and Schmidt operator $D = \sqrt{h}$, there is a normal realization with density h_0 such that, including multiplicities,*

$$\operatorname{spec}(h) = \operatorname{spec}\left(\operatorname{diag}(\mu_n) \otimes h_0 \otimes I_3/3 \otimes I_{d_{\max}}/d_{\max}\right). \tag{97}$$

Here h_0 is the reduced normalized source density after removing the qutrit and row factors; it differs from the source density on the full amplified Hilbert space in Lemma 4.5. Conversely every realization has a perfect strategy with this spectrum. In particular, in the extended nonnegative reals,

$$\operatorname{Tr} D = c_n^{\mathrm{amp}} \operatorname{Tr} \sqrt{h_0}, \qquad c_n^{\mathrm{amp}} = \sqrt{3 d_{\max}}\, \frac{1 + 4 \sum_{g=1}^{n} 2^{g/2}}{\sqrt{\mathfrak{c}_n}}. \tag{98}$$

Proof. Restrict to the original marginal support. Synchronization makes the certified scalar matrix algebras commute with the density. The named old projections commute with $\mathrm{Mat}_{d_{\max}}$; the bound parameters and graph units also provide the scalar Mat_3 frame. These two scalar algebras commute, so their exact matrix-unit decompositions give a density $I_3/3 \otimes I_{d_{\max}}/d_{\max} \otimes h_{\mathrm{base}}$ and factorizations of every named old synchronized projection.

The magic matrix units identify all four density blocks at each bridge. The U_g identify consecutive normalized selected density blocks, with ratio two. The cover therefore implies that the full density has the spectrum in (97), where h_0 is its normalized source block after removal of both scalar factors. It remains to verify that this h_0 is the density of a realization; simply compressing $v^* a_\alpha$ across $\mathrm{Mat}_{d_{\max}}$ would not prove that assertion.

The frame refinements make v act trivially on Mat_3. After removing that frame it is a unitary between the $d_{\max}$-amplifications of the source and far spaces, intertwining their normalized densities and every transported unitary label. In each positive density eigenspace these spaces are finite dimensional. Apply finite-amplification cancellation to their transported unitary tuples. This gives an unamplified unitary for each positive eigenvalue. Their Hilbert direct sum $\widetilde{v}$ intertwines h_0 with the normalized far density, and intertwines all transported labels simultaneously. The densities are faithful, so these eigenspaces exhaust both spaces.

All centralizer contractions a_α are polynomials in old labels and factor across both scalar algebras; write their reduced forms $\widetilde{a}_\alpha$. Define $\widetilde{A}_\alpha = \widetilde{v}^* \widetilde{a}_\alpha$. The reduced translated equations in Lemma 4.5 give every application identity. Intertwining of the normalized densities, together with the source/far weight 2^n, gives $h_0 \widetilde{A}_\alpha = 2^{-n} \widetilde{A}_\alpha h_0$. This proves the realization assertion. The new creations need not be restrictions of the original creations.

Completeness uses (27), followed by the two normalized maximally mixed factors, so has exactly the same spectrum. Taking the square root and the trace of this finite tensor product proves (98); nonnegative spectral sums justify it even when the answer is infinity. □

16.2 The tree at a general ratio

The tree part of Section 5 works for every $0 < \nu < 1/4$. For clarity we give the density calculation. For a word w in four letters, write A_w for the product of its creations and

$$Q_\ell = \sum_{|w|=\ell} A_w A_w^*.$$

The isometry relations and covariance $h_0 A_\alpha = \nu A_\alpha h_0$ imply

$$\operatorname{Tr}(h_0 Q_\ell) = (4\nu)^\ell, \qquad \operatorname{Tr}(h_0 P_\circ) = 1 - 4\nu.$$

The decreasing Q_ℓ have a strong limit of zero h_0-weight, hence zero by faithfulness. Their successive differences are $\sum_{|w|=\ell} A_w P_\circ A_w^*$. Thus, with $\mathcal{K} = P_\circ \mathcal{H} \neq 0$,

$$\mathcal{H} \simeq \ell^2(\mathcal{W}) \otimes \mathcal{K}, \quad h_0 = h_{F,\nu} \otimes \rho_{\mathrm{spec}}, \quad h_{F,\nu} = (1 - 4\nu) \sum_w \nu^{|w|} |w\rangle\langle w|, \tag{99}$$

where ρ_{spec} is a faithful density. The covariance makes every word-sector projection commute with h_0, so transporting the root block proves the operator equality. Depth ℓ has probability $(1 - 4\nu)(4\nu)^\ell$; $(4\nu)^\ell$ is the mass at depth at least ℓ. At $\nu = 1/16$ this is (32).

The coordinate, Weyl-clock, bounded period-reader and deterministic tableau equations of Sections 5–7 contain no Fock density ratio. Their proofs use only faithful normality, positive finite-dimensional density eigenspaces, the exact tree decomposition, and a positive weight at every finite depth. All hold in (99). Consequently their soundness and canonical completeness hold unchanged for every $\nu = 2^{-n} < 1/4$. In particular the Weyl determinant still forces a root phase of finite order: on a positive density eigenspace of dimension d on which $Z_{\mathrm{clk}} = \zeta$, the relation $J_{\mathrm{W}} K_{\mathrm{W}} = \zeta K_{\mathrm{W}} J_{\mathrm{W}}$ gives $\zeta^d = 1$.

16.3 Summability and its critical point

Equation (99) gives

$$\operatorname{Tr} \sqrt{h_{F,\nu}} = \sqrt{1-4\nu} \sum_{\ell=0}^{\infty} (4\sqrt{\nu})^{\ell} = \frac{\sqrt{1-4\nu}}{1-4\sqrt{\nu}} \tag{100}$$

when $\nu < 1/16$, and infinity when $1/16 \le \nu < 1/4$. For the canonical YES realization $\rho_{\mathrm{spec}} = I_{d_{\mathrm{clk}}}/d_{\mathrm{clk}}$ with clock order d_{clk}, Lemma 16.1 therefore gives

$$\operatorname{Tr} D = \sqrt{3 d_{\mathrm{clk}} d_{\max}}\, \frac{1 + 4\sum_{g=1}^{n} 2^{g/2}}{\sqrt{\mathfrak{c}_n}}\, \frac{\sqrt{1-2^{2-n}}}{1-2^{2-n/2}} \quad (n > 4). \tag{101}$$

For five bridges and the optional eight-bridge construction, the exact coefficients are

$$n = 5: \quad \operatorname{Tr} D = \frac{(81+53\sqrt{2})\sqrt{7}}{2\sqrt{83}} \sqrt{d_{\mathrm{clk}} d_{\max}}, \tag{102}$$

$$n = 8: \quad \operatorname{Tr} D = \frac{(121+60\sqrt{2})\sqrt{21}}{2\sqrt{2041}} \sqrt{d_{\mathrm{clk}} d_{\max}}. \tag{103}$$

They are approximately $22.65\sqrt{d_{\mathrm{clk}} d_{\max}}$ and $10.44\sqrt{d_{\mathrm{clk}} d_{\max}}$.

Remark (Four bridges are critical). At $n = 4$, every realization has $\operatorname{Tr} \sqrt{h_0} = \infty$, regardless of ρ_{spec}, because the per-depth amplitude sum in (100) is constant and nonzero. Lemma 16.1 then makes every perfect pure game strategy have infinite amplitude sum. Theorem 15.3 implies that the channel of every game G_e^{clk} of Theorem 7.1 has no k-message normal code, for every input index. Applying those games directly would therefore give an empty YES set, rather than the desired hardness reduction.

16.4 Upper membership

Convert a channel instance (N, m) to a finite game $\mathcal{G}_{\mathrm{code}}(N, m)$. Alice receives $i \in [m]$ and answers a channel input x; Bob receives a channel output y and answers $j \in [m]$. All question pairs have positive rational probability. A tuple loses exactly when $N(y \mid x) > 0$ and $j \ne i$. With the same resource and measurements,

$$\mathrm{ZE}_t(N, m) \iff \mathcal{G}_{\mathrm{code}}(N, m) \text{ has a perfect } t\text{-strategy}. \tag{104}$$

Indeed the error probability of the code is the finite sum of these nonnegative expectations weighted by $N(y \mid x)$. It vanishes exactly when every forbidden expectation vanishes. The equivalence preserves marginal entropy as well.

The compact-stratum predicate of Section 3 gives $\mathrm{PERF}_{\mathrm{qs}} = \bigcup_{\kappa} \mathrm{PERF}_{\mathrm{qs}}(\kappa)$ with each stratum Π_1^0, and mixed strategies reduce to pure perfect components, as in Lemma 15.1. Equation (104) therefore proves $\mathrm{ZE}_{\mathrm{qs}} \in \Sigma_2^0$. The entropy-budget strata of Section 3.4 prove $\mathrm{ZE}_{\mathrm{finS}} \in \Sigma_2^0$ in the same way. If a mixed perfect resource has finite entropy on one marginal, entropy concavity gives a positive-weight pure perfect component of finite entropy, and for a pure bipartite component the two marginal entropies agree. Hence using one or both finite marginal entropies leads to the same feasibility set.

16.5 Five bridges for normal hardness

Let $\mathcal{A}_{\mathrm{dec}} = \{e : \exists j\ \forall t\ \mathcal{R}_{\mathrm{dec}}(e, j, t)\}$ for any decidable predicate $\mathcal{R}_{\mathrm{dec}}$. Use the clock presentation of Sections 5–7, with its computation program hardwiring e, and change only its bridge count to five. Denote the resulting game by $G_e^{(5)}$. Its clock and computation mechanism is unchanged. At depth r the four-letter tree carries a 2^r-by-2^r coordinate square. A root Weyl pair commuting with the root density has a nonzero finite-order phase sector; its determinant on a positive density eigenspace forces this. On a phase of order d_{clk} the reader equations (45) force P_{ret} to be exactly the kernel projection of $W - I$, so the first positive return is precisely time d_{clk}. The tableau freezes a waiting front at that time, halts if d_{clk} is one or composite, and otherwise checks $\mathcal{R}_{\mathrm{dec}}(e, j, 0), \mathcal{R}_{\mathrm{dec}}(e, j, 1), \dots$ for $d_{\mathrm{clk}} = p_{j+1}$, halting at the first failure. A halt is forbidden by an operator equation, and every depth has positive normal weight.

If the game is perfect, Lemma 4.5 extracts this mechanism and therefore a witness j. Conversely a witness j uses the canonical tree, a d_{clk}-dimensional Weyl pair for $d_{\mathrm{clk}} = p_{j+1}$, the good primitive phase and bounded reader of Section 6, and its nonhalting classical tableau. The reader norm estimate depends only on the phase and the counter, so it holds for every ν. Its density is $h_{F,1/32} \otimes I_{d_{\mathrm{clk}}}/d_{\mathrm{clk}}$. This is the simultaneous canonical realization, so Lemma 4.5 supplies a perfect game strategy. Formula (102) makes its Schmidt amplitudes summable.

Now put

$$N_e = N_{\Gamma(G_e^{(5)})}, \qquad k_e = |\mathcal{Q}_{A,e}| + |\mathcal{Q}_{B,e}|,$$

where $\mathcal{Q}_{A,e}, \mathcal{Q}_{B,e}$ are the question sets of $G_e^{(5)}$. These data are finite and rational and are computed from e without knowing its witness. Theorem 15.3 and the preceding two implications give

$$\mathrm{ZE}_{\mathrm{qs}}(N_e, k_e) \iff e \in \mathcal{A}_{\mathrm{dec}}.$$

This proves Σ_2^0-hardness and the first part of Theorem C. No channel in this reduction has a finite-dimensional saturating code: otherwise the finite converse would extract four orthogonal isometries on a nonzero finite-dimensional root space, an impossibility. We make no claim here that a value-one promise is inherited from the source construction.

16.6 Finite entropy already at five bridges

Use exactly the same five-bridge reduction. The forward code marginal is the normalized square root of the canonical game marginal. Its tree factor is

$$\widehat{h}_5 = \frac{\sqrt{h_{F,1/32}}}{\mathrm{Tr}\,\sqrt{h_{F,1/32}}} = (1 - 2^{-1/2}) \sum_w 2^{-5|w|/2} |w\rangle\langle w|.$$

At depth ℓ there are 4^ℓ words, so the depth distribution is geometric with ratio $2^{-1/2}$. Its mean is $2^{-1/2}/(1 - 2^{-1/2}) = 1 + \sqrt{2}$. Each eigenvalue at depth ℓ has negative base-two logarithm $-\log_2(1 - 2^{-1/2}) + (5/2)\ell$. Taking its expectation proves

$$S(\widehat{h}_5) = -\log_2(1 - 2^{-1/2}) + \frac{5}{2}(1 + \sqrt{2}) \approx 7.807 \text{ bits}. \tag{105}$$

The full canonical code marginal is

$$\mathrm{diag}(\hat{\mu}_5) \otimes \widehat{h}_5 \otimes I_{d_{\mathrm{clk}}}/d_{\mathrm{clk}} \otimes I_3/3 \otimes I_{d_{\max}}/d_{\max}, \qquad \hat{\mu}_{5,i} = \frac{\sqrt{\mu_{5,i}}}{\sum_{i'} \sqrt{\mu_{5,i'}}}.$$

Thus its entropy is $H(\hat{\mu}_5) + S(\widehat{h}_5) + \log_2(3 d_{\mathrm{clk}} d_{\max})$, which is finite. The pure code has the same entropy on both sides. A NO input has no normal code at all, hence no finite-entropy code. The same map therefore proves the finite-entropy half of the first part of Theorem C.

Remark (Eight bridges). With $n = 8$, the normalized square root of $h_{F,1/256}$ is exactly $h_{F,1/16}$, with entropy $\log_2(4/3) + 4/3$. Its amplitude sum is (103). This is another finite-entropy construction; five bridges already suffice for hardness.

16.7 Finite-dimensional zero-error communication

For each fixed dimension, the channel code game in (104) has a rational semialgebraic perfect-strategy predicate: split matrix entries into real and imaginary parts, impose positivity, POVM sums, normalization and zero losing probabilities. Quantifier elimination decides it. Enumerating dimensions gives $\mathrm{ZE_q} \in \Sigma_1^0$.

For hardness, the perfect-completeness reduction of $\mathsf{MIP}^* = \mathsf{RE}$ gives, from a halting instance, a finite game with a perfect finite-dimensional strategy, and from a nonhalting instance a game of entangled value at most $1/2$ [34, Theorem 12.6]. Mančinska's Corollary 7 maps perfect finite-dimensional game feasibility to the saturating one-shot channel threshold [39]. Thus $\mathrm{ZE_q}$ is Σ_1^0-complete. This conclusion needs only her finite theorem. Bhattacharyya, Mehta and Zhao leave undecidability of $C_{0,E}^{(1)}$ open in [5, Section 1.4]. This corollary settles its one-shot threshold problem: it is undecidable and Σ_1^0-complete already for classical channels. Here E denotes unrestricted finite-dimensional entanglement. The unrestricted asymptotic capacity question remains open.

17 The commuting row

With commuting resources one-shot zero-error feasibility drops to the commuting level. The code game (104) identifies $\mathrm{ZE_{qc}}$ with perfect commuting play for a particular finite game. Commuting correlations form a compact set, and the NPA hierarchy converges from above to their value [45]. Its rational moment relaxations can be decided by real quantifier elimination. A NO instance eventually has an infeasible perfect-value relaxation; conversely a feasible relaxation at every order gives a commuting strategy by compactness of the full moment system. Hence $\mathrm{ZE_{qc}} \in \Pi_1^0$.

We prove the reduction for the lower bound, including codes with arbitrary commuting resources. Slofstra proved Π_1^0-hardness for linear system games [57]; with the NPA bound this is Π_1^0-completeness, which Mousavi, Nezhadi and Yuen reprove for synchronous games [44, Theorem 6.4]. The source is perfect commuting feasibility for finite synchronous games with positive diagonal question probabilities; see [44, Theorem 6.4 and Section 1.3]. Their synchronous framework and tracial representation are stated in [44, Theorem 1.6]. Such a game has the same question and answer sets on both sides, and distinct answers on an equal-question pair are losing.

17.1 Normal functionals from a commuting code

Represent a code on a Hilbert space with vector ξ, encoding effects $E_{i,x}$ in an algebra commuting with Bob's von Neumann algebra $\mathcal{N}$, and decoder effects in $\mathcal{N}$. Define normal positive functionals

$$\phi(T) = \langle \xi, T\xi \rangle, \qquad \phi_{i,x}(T) = \langle \xi, E_{i,x} T \xi \rangle \quad (T \in \mathcal{N}).$$

The two operators commute, so these functionals are positive and $\sum_x \phi_{i,x} = \phi$. Compress $\mathcal{N}$ to the support of ϕ. This makes ϕ faithful. It preserves all the functionals, their sums, and decoder tests, since each $\phi_{i,x} \le \phi$ is supported on the same support. The compressed encoders still commute with the compressed Bob algebra.

When $i \neq j$ and $x \sim x'$, the functionals $\phi_{i,x}, \phi_{j,x'}$ have orthogonal supports in $\mathcal{N}$. To see this, for any common output y their positively weighted functionals are summands of the two distinct-message output functionals. Perfect decoding makes the support of one lie in the eigenvalue-one space of a decoder effect, and that of the other in its kernel. These spectral spaces are orthogonal. Support inclusion for positive summands proves the assertion.

Lemma 17.1 (Functional saturation)**.** *Suppose the input graph is partitioned into k cliques $\mathcal{C}_c$ and a commuting code sends k messages. For*

$$\psi_{i,c} = \sum_{x \in \mathcal{C}_c} \phi_{i,x}, \qquad s_{i,c} = \operatorname{supp} \psi_{i,c},$$

the $s_{i,c}$ lie in the centralizer $\mathcal{N}_\phi = \{s : \phi(sT) = \phi(Ts) \text{ for all } T \in \mathcal{N}\}$ and satisfy

$$\psi_{i,c}(T) = \phi(s_{i,c}T) = \phi(Ts_{i,c}).$$

Their rows and columns are orthogonal projection partitions.

Proof. For each c the supports are orthogonal in i. Domination by ϕ gives

$$\sum_i \psi_{i,c}(1) \leq \sum_i \phi(s_{i,c}) \leq 1.$$

The sum over c is k, so every inequality is an equality. Faithfulness gives $\sum_i s_{i,c} = 1$. For each i, c, $\eta = \phi - \psi_{i,c}$ is positive and $\eta(s_{i,c}) = 0$. Cauchy–Schwarz for a positive functional gives $\eta(s_{i,c}T) = \eta(Ts_{i,c}) = 0$. The support property of $\psi_{i,c}$ gives $\psi_{i,c}(s_{i,c}T) = \psi_{i,c}(Ts_{i,c}) = \psi_{i,c}(T)$. Combining them proves the formula and centralizer membership.

Summing over c for fixed i gives $\phi((\sum_c s_{i,c} - 1)T) = 0$ for every T. Take $T = (\sum_c s_{i,c} - 1)^*$ and use faithfulness to obtain the row sum. A finite sum of projections equal to identity is orthogonal, as in Lemma 15.2. □

17.2 Synchrony makes the fine supports tracial

Apply the lemma to $\Gamma(G)$, with $k = 2|\mathcal{Q}_A|$, for a synchronous game G. Aggregate the fine functionals

$$\psi_{c,a} = \sum_i \phi_{i,(c,a)}.$$

They sum to ϕ for each tagged question c. Across a cross losing edge their supports are orthogonal: unequal message indices use the code condition, and equal message indices use distinct clique supports in one row of $(s_{i,c})$. Orthogonality passes to finite sums.

Fix an original question u. Write $\alpha_a = \psi_{(L,u),a}$, $\beta_a = \psi_{(R,u),a}$, and $p_a = \operatorname{supp} \alpha_a$. For $a \neq b$, synchrony gives $p_a \operatorname{supp} \beta_b = 0$. Hence

$$\alpha_a(1) \leq \phi(p_a) = \beta_a(p_a) \leq \beta_a(1).$$

Both outside families sum to one, so these inequalities are equalities for every a. The same Cauchy–Schwarz argument used in Lemma 17.1, now for $\phi - \alpha_a$ and $\phi - \beta_a$, gives

$$\alpha_a(T) = \beta_a(T) = \phi(p_a T) = \phi(Tp_a).$$

Faithfulness implies that their supports are the same p_a, and that $\sum_a p_a = 1$. Thus the fine outcomes are projections in $\mathcal{N}_\phi$.

The restriction $\tau = \phi|_{\mathcal{N}_\phi}$ is a faithful normal tracial state. For any original losing tuple (u, a, v, b) the preceding support orthogonality gives $p_{u,a}p_{v,b} = 0$. The left/right multiplication construction on $L^2(\mathcal{N}_\phi, \tau)$ therefore gives a perfect commuting strategy for G. This argument

starts from an arbitrary saturating code; it assumes neither projective encoders nor a tracial original resource.

Conversely a perfect commuting strategy for synchronous G has its usual tracial projection representation [50, Theorem 5.5]; see also [35, 44]. For each tagged question use its projection partition to encode its designated message in the left algebra of the tracial GNS space; use the right algebra to decode. The conditional functionals are $T \mapsto \tau(p_{c,a}T)$. For distinct designated messages and confusable vertices, their supports are orthogonal by the losing relation. Each output's distinct-message supports therefore give a decoding measurement. This is a commuting k-message code for $N_{\Gamma(G)}$.

Theorem 17.2. *The one-shot feasibility problem* $\mathrm{ZE}_{\mathrm{qc}}$ *for finite rational classical channels is* Π_1^0*-complete.*

Proof. For each finite synchronous game G with positive diagonal question probabilities, the two constructions give the effective equivalence

$$G \text{ is commuting-perfect} \iff \mathrm{ZE}_{\mathrm{qc}}(N_{\Gamma(G)}, 2|\mathcal{Q}_A|).$$

The source problem is Π_1^0-complete, so this proves hardness. The NPA upper predicate gives membership. □

18 A channel whose codes require infinite entropy

This section proves the second part of Theorem C. The channel comes from a four-bridge version of the Catalan assembly of [24]. Its game has finite-entropy perfect strategies, but the normalized square-root map of Theorem 15.3 turns every one of them into a code with infinite entropy.

18.1 The squared Catalan density

Let $\mathcal{W}_{\mathrm{Cat}}$ consist of the empty word and the legal chronological words in an up-step A and a down-step B. An up-step is always allowed; a down-step is allowed only at height at least two. Let $u(w)$ count up-steps. The number of words with j up-steps is the Catalan number Cat_j. On $\ell^2(\mathcal{W}_{\mathrm{Cat}})$ let A, B append the corresponding legal letter, with B zero when forbidden. Let C_{Cat} have eigenvalue $2^{-\mathrm{ht}(w)}$, let P project onto the empty word, and let Q project onto height one.

We require the local relations of [24, Section 3]:

$$A^*A = I, \qquad AA^* + BB^* + P = I, \qquad B^*B = I - P - Q, \tag{106}$$

$$C_{\mathrm{Cat}}P = P, \qquad C_{\mathrm{Cat}}Q = Q/2, \qquad 0 \le C_{\mathrm{Cat}} \le I, \tag{107}$$

$$C_{\mathrm{Cat}}A = AC_{\mathrm{Cat}}/2, \qquad C_{\mathrm{Cat}}B = 2BC_{\mathrm{Cat}}. \tag{108}$$

Here the modular ratio is squared:

$$h_0A = Ah_0/16, \qquad [h_0, B] = [h_0, C_{\mathrm{Cat}}] = [h_0, P] = [h_0, Q] = 0. \tag{109}$$

Here h_0 is a faithful normal density. In this section A, B, P, Q denote the Catalan labels just defined. The counter relation $C_{\mathrm{Cat}}A = AC_{\mathrm{Cat}}/2$ is unchanged.

Lemma 18.1 (Normal classification at ratio 1/16)**.** *Every normal realization of* (106)–(109) *is, up to a unitary,*

$$\mathcal{H} = \ell^2(\mathcal{W}_{\mathrm{Cat}}) \otimes \mathcal{K}_{\mathrm{Cat}}, \qquad h_0 = k_{\mathrm{Cat}} \otimes \rho_{\mathrm{spec}}, \qquad k_{\mathrm{Cat}} = \frac{1}{8 - 4\sqrt{3}} \sum_w 16^{-u(w)} |w\rangle\langle w|,$$

where ρ_{spec} is a faithful normal density. All five local operators act canonically on the first factor and as identity on the second.

Proof. We give the normality argument to check the ratio change in [24, Lemma 3.1]. The boundary equations put P, Q in the counter eigenspaces one and one half and give $PQ = 0$. The range projections AA^*, BB^*, P are orthogonal and sum to identity; B is a partial isometry with initial projection $I - P - Q$. All these projections commute with C_{Cat}.

There is no spectrum of C_{Cat} in $(1/2, 1)$, since there B is isometric and $C_{\mathrm{Cat}}B = 2BC_{\mathrm{Cat}}$ would transport it above one. The same reasoning at eigenvalue one gives $1_{\{1\}}(C_{\mathrm{Cat}}) = P$. At eigenvalue one half, B would transport the complement of Q into P, orthogonally to its own range, so $1_{\{1/2\}}(C_{\mathrm{Cat}}) = Q$. Iterating B excludes every interval $(2^{-j-1}, 2^{-j})$. Functional calculus for the bounded intertwining relation justifies these spectral transports.

The zero-counter space reduces the operators and h_0. On it $P = Q = 0$, so A, B are isometries with orthogonal ranges summing to identity. If its h_0-mass is t, cyclicity and (109) make the two range masses $t/16$ and t. Their sum is t, forcing $t = 0$. Faithfulness removes this space.

Apply legal words to $P\mathcal{H}$. Their sectors are orthogonal: different last letters have orthogonal ranges; equal last letters cancel with $A^*A = I$ or with $B^*B = I - P - Q$ on a legal predecessor; and a nonempty sector is orthogonal to P. Denote the word isometry by V_w^{Cat}. Covariance gives

$$h_0 V_w^{\mathrm{Cat}} = 16^{-u(w)} V_w^{\mathrm{Cat}} h_0.$$

The word-sector projections commute with h_0, and their closed span reduces all the operators. If its orthogonal complement were nonzero, the compact faithful density there would have a largest positive eigenvalue, with nonzero finite-dimensional eigenspace L. The adjoint covariance gives $A^*L = 0$. Since $P = 0$ on this complement, $BB^* = I$ on L. Commutation with h_0 makes L reduce B, so $B|_L$ is unitary. It also reduces C_{Cat}. Tracing $C_{\mathrm{Cat}}B = 2BC_{\mathrm{Cat}}$ on L forces $\operatorname{Tr}_L C_{\mathrm{Cat}} = 0$, hence $C_{\mathrm{Cat}}|_L = 0$, contrary to the absence of zero-counter mass. Thus the word sectors exhaust the support.

Their density blocks are $16^{-u(w)} V_w^{\mathrm{Cat}} (P h_0 P) (V_w^{\mathrm{Cat}})^*$. The word count is Cat_j: for $j \geq 1$, words with j up-steps and b down-steps, $0 \leq b \leq j - 1$, number

$$\binom{j+b-1}{b} - \binom{j+b-1}{b-1};$$

reflection of the first bad prefix gives this formula, and summing over b gives Cat_j. The empty word supplies $\mathrm{Cat}_0 = 1$. The generating function

$$F(z) = \sum_{j \geq 0} \mathrm{Cat}_j \, z^j = \frac{1 - \sqrt{1 - 4z}}{2z}$$

gives $F(1/16) = 8 - 4\sqrt{3}$. Normalization then gives the asserted tensor product. The same sector identifications fix the local operators. □

Define the critical Catalan density of [24] by

$$h_{\mathrm{Cat}} = \frac{1}{2} \sum_w 4^{-u(w)} |w\rangle\langle w|.$$

The generating function yields the exact identities

$$\operatorname{Tr} h_{\mathrm{Cat}}^2 = 2 - \sqrt{3}, \qquad k_{\mathrm{Cat}} = \frac{h_{\mathrm{Cat}}^2}{2 - \sqrt{3}}, \qquad \operatorname{Tr} \sqrt{k_{\mathrm{Cat}}} = \frac{\sqrt{6} + \sqrt{2}}{2}, \qquad \frac{\sqrt{k_{\mathrm{Cat}}}}{\operatorname{Tr} \sqrt{k_{\mathrm{Cat}}}} = h_{\mathrm{Cat}}. \tag{110}$$

Moreover

$$S(h_{\mathrm{Cat}}) = \sum_{j\geq 0} \frac{\mathrm{Cat}_j}{2\,4^j}(1+2j) = \infty, \tag{111}$$

since $\mathrm{Cat}_j\, 4^{-j} \sim (\sqrt{\pi} j^{3/2})^{-1}$. By contrast the squared Catalan factor has

$$S(k_{\mathrm{Cat}}) = \log_2(8-4\sqrt{3}) + \frac{4}{\sqrt{3}} - 2 \approx 0.4094 \text{ bits}.$$

Indeed, with F as above and $z = 1/16$, its mean up-step count is $zF'(z)/F(z) = 1/\sqrt{3} - 1/2$; averaging $\log_2 F(z) + 4u(w)$ gives the formula. Its multiplicity-weighted probabilities decay geometrically. The canonical perfect strategy of the game $G^{(4)}_{\mathrm{Cat}}$ below therefore has finite entropy: $S(k_{\mathrm{Cat}})$ is its infinite Catalan factor's contribution, and all its other canonical factors are finite dimensional. The channel of $G^{(4)}_{\mathrm{Cat}}$ instead forces the factor h_{Cat} and hence infinite entropy. The communication task itself creates the entropy demand through the square-root map.

18.2 A four-bridge Catalan assembly

The Catalan relations are not the four-creation syntax of Definition 4.4. We therefore specify their own assembly, extending [24, Section 6] from two bridges to four. Use the seventeen corners, twenty-five frame-bound parameters, cover $\mathfrak{c}_4 = 121$, axis equations, support-copy equations and block-unitary transport equations of Lemma 4.5. The transported unitary encodings are those of $B, C_{\mathrm{Cat}}, P, Q$: two unitaries b_+, b_- for B, one for the positive contraction C_{Cat}, and the reflections for P, Q. Thus $|\Lambda| = 5$. For each bridge use the six frame refinements and these five graph-label refinements, on both parties, retaining the original opposite FMS questions.

For $r = 0, \dots, 4$, denote the decoded corner copies by b_r, c_r, p_r, q_r. Impose the root reflection equations for p_0, q_0. Introduce one centralizer contraction $a = (a_+ + a_-)/2$. The remaining equations are

$$a = e_4 a e_0, \qquad a^*a = e_0, \qquad aa^* + b_4 b_4^* + p_4 = e_4, \tag{112}$$

$$b_0^* b_0 = e_0 - p_0 - q_0, \qquad p_0 q_0 = 0, \qquad c_0 p_0 = p_0, \tag{113}$$

$$c_0 q_0 = q_0/2, \qquad c_4 a = a c_0/2, \qquad c_0 b_0 = 2 b_0 c_0. \tag{114}$$

The range equation is imposed at the far corner because aa^* lies under e_4; conjugation by v below moves it to the source.

Let $v = \mathsf{U}_4\mathsf{U}_3\mathsf{U}_2\mathsf{U}_1$ be the bridge chain. Extraction on the source gives

$$A = v^*a, \qquad B = b_0, \qquad C_{\mathrm{Cat}} = c_0, \qquad P = p_0, \qquad Q = q_0.$$

Conjugate the far range equation by v, using the actual transported copies. This gives $AA^* + BB^* + P = e_0$. The other local equations follow directly; for example

$$C_{\mathrm{Cat}} A = c_0 v^* a = v^* c_4 a = v^* a c_0/2 = A C_{\mathrm{Cat}}/2.$$

Since a is a centralizer contraction and $hv = 16vh$, its source density has $h_0 A = A h_0/16$. The other decoded labels commute with it. After removal of the finite scalar factors, use exactly the finite-amplification cancellation proof of Lemma 16.1, now for these five transported unitary labels. It gives a realization of (106)–(109) and the same spectrum identity with μ_4.

Conversely use the canonical Catalan operators and density k_{Cat} in $\mathrm{diag}(\mu_4) \otimes k_{\mathrm{Cat}}$, put $a = |e_4\rangle\langle e_0| \otimes A$, and use identity transports. The ratio makes a commute with the scaffold density. The blockwise polar encodings, actual graphs, label refinements and finite Boolean-row extension used in Lemma 4.5 realize all the displayed equations simultaneously. Thus they compute a finite rational game $G^{(4)}_{\mathrm{Cat}}$ with a perfect normal strategy.

18.3 Every normal code has infinite entropy

Set $N = N_{\Gamma(G^{(4)}_{\mathrm{Cat}})}$ and let m be the number of tagged local questions of $G^{(4)}_{\mathrm{Cat}}$. The canonical strategy just constructed has finite amplitude sum, by (110) and its finite scaffold and scalar factors. Theorem 15.3 therefore gives a normal m-message code.

Consider any normal code, including a mixed resource, and let ρ_B be its Bob marginal. The assemblage converse and Theorem 15.3 construct a pure perfect game strategy with

$$h_{\mathrm{game}} = \frac{\rho_B^2}{\operatorname{Tr}\rho_B^2}, \qquad \rho_B = \frac{\sqrt{h_{\mathrm{game}}}}{\operatorname{Tr}\sqrt{h_{\mathrm{game}}}}.$$

The assembly spectrum identity and Lemma 18.1 give

$$h_{\mathrm{game}} \simeq \operatorname{diag}(\mu_4) \otimes k_{\mathrm{Cat}} \otimes \rho_{\mathrm{spec}} \otimes I_3/3 \otimes I_{d_{\max}}/d_{\max}.$$

Here $\simeq$ means unitary equivalence of positive operators. Trace summability forces $\operatorname{Tr}\sqrt{\rho_{\mathrm{spec}}} < \infty$. Taking the normalized square root gives

$$\begin{aligned} \rho_B &\simeq \operatorname{diag}(\hat{\mu}_4) \otimes h_{\mathrm{Cat}} \otimes \rho_{\mathrm{spec,code}} \otimes I_3/3 \otimes I_{d_{\max}}/d_{\max}, \\ \hat{\mu}_{4,i} &= \frac{\sqrt{\mu_{4,i}}}{\sum_j \sqrt{\mu_{4,j}}}, \quad \rho_{\mathrm{spec,code}} = \frac{\sqrt{\rho_{\mathrm{spec}}}}{\operatorname{Tr}\sqrt{\rho_{\mathrm{spec}}}}. \end{aligned} \tag{115}$$

The Catalan factor is exactly h_{Cat}; the full marginal also includes the finite factors and the possible spectator $\rho_{\mathrm{spec,code}}$. Equation (111) gives $S(\rho_B) = \infty$ for every code.

For a pure resource its two marginal spectra coincide, so Alice's entropy is infinite as well. For a mixed resource $\omega = \sum_j t_j |\psi_j\rangle\langle\psi_j|$, each positive-weight component is a perfect code under the same encoding and decoding measurements. This follows by applying positivity to each forbidden error expectation. Each such pure component has infinite entropy on both marginals. Extended entropy concavity gives

$$S(\omega_A) \geq t_j S((\psi_j)_A) = \infty,$$

and similarly for Bob. This proves the second part of Theorem C for all normal codes.

19 Discussion and open problems

The hierarchy separates several ways to ask for certainty. Finite-dimensional perfection has a finite existential witness. Commuting perfection is controlled by compactness. Value one asks for arbitrarily accurate finite witnesses. Spatial and semifinite perfection combine an unbounded finite choice with an infinite verification. In the spatial reduction the chosen object is the order of a clock phase; in the semifinite reduction it is the depth of a critical tree. Without a trace, as in every type-III factor, the choice disappears and the commuting level returns.

19.1 Entropy budgets

The entropy restriction does not lower the complexity. Each YES instance G^{clk}_e of the spatial reduction has a finite-entropy witness. Corollary 3.6 shows that the least sufficient integer entropy budget, maximized over games of bit length at most n, is a function of Turing degree $0''$ that eventually dominates every total $0'$-computable function. Corollary 3.7 proves the same profile for the least sufficient Schmidt stratum. Finite-entropy compactness also controls entire prescribed tables by Corollary 3.4. The finite-entropy and arbitrary-spatial decision sets have the same completeness degree, although the sets themselves differ: the entropy-game construction in [24] has spatial perfection and no finite-entropy perfect strategy, and Corollary 14.2 classifies their difference.

19.2 Prescribed tables and escaping mass

The largest-Schmidt compactness argument of Theorem 3.1 does not directly control a prescribed correlation table without a uniform tail bound. For a concrete obstruction, put

$$D_{\mathrm{esc},n} = \mathrm{diag}(1, \underbrace{n^{-1/2}, \ldots, n^{-1/2}}_{n \text{ entries}}, 0, \ldots).$$

Then $\operatorname{Tr} D_{\mathrm{esc},n}^2 = 2$ and $D_{\mathrm{esc},n} \to |0\rangle\langle 0|$ in operator norm. Alice measures $|0\rangle\langle 0|$ and its complement; Bob has one trivial outcome. Every normalized vectorization of $D_{\mathrm{esc},n}$ gives Alice probabilities $(1/2, 1/2)$. The normalized limit gives $(1, 0)$. The largest-amplitude normalization preserves zero equations while changing nonzero normalized probabilities.

For the finite-entropy realization class, Corollary 3.4 proves the Σ_2^0 upper bound for exact rational or real-algebraic tables.

19.3 Uniform membership and algebraic correlations

Two questions remain about prescribed spatial correlations. Is membership of real-algebraic tables in C_{qs} recursively enumerable uniformly over finite scenarios? Must every spatially perfect game have an algebraic perfect table? The two answers cannot both be yes: enumerating algebraic tables, checking their losing zeros, and dovetailing their membership searches would enumerate the YES instances of our Σ_2^0-complete family. The fixed-scenario remark of Fu, Miller and Slofstra [27] does not by itself settle the required uniformity.

19.4 Zero-error capacities

Theorem C concerns one use of a channel. A tensor power of the channel of a game has the strong-product confusability graph, which is not the confusability graph of the AND-repeated game, so one-shot hardness does not transfer to the entanglement-assisted zero-error capacity. Whether this capacity is computable, and whether normal entanglement can give a strictly larger capacity than finite-dimensional entanglement, remain open. For the squared-Catalan channel of Section 18, normal entanglement already reaches the clique-partition bound of Lemma 15.2 in one use; whether finite-dimensional entanglement reaches the same capacity asymptotically is unknown.

19.5 Other questions

The type languages are existence languages: the host and its normal state may be chosen together. They do not decide whether a prescribed factor and state realize a correlation, or determine the least resources of a strategy. Natural questions ask which syntactic classes force finite-entropy spatial perfection whenever spatial perfection is possible, and how quantitative entropy requirements interact with the difference classification.

Can the game compiler be made substantially smaller? Can the period reader be realized by a simpler predicate? What are the optimal effective bounds for prescribed tables when one supplies an entropy budget or a computable bound on the Schmidt tail? A classification of unrestricted prescribed-table membership would require additional control of the normalization that escapes in the example above.

How this work was done

Opus 5.5, GPT-6.1 Sol and GPT-6 assisted with derivations, checking, programming and writing. The authors selected the questions, directed the work and take responsibility for the

paper. The semifinite-but-not-spatial completion, the moment ladder and the nested compiler were derived with models from one developer and checked by a model from the other; these checks were automated.

Exact and numerical checks of the finite identities, counts and examples used in the proofs are in the repository,

https://github.com/Apsiape/perfect-play-arithmetical-hierarchy,

and Appendix C describes their scope. Appendix F describes the formal verification of Theorem A, together with the selected formalized lemmas for Theorems B–D.

A The nested assembly

A.1 Input, output, and the scope of the calibration

An input is a finite real rational $*$-polynomial presentation in typed centralizer labels and contractions $X_1, \dots, X_s$. A label has type projection, unitary, positive contraction, or contraction. The modular symbols have the prescribed common ratio

$$h_0 X_i = \tfrac{1}{2} X_i h_0. \tag{116}$$

Each monomial contains at most two modular symbols, counting adjoints; centralizer coefficients can occur before, between, and after them. Centralizer polynomial equations and degree-one endpoint equations are included. A normal realization has a nonzero Hilbert space and a faithful trace-class density h_0 of trace one, with all identities on its support. The ratio in (116) is part of the input convention; it is not an operator to be synchronized. Section B treats the common ratio 2^{-n}. Arbitrary unrelated real ratios are outside this stated lemma: the calibration used here supplies powers of two.

The proof uses two calibrated levels so that an entire inner activity, including off-diagonal lifted contractions, is transported by one outer map. A centralizer corner unitary then identifies equal-density corners. Products of the resulting modular operators can be translated into actual centralizer products. The whole-activity refinement is what prevents independent gauges from changing those products.

Theorem A.1 (Nested compiler)**.** *Every such presentation effectively determines a finite rational Boolean game. A perfect normal spatial strategy extracts a normal realization on a source corner of mass* $1/81$*. Every normal realization has one simultaneous perfect extension using a* 25*-corner scaffold, a qutrit frame, and a finite row ancilla. The extension preserves finiteness of every positive moment and of entropy. With types, centralizer conditions, and modular covariance exact, constructed approximate realizations obey*

$$1 - \omega \le C_{\mathrm{nest}} \sum_j \|r_j\|_{2,h_0}^2, \qquad \|r\|_{2,h_0}^2 = \mathrm{Tr}(h_0 r^* r), \tag{117}$$

where C_{nest} *depends only on the finite compiled presentation.*

A.2 Corners, parameters, and activities

Begin with nine synchronized corner projections

$$f,\ m_{00}, m_{01}, m_{10}, m_{11},\ E_{00}, E_{01}, E_{10}, E_{11}.$$

Name a synchronized activity $E_{\mathrm{in}}^{\mathrm{act}}$ and impose

$$E_{\mathrm{in}}^{\mathrm{act}} = f + \sum_{ab} m_{ab}, \qquad E_{\mathrm{in}}^{\mathrm{act}} + \sum_{ab} E_{ab} = I. \tag{118}$$

The inner conditional FMS has flag f, activity $E_{\mathrm{in}}^{\mathrm{act}}$, and selected corner $e = m_{00}$. The outer conditional FMS has flag *the whole* $E_{\mathrm{in}}^{\mathrm{act}}$, activity I, and selected corner $E = E_{00}$. Retain every original question, both orientations of each event binding, and every inactive answer. The inner inactive event is $I - E_{\mathrm{in}}^{\mathrm{act}}$. The outer inactive event is zero. In particular its column partition has flag $E_{\mathrm{in}}^{\mathrm{act}}$, four E_{ab} answers, and a zero inactive answer. No new inner flag is substituted for $E_{\mathrm{in}}^{\mathrm{act}}$ in that partition.

The frame-bound parameter list is exactly the nine corners, $E_{\mathrm{in}}^{\mathrm{act}}$, and the four positive ZI, IX events, two for each bridge: fourteen parameters. Use the parameter-binding rows of Section 4.2. Set $M_{\mathrm{in}} = E_{\mathrm{in}}^{\mathrm{act}} - f$ and $M_{\mathrm{out}} = I - E_{\mathrm{in}}^{\mathrm{act}}$. The four unitary axis labels are equated to $I - 2M + 2E_{\mathrm{axis},+}$ for their respective bridge. Let $r_{ab} = x_{\mathrm{in}}^{a} z_{\mathrm{in}}^{b} e$ and $R_{ab} = x_{\mathrm{out}}^{a} z_{\mathrm{out}}^{b} E$ denote the actual magic corner maps. These are centralizer expressions in the bound parameters and axes. The two calibrated transports satisfy

$$u^*u = f,\ uu^* = e,\ hu = 2uh; \qquad \mathsf{U}^*\mathsf{U} = E_{\mathrm{in}}^{\mathrm{act}},\ \mathsf{U}\mathsf{U}^* = E,\ h\mathsf{U} = 2\mathsf{U}h. \tag{119}$$

The soundness argument below derives the source mass $\mathrm{Tr}(hf) = 1/81$ from these relations and the cover.

A.3 The complete transport list and the whole-activity block

Encode a projection P by $2P-I$, a positive contraction C by a unitary with $C = (2I+U+U^*)/4$, a general contraction by the average of two unitaries, and a unitary directly. Identities in a supported corner use that corner's identity. The outer transported list consists of:

1. the two inner projections f, e, regarded as projections in $E_{\mathrm{in}}^{\mathrm{act}}$;
2. all source label encodings that occur at any endpoint (transporting all source labels is an allowed deterministic choice);
3. both unitary encodings of every lifted contraction $a_i = ea_if$;
4. the finitely many derived coefficients specified below when a coefficient occurs inside a quadratic word.

For every encoding name k, introduce global unitaries U_k, V_k with

$$(U_k - I)(I - E_{\mathrm{in}}^{\mathrm{act}}) = 0, \qquad (V_k - I)(I - E) = 0.$$

A root-supported encoding additionally fixes $I - f$. Its transported copy will then fix $E - f'$. Introduce a global unitary V_k^{blk} and impose

$$V_k^{\mathrm{blk}} = E_{\mathrm{in}}^{\mathrm{act}} U_k + \sum_{ab} R_{ab}(EV_k)R_{ab}^*. \tag{120}$$

There is no $I - E_{\mathrm{in}}^{\mathrm{act}} - M_{\mathrm{out}}$ term since the outer activity is I. In (120), $E_{\mathrm{in}}^{\mathrm{act}} U_k$ acts on the entire five-corner space $E_{\mathrm{in}}^{\mathrm{act}}\mathcal{H}$, including off-diagonal blocks between f and e. Replacing it by a sum of five unrelated diagonal encodings would lose the required common transport. The identity $(U_k - I)(I - E_{\mathrm{in}}^{\mathrm{act}}) = 0$ gives $U_k(I - E_{\mathrm{in}}^{\mathrm{act}}) = I - E_{\mathrm{in}}^{\mathrm{act}}$. Left multiplication by U_k^* gives $U_k^*(I - E_{\mathrm{in}}^{\mathrm{act}}) = I - E_{\mathrm{in}}^{\mathrm{act}}$; taking adjoints gives $(I - E_{\mathrm{in}}^{\mathrm{act}})U_k = I - E_{\mathrm{in}}^{\mathrm{act}}$. Thus $[U_k, E_{\mathrm{in}}^{\mathrm{act}}] = 0$. The same proof gives $[V_k, E] = 0$. Consequently $E_{\mathrm{in}}^{\mathrm{act}} U_k$ is a unitary on $E_{\mathrm{in}}^{\mathrm{act}}\mathcal{H}$, and each $R_{ab}(EV_k)R_{ab}^*$ is unitary on its own magic corner. Their supports are orthogonal and cover I, so (120) is block diagonal for $E_{\mathrm{in}}^{\mathrm{act}}\mathcal{H} \oplus \bigoplus_{ab} E_{ab}\mathcal{H}$ and is unitary.

For f use $U_f = I - 2E_{\mathrm{in}}^{\mathrm{act}} + 2f$, and for e use $U_e = I - 2E_{\mathrm{in}}^{\mathrm{act}} + 2e$; their outer unitary names determine

$$f' = \tfrac{1}{2}E(I + V_f), \qquad e' = \tfrac{1}{2}E(I + V_e).$$

For each bridge separately refine its retained signs for the six projections

$$F_1, \quad F_2, \quad F_3, \quad G_{12}(1), \quad G_{13}(1), \quad G_{23}(1).$$

For the outer bridge also refine them separately for every $G_{12}(V_k^{\mathrm{blk}})$. The unchanged opposite questions are retained. There are six refinements at the inner bridge and $6+|\Lambda|$ at the outer bridge, per retained question and per party.

Label refinement (Lemma 4.2) makes every retained sign commute with the six frame projections. In the frame coordinates these generate $\mathrm{Mat}_3 \otimes I$: the F_i give the diagonal matrix units and $F_i G_{ij}(1) F_j = e_{ij} \otimes I/2$ gives the others. Their commutant is $I_3 \otimes B(\mathcal{H}_{\mathrm{base}})$. Because the corners are frame-bound, the sign and its (c_0, f) block are frame scalar. Thus $\mathsf{U} = I_3 \otimes \widetilde{\mathsf{U}}$ (and likewise u). Writing the outer sign as $I_3 \otimes \widetilde{\mathcal{X}}_0$, the $(1,2)$ frame block of its commutator with

$$G_{12}(V_k^{\mathrm{blk}}) = \tfrac{1}{2}(F_1 + F_2 + e_{12} \otimes V_k^{\mathrm{blk}} + e_{21} \otimes V_k^{\mathrm{blk}*})$$

is $[\widetilde{\mathcal{X}}_0, V_k^{\mathrm{blk}}]/2$. On the outer sign support, with flag space $E_{\mathrm{in}}^{\mathrm{act}}\mathcal{H}$ and selected magic space $E\mathcal{H}$, the bridge lemma gives

$$\widetilde{\mathcal{X}}_0 = \begin{pmatrix} 0 & \widetilde{\mathsf{U}}^* \\ \widetilde{\mathsf{U}} & 0 \end{pmatrix}, \qquad V_k^{\mathrm{blk}} = \begin{pmatrix} E_{\mathrm{in}}^{\mathrm{act}} U_k|_{E_{\mathrm{in}}^{\mathrm{act}}\mathcal{H}} & 0 \\ 0 & EV_k|_{E\mathcal{H}} \end{pmatrix}.$$

Here c_0 means the selected outer magic direction E and f means its flag, which is the *whole* $E_{\mathrm{in}}^{\mathrm{act}}$ rather than the inner f. The (c_0, f) block of the commutator is exactly $\widetilde{\mathsf{U}}(E_{\mathrm{in}}^{\mathrm{act}} U_k) - (EV_k)\widetilde{\mathsf{U}}$, so

$$\mathsf{U}(E_{\mathrm{in}}^{\mathrm{act}} U_k) = (EV_k)\mathsf{U}, \tag{121}$$

with the same U for every k, including the two encodings of a contraction. Here $E_{\mathrm{in}}^{\mathrm{act}} U_k$ may mix all five inner corners: if $E_{\mathrm{in}}^{\mathrm{act}} U_k = (U_{rs})$ in inner-corner coordinates, (121) says $EV_k = \mathsf{U}(U_{rs})\mathsf{U}^*$. Multiplying (121) by the unitary adjoints gives transport also for U_k^*, V_k^*. Decoding the same polynomials in these unitaries therefore transports projections, positive contractions and averages of two unitaries. Hence $f' = \mathsf{U} f \mathsf{U}^*$, $e' = \mathsf{U} e \mathsf{U}^*$, and, with $\mathsf{U}_0 = \mathsf{U} f$, $\mathsf{U}_1 = \mathsf{U} e$,

$$a_i' = \mathsf{U}_1 a_i \mathsf{U}_0^*, \tag{122}$$

including both encodings of each a_i rather than just its diagonal compressions. Both sides are actual operators; neither is an independently gauged copy.

Introduce a centralizer contraction t by two global unitaries and impose

$$t = f' t e, \qquad t^* t = e, \qquad t t^* = f'. \tag{123}$$

The corner maps t and a_i are synchronized centralizer expressions; $u, \mathsf{U}, \mathsf{U}_0, \mathsf{U}_1$ are extracted transports and are never graph words. Set $\mathcal{H}_0 = f\mathcal{H}$ and $h_0 = 81 f h f|_{\mathcal{H}_0}$. The extracted contractions are

$$X_i = \mathsf{U}_0^* t a_i. \tag{124}$$

The soundness argument below shows that h_0 is a faithful trace-one density and that (124) satisfies (116).

A.4 All quadratic translations and arbitrary coefficients

Put $\ell = \mathsf{U}_0^* t \mathsf{U}_1^*$. It is a corner unitary from $e'\mathcal{H}$ to $f\mathcal{H}$. Direct multiplication, using only (122)–(123), gives

$$X_i^* X_j = a_i^* a_j, \tag{125}$$

$$\mathsf{U}_0 X_i X_j^* \mathsf{U}_0^* = t a_i a_j^* t^*, \tag{126}$$

$$X_i X_j = \ell a_i' t a_j. \tag{127}$$

For the last identity use $a_i \mathsf{U}_0^* = \mathsf{U}_1^* a_i'$; the intermediate projections are e, f', e' and cancel by their stated source/range rules. The following table is consequently an equivalence in both directions.

Root identity	Centralizer identity
$X_i^* X_j = R_{\mathrm{tar}}$	$a_i^* a_j = R_{\mathrm{tar}}$ at f
$\sum_i X_i X_i^* = R_{\mathrm{tar}}$	$\sum_i t a_i a_i^* t^* = {R_{\mathrm{tar}}}'$ at f'
$X_i X_j = X_k X_l$	$a_i' t a_j = a_k' t a_l$
$X_i^* X_j = X_k X_l^*$	$a_i'^* a_j' = t a_k a_l^* t^*$
$L X_i = X_j R$	$L' t a_i = t a_j R$

Here ${R_{\mathrm{tar}}}' = \mathsf{U}_0 R_{\mathrm{tar}} \mathsf{U}_0^*$ and $L' = \mathsf{U}_0 L \mathsf{U}_0^*$ are obtained from the transported encodings. The mixed quadratic row uses $\mathsf{U}_0 a_i^* a_j \mathsf{U}_0^* = a_i'^* a_j'$. Conjugation by U_0, or multiplication by ℓ, is injective on the specified corner. Thus these are operator equivalences rather than implications inferred from traces. In particular they give the displayed quadratic identities.

For completeness of the admissible class, here is a finite procedure for interior coefficients. Give X degree 1 and X^* degree -1. In a density eigenbasis, a degree-d word has matrix entries only where the output eigenvalue is 2^{-d} times the input eigenvalue. Distinct degrees are disjoint and orthogonal in $\|\cdot\|_{2,h_0}$; split a polynomial into degrees $-2, -1, 0, 1, 2$. Take adjoints of negative degrees. For centralizer coefficients K_L, K_M, K_R:

1. Degree one: $K_L X_i K_R$ lifts to ${K_L}' t a_i K_R$.
2. Degree zero: $K_L X_i K_M X_j^* K_R$ lifts to ${K_L}' t a_i K_M a_j^* t^* {K_R}'$ at f'. A term $K_L X_i^* K_M X_j K_R$ first becomes $K_L a_i^* t^* {K_M}' t a_j K_R$ at f; transport that whole supported centralizer expression to f' through its own derived encoding.
3. Degree two: put $H_{K_L} = t^* {K_L}' t$ at e and transport it to $H'_{K_L} = \mathsf{U}_1 H_{K_L} \mathsf{U}_1^*$ at e'. Then

$$K_L X_i K_M X_j K_R = \ell H'_{K_L} a_i' {K_M}' t a_j K_R.$$

   Indeed $\ell H'_{K_L} = K_L \ell$, by multiplication and $t t^* = f'$.

A derived coefficient with a known rational norm bound K is encoded after division by $\max(1, K)$, and its defining equation is added as a centralizer polynomial. Such a bound follows by multiplying and summing the declared contraction bounds. Only the two displayed kinds of derived coefficients are needed for degree two. Define the base labels and t first, the original transports next, and then the derived coefficients and their transports. Definitions involve no newly transported derived coefficient. Adding their separate refinements constrains the original U, and never extracts a fresh gauge. This is a finite, non-circular assembly for the whole stated class.

A.5 Soundness and the Boolean game

Start with a normal strategy, purify without querying the environment, and reduce to the original marginal supports by facts (N1)–(N8) of Section 4. Perfect matching precedes every use of projectivity. Boolean row tests give the rational identities on these supports. Graphs and triangles then give actual common unitary words. Parameter bindings identify the actual FMS events; they do not replace them by free labels. Apply the two conditional bridge theorems to those bound events. The frame refinements apply first; the block refinements then give (121). Decode labels, impose (123), and apply (124) and the translation procedure. This order uses neither the intended application nor its spectral classification in extracting the scaffold.

All cancellations of a coefficient operator use its dense range on the original local support, as proved in (N1)–(N8). On a mixed-state cut $A \mid BE$ the coefficient map need not be injective. All graph calculations are bounded block identities. The source density is faithful since f commutes with h and h is faithful on the reduced support.

Write $h = h_A$ on $P_A \mathcal{H}_A$. The synchronized corners are projections commuting with h; their sum identities make them orthogonal. Since $hu = 2uh$ and $h\mathsf{U} = 2\mathsf{U}h$, cyclicity of the

trace of a trace-class operator times bounded operators gives

$$\operatorname{Tr}(he) = \operatorname{Tr}(huu^*) = \operatorname{Tr}(u^*hu) = 2\operatorname{Tr}(hf),$$
$$\operatorname{Tr}(hE) = \operatorname{Tr}(h\mathsf{U}\mathsf{U}^*) = \operatorname{Tr}(\mathsf{U}^*h\mathsf{U}) = 2\operatorname{Tr}(hE_{\mathrm{in}}^{\mathrm{act}}).$$

The centralizer magic-corner maps give equal masses to the four inner selected copies and, separately, the four outer selected copies. Thus $\operatorname{Tr}(hE_{\mathrm{in}}^{\mathrm{act}}) = 9\operatorname{Tr}(hf)$ and $1 = \operatorname{Tr}(hE_{\mathrm{in}}^{\mathrm{act}}) + 4\operatorname{Tr}(hE) = 9\operatorname{Tr}(hE_{\mathrm{in}}^{\mathrm{act}})$, proving $\operatorname{Tr}(hf) = 1/81$. The bridge lemma holds also for a zero activity, so no nonvanishing sector is assumed here. It follows that $\mathcal{H}_0 = fP_A\mathcal{H}_A \neq 0$ and $h_0 = 81fhf|_{\mathcal{H}_0}$ is faithful, trace class and trace one.

A supported unitary commutes with its support by the calculation above. On $\mathcal{H}_0$ the four decoding types are therefore respectively U, $(I+U)/2$ with $U^2 = I$, $(2I+U+U^*)/4$, and $(U_+ + U_-)/2$. They are respectively unitary, projection, positive contraction, and contraction, and each commutes with h_0. The supported $a_i = ea_if$ is a contraction; (123) makes t a partial isometry from e onto f', and U_0 is one from f onto f'. Hence $X_i = \mathsf{U}_0^*ta_i$ maps $\mathcal{H}_0$ into itself with norm at most one. Since $h\mathsf{U}_0^* = \mathsf{U}_0^*h/2$ and all other factors centralize h,

$$h_0X_i = 81fh\mathsf{U}_0^*ta_i = \tfrac{81}{2}\mathsf{U}_0^*hta_i = \tfrac{81}{2}\mathsf{U}_0^*ta_ifhf = \tfrac{1}{2}X_ih_0.$$

Transport of decoded labels and their adjoints, followed by the bounded corner multiplications in Section A, now gives every imposed application identity on this actual $\mathcal{H}_0$ in both directions. The degree splitting uses the density eigenbasis on this support, never an inverse of h.

Bob's original marginal support is treated in the same way: matched coarse effects are its projective mirrors, the bridge and refinement lemmas apply on that support, and the same cyclicity gives source mass $1/81$. Mirroring products reverses their order, as in (N4a); no extra mirror equation is imposed. Alice's extracted realization already proves soundness; the symmetric support argument justifies all Bob-side bindings and refinements, including on a purified mixed-state cut with its original Bob marginal.

Expand the final finite centralizer polynomials in their real rational unitary words, cancel adjacent inverse tokens, and close under proper prefixes. Add all graphs, placement triangles, product triangles, inverse triangles, parameter bindings, and the off-diagonal polynomial rows. Apply the Boolean-row construction already proved above. Retain all FMS and refinement questions, pad the two question sets, and sample every pair uniformly with positive rational probability Q^{-2}. Accept every unspecified answer pair. Perfection gives zero probability for each individual rejecting rule, proving soundness.

A.6 One simultaneous completeness model

Let $\mu = \operatorname{diag}(1,2,2,2,2)/9$. On $\mathbb{C}^5_{\mathrm{out}} \otimes \mathbb{C}^5_{\mathrm{in}} \otimes \mathcal{H}_0$ take

$$h_{\mathrm{sc}} = \mu \otimes \mu \otimes h_0. \tag{128}$$

Outer index 0 is the inner activity; inner index 0 is f and inner index 1 is e. The selected outer index is 1. Thus

$$f = |0,0\rangle\langle 0,0| \otimes I, \qquad e = |0,1\rangle\langle 0,1| \otimes I,$$
$$f' = |1,0\rangle\langle 1,0| \otimes I, \qquad e' = |1,1\rangle\langle 1,1| \otimes I,$$
$$\mathsf{U} = |1\rangle\langle 0| \otimes I_5 \otimes I, \qquad t = |1,0\rangle\langle 0,1| \otimes I,$$
$$a_i = |0,1\rangle\langle 0,0| \otimes X_i, \qquad a_i' = |1,1\rangle\langle 1,0| \otimes X_i.$$

The density weights at e and f' are both $2h_0/81$, so t centralizes h_{sc}. Commutation of a_i with h_{sc} is exactly $2h_0X_i = X_ih_0$. All the copies use the identity outer identifications. The

inner and outer FMS effects are the real canonical ones on their respective five-dimensional factors, with the inner copy inactive off outer index 0. Their compressed sign vectors are $|f,f\rangle + (-1)^{xy}|c_x, r_y\rangle$, up to the common positive amplitude and spectator factors. The FMS sign completeness used here is the explicit calculation at the end of the proof of Lemma 4.5.

Every positive density eigenspace is finite-dimensional. For a centralizer contraction T, its polar partial isometry extends on each such eigenspace to a unitary V. Then

$$U_\pm = V\big(|T| \pm i\sqrt{I - |T|^2}\big), \qquad T = (U_+ + U_-)/2. \tag{129}$$

Direct sums give bounded centralizer unitaries. On a prescribed support such as $E_{\mathrm{in}}^{\mathrm{act}}$, do this inside that support and extend by identity outside. For a root positive contraction use $U = 2C - I + 2i\sqrt{C(I-C)}$, and for a projection use $2P - I$. Encode source choices once and transport precisely those choices into E; (120) now holds simultaneously for every label, even a lifted a_i with off-diagonal inner blocks. Encode the global t by (129).

Append $I_3/3$ and all graph, triangle, and parameter-binding projections. For every refinement use actual joint products: its block unitary commutes with its own FMS signs. Distinct labels require no mutual commutation. Finally append $I_{d_{\max}}/d_{\max}$ and the Boolean-row effects. Alice uses these operators and Bob their transposes in a product density eigenbasis. Facts (N1)–(N8) prove every matching, calibration, orthogonality, and row-zero test with this one state and assignment.

Both nested bridges have magic density twice the flag density. For the inner one the flag spectator density is h_0 (up to its scalar outer weight); for the outer one it is $\mu \otimes h_0$. Denote either normalized spectator by ρ, and the total flag mass by λ. The canonical magic directions satisfy $\langle c_x, r_y\rangle = \epsilon_{xy}/\sqrt{2}$, where $\epsilon_{xy} = (-1)^{xy}$. Compressing Ω_h to the two sign supports gives, omitting only the unchanged frame and row factors,

$$\Omega_{xy} = \sqrt{\lambda}(|f,f\rangle + \epsilon_{xy}|c_x, r_y\rangle) \otimes \Omega_\rho.$$

Alice's sign observable exchanges f, c_x and Bob's exchanges f, r_y. Applying the swaps gives $(\mathcal{X}_x \otimes I)\Omega_{xy} = \epsilon_{xy}(I \otimes \mathcal{Y}_y)\Omega_{xy}$. Their spectral sign answers therefore never violate $ab = \epsilon_{xy}$. The other column/row projections give the retained answers and the calibration rules exactly. The inner measurements have inactive effect $I - E_{\mathrm{in}}^{\mathrm{act}}$ off outer index 0; that sector is matched on the two sides. The outer bridge is active everywhere.

Under $E\mathcal{H} \cong \mathbb{C}_{\mathrm{in}}^5 \otimes \mathcal{H}_0$, let $\widehat{U}_k$ be the entire chosen unitary on $E_{\mathrm{in}}^{\mathrm{act}}\mathcal{H}$. Identity outer identifications make

$$V_k^{\mathrm{blk}} = I_{\mathrm{out}} \otimes \widehat{U}_k.$$

Every outer FMS effect, including its noncentralizer sign effects, acts on the outer factor alone and hence commutes with V_k^{blk}, even when $\widehat{U}_k$ has off-diagonal inner blocks. The inner bridge has only the six frame refinements, which commute because the frame is a separate scalar factor. Thus every labelling refinement is the actual positive joint product of commuting projections, sums to the original coarse measurement, and has the required graph marginal. Transposition in the product density eigenbasis preserves these commutators and products on Bob's side. All centralizer matching and orthogonality tests follow from (124), and the appended row factor gives all Boolean row tests simultaneously with the same state. This proves the claimed nested completeness assignment.

The complete marginal is a tensor product of h_0 with a fixed finite-dimensional density η. Thus $\mathrm{Tr}[(\eta \otimes h_0)^\alpha] = \mathrm{Tr}(\eta^\alpha)\,\mathrm{Tr}(h_0^\alpha)$ for every $\alpha > 0$, and its entropy is $S(\eta) + S(h_0)$. Conversely a finite full moment gives a finite moment of the extracted realization: the source corner f commutes with the marginal density h, so $\mathrm{Tr}[(hf)^\alpha] = \mathrm{Tr}(h^\alpha f) \le \mathrm{Tr}\, h^\alpha$, and normalization by the mass $1/81$ gives $\mathrm{Tr}\, h_0^\alpha \le 81^\alpha\, \mathrm{Tr}\, h^\alpha$. Likewise the block decomposition $h = hf \oplus h(I-f)$ gives $S(h_0) \le 81\, S(h)$. These observations give the stated resource preservation.

A.7 The loss constant, including the normalization factors

Keep exact structural hypotheses and choose all derived coefficient encodings from their actual expressions, so their defining rows stay exact. A homogeneous source residual r has a lifted centralizer residual p. For a degree-one or degree-two lowering residual the lift has input corner f, so

$$\|p\|^2_{2,h_{\mathrm{sc}}} = \frac{1}{81}\|r\|^2_{2,h_0}.$$

A degree-zero residual transported to f' instead has factor $2/81$. For negative degrees take the adjoint; if $h_0 r = 2^{-d} r h_0$, cyclicity gives $\|r^*\|^2_{2,h_0} = 2^{-d}\|r\|^2_{2,h_0}$. Therefore each chosen translation has factor $\beta_j/81$, with $\beta_j \le 4$. Splitting degrees preserves the sum of squared norms because their density matrix entries have disjoint ratios.

For the qutrit row Δ_j the two off-diagonal blocks are p_j, p_j^*. Since p_j is in the centralizer, trace cyclicity gives

$$\mathrm{Tr}(h_{\mathrm{old}}\Delta_j^2) = \frac{2\beta_j}{243}\|r_j\|^2_{2,h_0}.$$

Let L_j clear the rational row denominators and let d_j be its padded power-of-four column dimension. Equation (19) then gives, for one oriented row-zero test,

$$Q^{-2}p(\Pi_0^{\mathrm{row}}, \Pi_{d_j}^{\mathrm{ref}}) = \frac{8L_j^2\beta_j}{243 d_j^3 Q^2}\|r_j\|^2_{2,h_0}. \tag{130}$$

Sum these probabilities; overlapping rules can only reduce total loss. If both orientations are retained, include both in the sum (equivalently multiply the maximum constant by two). One may therefore take the maximum displayed coefficient, with the explicit orientation multiplicity, as C_{nest}. It is independent of the Hilbert space, density spectrum, witness period, and computation time. If just one forbidden projection has root probability p, its residual squared norm is p, and (117) reads loss at most $C_{\mathrm{nest}}p$. This proves the completeness estimate.

B The bridge-count and branching variants

Corollary B.1 (Common dyadic ratio and arbitrary finite branching)**.** *For every fixed $n \ge 1$, replace the inner bridge by an n-bridge chain and the outer bridge by an n-bridge chain with first flag the entire inner activity. The same degree-two compiler works with common ratio 2^{-n}. If $d_n = 1 + 4n$ and $Z_n = 1 + 4\sum_{g=1}^{n} 2^g$, the scaffold has d_n^2 scalar corners and source mass Z_n^{-2}. Any finite number b of creation labels may be used. For a full b-ary isometry presentation a normal realization requires $b2^{-n} < 1$ and contains the weighted Fock factor with weights $(1 - b2^{-n})2^{-n|w|}$.*

Proof. The explicit chain corner list is $e_0, m_{g,ab}$, $1 \le g \le n$, with $e_g = m_{g,00}$, activity $e_{g-1} + \sum_{ab} m_{g,ab}$, and cover $e_0 + \sum_{g,ab} m_{g,ab} = I$. The bridge transports U_g have ratio 2; their product has ratio 2^n. Use the block equation (120) at each link, including its inactive identity term when the link activity is not I. Transport the same encoding list through every link. This gives the actual common product transport, by induction on g. The inner chain density is

$$Z_n^{-1}\,\mathrm{diag}(1; 2^{\oplus 4}; 4^{\oplus 4}; \ldots; (2^n)^{\oplus 4}) \otimes h_0.$$

Take two such factors. The inner last selected corner and the outer last selected copy of the source both have density $2^n h_0/Z_n^2$, so the same t construction applies. Replace $\mathsf{U}_0, \mathsf{U}_1$ by restrictions of the common outer chain product. Every multiplication in the proof above remains valid, and the common quadratic factor still exists. The finite mass factors in (130) change accordingly and remain presentation-dependent constants.

The number b affects only the finite list of lifted a_i and its isometry, orthogonality, and range rows. No bridge has to possess b magic corners: its magic square still has four. If $A_i^* A_j = \delta_{ij} I$ and $hA_i = 2^{-n} A_i h$, cyclicity gives total range mass $b2^{-n}$. Pull a positive density eigenvector backwards through nonzero A_i^*; its eigenvalue increases by 2^n and so the process must terminate. The defect and all its word descendants exhaust the support. The word sectors are orthogonal, and their densities are $2^{-n|w|}$ times the root density. Normalization forces $b2^{-n} < 1$ and gives the stated factor. Its α-moment converges precisely when $b2^{-n\alpha} < 1$. This proves both the $n \neq 4$ and $b \neq 4$ variants. □

Exactly n bridges for the bare-tree presentations of Section 9. The bare-tree and degree-one endpoint presentations need only one chain, hence exactly n bridges, rather than the two chains needed for the quadratic extension. Here is its complete specialization. Use the preceding $1+4n$ corner list and cover, bind those corners and the $2n$ positive axis events to the frame, and set $v = \mathsf{U}_n \cdots \mathsf{U}_1$. For each transported encoding use the copies $u_{g,\ell}$ fixing $I - e_g$ and the block equation (25). Refine that link for the six frame projections and every $G_{12}(V_{g,\ell})$. This gives the same calibrated transport of every label through v, with no independent link gauges. Introduce $b_i = e_n b_i e_0$ and extract $A_i = v^* b_i$. The input rows are exactly

$$b_i^* b_j = \delta_{ij} e_0, \qquad \sum_i b_i b_i^* = e_n - P_n, \qquad L_n b_i = b_j R_0,$$

plus the root centralizer polynomial rows. These are respectively the isometry/orthogonality, range, and endpoint translations; multiply by v, v^* to obtain both implications. The cover fixes source mass Z_n^{-1}. The one-chain density displayed in the proof, the same finite-block encodings, and actual joint refinements give simultaneous completeness. The one-oriented loss constant is

$$\max_j \frac{8L_j^2 \beta_j}{3 Z_n d_j^3 Q^2}, \qquad \beta_j = 1 \text{ for source and endpoint rows}, \quad \beta_j = 2^n \text{ for range rows}.$$

For the two-chain quadratic version replace Z_n by Z_n^2 and use $\beta_j \leq 2^{2n}$. Any finite b is handled just by the displayed finite i, j lists. Thus the one-chain n-bridge compiler used in Section 9 and the nested degree-two extension are both specified.

C Software checks

The finite checks are in the directory `verification/` of the repository below, and the Lean development of Appendix F is in `lean/`. The proofs in this paper do not depend on the finite checks. They test finite instances of the identities, constants and constructions; `verification/DETAILS.md` records every check and count.

`https://github.com/Apsiape/perfect-play-arithmetical-hierarchy`

Checked exactly. Exact rational or algebraic arithmetic checks

- the tree, clock and reader identities on ten finite models (depths 0–4, clock orders 2 and 3), where exactly the four top-depth isometry rows fail, as (51) predicts;
- a compiled four-bridge instance with one modular isometry and no transported labels (748 rational rows, 27 991 questions per party), and the worked tableau of Table 4;
- the scaffold constants $4n+1$, $6n+1$ and $\mathfrak{c}_n$, the five- and eight-bridge constants (102) and (103), the code entropy (105) and the Catalan identities (110);
- finite binary trees, reader stand-ins and chain scaffold masses for the type rows, and the Boolean profiles of every reduction in Theorem 14.1;

- the finite selector models of Section 13, and the tree spectra and comb arithmetic of Sections 9–10;
- five literal instances of the nested compiler of Appendix A (three predicates covering both decision branches and two comb programs, with 84 402 to 103 474 rational rows each), every row evaluated in the rational free group ring, and a synthetic presentation with 518 601 rows that exercises every translation route;
- the rotation margin of Appendix D.

Checked in floating point. With explicit tolerances: the truncated Fock model with its Weyl pair and tableau; good primitive phases for every prime below 4000 (Lemma 6.1 is the proof for all primes); the trace-class correspondence on the magic square and the flagged magic square, including the negative control for the marginal D^2 mentioned after Theorem 15.3; the reader bounds and reductions of Section 13; the Fibonacci clock models of orders 3, 5 and 8; randomized group-law and nested transport identities.

Not checked. None of the finite checks proves infinite-dimensional existence or extraction, compactness, the general compiler, the Higman and HNN embedding theorems, the classification of type-III factors, or any hardness reduction. No full reduction game is written out; the questions of the full instance in Section 7 are only counted. The finite group stand-ins do not represent the groups $\mathcal{G}_e$ of Section 11.3.

How to run. In `verification/`, run `python -B run_all.py --output-dir DIR` with `DIR` new or empty; add `--full-counts` to repeat both question counts. The runner regenerates the nested instances, compares counts and hashes with the shipped records, and fails if any check fails. Its `README.md` lists dependencies and maps the program labels to the equations of this paper. The paper itself builds with `python -B build.py` in `paper/`.

Lean. Besides the proof of Theorem A, a set of local Lean results checks selected steps of the written proofs. Table 6 shows representative entries. WHOLE means that the stated local assertion is checked in full, and STEP that one step of the cited result is checked. Theorems B–D are not checked beyond these local lemmas.

Table 6: Scope of selected Lean results; WHOLE refers only to the stated local assertion.

Local result	Paper reference (stable label)	Extent
Good primitive phase, every prime	Lemma 6.1 (`lem:goodphase`)	WHOLE
Printed admissibility of the Theorem A application tables	Definition 4.4 (`def:admissible`); table (48)	WHOLE
Bridge component identity	Equation (12) (`eq:bridgevector`)	WHOLE
Magic-square rigidity and matrix units	Lemma 4.1 (`lem:bridge`)	STEP
Scalar Fock entropy series	Equation (33) (`eq:fockentropy`)	WHOLE
Conditional clock and reader implications	Equations (41), (44)–(47) (`eq:det`, `eq:coordinateclock`, `eq:reader`, `eq:return`, `eq:D`)	STEP

D An exact rotation margin

For $\vartheta = (\sqrt{5}-1)/2$ and $\zeta = e^{2\pi i\vartheta}$, use the lower bound of Section 14.2. Put $\gamma_y = 2^{y+1}|\zeta^y - 1|$. The global minimum of $\log_2 \gamma_y$ is attained at $y = 1$, and

$$2.89 < \log_2 \gamma_1 < 2.9, \qquad \log_2 \gamma_1 = 2.898452049757\ldots. \tag{131}$$

Here is a certificate, including the infinite tail. For $y \geq 3$, the preceding bound gives $\gamma_y \geq 2^{y+3}/(\sqrt{5}\,y + 1/2) > 8$: at $y = 3$ use $\sqrt{5} < 9/4$, and the ratio increases because $2(\sqrt{5}\,y + 1/2) > \sqrt{5}\,(y + 1) + 1/2$. For $y = 2$ the exact certificate described below gives $\gamma_2 > 8$. For $y = 1$ it gives $1 < \gamma_1 < 8$ and $\gamma_{1,-}^{100} > 2^{289}$, $\gamma_{1,+}^{10} < 2^{29}$ for rational bounds $\gamma_{1,-} < \gamma_1 < \gamma_{1,+}$. To construct them, enclose $\sqrt{5}$ by integer square roots at scale 10^{80}, and π by $16\arctan(1/5) - 4\arctan(1/239)$ with 60 alternating terms and the next-term remainder. Evaluate $\sin(\pi\vartheta)$ at the rational endpoints using 40 alternating terms and the next term; sine is decreasing on this interval in $(\pi/2, \pi)$. Round outward to denominator 10^{40} and compare the displayed integer powers. For $y = 2$, use $|\sin(2\pi\vartheta)| = \sin(\pi(2\vartheta - 1))$; its argument is in $(0, \pi/2)$, where sine increases, and the same series gives $16\sin(\pi(2\vartheta - 1)) > 8$. The exact-fraction routine in `verification/types/` carries out these steps. It certifies the $y = 1, 2$ enclosures and the rational base and coefficient inequality of the tail estimate. Together with the induction above these prove the minimum over all positive integers. The weaker positive margin of Section 14.2 is all that contractivity requires.

E Results used from other papers

Tables 7 and 8 list every load-bearing result imported from other papers, with the place where it is used. Results restated with proof here are marked as such.

Table 7: Imported results.

Result	Source	Used in
Perfect completeness of $\mathsf{MIP}^* = \mathsf{RE}$	[34, Theorem 12.6]	Sections 3, 14, 16.7
Commuting perfection is Π_1^0-complete	[57]; [44, Theorem 6.4]	Sections 12, 17
Finite-dimensional value one is Π_2^0-complete	[44, Theorem 6.9]	Table 1
Convergent moment hierarchy	[45]	Sections 3, 17
Flagged magic-square geometry	[39]; square [42, 53]	Section 4.1
Normal bridge, label refinement, rational rows, row loss	[24, Lemmas 4.1, 4.2, Theorem B, Eq. (32)]; restated with proof	Section 4
Catalan game, local relations and two-bridge assembly	[24, Theorem C, Sections 3 and 6, Lemma 3.1]	Sections 14, 18
Standard-form reduction of commuting strategies	[20, Lemma 1.6]; restated	Section 2.4
Commuting calibrated bridge	[20, Lemma 2.2]; proof recalled	Section 4.1.3
Expected-inclusion transfer	[58]; [20, Lemma 3.1]; proof given	Section 12
Tracial mechanism for linear-system games	[20, Proposition 7.4]	Section 14
Cuntz gauge KMS state and its factor	[47]; [31, Theorem 4.9, Example 4.13]; [20, Section 3.1]	Section 11.5
Bounded centralizer perturbation	[52, Lemma 4.5]	Section 4.1.3
Effective Higman embedding	[32]; [7, Theorem 2.2]; [15]	Section 11.3
HNN normal form	[33]	Section 11.3

Table 8: Imported results. (continued)

Result	Source	Used in
Asymptotic ratio set and S-invariant	[12, Section 3.6, Theorem 3.6.1]	Section 12
Absorption argument	[29, Lemma 3.8]	Section 12
Flow of weights classification	[14]	Section 12
Injective factor classification	[13]	Section 14.2
Linear-system games: nonclosure, solution groups, spatial implies finite	[56]; [8, Theorems 4–5]; [9]	Section 14
Assemblage characterization of zero-error codes	[16]	Section 15
Clique saturation, square-root map, Corollaries 7 and 8	[39, Lemma 5, Theorem 6, Corollaries 7–8]	Sections 15, 16.7
Douglas factorization	[17]	Section 15
Synchronous tracial representation	[50, Theorem 5.5]; [35, 44]	Section 17
Limit lemma	Shoenfield (1959) [55]	Section 3.4
Right-c.e.-in-$0'$ reals as computable limsups	[3, Lemma 4]	Section 10

The dependence is one way. This paper uses the entropy-game paper [24] and the type-game paper [20]; neither uses any result proved here.

F Lean verification of Theorem A

Theorem A is proved in Lean 4 with no hypotheses; the proof uses only Lean's three standard axioms. The comparator challenge `lean/Challenge.lean` states Theorem A as follows, with `TheoremAStatement` as defined in `lean/Sigma2/Statements.lean`:

```
theorem theoremA_paper : TheoremAStatement

def TheoremAStatement : Prop :=
  SigmaTwoComplete SpatialPerfect ∧ SigmaTwoComplete EntropyPerfect ∧
  PromiseHard SpatialPerfect PromisedGame ∧ PromiseHard EntropyPerfect PromisedGame
```

Here the languages consist of natural-number codes of finite rational Boolean two-player games, and invalid codes belong to none of them. `SpatialPerfect` is the set of games with a perfect normal spatial strategy (perfection is attained, not merely value one), and `EntropyPerfect` the set of games with such a strategy whose two reduced density operators have finite entropy; Section 2 explains why the one-marginal entropy convention gives the same language. `SigmaTwoComplete` means Σ^0_2 membership and hardness under total computable many-one reductions. `PromiseHard` says that each hardness reduction can be chosen so that every output lies in `PromisedGame`: it has finite-dimensional value one and no perfect finite-dimensional strategy. The same statement is proved as `theoremA` in `lean/Sigma2/Main.lean` and as `theoremA_paper`, through the printed admissibility predicate, in `lean/Sigma2/MainPaper.lean`. The theorem `theoremA_shared_family` in `lean/Sigma2/SharedFamily.lean` proves that a single total computable map is a promised reduction for both languages, which is the family of Theorem A.

F.1 What is checked

The first column uses this paper's numbering. The Lean names `lemma_4_4_strong_of_obligations` and `lemma_3_1_compact_sandwiches` refer to Lemma 4.5 at $n = 4$ and to Lemma 3.2. Several theorems in the table take as an argument a compiler specification, of type `CompilerSpecification.Specification`. The final theorems supply it from the proved obligations `CompilerObligations.obligations_all`, so it is not a hypothesis of any final theorem. The final theorems have no hypotheses. The formal Σ^0_2 upper bounds of

Theorem A use no decision procedure for real arithmetic. Each stage is a conjunction of polynomial equations and non-strict inequalities. Every satisfiable finite stage or finite entropy relaxation has a solution with all coordinates in $[-1, 1]$: amplitudes and probabilities lie in $[0, 1]$, and the formal stages encode each effect as $E = Y^*Y$, where a Gram factor Y of an effect in a POVM has entries of modulus at most one. On this box each constraint is Lipschitz in the maximum norm (each coordinate moved by at most the spacing), with a constant computed from its coefficients. A stage therefore fails exactly when, for some k, at every point of the grid of spacing $1/(k+1)$ some constraint is violated by more than its Lipschitz constant times $1/(k+1)$. Soundness is the Lipschitz estimate and completeness is compactness of the box. Each test is a finite exact rational computation, so the stages are uniformly Π^0_1 (`UpperGridCertificate.boxTruth_iff_cert`, `Upper.finite_stage_piOne`, `Upper.entropy_relaxation_piOne`).

Printed result	Lean name	Checked form
Theorem A	`Sigma2.theoremA`; `Sigma2.theoremA_paper`; `Sigma2.theoremA_shared_family`	Full: both completeness clauses, both promised-hardness clauses, and one computable hardness family for both languages.
Lemma 4.5, $n = 4$	`Sigma2.CompilerContract.lemma_4_4_strong_of_obligations`; `Sigma2.CompilerSpecification.strong_compiler`; `Sigma2.CompilerSpecification.four_bridge_compiler`; `Sigma2.CompilerObligations.obligations_all`	Partial: four-bridge exact compilation, with all seven compiler obligations proved; not general n or commuting hosts.
Theorem 3.1, membership clauses; Lemma 3.2	`Sigma2.Upper.uniform_strata_piOne`; `Sigma2.Upper.spatial_union`; `Sigma2.Upper.spatial_sigmaTwo`; `Sigma2.Upper.lemma_3_1_compact_sandwiches`	Partial: the first two clauses of Theorem 3.1; Lemma 3.2.
Section 3.2 and the relaxations of Proposition 3.3	`Sigma2.UpperGridCertificate.boxTruth_iff_cert`; `Sigma2.UpperGridCertificate.boxTruth_piOne`; `Sigma2.UpperBoxStages.stage_boxTruth`; `Sigma2.UpperBoxStages.entropy_boxTruth`; `Sigma2.Upper.finite_stage_piOne`	The finite predicates are uniformly Π^0_1 by exact rational grid certificates; quantifier elimination is not used.
Proposition 3.3; Corollary 3.4; equation (5)	`Sigma2.UpperEntropyAssembly.perfect_of_sequence`; `Sigma2.Upper.entropy_tail`; `Sigma2.Upper.entropy_stage_slack`; `Sigma2.Upper.entropy_scheduled_iff`; `Sigma2.Upper.entropy_relaxation_piOne`; `Sigma2.Upper.entropy_sigmaTwo`	Partial: perfect-game compactness and effective finite-entropy membership; not prescribed-table compactness, entropy infima or optimal entropy.
Sections 2 and 4, normal toolkit	`Sigma2.Preliminaries.normal_pure_perfect_equivalence`	Partial: the normal/pure perfection equivalence, not all of (N1)–(N8).
Lemma 4.1	`Sigma2.Compiler.lemma_4_1_normal_bridge`; `Sigma2.Compiler.calibrated_bridge_vector`	Partial: the spatial support interface and component identity; not commuting hosts.

Printed result	Lean name	Checked form
Lemma 4.2	`Sigma2.Compiler.lemma_4_2_label_refinement`; `Sigma2.Compiler.joint_refinement_iff`	Coarse-sign identification and the converse commuting refinement.
Lemma 4.3; equation (19)	`Sigma2.Compiler.lemma_4_3_boolean_rows`; `Sigma2.Compiler.row_loss_formula`	Conditional: perfect games, with the `RowsHold` and `NamedBinaryProjections` premises; (19) only as a trace identity under a slot-sum hypothesis, not an event probability.
Definition 4.4	`Sigma2.Presentation`; `Sigma2.Presentation.admissible`; `Sigma2.NormalRealization`; `Sigma2.Compiler.Compiled`; `Sigma2.Compiler.ApproxCompiled`; `Sigma2.PaperAdmissible`; `Sigma2.PaperAdmissibility.application_paper_admissible`; `Sigma2.PaperAdmissibility.saturation_bridge`	`Presentation.admissible`, the domain of the compiler theorems, omits two transport conditions of the printed definition. `PaperAdmissible` is the printed definition; it holds for the Theorem A input tables, and every admissible presentation saturates to a `PaperAdmissible` one with equivalent normal realizations and the same compiled game.
Section 4.7, equations (28)–(29)	`Sigma2.CompilerSpecification.approximate_compilation`; `Sigma2.CompilerApproxBase.approxCompiled_of_base15`	Partial: an existential dimension-independent loss bound under exact structural hypotheses; not the explicit coefficient (29), robust extraction, or preservation of an input witness or its finite dimension.
Section 5.1, equations (30)–(33)	`Sigma2.Tree.fock_decomposition`; `Sigma2.Tree.fock_density`; `Sigma2.Tree.fock_entropy`; `Sigma2.Tree.no_finite_realization`	Rooted Fock structure, weights, entropy and finite-dimensional obstruction.
Lemma 5.1	`Sigma2.Tree.lemma_5_1_definite_computation`; `Sigma2.ReductionSoundBridge.nonreject_of_data`	Classical tableau on a root-clock sector, with its concrete data from the row-extraction modules.

Printed result	Lean name	Checked form
Section 6, equations (40)–(46); Lemma 6.1	`Sigma2.Clock.finite_phase_sector`; `Sigma2.Clock.coordinate_clock`; `Sigma2.Clock.exact_return`; `Sigma2.Clock.lemma_6_1_good_primitive_phase`; `Sigma2.Clock.bounded_reader`	Conditional: `finite_phase_sector`, `coordinate_clock` and `exact_return` take `hP:P.admissible` and `D:Clock.DataR`; the last two also take a root vector, and exact return a primitive-phase eigenvector. `bounded_reader` assumes a primitive phase and its separation bound; Lemma 6.1 supplies a suitable phase and bound for every prime. `Reduction.realization_sound` supplies the clock data for Theorem A.
Section 7; Theorem 7.1	`Sigma2.Reduction.effective_program`; `Sigma2.Reduction.application_admissible`; `Sigma2.Reduction.presentation_codes_computable`; `Sigma2.Reduction.realization_sound`; `Sigma2.Reduction.realization_complete`; `Sigma2.CompiledReduction.theorem_7_1_reduction`	Effective input construction, both perfection equivalences and absence of any finite-dimensional perfect output.
Section 7.1, equations (50)–(54)	`Sigma2.CompiledFockApproximation.normal_approximation`; `Sigma2.CompiledFockApproximation.finite_approximation`; `Sigma2.CompiledValue.four_defect_approximation`; `Sigma2.CompiledValue.value_one`; `Sigma2.CompiledValue.reduction_promised`	Partial: geometric approximation and the promise on every input; the Q, L, d of `four_defect_approximation` are existential, not the actual row counts or denominators.
Other local results	Thin core, `lean/THIN-CORE.md`	Local statements only; see Table 6.

Not formalized: Theorems B–D beyond their thin-core lemmas, including the general semifinite and type classification and the communication classification; Theorem 8.1 beyond its winning-functional identities, and the last two clauses of Theorem 3.1; the general n-bridge and commuting-host compiler; the economical ten-label schedule, its size counts and equation (54) with actual row denominators (the formal compiler transports all typed labels); the reduction of strategies on arbitrary Hilbert spaces to $\ell^2(\mathbb{N})$ by compression; real quantifier elimination, which the formal proof does not need.
Conventions: both local spaces are $\ell^2(\mathbb{N})$, and normal states are countable ensembles of unit vectors in $\ell^2(\mathbb{N} \times \mathbb{N})$, their completed tensor product. A finite-dimensional strategy is a unit vector supported on finitely many basis vectors, with POVMs on $\ell^2(\mathbb{N})$. Every game has at least one question and one answer for each party.

F.2 Trust base

The proofs use Lean 4.35.0 and the Mathlib commit

`56a005855ea17e0b0958f73ec801ae140b0a1679`;

every pin is in `lean/`. Every declaration of the `Sigma2` library, including every theorem in the table, uses no axioms beyond Lean's three standard axioms:

```
'Sigma2.theoremA' depends on axioms: [propext, Classical.choice, Quot.sound]
'Sigma2.theoremA_paper' depends on axioms: [propext, Classical.choice, Quot.sound]
'Sigma2.theoremA_shared_family' depends on axioms: [propext, Classical.choice, Quot.sound]
```

The proof of Theorem A passes two independent kernel checkers, `leanchecker` and `nanoda`. The comparator challenge `lean/Challenge.lean` imports only Mathlib and defines the

statement vocabulary transparently. It states Theorem A as displayed above, the shared hardness family `theoremA_shared_family`, and the printed admissibility definition `paper_admissible_iff`. `lean/Solution.lean` imports the proofs, and `lean/comparator.json` permits only the three standard axioms. From `lean/`, on Linux with bubblewrap and with the dependencies placed as below, one command checks all three: `lake comparator --config comparator.json`. Records of the comparator run and the kernel checks are in `lean/evidence/`.

F.3 Reproduce

With the Lean 4.35.0 toolchain first on the path and the pinned dependency sources placed under `lean/vendor/` as described in `lean/README.md`, run from `lean/` (the first line is PowerShell syntax):

```
$env:LEAN_NUM_THREADS = '1'
lake exe cache get
lake build
lake env lean -j1 -M6144 Audit.lean
```

The first line selects one Lean thread, and `lake exe cache get` fetches the Mathlib build cache. `lake build` rebuilds every project module from source. It audits the axioms of every declaration of the library, including private declarations, and rejects admitted proofs and additional axioms; it audits the proofs of the two final Theorem A theorems and checks that every Lean name in the table exists. The last command prints the axioms of every theorem in the table; `lean/AXIOMS.txt` is the expected output.

References

[1] Scott Aaronson. The Busy Beaver Frontier. Technical Report TR20-115, Electronic Colloquium on Computational Complexity, 2020.

[2] Saugata Basu, Richard Pollack, and Marie-Françoise Roy. *Algorithms in Real Algebraic Geometry*. Springer, second edition, 2006.

[3] Verónica Becher, Yann Bugeaud, and Theodore A. Slaman. The irrationality exponents of computable numbers. *Proceedings of the American Mathematical Society*, 144(4):1509–1521, 2016.

[4] Salman Beigi. Entanglement-assisted zero-error capacity is upper bounded by the Lovász theta function. *Physical Review A*, 82:010303(R), 2010. arXiv:1002.2488v2.

[5] Archishna Bhattacharyya, Arthur Mehta, and Yuming Zhao. On the undecidability of quantum channel capacities. https://arxiv.org/abs/2601.22471v3, 2026. In Proceedings of the 67th Annual IEEE Symposium on Foundations of Computer Science (FOCS 2026), to appear.

[6] Ranyiliu Chen and Yuming Zhao. A non-robust quantum correlation self-test, 2026. arXiv:2609.25117v2.

[7] Maurice Chiodo. On torsion in finitely presented groups. *Groups, Complexity, Cryptology*, 6(1):1–8, 2014. https://doi.org/10.1515/gcc-2014-0001; arXiv:1107.1489v4.

[8] Richard Cleve, Li Liu, and William Slofstra. Perfect commuting-operator strategies for linear system games. *Journal of Mathematical Physics*, 58:012202, 2017. arXiv:1606.02278.

[9] Richard Cleve and Rajat Mittal. Characterization of binary constraint system games. In *Automata, Languages, and Programming (ICALP 2014)*, pages 320–331, 2014. arXiv:1209.2729.

[10] Andrea Coladangelo. The I_{3322} Bell inequality requires infinite dimensions. https://arxiv.org/abs/2609.06038, 2026.

[11] Andrea Coladangelo and Jalex Stark. An inherently infinite-dimensional quantum correlation. *Nature Communications*, 11:3335, 2020. Preprint arXiv:1804.05116.

[12] Alain Connes. Une classification des facteurs de type III. *Annales scientifiques de l'École Normale Supérieure*, 6(2):133–252, 1973.

[13] Alain Connes. Classification of injective factors: Cases II_1, II_∞, III_λ, $\lambda \neq 1$. *Annals of Mathematics*, 104(1):73–115, 1976.

[14] Alain Connes and Masamichi Takesaki. The flow of weights on factors of type III. *Tohoku Mathematical Journal*, 29(4):473–575, 1977. Parts 1 and 2.

[15] William Craig. On axiomatizability within a system. *The Journal of Symbolic Logic*, 18(1):30–32, 1953.

[16] Toby S. Cubitt, Debbie Leung, William Matthews, and Andreas Winter. Improving zero-error classical communication with entanglement. *Physical Review Letters*, 104:230503, 2010. arXiv:0911.5300.

[17] Ronald G. Douglas. On majorization, factorization, and range inclusion of operators on Hilbert space. *Proceedings of the American Mathematical Society*, 17(2):413–415, 1966.

[18] Seth Douglas. The I3322 quantum value is attained spatially but not in finite dimension. Zenodo, https://doi.org/10.5281/zenodo.21782008, 2026. All versions, first released August 2026; arXiv:2609.05555.

[19] Seth Douglas and Nidhal Mghirbi. A finite game that normal hyperfinite strategies approach but never win. https://doi.org/10.5281/zenodo.23816141, 2026. Version 1.0.0, preprint.

[20] Seth Douglas and Nidhal Mghirbi. Finite games that read the type of a von Neumann algebra. https://doi.org/10.5281/zenodo.23931154, 2026. Version 1.1.0, preprint.

[21] Seth Douglas and Nidhal Mghirbi. One-shot zero-error communication with normal entanglement is Σ_2^0-complete. Zenodo, https://doi.org/10.5281/zenodo.23924868, 2026.

[22] Seth Douglas and Nidhal Mghirbi. Perfect play has different arithmetical complexity in semifinite and type III algebras. Zenodo, https://doi.org/10.5281/zenodo.23925091, 2026.

[23] Seth Douglas and Nidhal Mghirbi. Perfect play in a binary-scoring nonlocal game requires infinite dimension but only finite energy. https://doi.org/10.5281/zenodo.23772890, 2026. Version 2.0.0, preprint.

[24] Seth Douglas and Nidhal Mghirbi. Perfect play in a finite nonlocal game requires infinite entanglement entropy. https://doi.org/10.5281/zenodo.23932017, 2026. Version 1.1.0, preprint.

[25] Seth Douglas and Nidhal Mghirbi. Perfect play with spatial quantum strategies is Σ_2^0-complete. Zenodo, https://doi.org/10.5281/zenodo.23910986, 2026.

[26] Tobias Fritz. Curious properties of free hypergraph C^*-algebras. *Journal of Operator Theory*, 83(2):423–445, 2020. arXiv:1808.09220.

[27] Honghao Fu, Carl A. Miller, and William Slofstra. The membership problem for constant-sized quantum correlations is undecidable. *Communications in Mathematical Physics*, 406:96, 2025. arXiv:2101.11087v5.

[28] Minbo Gao, Tianshi Yu, and Lihong Zhi. An explicit counterexample to Tsirelson's problem via a linear system game. https://arxiv.org/abs/2610.10248, 2026.

[29] Uffe Haagerup and Magdalena Musat. Classification of hyperfinite factors up to completely bounded isomorphism of their preduals. *Journal für die reine und angewandte Mathematik*, 630:141–176, 2009. arXiv:0706.3463.

[30] Samuel J. Harris. Universality of graph homomorphism games and the quantum coloring problem. *Annales Henri Poincaré*, 25(10):4321–4356, 2024. arXiv:2305.18116.

[31] Samuel J. Harris. Self-testing of exact entanglement embezzlement. https://arxiv.org/abs/2605.22713v2, 2026.

[32] Graham Higman. Subgroups of finitely presented groups. *Proceedings of the Royal Society of London. Series A*, 262(1311):455–475, 1961.

[33] Graham Higman, Bernhard H. Neumann, and Hanna Neumann. Embedding theorems for groups. *Journal of the London Mathematical Society*, s1-24(4):247–254, 1949.

[34] Zhengfeng Ji, Anand Natarajan, Thomas Vidick, John Wright, and Henry Yuen. MIP* = RE. *Annals of Mathematics*, 2026. Accepted; arXiv:2001.04383v4.

[35] Se-Jin Kim, Vern I. Paulsen, and Christopher Schafhauser. A synchronous game for binary constraint systems. *Journal of Mathematical Physics*, 59:032201, 2018. arXiv:1707.01016.

[36] Artus Krohn-Grimberghe. Certified exact identification of the I3322 quantum value. https://arxiv.org/abs/2609.29693, 2026.

[37] Junqiao Lin. $\mathsf{MIP}^{co} = \mathsf{coRE}$. https://arxiv.org/abs/2510.07162, 2025.

[38] Martino Lupini, Laura Mančinska, Vern I. Paulsen, David E. Roberson, Giannicola Scarpa, Simone Severini, Ivan G. Todorov, and Andreas Winter. Perfect strategies for non-signalling games. *Mathematical Physics, Analysis and Geometry*, 23:7, 2020. arXiv:1804.06151.

[39] Laura Mančinska. The right entanglement for perfect nonlocality and zero-error communication. https://arxiv.org/abs/2610.06512v1, 2026.

[40] Laura Mančinska and David E. Roberson. Quantum homomorphisms. *Journal of Combinatorial Theory, Series B*, 118:228–267, 2016. Preprint "Graph Homomorphisms for Quantum Players", arXiv:1212.1724.

[41] Laura Mančinska, David E. Roberson, and Antonios Varvitsiotis. On deciding the existence of perfect entangled strategies for nonlocal games. *Chicago Journal of Theoretical Computer Science*, 2016(5), 2016. arXiv:1506.07429.

[42] N. David Mermin. Simple unified form for the major no-hidden-variables theorems. *Physical Review Letters*, 65:3373–3376, 1990.

[43] Hamoon Mousavi, Seyed Sajjad Nezhadi, and Henry Yuen. On the complexity of zero gap MIP*. In *47th International Colloquium on Automata, Languages, and Programming (ICALP 2020)*, volume 168 of *LIPIcs*, pages 87:1–87:12, 2020. arXiv:2002.10490.

[44] Hamoon Mousavi, Seyed Sajjad Nezhadi, and Henry Yuen. Nonlocal games, compression theorems, and the arithmetical hierarchy. In *Proceedings of the 54th Annual ACM SIGACT Symposium on Theory of Computing*, pages 1–11, 2022. Full version arXiv:2110.04651v2.

[45] Miguel Navascués, Stefano Pironio, and Antonio Acín. A convergent hierarchy of semidefinite programs characterizing the set of quantum correlations. *New Journal of Physics*, 10:073013, 2008. arXiv:0803.4290.

[46] Seyed Sajjad Nezhadi. *Quantum Games, Graphs, and Gödel*. PhD thesis, University of Maryland, 2025.

[47] Dorte Olesen and Gert K. Pedersen. Some C^*-dynamical systems with a single KMS state. *Mathematica Scandinavica*, 42:111–118, 1978.

[48] Connor Paddock and Simon Schmidt. A commuting operator self-test for exact entanglement embezzlement. https://arxiv.org/abs/2609.38083, 2026.

[49] Károly F. Pál and Tamás Vértesi. Maximal violation of a bipartite three-setting, two-outcome Bell inequality using infinite-dimensional quantum systems. *Physical Review A*, 82:022116, 2010.

[50] Vern I. Paulsen, Simone Severini, Daniel Stahlke, Ivan G. Todorov, and Andreas Winter. Estimating quantum chromatic numbers. *Journal of Functional Analysis*, 270(6):2188–2222, 2016.

[51] Jef Pauwels. The quantum supremum of the I_{3322} Bell inequality is not attained in finite dimension. https://arxiv.org/abs/2608.29734, 2026.

[52] Gert K. Pedersen and Masamichi Takesaki. The Radon–Nikodym theorem for von Neumann algebras. *Acta Mathematica*, 130:53–87, 1973.

[53] Asher Peres. Incompatible results of quantum measurements. *Physics Letters A*, 151:107–108, 1990.

[54] Tibor Radó. On non-computable functions. *Bell System Technical Journal*, 41(3):877–884, 1962.

[55] Joseph R. Shoenfield. On degrees of unsolvability. *Annals of Mathematics*, 69(3):644–653, 1959.

[56] William Slofstra. The set of quantum correlations is not closed. *Forum of Mathematics, Pi*, 7:e1, 2019. arXiv:1703.08618.

[57] William Slofstra. Tsirelson's problem and an embedding theorem for groups arising from non-local games. *Journal of the American Mathematical Society*, 33:1–56, 2020. arXiv:1606.03140.

[58] Masamichi Takesaki. Conditional expectations in von Neumann algebras. *Journal of Functional Analysis*, 9(3):306–321, 1972.

[59] Lauritz van Luijk, Alexander Stottmeister, Reinhard F. Werner, and Henrik Wilming. Embezzlement of entanglement, quantum fields, and the classification of von Neumann algebras. https://arxiv.org/abs/2401.07299v4, 2024.